Applied passive acoustic monitoring of South Africa's most threatened marine mammal, the Indian Ocean humpback dolphin (Sousa plumbea)

by
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Dissertation presented for the degree of
Doctoral of Zoology in the Faculty of Science
at Stellenbosch University

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March 2025

The financial assistance of the National Research Foundation (NRF) towards this research is hereby acknowledged. Opinions expressed and conclusions arrived at, are those of the author and are not necessarily to be attributed to the NRF.

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THESIS ABSTRACT

Monitoring is crucial for conservation, but elusive species, including some species of coastal delphinids, represent particular challenges. Traditional visual surveys can often underestimate metrics such as abundance. For Indian Ocean humpback dolphins (*Sousa plumbea*) in South Africa, near-shore habitats and low encounter rates hinder photographic data collection. Passive acoustic monitoring (PAM) offers a non-invasive alternative for species monitoring. This thesis investigates the whistling behaviour of *S. plumbea* in South Africa, focusing on exploring the use of signature whistles for individual acoustic detection. Using boat-based recordings (2016–2023) from five sites throughout their South African range, it examines whistle variation across different geographic scales (including a comparison of whistle-contour time-frequency parameters across the extremes of the entire geographic range of the species) and the effect of group composition and behaviour on signature whistle production rates, aiming to optimise passive acoustic detection methods. The efficacy of individual acoustic detection using PAM was directly compared to visual detection (Photo ID) through a two-week survey period in Mossel Bay. Following this, an extended acoustic monitoring period along a portion of the south coast subpopulation investigated individual acoustic monitoring across three sites, using five hydrophones over six weeks, providing abundance estimates, as well as density estimates generated using spatially explicit capture-recapture (SECR). Macrogeographic variation in whistle contours was evident between South Africa and India, two populations at the furthest extent of the species' geographic range (Random Forest model correct classification rate; South Africa 84% (35.2% chance) and India 93% (64.8% chance)). Moderate variation within South Africa ranged from 39–90% across study sites, indicating increased acoustic variation with geographic distance. Group composition factors such as group size, group spread, calf presence, and behaviour had little impact on signature whistle production. These findings suggest that these factors are not crucial for optimising the detection of signature whistles in the acoustic monitoring of *S. plumbea*. Comparing monitoring methods, photographic identification was more cost-effective (6,400 ZAR.dolphin^{-1}) than acoustic methods (12,266.40 ZAR.dolphin^{-1}). However, acoustic detection was diurnal, identifying more individuals and signature whistles at night and across the entire survey area. Visual surveys detected dolphins at only two locations, emphasising PAM's effectiveness as an alternative method for detecting *S. plumbea*. Abundance estimates from signature whistle detected along a 120 km stretch of South Africa's south coast revealed limited individual movement, with low abundance (SECR: 23–26; POPAN: 26, 95% CI 23–30) and density (0.000061–0.000439 animals.ha^{-1}). This proof-of-concept study explored mooring configurations and model optimisation, demonstrating that expanding the geographic range of the acoustic array could be more effective for increasing unique recaptures than clustering acoustic receivers in a smaller area. This thesis demonstrates that PAM is a reliable alternative to photographic surveys to study the elusive coastal species *S. plumbea* and produces sufficient data to understand population abundances and movements using individual acoustic signatures. These techniques offer a scalable and reproducible approach to producing population estimates, which can equip wildlife practitioners and scientists with the tools to monitor this elusive species reliably over time and space.

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Sasha Dines, Ponta do Ouro, Mozambique February 2020

‘Herme’, the first humpback dolphin whistle I recorded underwater

This thesis is dedicated to my Grandad Paul,

who first inspired my love for all things aquatic.

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ACKNOWLEDGEMENTS

This work would not have been possible without the unwavering support of my supervisors, who cheered me on until the very last moment.

John, thank you for keeping me on track. You've literally written the book on 'How to write a PhD'. Your words of advice have been invaluable. Thank you for having the patience to spend hours pouring over paragraph structure with me. Simon, the voice of reason, thank you for your constant encouragement and wicked sense of humour throughout some very long office days. And Tess – I couldn't have picked a better adventure buddy. You've shown me science can be done anywhere, anytime, including in a coffee shop while juggling a newborn. It's been a very busy few years, and in that time, you've had a baby, and I managed to birth this PhD! Two great feats. Thank you for being a friend and a mentor.

I want to take a moment for my favourite Brazilian. I don't think any of us could have imagined the events of this last year to have happened. Gui will forever be a huge part of my journey over the last few years. His constant enthusiasm and positivity, mixed with endless thought-provoking questions, pushed me to be a better scientist throughout this process.

Ultimately, this PhD would not have been possible without the support and generosity of collaborators across southern Africa. Over the last four years, I've had the privilege of travelling and working with so many inspiring and supportive people. But to name a few...

The Sea Search team, particularly Rachel and Cat, were always there with words of encouragement, assistance, and advice. I am very grateful for your support and friendship throughout this journey.

And to Jack, who stepped up to be the best housemate one needs when in the throes of PhD writing. Thank you for the endless amounts of coffee, snacks, and chocolate in the last few weeks. This PhD really wouldn't have happened without all our very late-night dolphin discussions, so thank you. Now, it's time to hand over those coattails. Your turn next…

The whole SouSA consortium, for your support throughout the project. To Els and the Whale Unit team – for letting me join your boat back in 2019 and showing me the cetacean ropes. Gwen, sorry for sending you after the wrong mooring for several days, but your support and willingness to help saved a large portion of my data. Shanan, for being the most amazing campsite hostess and showing me the Richards Bay dolphins. As well as Ocean Odyssey Knysna and Offshore Adventures Plettenberg Bay, who willingly provided valuable boat time.

The Oceans Research team. Enrico, thank you for always believing in me from the very start. I'm sorry I switched to dolphins, but I hope you can see now why I fell in love with them. Justin, there's no skipper I would trust more to get up close and personal with these pesky dolphins in the breakers. We've spent hundreds of hours on the ocean together, and I'm looking forward to plenty more. To all the field specialists and students who have worked on the project over the years, and particularly to Lacey, you've helped to put the whistles of the Mossel Bay humpback dolphins on the map.

Most of all, thank you to my mum and dad. You've always pushed me to follow something I love, even if that meant living on the other side of the world, and for that, I'm forever grateful. You are my biggest supporters and advocates. I hope you enjoy reading.

Finally, thank you to my examiners for their comments and suggestions throughout this thesis process and to my funders: the South African National Research Foundation, the PADI Foundation, IUCN Save our Species and Sea the Future.

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Table 3.4. Encounter information across the 22 *Sousa plumbea* encounters showing whistle counts across each encounter. Encounter number refers to the numbered encounters on that day, as some days had multiple encounters recorded. Five study sites are included (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay and Richards Bay) with the date of recording (2016-2023). For each encounter, the total number of whistles recorded, the number of high signal-to-noise ratio whistles (rated 2 or 3), the total number of signature whistles (SWs) detected, and the total number of distinct signature whistle types (SWT), representing the number of individuals, are presented. A total of 20 unique SWTs were identified across all encounters. Encounter 20191016_01 (Mossel Bay) was detected as an outlier based on whistle production and, therefore, removed from further analysis, as indicated in italics.

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Table 3.5. Generalised linear mixed models for signature whistle types (SWT), signature whistles (SW) and inter signature whistle interval (ISWI) in *Sousa plumbea* across 21 encounters from five South African study sites (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay, Richards Bay, 2016-2023). The table includes the best fit model and individual class models that include each covariate modelled individually. The table also provides metrics to provide information about the goodness of fit and complexity of the model including Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), Log-Likelihood (LogLik), Deviance and Residual Degrees of Freedom (df residual). P values <0.01 are highlighted to show significant effects.

Table 4.1. Hydrophone deployment locations for *Sousa plumbea* in Mossel Bay, South Africa, from 19th March – 5th April 2021. Includes hydrophone model (SoundTrap300/500), start and end times of recordings and recording settings (Hz).

Table 4.2. Confusion matrix for the regionally trained species classifier using the whistle classifier module in PAMGuard for both *Sousa plumbea* and *Tursiops aduncus* based on test data throughout South Africa. The percentage of correct classification in the training model for *Sousa plumbea* is highlighted in bold.

Table 4.3. Summary of *Sousa plumbea* detections during the 18-day deployment period in Mossel Bay, South Africa, from March 19th to April 5th 2021. The table provides summaries of encounter durations and the time taken to detect all individuals in each encounter using both acoustic and photo ID methods, along with overall individual detection rates (total number of individuals/encounter duration in minutes). The individual detection rate was calculated by dividing the total number of individuals by the encounter duration. For each metric, the mean (standard deviation, SD), minimum, and maximum values are presented.

Table 4.4. Fieldwork and operational costs and total project hours incurred across both acoustic and boat-based recording methods for *Sousa plumbea* during during the 18-day deployment period in Mossel Bay, South Africa, from March 19th to April 5th 2021. Costs per individual *Sousa plumbea* were calculated as operational costs (total time X researcher time) plus total fieldwork costs (minus equipment) and based on a total of 10 signature whistle types (SWTs) and 16 Photo ID individuals detected during this survey period using either method.

Table 4.5. SWOT (strengths, weaknesses, obstacles and threats) analysis of both acoustic and photo ID methodologies to detect *Sousa plumbea* during the 18-day deployment period in Mossel Bay, South Africa, from March 19th to April 5th 2021.

Table 5.1. Deployment summaries for the five moorings deployed throughout the central portion of the south coast of South Africa to acoustically detect *Sousa plumbea*. Mooring codes follow from the nomenclature in Chapter 4. Therefore, Mossel Bay moorings are M1 and M3, and Buffelsbaai continues from M7. Recording start dates were all scheduled at the same time, and recording ends varied based on battery and retrieval dates.

Table 5.2. Summary table of *Sousa plumbea* whistles detected across the acoustic array, with a total number of signature whistles (SW) and signature whistle types (SWTs, individuals) detected at each mooring. The proportion of whistles analysed (using signal-to-noise ratio of 2 or 3) was calculated from total detections at each mooring and presented as a percentage.

Table 5.3. Estimated population size ($\widehat{N}$) of *Sousa plumbea* for both closed (Huggins) models for each bay individually and combined estimates as well as open (POPAN Jolly-Seber) model for the combined population estimate for *Sousa plumbea* throughout the south coast subpopulation range. Number of parameters (npar), standard error (SE), lower and upper limits of the 95% confidence interval (Lower CI and Upper CI).

Table 5.4. Full model outputs across different model combinations to estimate density for *Sousa plumbea* throughout the south coast survey area, South Africa. Model combinations are highlighted in grey. Combinations include: 1) Capture history truncations (Full detections and

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capped at 200 and 100 detections); 2) Mask objects (High, low and default resolutions); and Trap configurations (Full moorings - *CaptHit.Full*, removed two least active receivers – *CaptHist.Rem*, Combined the two moorings from the same bay – *CaptHist.Comb*). Model parameters and estimates are reported, as well as the detection function used and Akike Information Criterion (AIC) scores. Bolded values are best estimates of density (D) and sigma (σ).

Table 5.5. Predicted estimates of density and abundance for *Sousa plumbea* throughout the south coast survey area are shown for each of the model combinations. Estimates were calculated from the *estimated density * study area*. Sigma is an estimate of the average distance an individual moves between capture occasions and over the study period. All values were transformed from the log scale used by the model to real estimates.

Table 5.6. Summary of study effort and capture-recapture results obtained from previous studies on *Sousa plumbea* in southern South Africa: Mossel Bay (James et al., 2015; 2011-2013), Goukamma (Conry, 2017; 2014-2015), Plettenberg Bay (Jobson, 2006; 2002-2003), Plettenberg Bay (Greenwood, 2013; 2012-2013) and this study: Mossel Bay, Buffelsbaai, Plettenberg Bay, combined south coast (five weeks, 2022). Values not modelled are reported as na.

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1 GENERAL INTRODUCTION

ACOUSTIC COMMUNICATION IN DELPHINIDS

Communication is the transfer of information from signaller to receiver and can occur through visual, olfactory, chemical, acoustic senses, or, at very short distances, through touch (Labra et al., 1999; Marler, 1961). The complexity of a species' communication system is often linked to its level of social organisation, and understanding how species communicate can provide information on the species' relationship both to its conspecifics and environment (Klump, 2020). Signals can contain information on innate or fixed aspects such as size, age, and sex (Seyfarth et al., 2010) or variable aspects such as health, arousal, and rank (Bergman & Sheehan, 2013; Briefer, 2020; Taylor et al., 2010). The ability to recognise other species, groups or even specific individuals within the group can be indicative of social complexity and sophisticated communication processes within a species (Yorzinski, 2017). Animals can prioritise a certain type of communication system based on circumstance, such as proximity to conspecifics, or environmental conditions, such as aquatic or high-density habitats (Ey & Fischer, 2009). In such environments, acoustic signalling is beneficial due to the flexibility in the rapid transmission of signals and effective propagation over varied distances (Ladich & Winkler, 2017).

Delphinids, a family of over 35 species, have a range of social structures from large pods of 1000s of individuals with continuous fission-fusion dynamics (Richards et al., 2013) to strong matrilineal groups (Deecke et al., 2010) and are present throughout the global oceans (Best, 2007) leading to the development of a sophisticated acoustic repertoire (Janik, 2009; Luís et al., 2021). Delphinids can produce two main types of sound: clicks (including burst pulse sounds) and tonal frequency-modulated sounds (including low-frequency narrow-band sounds, LFNs, and whistles) (Jones et al., 2020; Luís et al., 2021; Popper, 1980). Clicks are short signals that reach into the ultrasonic frequency range and occur in trains with an inter-click interval of more than 10 ms (Au et al., 1974; Jones et al., 2020). They are most commonly used in echolocation to

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perceive the environment, especially during foraging (Herman & Tavolga, 1980). Burst pulse signals are rapid click trains, typically defined as having an inter-click interval of less than 10 ms (Luís et al., 2021). The frequency of click trains and burst pulses fluctuates depending on function and behavioural context. Burst pulses appear to be involved in communication interactions (Lammers 2003), as the inter-click interval exceeds the round-trip propagation time required to receive individual click echoes (Ivanov, 2004). In a review of sounds produced by bottlenose dolphins (*Tursiops truncatus*), Jones et al., (2020) describe low-frequency narrow band (LFN) sounds as having a fundamental frequency of less than 2 kHz (0.26 – 1.28 kHz) and durations less than 0.41 s (0.01 – 0.41 seconds). LFNs are rarely reported within the literature, and it is difficult to determine how widely used these sounds are (Jones et al., 2020).

Whistles are narrowband frequency-modulated sounds and one of the more commonly studied call types due to their dominant use by many species and ease of measurement (Herzing, 2014; Janik, 2009). Most whistling delphinid species focus their energy below 20 kHz (Popper, 1980), yet the maximum recorded fundamental frequencies can reach over 40 kHz in *T. truncatus* (41.80 kHz: Hiley et al., 2017). Whistles function as a communicative signal, and they are commonly produced under socialising contexts (La Manna et al., 2019; Marley et al., 2017; Papale et al., 2017). Whistles can convey arousal states and behaviour (Quick & Janik, 2008) and also function to convey identity (Sayigh et al., 1999) within the frequency modulation of the tonal sound (Kershenbaum et al., 2013; Sayigh et al., 1999). The fission-fusion group dynamics of delphinid species, such as common (*Delphinus* spp.) or bottlenose dolphins (*Tursiops* spp.), lead to a greater level of individual distinctiveness within their whistle repertoire to account for the fluidity of these associations and ensure the maintenance of complex social relationships and behaviour (Fearey et al., 2019; Reiss et al., 1997).

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SIGNATURE WHISTLES IN DELPHINIDS

Individual recognition in animals occurs through multiple senses in different species. However, in the marine environment, communication through acoustic channels is the most powerful and widespread (Ladich & Winkler, 2017). Unlike voice cues, which can affect all animal calls, bottlenose dolphins can encode individual identity information within the frequency modulation pattern of whistles (Janik & Slater, 1997; Sayigh et al., 2017). These individually unique whistles are termed 'signature whistles' (Janik, 2009). Caldwell and Caldwell (1965) first identified these individually distinctive frequency contours in bottlenose dolphins and hypothesised that these were used to transmit identity information. A signature whistle has since been defined more formally as a learned, individually distinctive whistle type in a dolphin's repertoire that broadcasts the identity of the whistle owner as a contact call (Sayigh et al., 2007). Individually distinct calls are extremely rare in the animal kingdom (Boughman & Moss, 2006). Signature whistles develop in the first year of a calf's life through vocal learning and the auditory experience of the environment and conspecifics therein (Sayigh et al., 1990). MacFarlane (2016) identified three potential functions for signature whistles: maintaining contact, regaining lost contact, and advertising identity. As signature whistles are developed through vocal learning, this ability means that whistle matching is also possible (Janik, 2000). Individuals are known to whistle match and copy the signature whistles of those with whom they form a strong social bond (King et al., 2014; Probert et al., 2021). However, this copying occurs in small numbers and usually when the copied individual is present (King et al., 2013), with the copied whistles identifiable through parameter variations within the copy (King et al., 2013).

Research on signature whistles has been conducted for over 60 years (Janik & Sayigh, 2013), mainly in *T. truncatus*, using either captive (Janik & Slater, 1998) or temporarily restrained wild individuals (Esch et al., 2009; Sayigh et al., 1990, 1999). As signature whistles act predominantly as cohesion calls, they can constitute up to 100% of whistles produced during these separation events (when isolated or restrained from conspecifics) (Janik & Slater, 1998). However, this value

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is considerably reduced in recordings from non-restrained wild populations (38% - 70% (Cook et al., 2004). Functioning primarily as contact calls in these freely interacting contexts, signature whistles are produced at higher rates when animals meet at sea (Quick & Janik, 2008), mother and calves reunite (Smolker et al., 1993) and when engaged in certain behaviours, such as socialising, that require a form of individual recognition (Sayigh et al., 1999). In bottlenose dolphins, these contact calls were identified as bouts of stereotyped whistles emitted within 1 to 10 seconds of each other (Janik, 2013). Although most widely documented in bottlenose dolphins, signature whistles have been reported in other odontocete species, such as narwhals (*Monodon monoceros*) (Shapiro, 2006), short-finned pilot whales (*Globicephala macrorhynchus*) (Sayigh et al., 2013), melon-headed whales (*Peponocephala electra)* (Kaplan et al., 2014) and common dolphins (*Delphinus delphis*) (Fearey et al., 2019). Some studies have also documented likely signature whistle use within the genus *Sousa* using opportunistic recording events (Cheng et al., 2017; Van Parijs & Corkeron, 2001a).

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PRACTICAL USES OF SIGNATURE WHISTLES FOR RESEARCH

Research approaches using individual identification (capture-recapture) can provide long-term metrics, from the population level down to the individual, such as abundance, habitat range or movement, social networks, and group size (Hammond, 1986; Karlsson et al., 2005). Individual identification from naturally occurring markings is useful when more invasive marking techniques are challenging to perform, detect, or are ethically inappropriate (Terry et al., 2005). Furthermore, individually distinctive acoustic signals are less invasive and have been suggested as a suitable alternative to active marking or photographic identification (Janik et al., 2013). Signature whistles are naturally occurring, individually distinctive and stable over time (up to 43 years, Sayigh et al., 2022) and overcome many of the biases inherent in using artificial marks or marks from scarring or injuries (James et al., 2022). Although previously suggested by Janik et al., (2013), Longden et al., (2020) first identified signature whistles as a potentially suitable technique to generate abundance estimates in a Namibian population of bottlenose dolphins (*Tursiops truncatus*) by using signature whistles in a capture-recapture framework where the results compared favourably with established photographic methods (Friday et al., 2008; Urian et al., 2015) . Further work has also investigated the role of recording duty cycles and fine-scale hydrophone placement on the use of signature whistles in a capture-recapture framework (Fearey et al., 2022). Forming part of the SWORD project (Signature Whistles for Occurrence, Recapture and Density, Fearey et al., 2021), this proposed methodology, using signature whistles in a capture-recapture framework for acoustic-based monitoring, has yet to be applied outside the genus *Tursiops*.

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ACOUSTIC BEHAVIOUR OF *SOUSA* SPP

There are around 50 studies on the acoustic behaviour across all four species in the genus *Sousa*, with *Sousa chinensis* the most widely published, around Malaysia and China (Hoffman et al., 2015; Sims et al., 2012; Wang et al., 2013), then *Sousa sahulensis* in Australia (de Freitas et al., 2015; Hoffman et al., 2017; Schultz & Corkeron, 1994a; Van Parijs & Corkeron, 2001b), as well as a single publication on the acoustic behaviour of the IUCN Critically Endangered *Sousa teuszii* in western Africa (Weir, 2010) (Table 1.1). *S. chinensis* has the most widely defined whistle repertoire with maximum reported whistle frequencies of 33 kHz (Wang et al., 2013). Whistle measurements vary across studies, with some studies presenting limited information (Sims et al., 2012; Zbinden et al., 1977) and some are limited by the bandwidth of recording equipment (Schultz & Corkeron, 1994a; Van Parijs & Corkeron, 2001b; Weir, 2010).

To date, three published studies have examined the acoustic characteristics of *Sousa plumbea* in Pakistan, India and Zanzibar. Zbinden et al., (1977) published the first acoustic study on *S. chinensis* (now *S. plumbea*) from the Indus River delta region, Pakistan. The study was limited by equipment frequency range but demonstrated a long whistle duration of up to 2.5 seconds and call combinations of burst pulse trains and whistle production (screams). Nearly four decades later, Bopardikar et al., (2018) published a full description of the whistle repertoire (single element tonal signals) for *S. plumbea* along the Maharashtra coast, India, with a frequency range of up to 33 kHz and durations of up to 1.6 seconds. The first click analysis for the genus was published by Yang et al., (2020), who showed a high rate of species classification of *S. plumbea* compared to sympatric *T. aduncus*. Across all studies, caution should be taken in extrapolating findings to other *S. plumbea* populations, as interspecific and intraspecific differences in time frequency features across regions have been documented in *Sousa* spp. (Hoffman et al., 2015; Wang et al., 2013).

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Table 1.1. Summary of_literature review of existing acoustic repertoire data across the genus *Sousa.* Key publications covering all four call types (whistles, clicks, burst pulses and low-frequency narrowband calls (LFNs)). Whistle repertoire literature review for the *Sousa* genus is presented in Table 2.11.

Reference	**Species**	**Location**	**Whistles**	**Broadband - clicks**	**Burst Pulse**	**LFNs**	**Comments**
Goold & Jefferson, 2005	*S. chinensis*	Hong Kong		Delphinid in waveshape - spectral energy extending up to 30 - >200 kHz total pulse duration is a few tens of microseconds			Opportunistic recordings from a group of 10 individuals. The orientation of animals to hydrophone is unknown, and reverberation is present in the recordings.
Seekings et al., 2010	*S. chinensis*	Singapore	Duration <1s and common centre frequency 3.8 kHz	Click trains are recorded as vocalisations towards a target	Recorded from an adult male (Barks) but outside the scope of the paper		Captive population.
Xu et al., 2012	*S. chinensis*	Leizhou Bay, China	Mean Values Duration 0.16 s Mean: SF 4.26 kHz EF 4.86 kHz Min freq 3.9 kHz Max Freq 5.31 kHz				
Sims et al., 2012	*S. chinensis*	Eest Hong Kong	Max fundamental freq 24kHz	Frequencies extended outside of the flat frequency response range and, therefore, were excluded	4.1 kHz -24.9 kHz (Barks)	Absent – likely masking due to ambient noise	Limited data. Whistles more similar to Zbinden et al. 1977 than Van Parijs and Corkeron (2001a).

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				from frequency analysis			
Wang et al., 2013	*S. chinensis*	Southern China	Duration (mean ± SD: 370.19 ± 285.61 ms; range: 29–2923 ms broad frequency range (fundamental contour ranged from 0.52 - 33 kHz				Similar to Atlantic *S.teuszii.*
Fang et al., 2017	*S. chinensis*	Sanniang Bay, China		The mean peak frequency was 109.0 kHz, with a 3-dB bandwidth of 50.3 kHz and a 95% energy duration of 22 µs.			Full bandwidth and on-axis recordings and include comparison to wild and captive individuals and other species.
Hoffman et al., 2015	*S. chinensis*	Malaysia	Malaysian humpback dolphins covered frequencies from 1.2 to 27.1 kHz with durations from 0.010 – 1.575 s				First acoustic recordings extending this far south in SE Asia. Significant differences were found between the two Malaysian populations in whistle duration, frequency modulation, and all frequency variables except for minimum frequency, which is likely under

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							morphological constraints.
Dong et al., 2019	*S. chinensis*	Southern, China	Range 1.8 - 21.7 kHz Mean duration 419 ms				This study and Fang et al. (2015) found significant differences in frequency parameters in Sanniang Bay, China.
Zbinden et al., 1977	*S. plumbea*	Indus Delta, Pakistan	3 - 20 kHz 0.04 - 1.05 s, both single and in series	>500Hz 100-150µsec 1.5 -3.5 periods between pulse. Energy around 20 - 25kHz (Creaks and screams)			Limited data and limitations to recording equipment. no detailed time-frequency features reported.
Bopardikar et al., 2018	*S. plumbea*	Sindhudurg coast, Maharashtra, India	Freq Range 2.3 - 33 kHz Duration Range 0.01-1.6 s				First detailed reporting of time-frequency features in *S. plumbea.*
Yang et al., 2020	*S. plumbea*	Zanzibar, East Africa		Peak Freq 91 ± 22 kHz, centroid freq 87 ± 12 kHz 16 s -10 dB duration			All click parameters measured were higher in *S. plumbea* than in sympatric *T. aduncus*
Van Parijs & Corkeron, 2001b	*S. chinensis (now S. sahulensis)*	Eastern Australia	0.9-22 kHz	8 kHz - > 22 kHz Freq	Barks; 0.6 kHz - >22 kHz 0.1-8 s Quacks: 0.6-3.7 kHz, 0.08 – 2.7 s	0.5-2.6 kHz 0.006-2 s	Limited by recording equipment.
Van Parijs & Corkeron, 2001a	*S. chinensis (now S. sahulensis)*	Eastern Australia	Evidence for SW Repeated looped whistle (up to 4 loops) Mean 0.78 s	4 to 16 s 2 to 10 kHz Min Freq 20.8 to >22 kHz Max Freq			Opportunistic recordings from a live stranded animal (recorded in a holding tank).

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			SF 6.26 kHz EF 7.25 kHz Min freq 5.59 kHz Max Freq 14.46 kHz				Limited by recording equipment.
Schultz & Corkeron, 1994	*S. sahulensis*	Eastern Australia	Freq Range 1.2-16 kHz Duration range 0.04 - 0.36 s				Limited by recording equipment and available data (667 whistles).
Berg Soto et al., 2014	*S. sahulensis*	Eastern Australia	(n = 742) Duration: 0.3 s - 4.6 s. mean min and max frequency: 6.35±2 - 12.3±4 kHz	Mean Min Freq 4±3 kHz Max Freq >22 kHz Duration 2.79±2.66 s	Mean duration 0.86±0.84 s. Mean Min Freq 5.8±3 kHz		Classification of Australian Snubfin dolphins Limited click train analysis due to limitations to recording equipment.
de Freitas et al., 2015	*S. sahulensis*	Western Australia		Mean source level 199 ± 3dB mean centroid freq 106 ± 11 kHz ICI 79 ± 33 ms			Similar to sympatric *T. aduncus* - click parameters independent of sympatric delphinids – likely driven by body size.
Weir, 2010	*S. teuszii*	Angola, west Africa	4.8-8.2 kHz 85% single inflection point				Only published acoustic data on this species. Limited time-frequency features reported.

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THE STUDY SPECIES: *SOUSA PLUMBEA*

Sousa plumbea was recognised as a distinct species from *S. chinensis* in 2014 based on molecular analysis, skeletal morphology, external morphology and colouration (Jefferson & Rosenbaum, 2014; Mendez et al., 2013). The range *of S. plumbea* extends from their most southwestern point in False Bay, South Africa, along the southern and eastern African coastlines to as far as the west coast of India (Plön et al., 2015). In the southern African subregion, *S. plumbea* has been recorded from South Africa, Mozambique, Tanzania (including Zanzibar), Kenya, the Comoro Islands and Madagascar (Best, 2007).

Globally, *S. plumbea* is not highly abundant anywhere in its range, with global abundance likely in the low ten thousands (Braulik et al., 2015). Previous research in East Africa on *S. plumbea* has shown small, declining populations from several localised studies but a region wide assessment has not been done. Research in Maputo Bay and Bazaruto Bay, southern Mozambique, indicated low densities of 0.47 dolphins per km^2 (Peddemors, 1999) and a population in the Maputo Bay area of ~105 individuals in the late 1990s (Guissamulo & Cockcroft, 2005). In Zanzibar, further along the east coast of Africa, there has been a 63% reduction in the population size since 2002, with the last population abundance estimate of 19 (95% CI 14-25) in the Menai Bay conservation area in 2015 (Sharpe & Berggren, 2019). Limited information on population density is available across the rest of their African range, especially at a broader scale (Braulik et al., 2015). In South Africa, group size in *S. plumbea* is typically small, as previous studies have only seen a maximum of 30 individuals in a group, but groups are more commonly less than 10 (Plön et al., 2015). Several population estimates have been made throughout their South Africa range, mostly in localised areas such as in one or two bays, and from ~ 20 to 96 individuals (Table 1.2). A collection of photo identification catalogues from previous studies collected around South Africa from 2000 to 2016 identified 247 unique individuals along the whole South African distribution, suggesting a total South African population of fewer than 500 animals (Vermeulen et al., 2018), substantially

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lower than the 1000 estimated as the total country population by Karczmarski (1996) from photo ID data available in the late 1990s.

The general life history of *S. plumbea* in South Africa reflects that of a medium-sized coastal delphinid that occupies a very narrow habitat niche, typically less than 3 km from shore and in waters shallower than 25 m. Populations occur around a variety of coastal habitats, including rocky reefs, coastal lagoons, and shallow, protected bays (Braulik et al., 2015). Diet records for *S. plumbea* have been mainly restricted to specimens accidentally caught in shark nets along the Kwazulu-Natal (KZN) east coast, with little data available for the south coast region. The few records on their diet show them feeding mainly on estuarine and demersal fish (Peddemors, 1999). Their coastal habitat influences the feeding methods they adopt. This habitat means they are often associated with large estuarine systems and harbours. Although considered a marine dolphin species, there are a few reports of dolphins entering river systems, travelling up to 1 km within the Swartkops and Bushmans rivers in the Eastern Cape and into the Knysna estuary in the Western Cape (Plön et al., 2015). These sightings are often associated with brackish effluent in the lower estuarine environment during high tides. However, in the Breede River, Western Cape, there have been confirmed sightings and reports of *S. plumbea* as far as 20 km upriver (Sea Search, unpublished data).

MOVEMENT OF *SOUSA PLUMBEA*

Sousa plumbea are the most threatened marine mammal species in South Africa (Plön et al., 2015), making it essential for researchers to conduct comprehensive and effective long-term conservation assessments. Understanding the degree of dispersal and individual movement can highlight ecologically sensitive areas that require protection (Gormley et al., 2012). Previous photographic capture-recapture studies that assessed residency patterns or movement have largely focused on discrete regional areas around South Africa, which have been inadequate to

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provide a comprehensive assessment of the entire population (Bouveroux et al., 2018; James et al., 2015; Vermeulen et al., 2018) (Table 1.2). In all these assessments, connectivity between locations was poorly understood based on the limited data available (Durham, 1994; Jobson, 2006; Karczmarski et al., 1999; Keith et al., 2013; Saayman & Tayler, 1973).

In South Africa, *S. plumbea* is thought to be split into two distinct populations, one off the south coast and one off the east coast (Figure 1.1). The connectivity between these two regions is minimal, as indicated by low encounter rates (Durham, 1994; Keith et al., 2002), few strandings (Findlay et al., 1992), limited sightings during aerial surveys (Cockcroft et al., 1989), and infrequent captures in shark nets along southern KZN and the Transkei region (Atkins et al., 2013). This limited connectivity is further supported by the absence of matches in photographic identification catalogues between Algoa Bay and KZN, regions approximately 1050 km apart, reinforcing the existence of two geographically distinct populations (Karczmarski et al., 1999; Vermeulen et al., 2018). The south coast population is less well studied, with most information available from studies within specific bays, notably Algoa Bay, which was well studied in the early 1990s and again 16 years later, showing significant decreases in sightings rates, group sizes, the number of overall identified animals and an increase in the number of solitary animals (Bouveroux et al., 2018; Karczmarski et al., 1999; Karczmarski & Cockcroft, 1998; Koper et al., 2016). A series of student projects in Plettenberg Bay described similar trends throughout the region (Conry, 2017; Greenwood, 2013; Jobson, 2006) as well as Bouveroux et al.,'s (2019) study on social networks and some early shore-based observations throughout the southeastern coastline (Saayman & Tayler, 1973; Saayman, 1972). West of Plettenberg Bay, only a single published estimate exists on population size in Mossel Bay (James et al., 2015) and a general description of sighting rates and seasonality from whale-watching data (Vinding, 2016).

Only one published paper has studied the movement of *S. plumbea* individuals at a national scale along the South African coastline using available photo ID data from 2000-2016 (Vermeulen et

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al., 2018). The study found the average number of re-sightings of individuals across this period was six for the whole distribution, decreasing to 3.3 for individuals occurring in the south coast population only. Median range estimates for individuals were 120 km, with an overall range of movement of 30 km to 500 km, but most movements were less than 200 km suggesting that this is a typical home range or along-shore range. The study highlighted general 'hotspots' for the species along the coastline. However, it indicated that, due to their reported range distance, the protection of existing discrete marine protected areas (MPAs) may not be sufficient to protect this species.

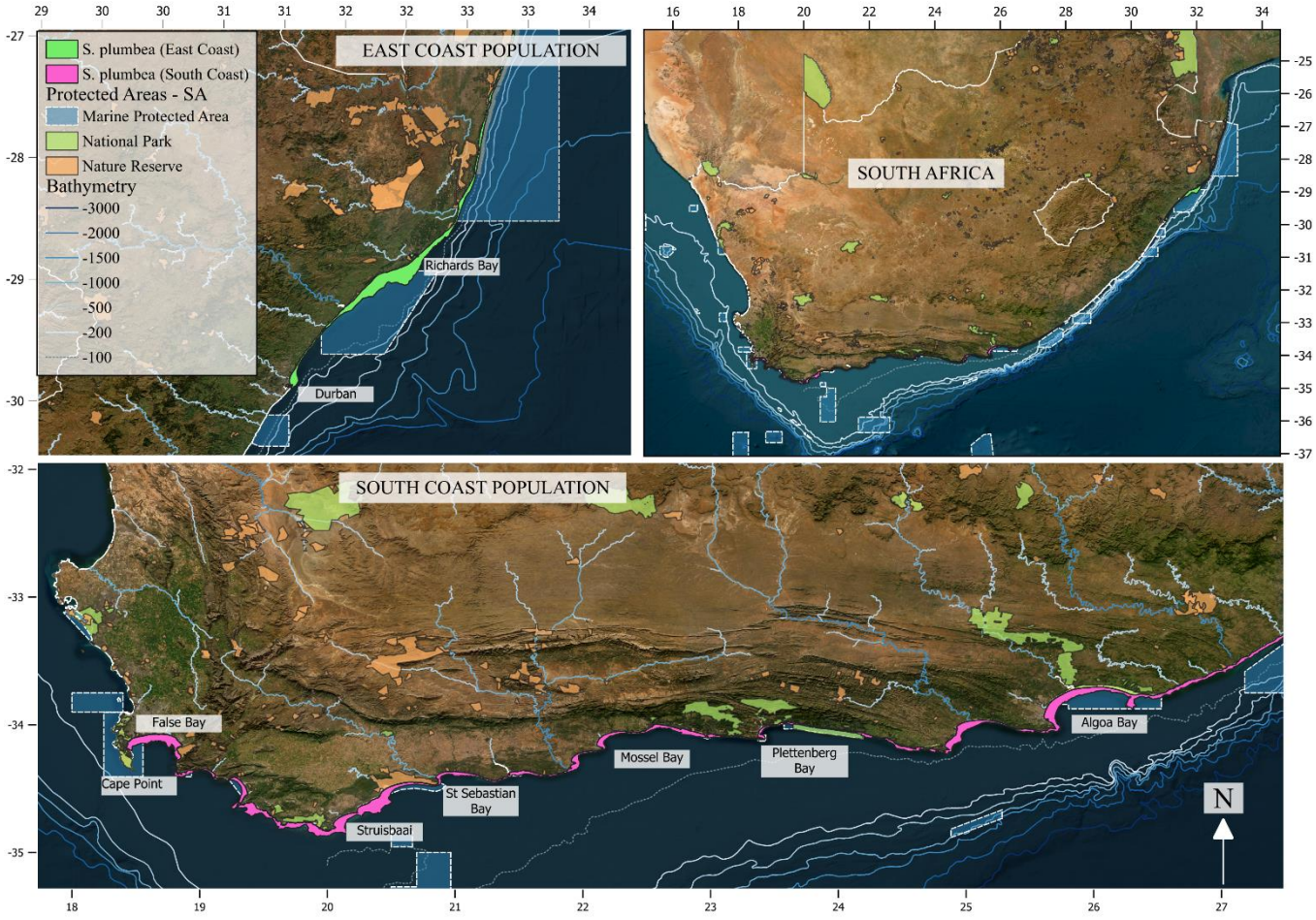


Figure 1.1. *Sousa plumbea* distribution throughout South Africa indicates two regions of occurrence, namely, the populations of the south and east coasts. Nationally, *S. plumbea* have an extreme coastal range, limited to less than 20 m water depth and not extending more than 2-5 km off the coastline, creating high spatial use overlap with Marine Protected areas throughout their coastal range. Spatial data sources: Environmental Systems Research Institute (ESRI, 2021), South African National Biodiversity Institute (SANBI, 2021), and IUCN Red List Assessment (Braulik et al., 2015).

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Table 1.2. Summary of study design and capture-recapture results from previous studies on *Sousa plumbea* in South Africa (adapted and updated from James et al., 2015).

Study site	Duration of study (months)	Number of surveys	Number of encounters	Group size (ind.; range [mean])	Max. individual distance travelled (km)	Catalogue size (ind.)	Animals seen more than once (ind.)	Distinct animals (%)	Closed model abundance estimate (ind. [95% CI])	Open model abundance estimates (ind. [95% CI])	Year of study	Reference
Mossel Bay	32	81	31	1-15 (4.9)	139	67	34	94	46* (38-56)	125 (61-260)	2011-2013	James et al., 2015
Algoa Bay	36	60	104**	3-24 (7)	110	70	37	92	NA	466 (447-485)	1991-1994	Karczmarski 1997; Karczmarski et al., 1999
Algoa Bay	3 years	106	50	1-23 (3.1)	NA	50	24	NA	NA	NA	2008-2011	Bouveroux et al., 2017
Goukamma - Tsitsikamma MPA	16	101 (231h.)	98	1-12 (3.9)	147	65	47	96	84 (72-98)	84 (72-115)	2014-2015	Conry 2017
Plettenberg Bay	24	87 (289 h.)	35	2-20 (7.2***)	32	63	36	77	112 (75-133)	727	2002-2003	Jobson 2006
Plettenberg Bay	17	20 38 h.	26	1-13 (4.7)	NA	41	26	NA	42 (42-55)	41 (28-54)	2012-2013	Greenwood 2013
KwaZulu-Natal (entire)	18	136	56	1-18 (6.7)	120	96	45	68	165 (134-229)	161 (81-240)	1991-1993	Durham 1994
Richards Bay	18	41	30	1-20 (5.1)	70	45	23	31	NA	38 (19-56)	1991-1993	Durham 1994
Richards Bay	7	73	56	1-20 (NA)	140	24 (181KZN)	56	76	NA	74 (60-88)	1998	Keith et al., 2002
Richards Bay	25	125	NA	NA	NA	92	56	54	244 (217-287)	170 (112-230)	1998-2000	Atkins 2002
National assessment ****	NA	37	NA	NA	NA	121	NA	NA	NA	NA	2016-2021	Frainer et al., 2022

*Mean closed population size **The high number of encounters made in Algoa Bay compared to other studies can be attributed to differences in survey methods, as the research boat was launched only in response to shore-based sightings of *Sousa plumbea* (Karczmarski et al., 1999). ***Mode: 5 (Jobson, 2006); Average 7.2 (Greenwood, 2013) **** National assessment of Rostrum Abnormalities. Data collated from dedicated and opportunistic encounter

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SOCIALITY IN *SOUSA PLUMBEA*

Group structures within cetaceans range from stable matriarchal family pods, as seen in orcas (*Orcinus orca*) (Deecke et al., 2010), to the variable and complex fission-fusion groups of bottlenose dolphins (*Tursiops* spp.), where group membership can change every 10-20 mins (Connor & Krützen, 2015; Galezo et al., 2018). The drivers of fission-fusion assortment are based on the benefits and constraints of the group over time and space (Aureli et al., 2008). However, group structure within this is far more complex, and it is essential to understand the variability in individual relationships within a group, which are key to social complexity (Hunt et al., 2019). It has been shown that the same individuals can display different social structures in different conditions, for example, when they compete to increase their reproductive success (Gowans, 2019). Aureli et al., (2008) demonstrated that individual variation in habitat, affiliations and group size are essential factors to consider in assessing the social structure of a population.

In *S. plumbea,* there has been relatively little published research regarding their social organisation (Bouveroux et al., 2019; Karczmarski et al., 1999; Saayman & Tayler, 1973). In South Africa, only 10s of animals have been identified a handful of times each in previous studies (Bouveroux et al., 2019; James et al., 2015; Koper et al., 2016; Vermeulen et al., 2018) (Table 1.2). In Algoa Bay *S. plumbea* had social bonds that were fluid and mostly of short-duration affiliations (Karczmarski et al., 1999). A more recent study in the Plettenberg Bay area (Bouveroux et al., 2019), detected two occurrences of stronger pairwise affiliations. Bouveroux et al., (2019) demonstrated a generally low level of gregariousness, with a well-differentiated social network with half the sampled dolphins (14 of 29 seen three or more times) showing strong social bonds between individuals, mostly as pairs. They suggest that the higher social bonds seen in this study could be a result of the decreasing group sizes and overall population in the area.

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Along the south coast of South Africa, a substantial decrease in the average group size of *S. plumbea* has been observed in several locations. For example, in Plettenberg Bay, average group sizes dropped from seven to four (Conry, 2017; Jobson, 2006), while in Algoa Bay, the average group size dropped from seven to three (Bouveroux et al., 2018; Karczmarski et al., 1999). General sociality in *S. plumbea* in South Africa is thought to be reflective of their small group sizes and an increase in the number of solitary individuals (Koper et al., 2016), limiting the number of potential associations (Gowans et al., 2007).

CURRENT CONSERVATION STATUS OF *SOUSA PLUMBEA* IN SOUTH AFRICA

Research on *S. plumbea* in South Africa since the late 1960s (see review by Elwen et al., 2011) has covered a wide range of topics, including socio-ecology (Karczmarski et al., 1999; Koper et al., 2016; Saayman, 1972), diet (Barros & Cockroft, 1999), anatomy (Frainer et al., 2019; Plön et al., 2012), regional distribution, habitat use, and movements (Atkins et al., 2005; Bouveroux et al., 2018; Koper et al., 2016; Vermeulen et al., 2018), local population abundance estimates (James et al., 2015; Keith et al., 2002), vulnerability to direct human impacts such as bycatch in shark nets (Atkins et al., 2013, 2016; Cockcroft, 1990, 1994), vessel traffic (Karczmarski et al., 1997; Koper et al., 2016) and health assessment, and pollutant loads (Aznar-Alemany et al., 2019; Cockcroft, 1999; Gui et al., 2016; Lane et al., 2014). However, these regional studies are limited in their capacity to provide insights into broader population health and dynamics. These studies have provided crucial baseline information and identified potential threats to the species. Still, they lack key information, including a reliable national abundance estimate, which is necessary to document future population trends.

In South Africa, although no empirical data are available on population trends at a broader national scale, Koper et al., (2016) and Bouveroux et al., (2018) suggested a significant decline in the frequency of occurrence, average group size, number of calves and the number of identified

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individuals when compared to a similar study in the early 1990s in Algoa Bay (Karczmarski et al., 1999). For Plettenberg Bay, a decrease in population size of 50% between 2002 and 2012 has been suggested, although this was based on limited sampling effort (Greenwood, 2013). In Richard's Bay, shark net mortalities have contributed to an unsustainable loss of *S. plumbea* (4.3 individuals or 5% - 10% of the population per year; (Atkins et al., 2013, 2016). These data strongly indicate an overall population decline along South Africa's coastline. Considering these inferred population declines, *S. plumbea* was up-listed to "Endangered" in South Africa on the IUCN Red List (Plön et al., 2015). This was based on an estimated extent of occurrence of < 5,000 km^2, a continuing decline in habitat quality (and likely area of occupancy) in key resource areas correlating with coastal development, the fragmentation of populations, and continuing adult mortality from anthropogenic disturbances (especially shark-nets; Plön et al., 2015). A quarter of the photographed South African population has been recorded with injured or abnormal rostrums, possibly caused by either fishing line entanglements, extreme feeding behaviours or genetic depression, leading to apparent conditions such as broken or misaligned jaws (Frainer et al., 2023). Most observed individuals were adults; therefore, it is difficult to know the extent of such impact on the population, as it is unknown how many juveniles did not survive these injuries or abnormalities. However, the wide prevalence throughout the South African range is likely driven by their small, fragmented population (Frainer et al., 2023).

Overall, there are limited population estimates throughout the species' full South African range. As a result, population trends across their range must often be inferred by assessing the extent of mortality or the degree of decline in relative sighting rates or group sizes. However, where data are available to evaluate changes over time, there are clear indications of significant declines in abundance, relative sighting rates, and/or group sizes. The reported levels of threats due to their restricted habitat and high reported mortality in portions of their range means that *S. plumbea* population size is almost definitely decreasing in South Africa (Braulik et al., 2015).

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STUDY LOCATION: SOUTH AFRICA

South Africa has a long, diverse coastline spanning 2,798 km, from rocky reefs to sandy beaches and estuaries. Mixing between the warmer Agulhas current and the cool Benguela current creates dynamic ocean currents, causing highly fluctuating water temperatures. The most westerly range of *S. plumbea* is at False Bay, at the most south-westerly tip of the continent, with no range extension into the westerly cooler Benguela region. The mean water temperatures along *S. plumbea's* coastal distribution range from 19 °C – 26 °C in the northeast, while the cooler south coast region has mean water temperatures of 14 °C – 20 °C (Dallas, 2008). Several major estuaries occur along the coastline, with many under regular management (Whitfield et al., 2012). Estuaries have been shown to be a likely habitat for the species, overlapping with many anthropogenic impacts, with some individuals being seen upriver in brackish, estuarine regions (Plön et al., 2015). South Africa's coastline is highly developed, with the vast majority of the coastline affected by human structures, including large coastal developments, active harbours, urbanisation, and development in estuaries (Claassens et al., 2023). However, a significant section of the *S. plumbea* range in the Western Cape falls within protected areas such as De Hoop, Stilbaai, Goukamma, Robberg, and Tsikamma marine protected areas (Mann-Lang et al., 2021) (Figure 1.1).

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THE SOUSA CONSORTIUM

A lack of scientific knowledge has been recognised as a major threat to *S. plumbea* conservation actions in South Africa, with many aspects of the species' life history and current changes to its population poorly understood. Given the lack of data on this species, a nationwide research collaboration, the SouSA consortium, was established in May 2016 (Vermeulen et al., 2018). The consortium aims to address the need for more knowledge on the conservation biology of *S. plumbea* in South Africa nationally and work as a collective group to achieve these aims. The consortium was recognised at the World Marine Mammal Conference in 2019 as a leading example in collaboration (S. Dines Judges Discretionary Award) and has collectively published several publications on *S. plumbea,* collating data at a national scale (Frainer et al., 2023; Plön et al., 2021; Vermeulen et al., 2018). The SouSA consortium is currently working with public, private and international stakeholders, such as the IUCN SOS group, to develop an effective biodiversity action plan for *S. plumbea* in South Africa. In 2024, the Indian Ocean Humpback Dolphin Conservation Network (HuDoNET) was established. HuDoNET was modelled on the successful initiatives of the Conservation Consortium of the Atlantic humpback dolphin (CCAHD) and the SouSA Consortium, with the aim to develop a build connections and collaboration among stakeholders throughout the species' range to encourage coordinated conservation actions (Atkins et al., 2024).

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THESIS OVERVIEW: CONCEPTUAL THESIS FRAMEWORK

The limitations of visual and olfactory senses in the marine environment have led to the evolution of sophisticated acoustic communication systems in cetaceans (Herman & Tavolga, 1980). Dolphins are known for their high vocal activity, producing a variety of acoustic signals, including frequency-modulated whistles used for communication (Jones et al., 2020). Among these, signature whistles are unique signals that are highly stable and stereotyped, serving as identification markers and contact calls in specific contexts (Caldwell & Caldwell, 1965; Sayigh et al., 2007). Evidence for signature whistle use has been presented across several species of odontocetes (see review in Janik, 2009), including preliminary evidence in *Sousa* spp. (Cheng et al., 2017; Gridley et al., Appendix 1; Van Parijs & Corkeron, 2001a). This thesis aims to understand the whistling behaviour of *S. plumbea* in South Africa and explore how signature whistles can be effectively used for acoustic monitoring of this elusive coastal species. Here, I present a conceptual framework for my thesis (Figure 1.2). The framework describes each data chapter (Chapters 2-5) and the main themes within. The figure presents how the chapters are linked, from investigating general whistle characteristics and variability across spatial scales in Chapter 2 to a focus on the behavioural effects on signature whistle production in Chapter 3. In Chapter 4, the use of passive acoustic monitoring to recapture signature whistles is compared to a more standard approach of boat-based photo ID over a short field study period. Finally, individual detection over time and space for capture-recapture models is investigated in Chapter 5. Using signature whistles for occurrence, recapture and density (SWORD, Fearey et al., 2021), the thesis provides baseline data on how individual passive acoustic monitoring can best be utilised for this threatened species.

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Figure 1.2. Thesis overview diagram to demonstrate the conceptual framework of the thesis. The framework shows each data chapter (2-5), including chapter themes and datasets used and how they build on one another throughout the thesis.

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2 MICRO AND MACROGEOGRAPHIC VARIATION WITHIN THE WHISTLE REPERTOIRE OF INDIAN OCEAN HUMPBACK DOLPHINS, *SOUSA PLUMBEA*

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Prepared for publication in Marine Mammal Science

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ABSTRACT

Understanding geographic variation in a species' acoustic repertoire can provide information on the degrees of mixing, isolation and plasticity in species that vocally learn. This is the first study to describe the acoustic behaviour and geographic variation of *Sousa plumbea* whistles across different geographic scales. Acoustic characteristics of whistle contours were compared from the extreme west and east of the species' global range (South Africa and Maharashtra, India). In South Africa, 1181 whistle contours were analysed from 21 encounters across five different study sites (19 hrs 17 mins of recordings), spanning two geographically isolated regions across the south and east coasts. In India, 2260 whistle contours were analysed from 11 encounters in one region (3 hrs 6 mins of recordings). Contours were described based on seven temporal and spectral measurements. Comparisons were made between 1) South Africa and India (macrogeographic variation) and 2) between study sites in South Africa, the south and east coast (macrogeographic variation), and between south coast study sites (microgeographic variation). Principal component analysis (PCA) was used as a data reduction technique to visualize the collinear relationships among the temporal and frequency whistle characteristics. Random forest algorithms (RF) were then used to investigate fine-scale differences in whistle measurements across groups at different geographical scales. There was considerable overlap in the acoustic properties of whistles from *S. plumbea* recorded at their geographical range limits. However, overall classification with random forest was good (89.7% overall accuracy). In South Africa, classification to region was achieved with moderate success (64.9% overall classification). On inspection, Richards Bay (east coast) was distinct from the south coast study sites and whistles contours were classified with high accuracy (90% classification rate). Across the four south coast study sites, there was also considerable overlap and no evidence of isolation through distance. Overall, the whistle characteristics of *S. plumbea* recorded in South Africa were within the species range for most acoustic parameters, but occasionally exceeded the maximum frequency previously documented. The high degree of overlap of whistle features and low classification rates across the south coast sites indicate little microgeographic variation within the south coast of South Africa. Although there were whistle features that allowed for a degree of classification across these geographic scales, there were broad species-specific characteristics conserved across their range, which could assist with species ID classification.

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INTRODUCTION

Acoustic signals serve a crucial communicative function within animal societies, providing information about the signaller's identity, size, health, reproductive fitness (static information), or information on external factors such as predator threats and senders' motivational state (dynamic information) (Erbe & Thomas, 2022). The transfer of information within these signals between individuals is strongly linked to survival and reproductive strategies. Variation in these acoustic signals can be seen between and within populations and has been documented in taxa, including frogs (Morais et al., 2012), birds (Hamao et al., 2016), insects (Pinto-Juma et al., 2005) and mammals (Lameira et al., 2010).

Acoustic variation can result from genetic differences (Collins & Terhune, 2007), environmental effects (see acoustic adaption hypothesis Morton, 1975) and learning (Garland et al., 2020). Morphological features of the vocal apparatus can drive variation, along with an individual's age and body size (Fischer et al., 2004). Properties of the environment can also promote acoustic variation due to the influences of sound propagation to limit signal degradation (Morton, 1975). Closed habitats, such as forests, with a high density of vegetation have a greater number of surfaces for reverberation and absorption of signals. Here, birds adapt to concentrate energy in lower frequency, narrow band sounds for better propagation of signals (Dabelsteen et al., 2005; Ey & Fischer, 2009). Vocal production learning is the ability to produce new sounds by reference to auditory feedback and is an important consideration for acoustic variation, documented in six animal groups (Humans, bats, cetaceans, pinnipeds, elephants and birds including songbirds, parrots and hummingbirds; see Petkov & Jarvis, 2012). As individuals mix across social groups, signals can converge or differ from more distant individuals (Tyack & Sayigh, 1997). This leads to the behavioural matching of acoustic signals, accrued through copying or improvisation (King & McGregor, 2016).

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Intra-specifically, acoustic variation between populations can be distinguished at two spatial scales: macro- and microgeographic variation (Mundinger & Rumbaugh, 1982). Macrogeographic variation occurs over long distances and between geographically separated populations (Krebs & Kroodsma, 1980). In delphinids, for example, killer whales (*Orcinus orca*) in the North Pacific and Northeast Atlantic show macrogeographic variation in high-frequency whistle features between populations with little to no mixing of individuals (Samarra et al., 2015). Microgeographic variation is defined as variation between populations with the potential for mixing in a single or near-continuous population (Krebs & Kroodsma, 1980). For example, Rossi-Santos and Podos, (2006) demonstrated evidence of microgeographic variation in the whistles of the estuarine dolphin, *Sotalia fluviatilis*, with higher start and minimum frequencies of whistles to the north of their Brazilian coastal range and greatest differences occurring between the most distant sites, highlighting discontinuity in acoustic behaviour along a seemingly continuous range.

Delphinid acoustic signals can be categorised into clicks and whistles (including burst pulse sounds), with differences in levels of variation between these types (Janik, 2009). Clicks, often produced in click trains, are short signals which can reach very high-frequency ranges, are often used for echolocation, and can also serve a communicative function in some species (Arranz et al., 2016). Burst pulses are rapid click trains used for short-range and long-distance communication (Deecke et al., 2000; Lammers et al., 2006). Whistles are frequency-modulated tonal sounds, mostly used as communicative signals (Popper, 1980). Whereas clicks are innate and vary little between individuals and within a species (though, for an example of click variation in white-sided dolphins, see Soldevilla et al., 2010), whistles are acquired through vocal production learning (Fripp et al., 2005) and can vary across geographic scales, especially in isolated or disjunct populations (Janik, 2009).

The geographic variation in whistle diversity within a species can be investigated in multiple ways. Studies can investigate acoustic variation using amplitude, measuring spectral density

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(Morisaka et al., 2005) or changes to the frequency modulation curves of contours (Rendell et al., 1999). Spectral and temporal features of whistle contours provide information on the acoustic characteristics of the species, with standardised measurements allowing for comparison across studies (Bazúa-Durán et al., 2002; Gridley et al., 2012; Morisaka et al., 2005; Oswald et al., 2003; Rendell et al., 1999) In dolphins, whistles are often characterised by the start and end frequency, minimum and maximum frequency, as well as the duration of signal and the number of inflection points, as these features are thought to carry information on species-specific signalling (Gridley et al., 2015; Jones et al., 2020; Morisaka et al., 2005).

Sousa plumbea are listed as globally endangered on the IUCN Red List, with a global range from the Bay of Bengal, India, furthest east through to False Bay, South Africa, furthest west (Braulik et al., 2015). In South Africa, they are also listed as locally endangered, with a population of fewer than 500 individuals (Plön et al., 2016; Vermeulen et al., 2018). The species theoretically ranges from False Bay in the west (~ 18.47 ˚E) to the eastern South Africa-Mozambique border (~ 32.90 ˚E) but with animals split into two geographically distinct populations, one along the south coast and the other in KwaZulu-Natal (KZN) on the east coast, mostly centred around Richards Bay (Keith et al., 2013). A clear gap in sightings and strandings along a 245 km exposed rocky coast of the "Wild Coast" of the Eastern Cape and southern KZN coastline indicates individuals are unlikely to regularly cross this area, especially when considering the species median range of 120 km (Vermeulen et al., 2018). This has led to the 'two - population hypothesis' (Braulik et al., 2015; Caputo et al., 2020; Plön et al., 2015; Vermeulen et al., 2018) that these south and east coasts may represent different sub-populations and, consequently, management units (Plön et al., 2021). Yet, a recent genetic study based on samples collected from 1979 - 2017, comparing the two proposed subpopulations, found extremely low mitochondrial diversity, suggesting that South African humpback dolphins form a single genetic population (Lampert et al., 2021). However, behavioural adaptations often proceed genetics, especially with mtDNA and other regional differences in acoustic signals, which may facilitate intraspecific recognition (Tibbetts & Dale,

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2007). However, this has yet to be explored in *S. plumbea* along their South African range. Acoustic variation has only previously been explored in other *Sousa* spp., with Hoffman et al., (2015) finding variation in *S. chinensis* frequency variables between two locations along the western peninsular Malaysia coastline, suggesting the differences in whistles may be related to adaptations to the local acoustic habitat or unique whistles developing due to social interactions within each location. Monitoring of acoustic behaviour in *S. plumbea* can provide information on behavioural adaptation at timescales faster than genetics and may be useful to help elucidate isolation between these two populations in South Africa as well as behavioural barriers to interbreeding.

This study compares the acoustic characteristics of *S. plumbea* whistles across geographic ranges to investigate the degree of variation in their whistle features. It investigates variation between South Africa and India, two populations at the furthest western and eastern limits of the species' global distribution, and between populations within South Africa to investigate levels of both macro and microgeographic variation.

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Hypotheses

1) Variation between South Africa and India.

   – The highest level of variation will be seen between these two populations due to macrogeographic variation (no mixing between these two geographically isolated populations)

2) Variation within South Africa.

   – The south and east coast study sites will show macrogeographic variation due to the lack of continuous mixing of individuals between these two populations, supporting a two-population hypothesis in South Africa.

   – The least variation will be seen amongst the four south coast study sites (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay), which have known mixing of individuals along a continuous range and, therefore will show no or little microgeographic variation.

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METHODS

Data collection

In South Africa, data were collected from five study sites (four south coast sites: False Bay, Struisbaai, St Sebastian Bay and Mossel Bay and one east coast site, Richards Bay) between 2016-2023 (Table 2.1). Recordings were collected during boat-based surveys focused on dedicated *S. plumbea* encounters (south coast survey site distance False Bay – Mossel Bay 140 km and distance from Mossel Bay – Richards Bay 1260 km). Data from India were collected along the Sindhudurg coast of Maharashtra, the eastern species range during transects covering a distance of 121 km. (Bopardikar et al., 2018) (Figure 2.1).

In South Africa, locations were chosen based on the known *S. plumbea* range and sightings from previous studies (Keith et al., 2013; James et al., 2015; Vermeulen et al., 2018). Boat surveys were conducted during daylight hours with favourable weather conditions within 5 km of the shoreline (Beaufort sea state ≤ 3, swell < 2 meters). The boats used were either a 10 m Butt Cat with a raised viewing platform for observation (Mossel Bay), a 6.2 m Rigid Hulled Inflatable Boat (RHIB) (False Bay, Struisbaai, St Sebastian Bay) or a 5.5 m RHIB (Richards Bay), all with four-stroke engines for minimal animal disturbance from engine noise or boat proximity. Encounters began when *S. plumbea* were first sighted within 50 m of the boat. The boat approached from behind or to the side of the group, and for acoustic recording the engines and echosounder were turned off when safe to do so, and the boat drifted alongside or towards the group of dolphins. In some events, due to the proximity to the surf zone, the engine would be kept idle or the boat would slowly move alongside the group (less than 5 knots) while the hydrophone was still in the water. Acoustic recordings were made using a SoundTrap 300 HF (flat frequency response of 20 Hz - 150 kHz ± 3 dB; Ocean Instruments Inc., New Zealand), or HTI-96-MIN hydrophones (flat frequency response of 2 Hz – 30 kHz ± 1 dB; High Tech Inc., U.S.) attached to a TASCAM DR 680 recorder (TASCAM, U.S.) (Table 2.1). An acoustic encounter started when the hydrophone was placed in the water and continued until the dolphins were out of acoustic range (< 1 km Jensen et al., 2012).

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During acoustic encounters, data on group sizes and surface behaviours were recorded to ensure that whistle repertoire was sampled over various behavioural contexts and group compositions (Table 2.2). During acoustic encounters, continuous scans of other dolphin species and *S. plumbea* groups were made within visual range to ensure the recording data was true to the focal group. Data used in this study were restricted to single-species recordings (i.e. visual confirmation of species during the encounter and secondary assessment of species within recordings (e.g. presence of SNR 1 *Delphinus delphis* removed whole encounter) to reduce any effects of interspecific communication.

In India, data were collected during visual surveys conducted in the months of May and October 2014, December 2015, and January 2016 using a small, 11-meter-long survey vessel, within 5 km of the shoreline. All recordings were made only during daylight hours and low sea states (Beaufort ≤ 3) over 121 km of coastline. A Sensor Technology SQ26-08 hydrophone (Sensor Technology Ltd., Collingwood, Canada) with a sensitivity of -169 dB re 1 V/µPa and a M-Audio Track II (Cetacean Research Technology Inc., Seattle, WA) was used to collect recordings in May and October 2014. The same hydrophone was connected to a Zoom H1 handheld recorder (Zoom Corp., Tokyo, Japan) to collect acoustic data in December 2015 and January 2016. The hydrophone model had a flat frequency response (62 dB) in the relevant frequency range (1 – 34 kHz). All recordings were made using a sampling rate of 96 kHz and 24-bit resolution in the standard WAV file format. For each recording, the hydrophone was deployed at a depth of 1.5 – 2.0 m, and the engine of the survey vessel was turned off to minimise interference with the acoustic recordings.

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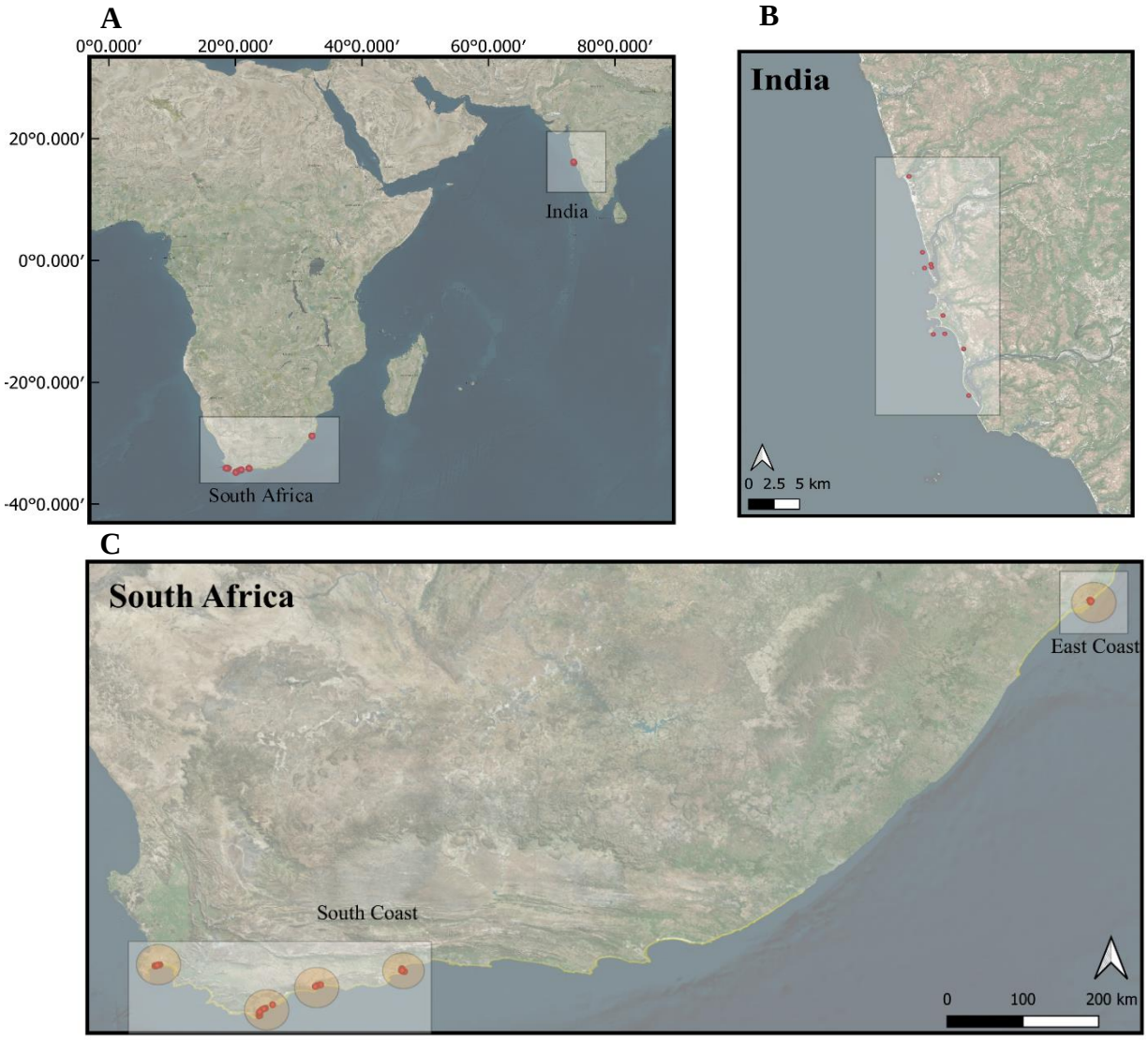


Figure 2.1. *Sousa plumbea* study sites distributed throughout South Africa and India. A) The red dots indicate encounters where *Sousa. plumbea* were recorded. B) The Indian population were recorded throughout the Maharashtra region (Bopardikar et al., 2018). C) The South African population were recorded in both the south and east coast regions. Within South Africa, five separate study sites were included, including four on the south coast (West to East: False Bay, Struisbaai, St Sebastian Bay and Mossel Bay) and one on the east coast (Richards Bay), as indicated by yellow circles on the figure.

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Data processing

South African data were visually analysed in Raven Pro v1.6 (Center for Conservation Bioacoustics, 2019) for *S. plumbea* whistle contours. Whistle contours were defined as continuous frequency-modulated tonal signals (Janik et al., 2013). The fast Fourier transform (FFT) parameters used to examine the data were 1024 point Hann window and 90% overlap. Contours were graded according to a visual assessment of each contour's signal-to-noise ratio (SNR). SNR was assigned from 1 – 3, 1) the signal was faint or partially masked but visible on the spectrogram 2) the signal was clear and unambiguous and 3) the signal was prominent and dominated the spectrogram (Gridley et al., 2012). Only good-quality contours (those graded SNR 2 or 3, not masked) were used in further analysis. Contours were manually annotated, and parameters from the lowest frequency signal, also known as the fundamental frequency, were measured in Raven Pro v1.6 (Table 2.3 and Figure 2.2a). A comparable approach to analysis was undertaken for acoustic data collected from *S. Plumbea* in India by Bopardikar et al., (2018), enabling comparison between sites. Cross-checking between manual acoustic analysis (S. Dines and I. Bopardikar) was conducted to ensure consistency in approaches. Although many parameters can be measured from acoustic signals in Raven Pro v1.6, several are considered standard in whistle acoustic analysis, and these were chosen to compare between sites (seven whistle parameters) (Figure 2.2b). To ensure data quality, only contours considered good SNR (2 or 3) were used from South Africa and > 6 dB signal-to-noise threshold in India (Table 2.1). A description of these measurements followed comparable studies of the genus *Sousa*, and include minimum, maximum, median (with interquartile range expressed as Q1 – Q3), mean (standard deviation), and coefficient of variation (CoV = (SD/mean) * 100) to describe inter and intra-population variability (Bopardikar et al., 2018; Dong et al., 2019; Yuan et al., 2021). Ultrasonic frequencies were recorded and classed as contours with maximum frequencies > 20 kHz (Oswald et al., 2004; Samarra et al., 2010).

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Table 2.1. Summary table of the six study sites from South Africa and India, detailing the recording equipment, sampling and bit rates, as well as the total number of field days at each location. The Euclidean distances (in km) between neighbouring recording sites are also provided, with False Bay, the westernmost site, serving as the starting point (0 km).

Distance to the previous site	**Country**	**Region**	**Study Sites**	**Recording equipment**	**Sampling Rate (Hz) and Bitrate**	**Field days**
0	South Africa	South Coast	False Bay	HTI -*96-MIN; High Tech Inc., United States* and Tascam digital recorder	96000 16 and 24	3
215	South Africa	South Coast	Struisbaai	HTI -*96-MIN; High Tech Inc., United States* and Tascam digital recorder	96000 24	4
125	South Africa	South Coast	St Sebastian Bay	Sound Trap 300 HF; Ocean Instruments, New Zealand;	576000 24	1
140	South Africa	South Coast	Mossel Bay	HTI-96-MIN; High Tech Inc., United States Tas- cam DR-680; TEAC America Inc., United States; or Zoom H1	96000 16	9
1400	South Africa	East Coast	Richards Bay	Sound Trap 300 HF; Ocean Instruments, New Zealand;	576000 24	2
			Total			19
6700	India		Maharashtra	Sensor Technology SQ26-08 hydrophone (Sensor Technology Ltd., Collingwood, Canada) with a M-Audio Track II (Cetacean Research Technology Inc., Seattle, WA)	96000 24	11
			Total			30

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Table 2.2. Summary of *Sousa plumbea* encounters recorded across the six South African and Indian study sites. Each row indicates a unique acoustic encounter, defined as starting when the hydrophone was placed in the water and continuing until the dolphins were out of acoustic range. Encounter information includes the best group size, primary and (if apparent) secondary behaviours (see Table 3.3).

Population	**Study site**	**Date**	**Best Group size**	**Primary Group behaviour**	**Secondary Group behaviour**
South Africa	False Bay	17/05/2016	8	Travelling	Milling
		23/06/2016	3	Milling	Socialising
		27/11/2016	5	Socialising	Travelling
	Struisbaai	01/03/2017	10	Socialising	Foraging
		02/03/2017	5	Travelling	Socialising
		04/03/2017	7	Travelling	Socialising
		07/03/2017	3	Milling	
	St Sebastian Bay	17/09/2019	8	Socialising	Milling
		17/09/2019	3	Travelling	Milling
		17/09/2019	8	Socialising	Milling
	Mossel Bay	30/09/2019	6	Socialising	
		16/10/2019	8	Travelling	Milling
		06/03/2020	3	Milling	
		17/03/2021	1	Socialising	
		18/03/2021	5	Milling	
		24/03/2021	4	Travelling	
		28/03/2021	8	Travelling	

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	Richards Bay	10/03/2020	4	Milling	
		10/03/2020	9	Travelling	Socialising
		10/03/2020	4	Travelling	Socialising
		19/03/2020	9	Travelling	Socialising
India	Maharashtra	03/05/2014	10	Socialising	
		08/05/2014	6	Travelling	
		12/05/2014	26	Foraging	
		20/10/2014	24	Foraging	
		22/10/2014	30	Socialising	
		20/12/2015	40	Milling	Foraging, Socialising
		21/12/2015	35	Foraging	Socialising
		30/12/2015	110	Socialising	Foraging, Milling
		09/01/2016	4	Milling	Foraging

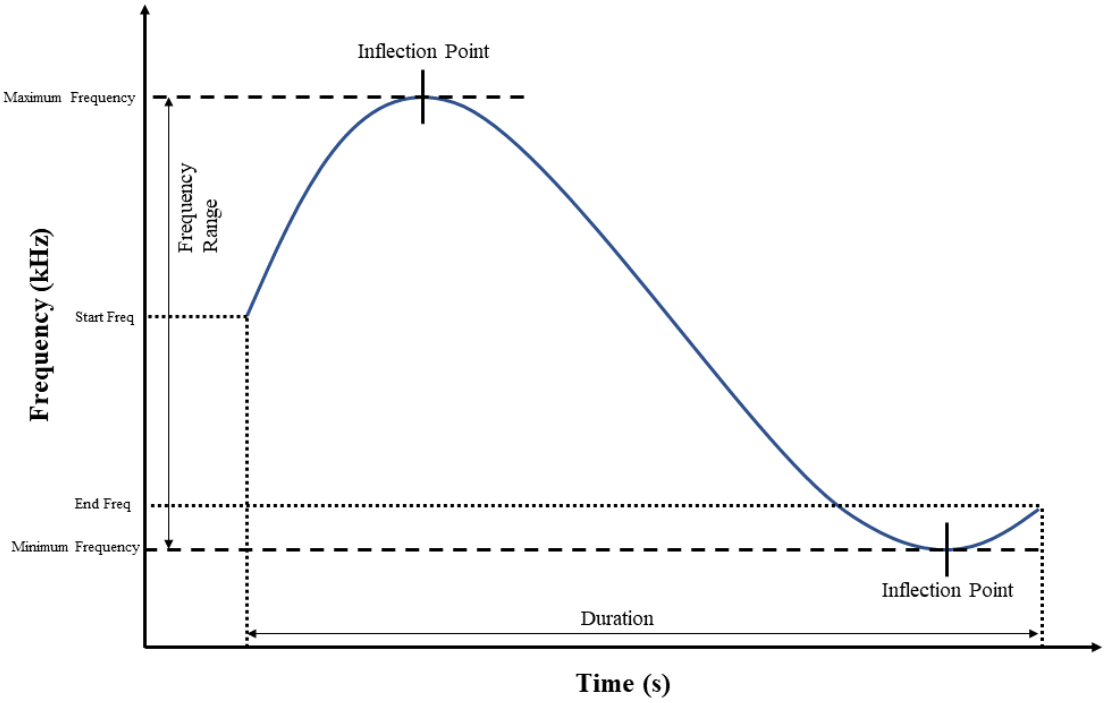
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Table 2.3. Descriptions of the seven contour measurements used for analysis (Bopardikar et al., 2018; Wang et al., 2013) . All frequency measurements were measured in kilohertz (kHz) and time measurements in seconds (s). Measurements were all calculated in Raven Pro v1.6 (Center for Conservation Bioacoustics, 2019) The first four measurements were calculated automatically from the annotation box function. Start frequency, end frequency and inflection points were manually measured.

Contour measurement	**Definition**
Minimum Frequency	Minimum of the fundamental frequency of the contour
Maximum Frequency	Maximum of the fundamental frequency of the contour
Start Frequency	Frequency at the start point of the contour
End Frequency	Frequency at the end point of the contour
Frequency Range	Maximum frequency of contour minus the minimum frequency of the contour
Duration	Duration of the contour, calculated as end time minus the start time
Inflection points	Count of local minima or local maxima where slope equals zero and point of change of curvature

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A



B

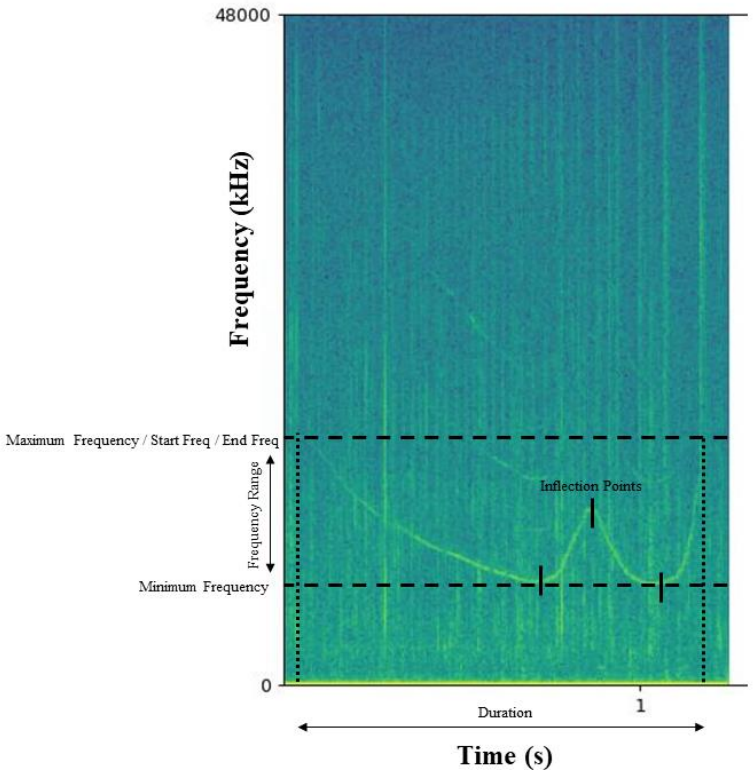


Figure 2.2. A) Diagram of acoustic measurements on a spectrogram, illustrating an example whistle contour with time (s) on the X-axis and frequency (kHz) on the Y-axis. Measurements shown are 1) Maximum frequency, 2) Minimum frequency, 3) Frequency range, 4) Start frequency, 5) End frequency, 6) Duration, and 7) Inflection points. B) Diagram of acoustic measurements on a spectrogram, using a *Sousa plumbea* whistle contour example from the south coast, South Africa. In this example, the maximum frequency, start frequency and end frequency are all equal.

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Statistical analysis

Statistical analyses were conducted using R (version 4.0.2; R Core Team, 2020) to investigate whistle variability 1) between South Africa (data between all five study sites pooled) and India, and 2) within South Africa, between all five study sites and between south and east coast regions.

As an initial step, the seven whistle measurements were tested for collinearity at both the regional and site level, with high collinearity values detected between nearly all the frequency-based measurements (r = > 0.7). Due to the high collinearity between frequency characteristics, a principal component analysis (PCA) was applied to the data (R function: prcomp, package: stats) as a data reduction technique to visualize the collinear relationships among the temporal and frequency whistle characteristics. The PCA converted parameter measurements into uncorrelated axes called principal components (PCs), with each successive PC accounting for as much variance as possible in the dataset. PCA assumptions (linear relation between variables, sampling adequacy, and absence of outliers) were verified. PCA does not require balanced sample sizes across different groups as it is a dimensionality reduction technique that works on the entire dataset.

Random forest algorithms (RF) were then used to investigate fine-scale differences in whistle measurements across groups at different geographical scales. RF is a supervised classification method based on averaging multiple decision trees (Breiman, 2001). Using a tuned process, RF generated trees by randomly subsampling training data (75% for each model). Precision was assessed by the number of correct predictions per group (i.e. whistle contours assigned to the correct site or population, expressed as a percentage) and compared to the prediction explained by chance, Cohen's Kappa statistic. The global accuracy of the model precision is calculated as a proportion of the total correct classification from an RF procedure. Random forest algorithms were conducted using the package *randomForest* (Breiman, 2001). The number of trees was set

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to 1000 because the global error rate generally stabilised from this value. The number of variables selected at each node was set at the square root of the total number of variables included in the RF. To understand the contribution of each variable to the model prediction, the Gini variable importance index was used (Breiman, 2001) and to understand what variables provided more accuracy to the model the mean decrease in accuracy was used. The higher the value of mean decrease accuracy or mean decrease Gini score, the higher the importance of the variable in the model (Breiman, 2001). Both levels of geographic variation investigation were performed using this general method.

The RF was conducted on the full datasets from South Africa and India. RF analysis between the five South African study sites was conducted on the full whistle contour data sets for each site, except for Mossel Bay, which had a significantly higher sample size. Therefore, Mossel Bay was randomly subsampled to 155 contours based on the mean full sample sizes of the other four study sites to provide a more equal sample size across sites and to avoid overfitting the model based on Mossel Bay (R package tidyverse; dplyr Hadley & Garrett, 2016).

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RESULTS

In total, 207 boat-based surveys were conducted in South Africa from 2016 to 2023. Of these, *S. plumbea* were encountered on 83, i.e. 40% of occasions, and acoustic data were available for 64 (77%) of these. However, removing encounters where dolphins were silent, undetected, or potentially mixed with other species dramatically reduced usable data. Acoustic data from 21 (25%) encounters were subsequently used in this study. In total, 2298 contours were identified from the five study sites in South Africa, with 1181 (51%) of these visually assessed as good quality (SNR 2 or 3) and used for further analysis. Contours analysed from along the Sindhudurg coast, India, totalled 2260 across 11 encounters (Table 2.4). There were notable differences between group dynamics observed between both South African and Indian encounters. In South Africa, *S. plumbea* were recorded in small single species groups with sizes of 1-10 individuals, occurring with several sympatric whistling delphinid species (*Tursiops* spp. and *D. delphinus*) (Frainer et al., 2023) as well as predation threats from species such as white sharks (*Carcharodon carcharias*) along their distribution (Jewell et al., 2014; Plön et al., 2021). In India, *S. plumbea* were recorded in larger group sizes (up to a maximum group size of 110, Table 2.2), recorded in single species groups and occurring only with the sympatric finless porpoise (*Neophocaena phocaenoides*), a non-whistling species (Morisaka, 2012).

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Table 2.4. Summary of all acoustic encounters and the total analysed contours across all six study sites in South Africa and India for *Sousa plumbea*. Only contours with a good signal-to-noise ratio of 2 or 3 or > 6 dB signal-to-noise threshold were used for further analysis (presented in % of contours analysed). Totals are accumulated throughout the table, except for the % of contours analysed which are separated based on the South African and Indian datasets.

Country	Region	Study Sites	Encounters analysed	Total Recording length	Total contours	SNR S/3 contours	% contours analysed
South Africa	South Coast	False Bay	3	06:48:35	361	164	45
South Africa	South Coast	Struisbaai	4	02:44:20	223	117	52
South Africa	South Coast	St Sebastian Bay	2	04:29:51	263	158	60
South Africa	South Coast	Mossel Bay	9	03:49:29	1172	562	48
South Africa	East Coast	Richards Bay	3	01:25:39	279	180	65
		Total	**21**	**19:17:54**	**2298**	**1181**	**51**
India		Maharashtra	11	03:06:00	2260	2260	100
		Total	**32**	**22:23:54**	**4558**	**3441**	**100**

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South Africa and India

Contours produced by *S. plumbea* in South Africa had a larger mean frequency range than India (South Africa = 4.9 (± 2.7) kHz, India = 2.7 (± 2.2) kHz) (Figure 2.5). The maximum frequency contour recorded in South Africa reached 46.6 kHz, more than 13 kHz higher than the maximum frequency contour from India (India Max Freq = 33 kHz). The prevalence of ultrasonic frequency contours (> 20 kHz) was low across both populations. However, South Africa had a greater proportion of ultrasonic contours, contributing to 3.2% of contours in South Africa compared to 1.1% in India. In general, most contours from *S. plumbea* were short (< 1 sec), with only 4.7% of contours exceeding 1 sec duration in South Africa and 0.7%. in India. Chirps (contours with a duration of < 0.1s) were most prevalent in India (41%) compared to South Africa (7.3%) and in both populations the majority of contours were produced with 1-2 inflection points (South Africa = 88.7%, India = 85.2%). Complex contours, with four or more inflection points were more prevalent in India (India = 10%, South Africa = 3.7%) (Figure 2.3).

Table 2.5. Descriptive statistics were calculated for the seven whistle contour parameters of *Sousa plumbea*, measured for both South Africa (with all five study sites combined) and India. For each measurement, the minimum and maximum values, median (inter quartile range, IQR), mean (standard deviation, SD), and coefficient of variance (CoV)% were calculated. Frequency parameters were measured in kilohertz (kHz) and duration was measured in seconds (s).

		South Africa	**India**
(kHz)	n	1181	2260
Min Freq	Minimum	1.4	2.3
	Maximum	32.8	32.7
	Median (IQR)	6.0 (4.8 - 7.0)	7.3 (6.2 - 8.5)
	Mean (± SD)	6.4 (± 2.8)	7.6 (± 2.8)
	CoV	44	37.3
Max Freq	Minimum	3.4	3.1
	Maximum	46.6	33
	Median (IQR)	10.3 (8.7 - 12.7)	9.9 (8.1 – 12.2)
	Mean (± SD)	11.3 (± 4.3)	10.2 (± 3.3)
	CoV	38	33

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Start Freq	Minimum	1.8	2.3
	Maximum	33.9	33
	Median (IQR)	8.5 (5.8 - 11.4)	8.9 (7.1 – 11.0)
	Mean (± SD)	9.1 (± 4.0)	9.2 (±3.3)
	CoV	44.4	35.9
End Freq	Minimum	2.3	2.8
	Maximum	45.7	32.8
	Median (IQR)	8.9 (7.6 - 10.3)	9.3 (7.5 – 11.7)
	Mean (± SD)	9.5 (± 4.2)	9.6 (± 3.4)
	CoV	43.7	35.4
Freq Range	Minimum	0.7	0
	Maximum	21.8	14.9
	Median (IQR)	4.3 (3.1 - 5.6)	2.0 (0.9 - 4.0)
	Mean (± SD)	4.9 (± 2.7)	2.7 (± 2.2)
	CoV	55.4	83.9
Duration	Minimum	0.04	0.01
	Maximum	2.1	1.6
	Median (IQR)	0.3 (0.2 - 0.6)	0.1 (0.1 - 0.2)
	Mean (± SD)	0.4 (± 0.3)	0.1 (± 0.1)
	CoV	76	101.8
Inflection	Minimum	0	0
	Maximum	11	19
	Median (IQR)	1.0 (1 - 2)	0.0 (0 - 2)
	Mean (± SD)	1.2 (± 1.1)	1.8 (± 1.7)
	CoV	94	172.9

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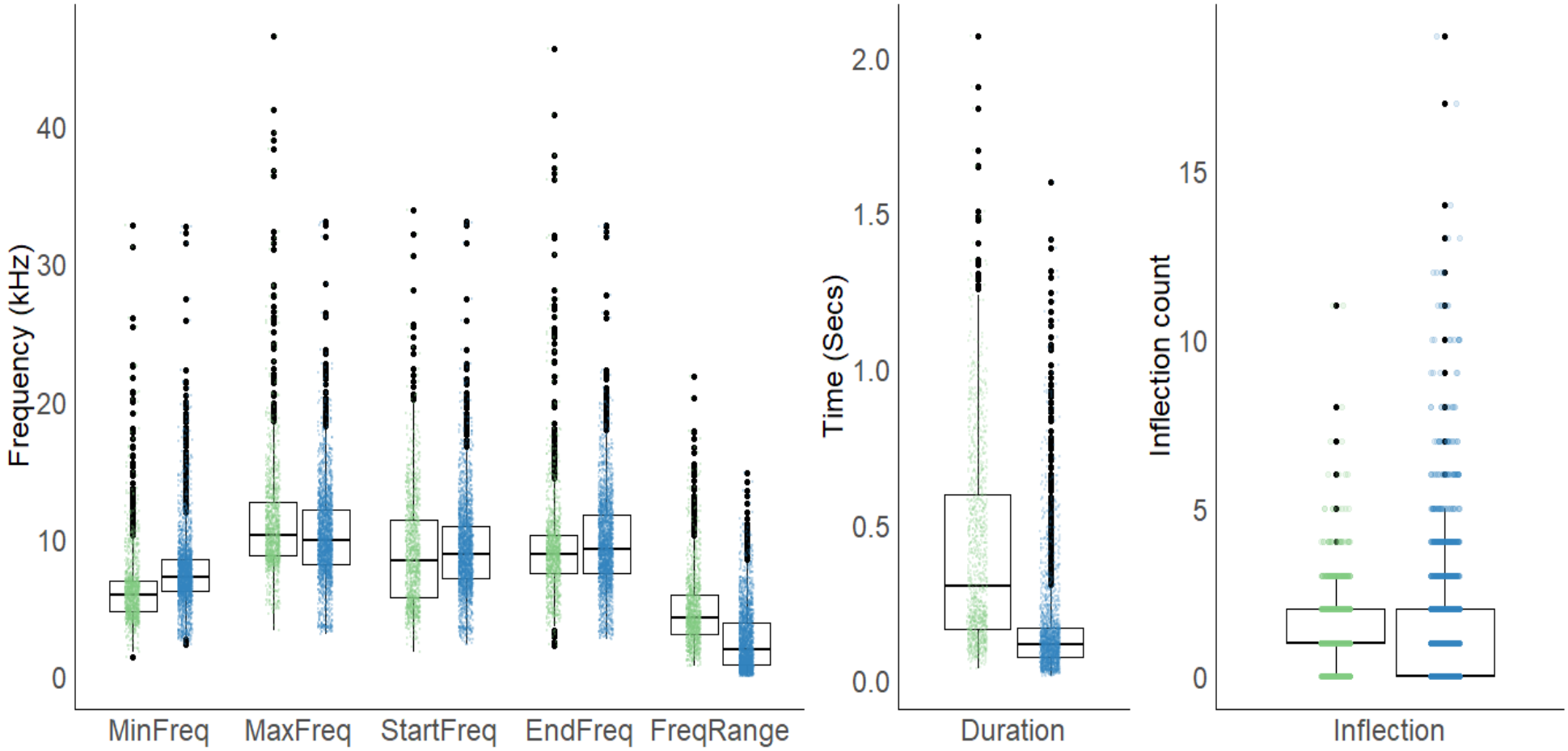


Figure 2.3. Variation of *Sousa plumbea* whistle parameters measured between South African and Indian datasets. All five sites in South Africa were pooled together (green, n = 1181) and compared to India (blue, n = 2260). Coloured dots show the distribution of the data for each population. In each plot, the box represents the inter-quartile range (IQR, middle 50%) of each data set, with centre lines representing median values, and the lines extending from the box (whiskers) represent the range within 1.5 times the IQR from the lower and upper quartiles. Points outside this range are considered outliers and shown as black dots.

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The first two principal components in the comparison between South Africa and India accounted for 86.1% of the variation in the contour measurements (Table 2.6). Principal component one accounted for over half of the total variability (59.7% of variance) and had high factor loadings (> 0.5) for the maximum frequency (0.56), start and end frequency (SF 0.5, EF 0.53), representing an effect of frequency characteristics in the first principal component (Figure 2.4). Principal component two accounted for 26.4% of the total variance and had high factor loadings for frequency range (-0.78) and minimum frequency (0.53).

Table 2.6. Principal Component loadings for the seven contour measurements of *Sousa plumbea* whistle contours recorded in South Africa (with the five study sites pooled together) and India. Values in bold had loadings greater than 0.5 or less than -0.5 (high factor loadings). Principal component one was responsible for 59.7% of the overall variance, with the first two principal components accounting for 86.1% of the variance.

	Components	
	PC1	PC2
Minimum Frequency	0.36	**0.53**
Maximum Frequency	**0.56**	-0.25
Start Frequency	**0.50**	0.12
End Frequency	**0.53**	0.06
Frequency Range	0.19	**-0.78**
Duration	0.00	-0.05
Inflection	-0.02	-0.19
Eigen Value	3.59	1.84
% Variance explained	59.70%	26.40%

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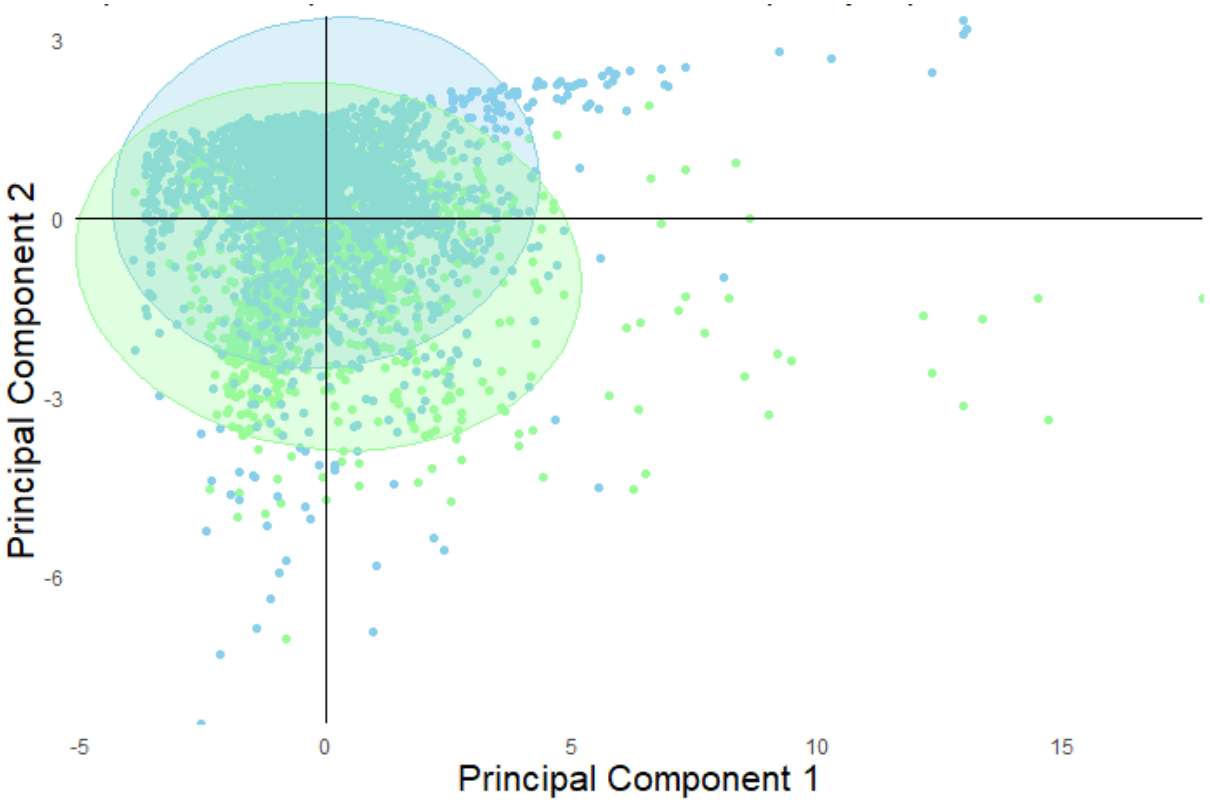


Figure 2.4. Principal component plot showing the first two components based on *Sousa plumbea* whistle contour measurements, derived from the seven measured parameters across the six pooled study sites in South Africa and the study site in India. Data points for South Africa are in green and India data points are in blue. The coloured ellipses contain 95% of data points for each population. A high degree of overlap (similarity) is found between these geographically separated populations based on the first two principal components. Principal component one explains 59.7% of the variance and component two explains 26.4% of the variance.

The random forest (RF) model for South Africa and India was conducted on the full data set for both populations (n = 3441 whistle contours). The RF was, therefore, based on a training dataset of 2581 whistle contours and a test dataset of 860 whistle contours. The tuned RF analysis supported a strong model performance with 89.7% model accuracy. South Africa had a high correct classification rate of 84% (compared to a chance rate of 35.2%), and India had a correct classification rate of 93% (chance rate of 64.8%), indicating that South Africa had a stronger classification with a correct classification of more than 50% more than chance (Table 2.7). The models' Cohens Kappa statistic of 77% indicates substantial agreement between the predicted

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and actual classification showing good model performance between the two populations. The most important variables for classification were duration, frequency range and inflection count.

Table 2.7. Confusion matrix for *Sousa plumbea* from both South Africa and India using a Random Forest algorithm based on the seven contour measurements. Overall model accuracy between South Africa and India was 89.7%. Classification of the test data is presented in percentages (%). Percentages in bold represent values of correct contour classification. Levels of chance for correct classification are presented alongside observed locations.

		Predicted	
		South Africa	India
Observed	South Africa (35.2%)	**84**	16
	India (64.8%)	7	**93**

1) Within South Africa

Contours from five study sites within South Africa, across two regions (south coast and east coast), were compared to investigate microgeographic and macrogeographic variation in acoustic behaviour across their South African range (Table 2.8). The highest maximum frequency was recorded in Mossel Bay (46.6 kHz), and the lowest minimum frequency was recorded in False Bay (1.4 kHz). Median start frequencies were all lower than their respective end frequencies, except for Richards Bay (median start frequency 9.8 kHz, median end frequency 9.2 kHz). The maximum whistle contour duration was recorded in Mossel Bay (2.1 s); however, there was little overall variation in median duration values across the sites (0.2 -0.4 s across the five sites). The number of inflection points was also similar across sites with median values of 0 or 1 across the sites, with the greatest number of inflection points recorded in Mossel Bay (11) (Figure 2.5).

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Table 2.8. Descriptive statistics were calculated for the seven whistle contour parameters of *Sousa plumbea*, measured across the five South African study sites. South coast study sites included False Bay, Struisbaai, St Sebastian Bay and Mossel Bay, and the east coast study site included Richards Bay. For each measurement, the minimum and maximum values, median (interquartile range, IQR), mean (standard deviation, SD), and coefficient of variance (CoV) % were calculated. Frequency parameters are in kHz and duration measured in seconds (s).

		South Coast				**East Coast**
		False Bay	**Struisbaai**	**St Sebastian Bay**	**Mossel Bay**	**Richards Bay**
(kHz)	n	164	117	158	562	180
Min Freq	Minimum	1.4	3.3	1.8	2.3	1.9
	Maximum	25.4	13.6	31.2	32.8	16.6
	Median (IQR)	5.8 (4.6 - 6.9)	5.1 (4.3 - 6.2)	7.1 (4.2 - 9.2)	6.1 (4.9 - 6.7)	6.5 (5.0 - 9.8)
	Mean (± SD)	6.5 (± 3.4)	5.5 (± 1.7)	7.5 (± 4.2)	6.0 (± 2.0)	7.3 (± 3)
	CoV	52.5	30.7	56.4	33.5	41.9
Max Freq	Minimum	3.4	5.6	3.7	4.7	5.9
	Maximum	39	21.5	41.2	46.6	20.5
	Median (IQR)	10.5 (9.1 - 12.6)	9.9 (9.5 - 10.5)	11.2 (6.5 - 12.4)	9.8 (8.8 - 12.4)	12.5 (8.1 - 15. 7)
	Mean (± SD)	11.9 (± 5.4)	10.3 (± 2.4)	11.5 (± 6.2)	10.9 (± 3.5)	12.3 (± 4.2)
	CoV	45.1	23.5	53.9	31.7	34
Start Freq	Minimum	1.8	3.2	3	2.8	2.1
	Maximum	25.4	21.5	32.2	33.9	21.1
	Median (IQR)	9.5 (6.9 - 11.6)	6.6 (4.7 - 9.2)	8.9 (5.2 - 11.0)	8.2 (6.4 - 11.0)	9.8 (5.4 - 14.7)
	Mean (± SD)	9.7 (± 3.9)	7.3 (± 3.4)	9.1 (± 4.8)	8.9 (± 3.3)	10.1 (± 5.2)
	CoV	40.7	46.4	53.1	36.8	51.8

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End Freq	Minimum	2.3	5.5	2.9	2.6	3.1
	Maximum	37.9	21.5	40.8	45.7	18.4
	Median (IQR)	9 (7. 8- 10.1)	9.4 (8.9 - 10.1)	10.1 (5.3 - 11.2)	8.6 (7.6 - 9.7)	9.2 (7.4 - 14.1)
	Mean (± SD)	10.0 (± 5.7)	9.7 (± 2.2)	10.0 (± 5.8)	8.9 (± 3.3)	10.5 (± 3.7)
	CoV	56.8	22.3	58.5	37.7	35.7
Freq Range	Minimum	1.4	1.3	0.7	0.8	1.5
	Maximum	20.3	12.1	16.3	21.8	12.9
	Median (IQR)	4.6 (3.3 - 6.4)	4.7 (3.8 - 5.6)	2.6 (1.7 - 5.1)	4.3 (3.4 - 6.0)	4.4 (3.0 - 6.3)
	Mean (± SD)	5.4 (± 3.2)	4.9 (± 1.8)	4.1 (± 3.6)	4.9 (± 2.5)	5 (± 2.6)
	CoV	58.8	37.6	88.1	48.5	52
Duration	Minimum	0.1	0.04	0.1	0.1	0.1
	Maximum	1.3	0.8	1.3	2.1	1.8
	Median (IQR)	0.2 (0.1 - 0.4)	0.3 (0.2 - 0.5)	0.2 (0.1 - 0.5)	0.4 (0.2 - 0.7)	0.2 (0.1 - 0.6)
	Mean (± SD)	0.3 (± 0.2)	0.3 (± 0.2)	0.4 (± 0.3)	0.5 (± 0.3)	0.3 (± 0.3)
	CoV	75.9	55.3	84.8	68.9	86.6
Inflection	Minimum	0	0	0	0	0
	Maximum	6	3	5	11	7
	Median (IQR)	1 (1 - 1)	1 (1 - 2)	0 (0 - 1)	1 (1 - 2)	1 (0 - 1)
	Mean (± SD)	1.1 (± 0.9)	1.3 (± 0.8)	0.9 (± 1.2)	1.4 (± 1.2)	1 (± 1.2)
	CoV	81.2	66.8	130.6	87.2	115.9

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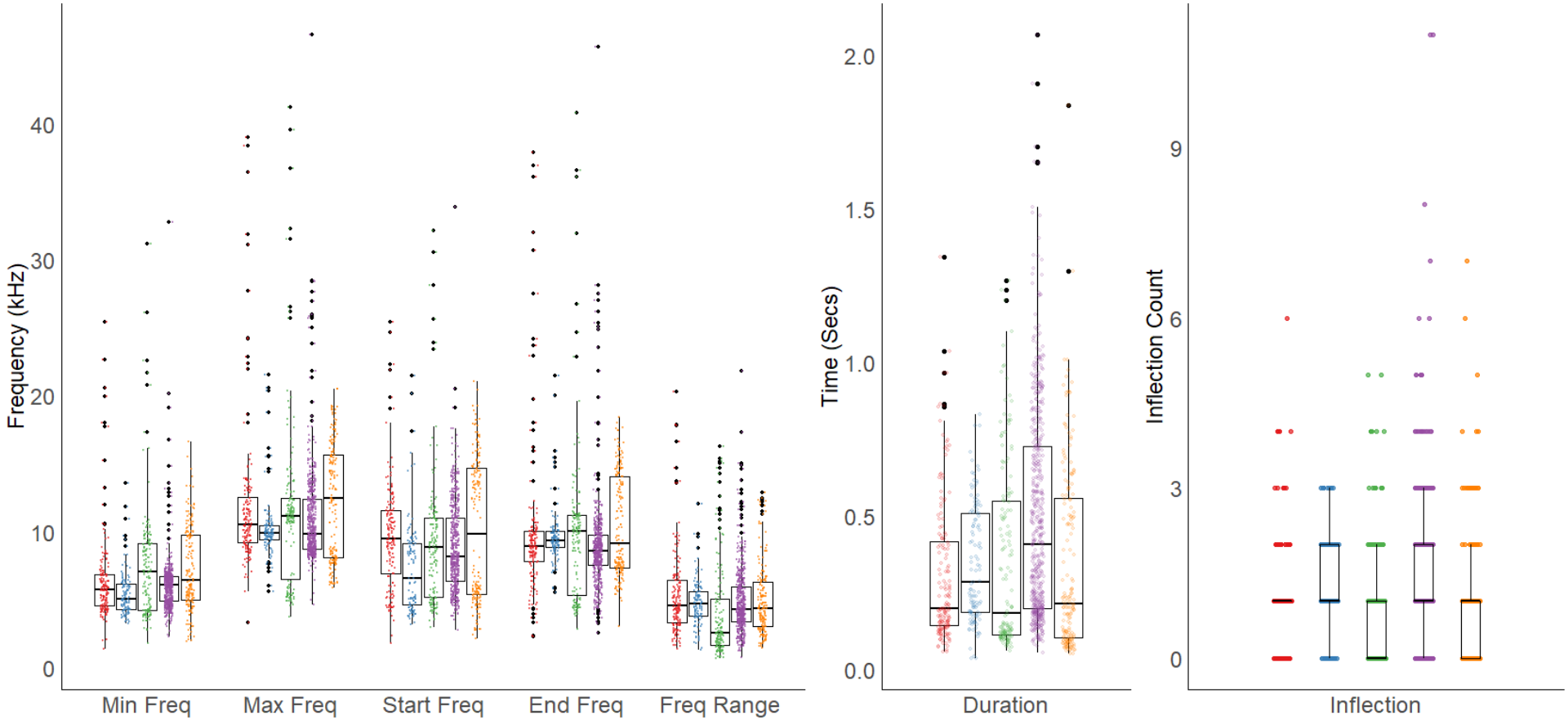
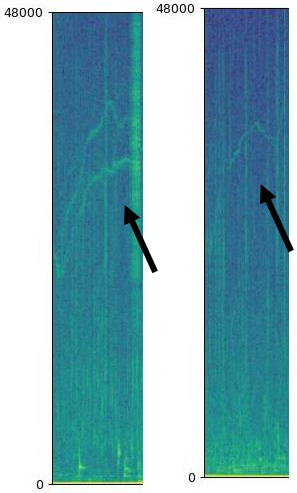


Figure 2.5. Variation of *Sousa plumbea* whistle parameters measured across the five South African study sites (From left to right and west to east; False Bay Red, Struisbaai Blue, St Sebastian Bay Green, Mossel Bay Purple and Richards Bay Orange). Coloured dots show the distribution of the data for each region. The five frequency-based measurements (minimum and maximum frequency, start and end frequency and frequency range) are presented on the left-hand side graph, duration (measured in seconds) is presented in the middle graph, and the Inflection count is on the right-side graph. In each plot, the box represents the inter-quartile range (IQR, middle 50%) of each data set, with centre lines representing median values, and the lines extending from the box (whiskers) represent the range within 1.5 times the IQR from the lower and upper quartiles. Points outside this range are considered outliers and shown as black dots.

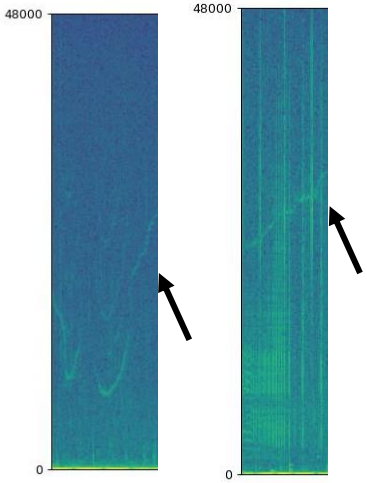
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Although the overall prevalence was low at 3.5% of all whistle contours across South Africa, ultrasonic contours were recorded at all five study sites and during most of the data collection period (2016 - 2021) (Figure 2.6). The maximum recorded frequency was recorded in Mossel Bay (46.6 kHz), with Mossel Bay and False Bay having the greatest proportion of ultrasonic whistle contours recorded (2.1% and 6.7%, respectively). Only two contours from Richards Bay were recorded as ultrasonic. Whistle contours with a maximum frequency of more than 30 kHz were recorded in False Bay, St Sebastian Bay and Mossel Bay.

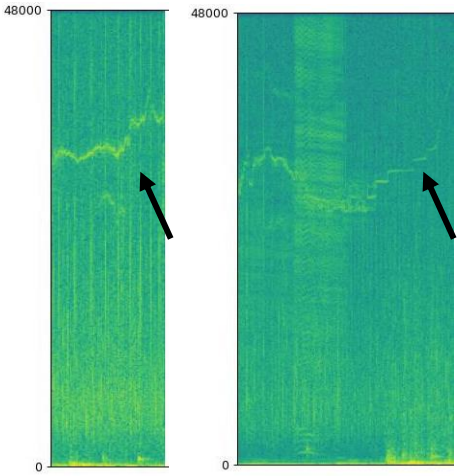
False Bay



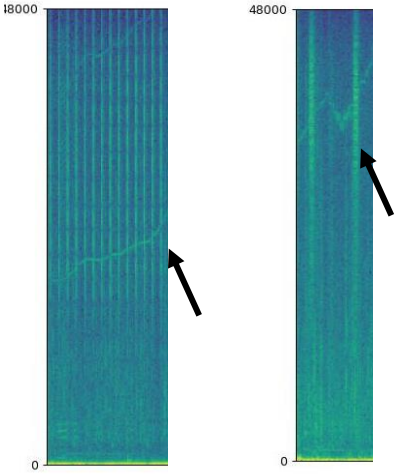
Struisbaai



St Sebastian Bay



Mossel Bay



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Richards Bay

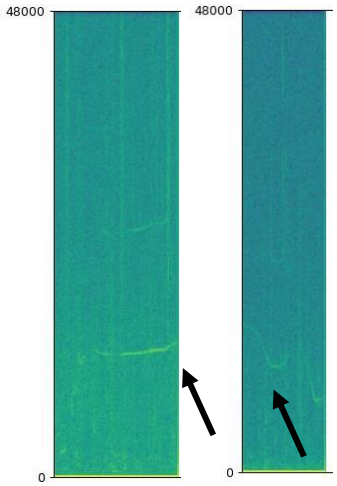


Figure 2.6. Two examples of ultrasonic contours in *Sousa plumbea* (defined as maximum frequencies exceeding 20 kHz), recorded from each of the five study sites in South Africa. For each example, the Y axes is 0- 48 kHz for all spectrogram examples. Spectrogram settings were brightness and contrast 50%, fast fourier transform (FFT) 1024 point Hann window.

The principal component analysis from within South Africa found the first two principal components accounted for 88.11% of the variation in the contour measurements. The first principal component accounted for a large proportion of the variation, 78.06% and as in the South Africa/India analysis, the maximum frequency had a high factor loading (0.58), along with end frequency (0.52). Similarly, in principal component two (accounting for 10.05% of variation) again, minimum frequency and frequency range had high factor loadings (0.5 and - 0.71 respectively) (Figure 2.7).

Table 2.9. Principal Component loadings for the seven contour measurements of *Sousa plumbea* whistle contours recorded across five study sites in South Africa (south coast; False Bay, Struisbaai, St Sebastian Bay, Mossel Bay and east coast; Richards Bay). Values in bold had loadings greater than 0.5 or less than - 0.5 Principal component one was responsible for 78.06% of the overall variance, with the first two principal components accounting for 88.11% of the overall variance.

	Components	
	PC1	PC2
Minimum Frequency	0.33	**0.50**
Maximum Frequency	**0.58**	-0.21
Start Frequency	0.48	-0.15
End Frequency	**0.52**	0.40
Frequency Range	0.25	**-0.71**
Duration	-0.01	-0.04
Inflection	0.01	-0.13
Eigen Value	3.76	1.54
% Variance explained	78.06%	10.05%

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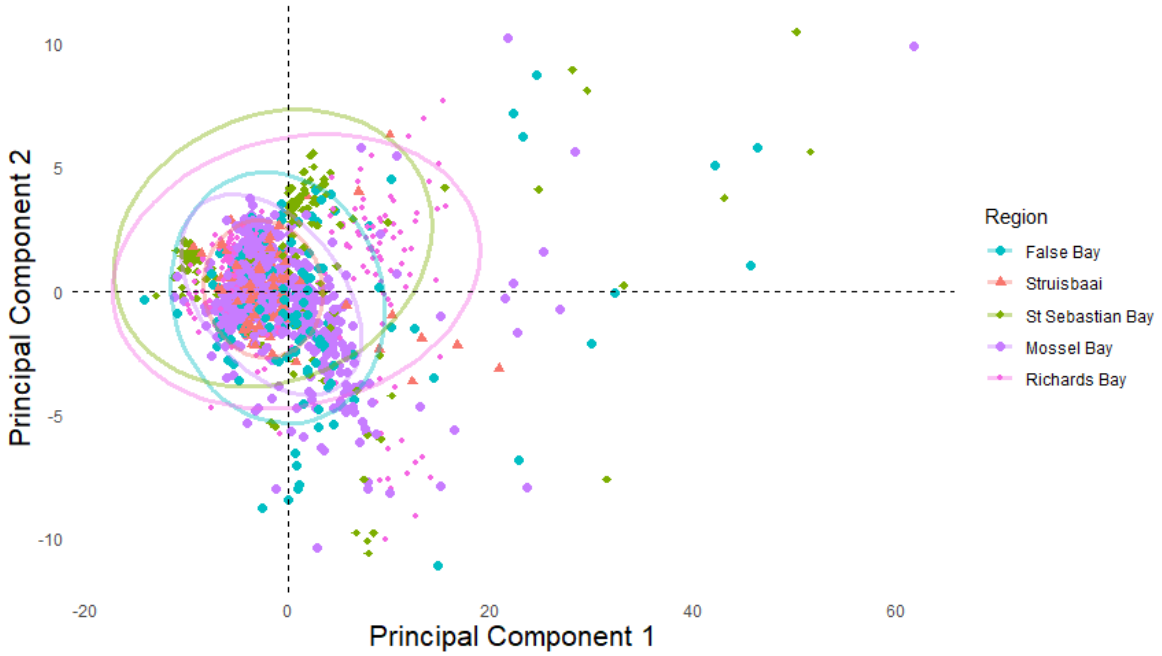


Figure 2.7. Principal component plot showing the first two components based on *Sousa plumbea* whistle contour measurements, derived from the seven measured parameters across the five South African study sites (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay and Richards Bay). The coloured ellipses contain contain 95% of data points for each population. A high degree of overlap (similarity) is found between study sites based on the first two principal components. Principal component one explains 78.06% of variance and component two explains 10.05%.

Using the subsampled Mossel Bay data, along with the full dataset from the other study sites, the total South African dataset used for the RF algorithm was 774 whistle contours. The random forest analysis was, therefore, based on a training dataset of 580 whistle contours and a test dataset of 194 whistle contours. The tuned random forest found a moderate overall accuracy for the model (65%). Correct classification was imbalanced across South Africa, with False Bay having the lowest correct classification rate (39%), while correct classification ranged from 58% - 68% for the other three south coast sites (Table 2.10). Misclassification rates were high across the south coast sites with misclassification highest in non-neighbouring sites for all four south coast sites. Richards Bay had the highest correct classification rate at 90%. These suggest that Richards Bay contours are more distinct, while south coast sites have a greater level of misclassification, (such as False Bay; Struisbaai (11%) and Mossel Bay (41%)). A moderate Cohens Kappa test statistic (55%) further indicates low agreement between the predicted and actual classes in the test dataset. Whistle characteristics with high mean decrease accuracy (i.e. maximum frequency, end frequency and duration) contributed to differentiating between South African study sites the most.

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Table 2.10. Confusion matrix from the Random Forest algorithm for *Sousa plumbea* across the five South African study sites (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay and Richards Bay). Values in bold represent the percentage of correct classification of contours per study site. Distances between neighbouring study sites are included from west to east (km). Overall model accuracy was 65%. Classification is presented as percentages (%). Levels of chance for correct classification are presented alongside observed locations.

		Predicted				
	Distance from the western site	**0 km**	**~215 km**	**~125 km**	**~140 km**	**~1400 km**
Observed		FB	SB	SSB	MB	RB
	FB (24%)	**39**	11	2	41	7
	SB (13%)	4	**58**	8	8	23
	SSB (19%)	14	6	**67**	11	3
	MB (19%)	14	0	8	**68**	11
	RB (25%)	2	0	4	4	**90**

In comparison to published literature on whistle parameters in the genus *Sousa*, the mean maximum frequencies for South Africa (11.3 kHz ± 4.3 SD) are higher than *S. teuszii* (8.2 kHz ± 3.1 SD Weir, 2010), and *S. chinensis* in China (7.26 kHz ± 4.13 SD Dong et al., 2021 and 7.69 kHz ± 3.77 SD Wang et al., 2013) (Table 2.11). Only *S. chinensis* in Malaysia and *S. sahulensis* in Australia have reported higher mean maximum frequencies than South Africa (13.4 kHz ± 3.7 SD Hoffman et al., 2015 and 14.38 kHz ± 14.38 SD Van Parijs & Corkeron, 2001b, 12.3 kHz ± 4.1 SD Berg Soto et al., 2014). Reported mean durations throughout the genus are similar, ranging from 0.16 secs (*S. sahulensis*, Van Parijs and Cockeron, 2001b) – 0.76 secs (*S. teuszii* Weir, 2010). Inflection also shows little variation across the genus with South Africa, yet the largest mean inflection points were reported from *S. plumbea* (1.2 ± 1.1 SD This study and 1.8 ± 1.7 SD Bopardikar et al., 2018). General conclusions from comparisons between studies are limited due to variations in sampling rate, scope of studies and survey effort.

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Table 2.11. Literature summary of whistle measurements across the genus *Sousa*. Studies range in their sample size, sampling rate and scope of measurements reported but available values are presented. Seven whistle measurements are included for general comparison across species. These measurements include maximum frequency (Max Freq), minimum frequency (Min Freq), frequency range (Freq Range), start frequency (Start Freq) and end frequency (End Freq), measured in kilohertz (kHz). Duration is measured in seconds (s) and inflection is measured in count of inflection points.

		Sousa teuszii	***Sousa plumbea***			***Sousa chinensis***							***Sousa sahulensis***			
	Location	Angola	South Africa	India	Pakistan	Matang, Malaysia	Langkawi, Malaysia	China	China	Singapore	China	China	Australia	Australia	Australia	Australia
	Publication	Weir, 2010	Dines, This study	Bopardikar et al., 2018	Zbinden et al., 1977	Hoffman et al., 2015	Hoffman et al., 2015	Dong et al., 2021	Wang et al., 2013	Seekings et al., 2010	Sims et al., 2012	Dong et al., 2019	Schultz & Corkeron, 1994	Van Parijs & Corkeron, 2001b	Van Parijs & Corkeron, 2001a	Berg Soto et al., 2014
	Sample Rate	<44 kHz	>96 kHz	96 kHz	70 kHz	96 kHz	96 kHz	128 kHz	20 kHz - 80 kHz	up to 24 kHz	20 kHz - 80 kHz	128 kHz	30 kHz - 16 kHz	20 kHz - 22 kHz	20 kHz - 22 kHz	-
	n	86	1181	2260	117	960	823	226	2651	1166	195	529	667	320-329	214	1024
Max Freq	Range	5.1-23.4	3.4-46.6	3.1-33	25 (Max)	3.7-23.2	1.8-27.1	1.73-21.35	0.56-33	-	24.9 (Max)	1.92-21.68	-	1.8-22	-	4.18-21.78
	mean	8.2	11.3	10.2	-	13.4	11.8	7.26	7.69	-	-	7.83	-	10.58	14.38	12.3
	SD	3.1	4.3	3.3	-	3.7	3.1	4.13	3.77	-	-	3.49	-	4.25	1.62	4.1
	Cov	-	38	33	-	28.01	26.28	57	49.4	-	-	-	-	-	-	-
Min Freq	Range	2.5-12.6	1.4-32.8	2.3-32.7	4 (Max)	3.0-21.2	1.2-26.5	0.76-12.17	0.52-21.19	-	4.1 (Max)	1.79-17.63	1.2 (Max)	0.9-20	-	1.66-16.2
	mean	4.8	6.4	7.6	-	8.8	8.4	4.2	5.07	-	-	5.55	-	7.62	5.7	6.4
	SD	2.3	2.8	2.8	-	3.3	2.8	1.64	2.18	-	-	2.63	-	3.35	0.89	2
	Cov	-	44	37.3	-	37.86	33.32	39	43.11	-	-	-	-	-	-	-
Freq Range	min-max	1-14.8	0.7-21.8	0-14.9	-	0.3-15.2	0.2-13.9	0.17-15.8	0-21.25	-	-	0-12.16	-	-	-	-
	mean	3.4	4.9	2.7	-	4.5	3.4	3.06	2.63	-	-	2.28	-	-	-	-
	SD	1.8	2.7	2.2	-	3	2.7	3.46	2.88	-	-	2	-	-	-	-
	Cov	-	55.4	83.9	-	66.47	80.4	113.1	109.8	-	-	-	-	-	-	-
Duration	min	0.05	0.04	0.01	0.04	0.01	0.02	60	0.02	0.16	-	0.03	0.04	0.2	-	0.03
	max	2.21	2.1	1.6	1.2	1.6	1.3	2222	2.9	1.36	-	1.9	0.36	1.3	-	1.26
	mean	0.76	0.4	0.1	0.2	0.2	0.2	472.39	0.37	0.8	-	0.42	-	0.16	4370	0.25

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	SD	0.47	0.3	0.1	0.1	0.2	0.2	339.7	0.29	-	-	0.3	-	0.13	2110	0.2
	Cov	-	76	101.8	-	96.71	94.85	71.9	77.15	-	-	-	-	-	-	-
Start Freq	min	2.8	1.8	2.3	-	14.2	1.6	0.71	0.47	-	-	1.79	-	0.9	-	-
	max	21	33.9	33	-	22.6	26.5	17.3	24.38	-	-	19.33	-	21	-	-
	mean	6	9.1	9.2	-	10.2	10.9	5.19	6.51	-	-	6.75	-	8.51	5.89	-
	SD	3.4	4	3.3	-	3.8	3.3	2.45	3.5	-	-	3.36	-	3.77	0.88	-
	Cov	-	44.4	35.9	-	37.86	30.73	47.2	53.76	-	-	-	-	-	-	-
End Freq	min	2.5	2.3	2.8	-	3.3	0.5	1.04	0.56	-	-	1.92	-	1.8	-	-
	max	23.4	45.7	32.8	-	23.2	27.2	19.22	25	-	-	20.73	-	22	-	-
	mean	6.3	9.5	9.6	-	12	10.3	6.27	6.13	-	-	6.54	-	10.1	7.49	-
	SD	3.9	4.2	3.4	-	4	2.9	3.8	3.01	-	-	3.13	-	3.91	2.99	-
	Cov	-	43.7	35.4	-	33.59	28.13	60.7	49.21	-	-	-	-	-	-	-
Inflection	Range	1-7	0-11	0-19	-	0-15	0-12	0-6	0-9	-	-	0-7	-	-	-	0-7
	mean	1.3	1.2	1.8	-	1.1	1.1	0.66	0.6	-	-	0.75	-	0.8	-	0.85
	SD	0.9	1.1	1.7	-	1.5	1.3	1.27	1	-	-	1.29	-	0.8	-	0.88
	Cov	-	94	172.9	-	141.63	111.17	193.1	172.04	-	-	-	-	-	-	-

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DISCUSSION

This study describes the general characteristics of *Sousa plumbea* whistle contours within their Indian Ocean range (South Africa and India) and compares these between South African regions to investigate the degree of macro and microgeographic variation within the species. The whistle contour measurements for South Africa fall within the range of those described throughout the *Sousa* genus globally. Although there is a strong overlap in variability in whistle contour time-frequency characteristics between each range, there was evidence of macrogeographic variation between South Africa and India and moderate evidence of macrogeographic variation between the south and east coasts of South Africa. Whereas there was very little microgeographic variation along the south coast of South Africa. Although there were distinctions between the populations, the strong overlap between whistle measurements indicates characteristics constrained by morphology with acoustic behavioural adaptations based on environmental effects.

South Africa and India are two populations at the furthest west and east of the species' Indian Ocean range, over 1400 km apart, showing the greatest level of distinction between the different geographic scales. Frequency parameters (maximum frequency, start and end frequency) accounted for the highest level of variability in the data, yet there was a strong overlap in general whistle unit measurements. Inflection, duration and frequency range were the most important measurements for accurate classification. Frequency parameters are more innate and species-specific, limited by morphological constraints and therefore variability in these measurements is recorded within the range expected within the genus (Frainer et al., 2019, Table 2.8). Inflection and duration, however, are more variable whistle contour features, reflecting differences in whistle characteristics due to behavioural states (Esch et al., 2009) or group composition during encounters (Ansmann et al., 2007). In South Africa, *S. plumbea* were recorded in small, single species groups with groups of 1-10 individuals, while Indian group sizes were larger, around 20-30, with a maximum group size of 110. Encounters in India were driven around foraging groups within the lagoon and estuarine environments (Table 2.2). Higher modulations and

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complexity in tonal sounds show a stronger correlation to group size and social structure (May-Collado & Wartzok, 2008). Therefore, the distinctiveness between South Africa and India is likely due to differences in group composition and behaviour driving changes to contour duration and whistle complexity (inflection point count).

Comparison between the south and east coast sites indicated that Richards Bay was more distinct than the south coast sites with maximum frequency an important variable for both model accuracy and classification. This supports the two-population hypothesis in that acoustic variation is greater between the south and east coast populations than seen within the south coast population, likely due to increased geographical isolation. Using mtDNA, Lampert et al., (2021) have shown these two populations to form a single population historically, however, this study was limited by a lack of samples from the south coast and more recent genetic structuring may be resolved should other genetic markers be investigated. Range contraction and habitat fragmentation likely split these into geographically distinct populations (Vermeulen et al., 2018). Further population declines have continued to separate these two populations, with little mixing of the two existing today (Caputo et al., 2017). Frequency measurements (maximum frequency, end frequency and start frequency) and duration were important variables for group classification. Maximum frequency also had a higher degree of variability which can be seen in the box and whisker plots, with the maximum frequency medians the most diverse of all the whistle measurements included. The acoustic adaptation hypothesis could explain geographic variation between the south and east coast regions, where acoustic signals are adapted to their environment to minimise degradation and maximise signal transmission and long-range communication (Hardt & Benedict, 2020). Environmental factors such as water depth, temperature and salinity can all affect the transmission and receiving of sound, which can influence the transmission of whistles (Mercado & Frazer, 1998). Richards Bay is an enclosed bay with an industrialised harbour and is a particularly important area for *S. plumbea* in KwaZulu-Natal (KZN), with sightings rates reported to be higher than in other areas in KZN (Durham, 1994; Keith et al., 2013). This could explain the differences between these south and east coast

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populations based on the habitat characteristics such as noise and depth that differ between these regions. Maximum frequencies were one of the strongest discriminating variables between the two regions. Populations that frequent shallower waters, with little wave action, have shown acoustic adaptation to favour higher frequency parameters (Forrest et al., 1993). Water temperature and salinity, although not measured within this study, likely also affect whistle transmission, with Richards Bay situated in a subtropical region along the east coast with warmer waters and higher salinity than the south coast (Van Niekerk et al., 2022). As well as habitat differences, the presence of boat activity has shown to cause dolphins to modify their vocalisations and acoustic behaviour (May-Collado & Quiñones-Lebrón, 2014; Perez-Ortega et al., 2021). Delphinids can adapt to underwater noise by 1) increasing source levels, or 2) shifting frequency or temporal parameters of signals to avoid masking of signals (Tyack & Janik, 2013). Boat activity has been seen to cause an upward shift in whistle frequency in the presence of boats (Heiler et al., 2016; Papale et al., 2015; Perez-Ortega et al., 2021). Guerra et al., (2014) also found bottlenose dolphins shifted to shorter whistle durations in the presence of boats when calves were absent from the group. Likely the high use of the harbour environment and habitat characteristics contributed to the distinctiveness between the east and south coast populations.

Within South Africa, the known mixing of individuals along the south coast likely explains the lack of microgeographic variation, supported by the acoustic similarity seen amongst south coast study sites. Individuals are known to move between False Bay and Mossel Bay (Vermeulen et al., 2018), with the same individuals detected across multi-year encounters (presented in Appendix 1). The known detection of individuals between south coast study sites may have been a cause for increased similarity between these sites with certain repeated contours driving levels of similarity between sites. This could be through both recording the same individuals or in that they share an acoustic environment, leading to vocal learning of their whistle features (Janik & Slater, 1997). Vermeulen et al., (2018) investigated movements of *S. plumbea* in their South African coastal range using photo ID and found dolphins were observed moving throughout the south coastline, but most matches were found in the central area between Mossel

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Bay and Plettenberg Bay, a site further east. The lack of south coast study sites further east of Mossel Bay, restricts this study's ability to investigate the effects of this further. This degree of movement could explain the higher misclassification rate amongst south coast sites, yet the degree of misclassification of south study sites with Richards Bay indicates a degree of overlap in general throughout the two populations, likely due to general species-specific whistle features.

In describing whistle contour characteristics throughout the South African population of *S. plumbea*, the study recorded ultrasonic whistles across their South African range, exceeding the maximum frequencies published across other studies within the *Sousa* genus. High-frequency contour production is a characteristic across the genus with ultrasonic frequencies (>20 kHz) in species such as *S. teuszii* (Weir, 2010), *S. chinensis* (Dong et al., 2019; Sims et al., 2012) and *S. sahulensis* (Van Parijs & Corkeron, 2001b). These results extend this range to *S. plumbea's* furthest westerly range in South Africa, and the study reports the highest frequency of *Sousa* whistles to date. Although infrequent, high-frequency contours (> 40 kHz) were recorded more frequently in South Africa than India and recorded from all five sites in South Africa. The maximum recorded contour frequency reached 46.6 kHz and was recorded in Mossel Bay in 2019. Morphologically, *Sousa plumbea* has a specialised structure next to the melon's small left posterior branch, associated with higher frequencies and sound directionality (Frainer et al., 2019). This structure is present in several other species that also feature high-frequency characteristics, including Spinner dolphins (*Stenella longirostris*), Risso's dolphins (*Grampus griseus*), white-beaked dolphins (*Lagenorhynchus albirostris*), and Franciscana dolphins (*Pontoporia blainvillei*) (Frainer et al., 2019). Equipment limitations and limited encounter durations in previous studies may mean these upper bandwidth capabilities are widespread within this genus but were either not recorded (Schultz & Corkeron, 1994; Van Parijs & Corkeron, 2001b) or investigated across other studies (Sims et al., 2012; Zbinden et al., 1977). Although ultrasonic capabilities are present across the genus, general frequency parameters vary, with *S. chinensis* in China (Dong et al., 2019; Sims et al., 2012; Wang et al., 2013) having lower mean maximum frequencies (7.69 kHz – 7.83 kHz) than *S. chinensis* in Malaysia (11.8 kHz – 13.4

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kHz, (Hoffman et al., 2015). Ultrasonic whistles have been recorded in other species, with whistles ranging up to 75 kHz in Northeast Atlantic killer whales (*Orcinus orca*) (Samarra et al., 2010), with the occurrence of these ultrasonic whistles attributed to both social and non-social behaviour or random individual variation (May-Collado & Wartzok, 2008; Samarra et al., 2015). Ultrasonic whistles in the freshwater dolphin, the Boto (*Inia geoffrensis geoffrensis*), with maximum frequencies up to 48.10 kHz, are produced within groups when engaged in social or feeding activities (May-Collado & Wartzok, 2007). Due to available recording equipment, the presence of ultrasonic whistles throughout South Africa is likely a common feature within the genus' morphological capabilities, with limited detection in previous studies. Further investigation is needed into the behavioural and group composition around the production of these whistles in *S. plumbea*.

The limited number of *S. plumbea* encounters and variation in sampling efforts across sites likely limited the degree of variation seen across South Africa. Across 207 surveys, *S. plumbea* was encountered on 83 occasions, with acoustic data available for 64, which was further reduced to 21 encounters based on the quality of acoustic recordings (including only using single-species encounters). The thresholds used to include acoustic encounters in the study may have diminished the diversity of acoustic encounters. Variation in the acoustic repertoire is known to be influenced by group size and behaviours in other species (Bazúa-Durán & Au, 2002; May-Collado & Wartzok; 2008). Overall, very high research effort provided data where only a quarter of encounters provided adequate acoustic recordings, and only ~ 50% of whistles recorded were suitable for further analysis. Low-quality whistle contours were present in the data due to low signal-to-noise ratios of contours (distant individuals, low amplitude calls) and signal masking due to snapping shrimp, reef noise and ambient noise (such as bouy chains). Therefore, applying a minimum threshold for contours was necessary to ensure the accuracy of the data. This high effort, mixed with the threshold for whistle contour quality, demonstrates the substantial effort required to research this endangered species and is likely a contributing factor to the limited data available on their acoustic repertoire throughout their distribution.

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In conclusion, this is the first study to investigate geographical variation in whistle contours from *S. plumbea* at both macro and microgeographic scales in South Africa and India. Results indicate a degree of increasing variation with increasing geographic isolation throughout the study range. The greatest variation between South Africa and India was observed in frequency measurements, particularly in the maximum recorded frequencies. The least variation was demonstrated amongst the four south coast sites in South Africa, which is likely attributed to the regular mixing of individuals throughout this range. A general overlap in whistle features throughout their South African range supports the use of whistle characteristics for acoustic monitoring.

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3 GROUP COMPOSITION AND BEHAVIOUR EFFECTS ON SIGNATURE WHISTLE PRODUCTION IN THE INDIAN OCEAN HUMPBACK DOLPHIN, *SOUSA PLUMBEA*

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Prepared for publication in Bioacoustics Journal

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ABSTRACT

In South Africa, the Indian Ocean Humpback dolphin, *Sousa plumbea,* is the country's most threatened marine mammal, with a small, fragmented distribution along their coastal range. Effective passive acoustic monitoring of this species requires understanding their individual acoustic behaviour and how variables such as group composition and behavioural states might affect their detection. Over seven years (2016 – 2023), boat-based focal follows were conducted of coastal dolphins, including *S. plumbea,* across five South African study sites (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay and Richards Bay). Of the 22 *S. plumbea* encounters with concurrent behavioural and GPS track data, 21 were suitable for further analysis (after the removal of one outlier encounter). The study investigated whether 1) number of signature whistle types (SWTs, number of individuals), 2) the total count of signature whistles (overall SW production) and 3) temporal production (inter-signature whistle intervals, ISWI) varied with group behaviour and composition, as these changes could impact individual detection probability within a capture-recapture framework. Of the 1382 total *S. plumbea* whistles identified, 791 had moderate or good signal-to-noise ratio. These were further analysed in a signature whistle framework using a bout analysis approach to identify 20 SWTs (proxy for individuals). The median rate of SWTs detected in an encounter was 0.03 SWTs/min (IQR 0-0.06). Across these encounters, 392 signature whistles were detected, with a median SW production rate of 0.14 SW/min (IQR 0-0.49) per encounter. Generalised linear mixed models found no effect of behaviour or group composition on SWT occurrence or SW production. Only ISWI showed a temporal effect, with group size and group spread showing a decreased signature whistle production, whereas calf presence increased signature whistle production. These results indicate little inference on behaviour and group composition based on whistle production. However, a broader sample size of SWT and SWs would likely provide stronger results. When designing capture-recapture studies using signature whistles in this species, monitoring group composition and behavioural effects are likely less important considerations when considering optimal detection design. Although behaviour and context are demonstrated to affect acoustic behaviour and SW detection in other dolphin species, similar effects are either not present, sample size-dependent, or masked by the overall low whistle production rates in *S. plumbea*.

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INTRODUCTION

Animal behaviour is known to be influenced by activity and group composition, with individuals adapting acoustically to these dynamic settings (Manning & Dawkins, 2012). Monitoring changes to acoustic behaviour under different contexts can inform us about changes to the level of stress or arousal and these have been used as a marker for welfare in domesticated and captive animals (Esch et al., 2009a; Laurijs et al., 2021, Probert et al., 2021). Variations in acoustic behaviour can include changes to signal amplitude (Nemeth & Brumm, 2009), the addition of spectral characteristics (Townsend & Manser, 2011) or a change in temporal production rate (Digby et al., 2013). In odontocetes, behavioural state, group composition and group size (Nowacek, 2005; Quick & Janik, 2008; Scarpaci et al., 2000) are known to influence whistle production.

Traditional capture-recapture studies investigate the population size of species through trapping and marking individuals to understand the rate of recaptures over several sampling occasions (Amstrup, 2010). The effectiveness of different markers used in these studies can often depend on the characteristics and behaviour of the species (Carter et al., 2012; Wegge et al., 2004). The use of physical tags requires capturing individuals, and for rare and elusive species, this is difficult and time-consuming. In recent years, acoustic monitoring of animals using individually distinct calls has been suggested as an alternative to tagging or photographic methods for capture-recapture studies (Longden et al., 2020; Terry et al., 2005). However, as with all methods, using naturally occurring sounds as detection markers has inherent biases, such as varying levels of detection probability between species, groups, and individuals. The detection probability of an individual, based on the production of individual calls, may be influenced by both intrinsic (arousal, hunger, health) and extrinsic (group behaviour and composition, ambient noise, weather), as well as life history stage and individual personality (Briefer, 2020). Understanding what influences individual acoustic production is key when designing the timing and locations of long-term passive acoustic arrays.

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In some delphinids, identity is encoded in unique signature whistles and is best studied in bottlenose dolphins (Janik, 2009). Signature whistles are developed through vocal learning and experience within an individual's environment (Janik, 2009). Signature whistles have a high level of individual distinctiveness in frequency modulation patterns, independent of voice cues (Janik et al., 2006). They are formed during the first few weeks as a calf and are shown to have long-term stability of up to at least 43 years (Sayigh et al., 2022). Signature whistles are behaviour-dependent and act as contact calls between conspecifics, used to maintain long-term associations and group cohesion within a fission-fusion society (Janik & Slater, 2000). Signature whistles are most widely documented in *Tursiops* species, with over 60 years of research (Caldwell & Caldwell, 1965; Janik & Sayigh, 2013). Over the decades, evidence has accumulated for signature whistle use across other odontocete species (*Tursiops aduncus* Gridley et al., 2014, *Stenella longirostris* Rio, 2023, *Stenella plagiodon* Caldwell et al., 1973, *Delphinus delphis* Fearey et al., 2019, *Sousa chinensis* Van Parijs & Corkeron, 2001, *Pseudorca crassidens* Rio & Rosales-Nanduca, 2024, *Lagenorhynchus obliquidens* Caldwell et al., 1971, *Sotalia guianensis* Lima & Le Pendu, 2014, and *Monodon monoceros* Shapiro, 2006).

Identification of an individuals' unique signature whistle type is best done using recordings of a particular animal during temporary capture events or isolation in captivity, where signature whistles are often produced at a high rate and operationally defined as the most produced whistle contour in this context (Janik, 2009b). In these circumstances, signature whistles account for up to 80 - 100% of the whistle repertoire (Caldwell et al., 1990). In free-ranging dolphins, identification of signature whistles is more difficult to ascertain due to the difficulty in identification of signallers. In the wild, signature whistles only contribute between 38% and 70% of whistles in freely interacting *Tursiops* groups (Janik, 2009a). Identifying signature whistles can be particularly challenging, especially in freely interacting wild populations (Janik, 2013). As dolphins produce both signature whistles and a variety of non-signature whistles, linking individual whistles to the signaller in a wild context and identifying which whistles are signatures is difficult without the use of localisation (Quick & Janik, 2008). Janik et al., (2013) developed

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a bout analysis method termed SIGnature IDentification (SIGID) to describe likely signature whistles in wild dolphins through the use of a single hydrophone. Using this method, developed for *T. truncatus*, a whistle type is classified as a signature whistle when 4 or more highly similar whistles recorded are from the same encounter, where on at least one occasion, the whistles were produced in a sequence with 75% or more (i.e. minimum 3 out of 4) occurring within 1 to 10 s (Janik et al., 2013; Kriesell et al., 2014). As acoustic behaviour differs between species (Kaplan & Reiss, 2017), the application of SIGID (i.e. bout criteria) may need to be relaxed or made more stringent when working with other species, and this is an important consideration when identifying new signature whistles in freely interacting groups (see Fearey et al., 2019, *D. delphis*).

Signature whistles can be used to communicate identity and maintain individual relationships in fission-fusion social structures, meaning signature whistle production is often linked to group size (Janik, 2009a). For smaller delphinid groups in the wild, whistle production generally increases as group size increases (Cook et al., 2004; Jones & Sayigh, 2002) up to about 10 animals in a group, after which individual whistle production starts to decrease in groups above 15 individuals due to increased potential masking by other whistles (Quick & Janik, 2008). However, this is not true across all studies. In Portugal, dos Santos et al., (2005) found no effect of changes to group size on signature whistle production per individual**,** indicating differences in effects on production across *Tursiops* spp. populations. In larger delphinid groups, whistle production varies, with species adopting a greater variety of communication methods in acoustically complex soundscapes. In Spinner dolphins (*Stenella longirostris*), larger pods have one individual within a subgroup who will call at a time and favour short-range burst pulses (Lammers et al., 2006). In some environments, adaptation is not from the signaller but rather from the signal receiver. Common dolphins (*Delphinus delphis*), which have group sizes of 500+ individuals, create an acoustically cluttered environment where masking of calls is likely (Fearey et al., 2019). Nityananda & Bee, (2011) refer to this as the 'cocktail party effect' and describe the ability of individuals to differentiate a single auditory signal in the presence of a high amount of background noise. King penguin

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(*Aptenodytes patagonicus*) chicks have also been shown to be able to perceive parental identity information amongst large colonies within a very noisy soundscape (Aubin & Jouventin, 1998).

With signature whistles used to communicate identity and maintain individual relationships in fission-fusion social structures, they are often linked to production under different group compositions (Quick & Janik, 2008). Whistles, in general, are well suited to longer-range communication as they are relatively omnidirectional in comparison to other odontocete call types and can propagate over longer distances than clicks or burst pulses (Jones et al., 2020). This makes whistles more likely to be used in larger groups where animals have the potential to be more dispersed (Cook et al., 2004). This idea was supported by Lammers et al., (2006), who found that in Hawaiian spinner dolphins, whistles were more commonly produced by dolphins spread further apart, suggesting whistles were used to maintain contact in more dispersed groups. Specifically looking at signature whistle use, in their role as a cohesion call, *Tursiops* spp. produce signature whistles when different groups meet at sea and when they join to form single groups or exchange individuals, reflective of their fission-fusion group structure. However, (Quick & Janik, 2012) noted that typically, only one individual from either group would call, leading to potential variability in the effect of group composition across species.

One of the most important functions of signature whistles is in mother-calf pairs to maintain contact and facilitate recognition when separated (Sayigh et al., 1990; Smolker et al., 1993; Tyack, 1997). In *Tursiops* species, higher call rates are seen in groups with calves present (King et al., 2016; Kriesell et al., 2014). Mothers significantly increase their signature whistle production in the first few weeks of the calf's life to initiate the recognition by calves within a group context (King et al., 2016). Production rates are also seen to increase during the separation of mother-calf pairs, with the rate of calls increasing with the distance of separation (King et al., 2016), as demonstrated in both captive or restrained (King et al., 2016; Sayigh et al., 1990) contexts and in wild, and freely ranging *T. aduncus* (Smolker et al., 1993). Mothers and calves are strongly bonded and when separate would need to remain in contact/ reunite once

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separated and, therefore, determine one another's positions quickly, i.e. under any condition, such as at night, in low visibility, or when threatened by predators, supporting the use of signature whistles to locate and maintain contact with others (Mello et al., 2005).

Lastly, group behaviour effects on signature whistle production can be evidenced in both captive and wild environments (Cook et al., 2004). In captivity, higher arousal levels from human-induced activity can affect signature whistle production (Probert et al., 2021b). While natural variability, as seen in the wild, due to cooperative behaviours, such as socialising and feeding, can also effect signature whistle production (La Manna et al., 2019). Dolphins produce signature whistles at higher rates when under stress, presumably as animals attempt to alert conspecifics to changes such as changing water levels in pools (Rachinas-Lopes et al., 2017) or during temporary restraint (Esch, Sayigh, Blum, et al., 2009b). Higher production rates are also seen due to arousal from scheduled activities and associated activity cues, such as feeding and training times (Probert et al., 2021b). During entrapment events, whistle call rates in *Tursiops* species ranged from 5 - 32.9 wh/min (Gridley et al., 2014), while other capture events indicate median call rates of 10.9 (King & Janik, 2013) and 14.3 wh/min (Esch, Sayigh, & Wells, 2009b). Outside of these specific, high arousal contexts, in free-swimming bottlenose dolphins, Quick & Janik (2008), found the mean general whistle production rate for individuals to be lower at 1 wh/2 min when travelling or 9.8 wh/2 min when engaged in social contexts. Higher whistle production rates have been recorded in the wild as groups meet at sea, commonly seen in species with fission-fusion social systems (Quick & Janik, 2012). Direct comparison between studies is difficult due to different group sizes, environments, and behaviours, as well as studies measuring general whistles or signature whistles. However, across studies, increased arousal (including stress) or socially cooperative events (such as co-operative feeding) elevate whistling rates compared to general polarised movements such as travel (Oswald et al., 2008).

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In South Africa, the Indian Ocean humpback dolphin (*Sousa plumbea*) exhibits elusive life history traits (Karczmarski et al., 1997; Koper et al., 2016). They are split into two geographically distinct populations, with a total population of less than 500 and average movements of less than 200 km (Vermeulen et al., 2018). Their limited range is extremely coastal, only extending up to 5 km offshore, and a maximum of 20 m water depth (Saayman & Tayler, 1973). Group sizes are typically small, especially compared with sympatric delphinid species such as *T. aduncus* and *D. delphis* (Gridley, 2010; Reisinger & Karczmarski, 2010). Group sizes have decreased over the past decade with the number of solitary animals increasing (Karczmarski, 1999). Although they occur in sympatry with several other whistling delphinid species (*T. aduncus, D. delphis, O. orca*), direct interactions are rare (Frainer et al., 2023). Conservation efforts for this species in South Africa are hampered by the difficulties in monitoring this species using standard boat-based methods. Their small population, low group sizes, and their propensity for shallow surf zones make boat-based focal follows difficult. Capture-recapture using individually distinctive calls has therefore been suggested to acoustically monitor elusive species (Janik et al., 2013; Terry et al., 2005). Signature whistles have previously been recognised in other *Sousa* species under restrained or stranded contexts (Cheng et al., 2017; Van Parijs & Corkeron, 2001) with further evidence of signature whistles in *S. plumbea* from Gridley et al., (Appendix 1) and Dines et al., (Appendix 2). If using signature whistles as a proxy for individual detection and monitoring along coastal habitats is possible, understanding variation in whistle production could be key to designing the locations and timing (such as duty cycling) within a passive acoustic array for effective detection probability.

This study explores individual whistle production of *S. plumbea* to understand which group composition and behavioural contexts influence individual whistling behaviour in this species. Whistle production was investigated in three ways 1) SWT; count of signature whistle types, effectively the number of individuals detected, 2) SW; count of total signature whistles produced, effectively the total group signature whistle production in each encounter, and 3) IWSI; inter signature whistle intervals, the temporal production rate of signature whistles for each individual (for whistle terms used throughout

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study see Table 3.1). The study uses group size, group spread, calf presence and surface behaviour as covariates to investigate how these may influence signature whistle production.

Hypotheses

1) Group Size — As group sizes increase, the count of signature whistle types present during an encounter are expected to increase (i.e., the number of individuals calling), as are the number of whistles produced and their temporal production.
2) Group spread—Dispersed and subgroup group spread will have a larger effect on signature whistle type production (number of individuals calling), signature whistle count (overall acoustic production),
3) Calves – With the presence of calves in a group, signature whistle production will increase, however signature whistle type (number of individuals) will not significantly increase as communication will only increase between mother-calf pairs.
4) Behaviour—Socialising will have the greatest effect on both signature whistle type production (the number of individuals calling) and signature whistle count (overall acoustic production).

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Table 3.1. Glossary of whistle terms used throughout the study (Janik, 2013; Janik & Sayigh, 2013; Kriesell et al., 2014) .

Term	**Definition**
Contour	A continuous frequency-modulated tonal signal
Whistle	A frequency-modulated tonal signal with a unit of 1 continuous contour (either a single element or connected multiloop whistle) or 2 or more repeated contours (loops) separated by a period of silence between 0.03 s and 0.25 s in duration (disconnected multi-loop whistle)
Whistle type	All repeated whistles of a particular frequency modulation pattern as determined by visual categorization
Signature whistle (SW)	A singular emission of a whistle of a particular signature whistle type
Signature whistle type (SWT)	Whistle type with distinct frequency modulation patterns, which meets (or exceeds) the bout criteria threshold of SIGID (Janik et al., 2013) in the current or previous studies of *S. plumbea* and is therefore logged within the part of the national *S. plumbea* Signature Whistle catalogue
Inter signature whistle interval (ISWI)	The time taken (s) between the start times of consecutive whistles of the same type in a bout
Whistle bout	Sequence of emissions of the same whistle type. Bouts separated by an ISWI of > 15 min are treated as separate

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METHODS

Data collection

The study was conducted off the south coast of South Africa from 2016 – 2023 (22 encounters over 21 recording days) using boat-based focal follows (five sites: False Bay, Struisbaai, St Sebastian Bay, Mossel Bay, Richards Bay Figures 3.1, 3.2). All boat-based recording locations were within 5 km of the shoreline and at a maximum 20 m water depth. Boat trips were conducted during daylight hours with favourable weather conditions (Beaufort sea scale < 4). Research was conducted on either a 30 ft Buttcat (Mossel Bay) or 6.2 m Rhib (False Bay, Struisbaai, St Sebastian Bay, Richards Bay), both with four-stroke engines for minimal animal disturbance from engine noise or boat proximity. The boat-based focal group follows began when *S. plumbea* was within 50 m of the boat. Engines were either turned off, if safe to do so, or left idle while continuous scans for other dolphin groups continued. Acoustic data were collected via dipping hydrophones (HTI-96-MIN; High Tech Inc., United States; flat frequency response of 2 Hz– 30 kHz ± 1 dB) connected to a digital recorder (Tascam DR-680; TEAC America Inc., United States; sampling at 96 kHz and sample depth 24-bit stereo or Zoom H1) or via a SoundTrap, an autonomous acoustic recorder, secured to a weighted line (Sound Trap 300 HF; Ocean Instruments, and New Zealand; frequency response: 20 Hz - 150 kHz ± 3 dB, sensitivity: 183.3 dB re. 1 µPa). Both hydrophone setups were suspended from the side of the boat ~3 m below water level. During recordings, group size, group spread, calf presence, and surface behaviours were either documented through point sampling every 3 minutes (Mann, 1999) or through continuous vocal notes (Table 3.2). Acoustic encounters were defined as the start of *S. plumbea* sightings when the hydrophone entered the water, until the end of the encounter, whereby the group either moved away from the boat or lost the group out of visual range, and the hydrophone was removed from the water. Only encounters where both acoustic recordings and concurrent behaviour and group composition data were collected were used in this study.

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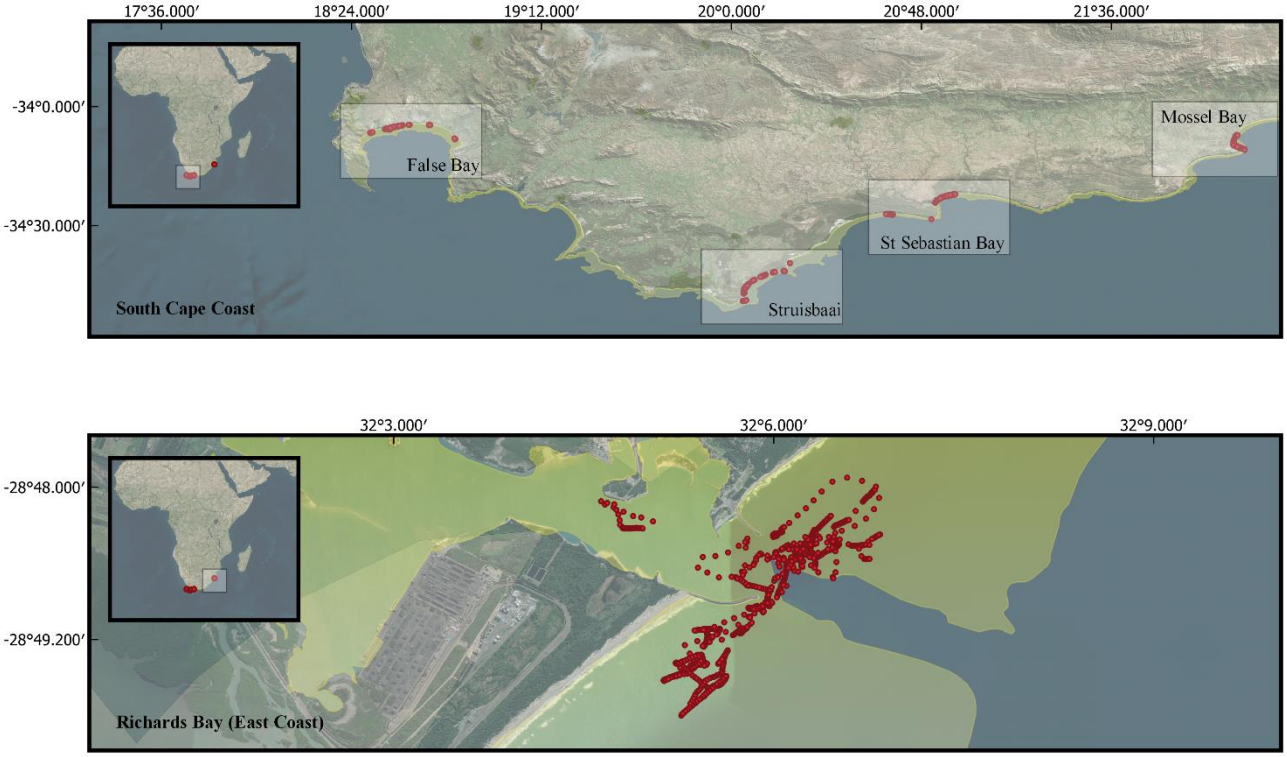


Figure 3.1. Track locations of *Sousa plumbea* boat-based acoustic recordings in south coast (top) and east coast (bottom) collected from five sites along the South Africa coast 2016-2023. A water depth of 20 m is indicated by a yellow range area, and red dots indicate boat-based tracks during encounters.

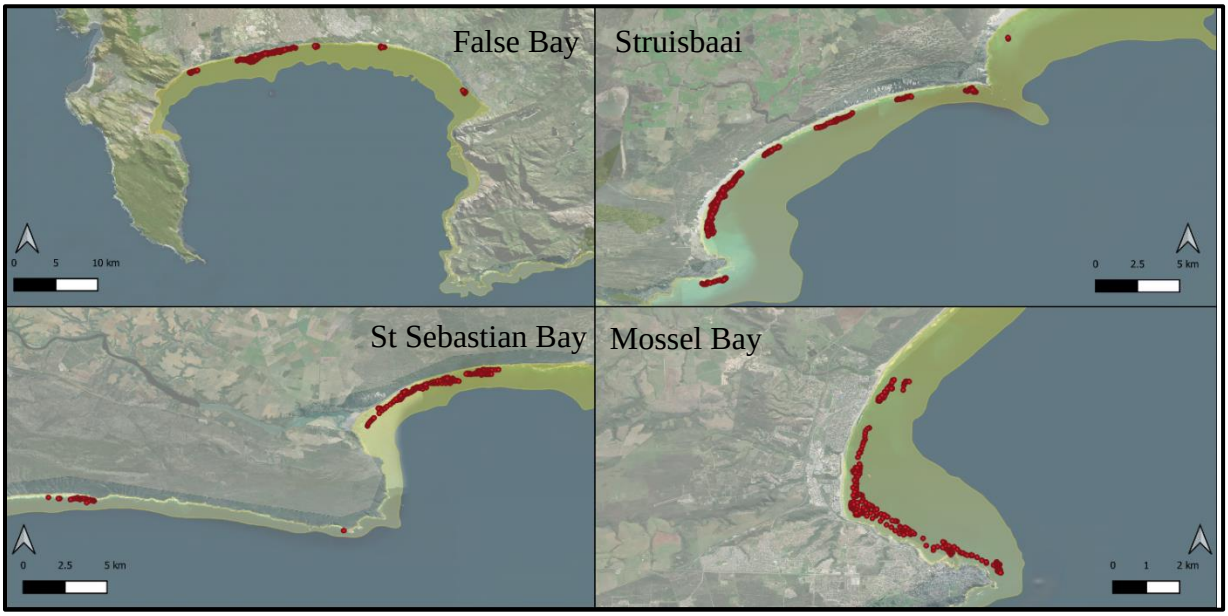


Figure 3.2. The four south coast study sites (False Bay, Struisbaai, St Sebastian Bay and Mossel Bay). Yellow range indicates 20 m water depth range highlighting the inshore encounters of *Sousa plumbea*. Red dots indicate tracks during recording encounters with *Sousa plumbea*.

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Table 3.2. Definitions of the behaviour and group spread categories recorded during vocal notes and three-minute point samples of boat-based focal follow encounters of *Sousa plumbea* across South African study sites (Henderson et al., 2012).

Behaviour	**Description**
Travelling	The group is moving in the same direction at a steady or constant speed
Milling	Group moving at a slow speed with variable direction of travel. Individuals will stay near the surface and remain in one area. Also known as resting
Socialising	Possible change of direction within the group and individuals will exhibit proximity with frequent active behaviours such as leaps, tail slaps, and body slaps.
Foraging	Variable direction of movement by individuals and the individuals in the group generally stay in the same area. Will be visible fish chasing and rapid bursts of sudden changes of direction linked to feeding behaviours
Group Spread	
Bunched	Swimming closely side by side or in a compact formation. The individuals are in close proximity to one another
Sub Grouped	Dolphins within the overall group are not evenly distributed; instead, they form smaller clusters or sub-groups within the larger group. These sub-groups may exhibit different behaviours, orientations, or interactions compared to the group as a whole.
Dispersed	Individuals are swimming at greater distances from each other, and there is a lack of close clustering and reduced proximity to conspecifics.

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Data processing

Only single species encounters with concurrent behavioural and group composition data were used for analysis. Encounters where concurrent behavioural and group composition data were available, but no signature whistles were recorded were still included in the analysis. Data were visually analysed in Raven Pro v1.6 (Center for Conservation Bioacoustics, 2019). The fast Fourier transformation parameters used to examine the data were a 1024-point Hamming window and 75% overlap.

Acoustic encounters were visually scanned for *S. plumbea* whistles, and whistles were manually annotated and categorised. A whistle was defined as one continuous contour or two or more repeated contours (a contour being a continuous tonal signal) separated by a period of silence between 0.03 s and 0.25 s in duration (Kriesell et al., 2014). This definition was in line with the definition confirmed in *S. plumbea* in Gridley et al., (Appendix 1). Each contour was graded visually based on the signal-to-noise ratio (SNR) for each contour. SNR was assigned from 1 – 3: 1) the signal was faint or partially masked but visible on the spectrogram, 2) the signal was clear and unambiguous, and 3) the signal was prominent and dominated the spectrogram (Gridley et al., 2015). Only good-quality whistles (those graded SNR 2 or 3) were used in the analysis (Table 3.3). These whistles were then visually categorised into signature whistle types, based on the *Sousa* signature whistle catalogue (Figure 3.3, collated for Gridley et al., (Appendix 1)) with consensus on whistle categorisation reached by two observers (Authors SD and TG). In this catalogue, stereotyped whistles were visually categorised into groups of highly similar contour shapes. Likely signature whistle types (SWTs) were identified through a bout analysis approach, termed SIGID, whereby whistles of a type repeated within 1 to 10 seconds are highly likely to be a signature whistle (Janik et al 2013). The minimum threshold of three out of four whistles occurring within 1-10 seconds has been verified for *T. truncatus* (Janik et al., 2013). SIGID has been extensively used to identify likely signature whistles from populations and species of free-ranging dolphins (Fearey et al., 2019; Kriesell et al., 2014; Quick & Janik, 2012). During analysis, each whistle of SNR 2/3 was assessed and compared to the signature whistle type catalogue for South Africa containing 21 types. Where contours

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matched, the whistle was assigned the catalogue ID. All whistle types unique to this current study (i.e. not previously catalogued in Gridley et al., (Appendix 1)) were assessed using the SIGID bout criterion. If met, these were assigned a novel SWT identification number with a representative contour added to the *S. plumbea* SWT catalogue.

Across each acoustic encounter, dominant group composition and behaviour variables were assigned based on point sample data and vocal track notes (Table 3.3). The total number of signature whistle types (SWT, number of individuals calling) and signature whistle (SW) emissions were calculated for each encounter. When calculating the inter-signature whistle interval (ISWI) for temporal production, each signature whistle was sorted by date and then by signature whistle type to be arranged in consecutive order of production with the nearest corresponding covariates associated with a whistle start time. The ISWI was calculated as between the start times of consecutive whistles in a bout. A new whistle bout was classified after an ISWI of 15 or more minutes, set to account for a dominant behavioural or group composition state across bouts (Lusseau, 2003). ISWI produces an inverse measure, with a greater ISWI indicating a slower whistle production rate.

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Table 3.3. Encounter information across the 22 *Sousa plumbea* encounters. Encounter number refers to the numbered encounters on that day, as some days had multiple encounters recorded. Encounters were only used if they had concurrent behaviour and track data collected alongside boat-based acoustic recordings but not all encounters had signature whistles (SWs) detected. Five study sites are included (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay and Richards Bay) with the date of recording, recording length of encounter and total cumulative recording time calculated (2016-2023). Recording equipment used, sample rate (Hz) and bit rate are included. Encounter group composition and behaviour are presented. Encounter 20191016_01 (Mossel Bay) was detected as an outlier based on whistle production and, therefore, removed from further analysis, as indicated in italics.

Region	Recording Equipment	Sample rate (Hz) Bit Rate	Encounter Date	Encounter	Recording Time hh : mm : ss	Group Size	Calves	Group Spread	Behaviour
False Bay	HTI and Tascam Recorder	96000 16/24	2016/05/17	1	01:32:08	8	Absent	Subgrouped	Travelling
			2016/06/23	1	01:29:59	3	Present	Bunched	Travelling
			2016/11/27	1	01:09:00	8	Present	Dispersed	Socialising
			2017/02/23	1	00:31:24	5	Present	Bunched	Milling
			2017/03/01	1	00:15:09	10	Absent	Bunched	Milling
			2017/03/01	2	00:31:21	10	Absent	Subgrouped	Socialising
			2017/03/30	1	00:17:48	1	Absent	Subgrouped	Travelling
			2017/04/18	1	01:06:14	3	Present	Bunched	Milling
			2017/04/20	1	00:42:02	1	Absent	Bunched	Milling
Struisbaai	HTI and Tascam Recorder	96000 24	2017/03/02	1	00:31:55	6	Present	Bunched	Travelling
			2017/03/04	1	00:32:40	7	Present	Bunched	Milling
			2017/03/07	1	00:53:15	3	Absent	Bunched	Milling
St. Sebastian Bay	SoundTrap	576000 24	2019/09/17	1	02:29:55	8	Absent	Subgrouped	Socialising
			2019/09/17	2	01:59:56	6	Absent	Bunched	Socialising
Mossel Bay	HTI Zoom H1 recorder HTI Tascam recorder	96000 16	*2019/10/16*	*1*	*01:01:20*	*8*	*Absent*	*Subgrouped*	*Travelling*
			2021/03/28	1	00:24:25	8	Present	Subgrouped	Travelling
			2023/07/18	1	01:28:00	2	Absent	Subgrouped	Travelling
			2023/07/21	1	00:18:00	3	Absent	Bunched	Milling
			2023/07/26	1	00:35:00	1	Absent	Bunched	Travelling
Richards Bay	SoundTrap	576000 24	2020/03/10	1	00:12:15	2	Present	Bunched	Milling
			2020/03/10	2	00:17:42	9	Present	Bunched	Milling
			2020/03/19	1	00:55:42	9	Absent	Bunched	Milling
					19:15:10 *Outlier rem 18:13:50*				

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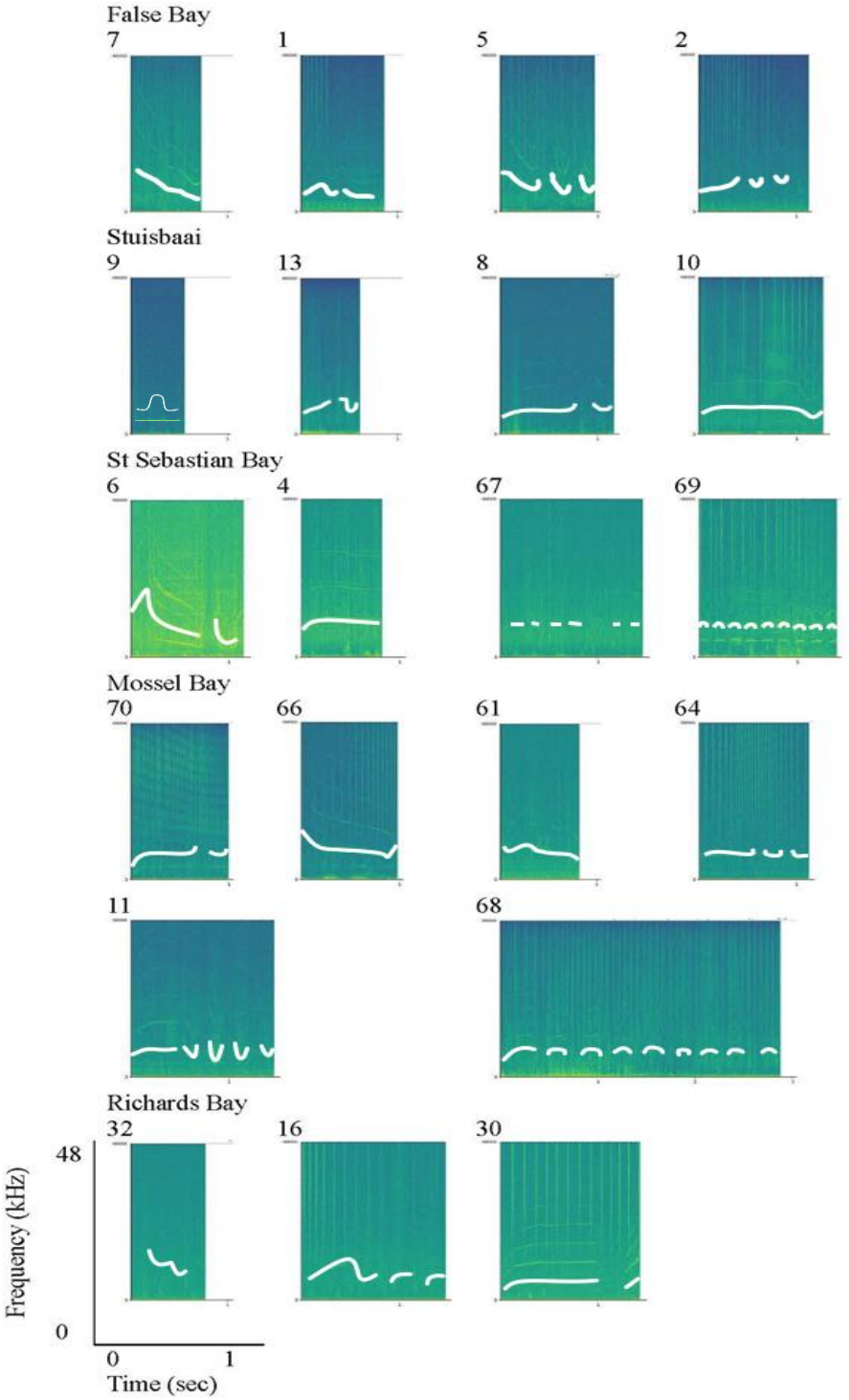


Figure 3.3. *Sousa plumbea* signature whistle catalogue, collated for Gridley et al., (Appendix 1). Signature whistle types (SWTs) are traced in white to highlight the frequency modulation pattern more clearly in each example. Stereotyped whistles detected in this study required one clear match to a whistle type in the catalogue.

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Statistical Analysis

Multiple analyses were performed to investigate the effect of various covariates on signature whistle production using generalised linear mixed models (GLMMs) R (version 4.0.2; R Core Team, 2020) in R package *lme4* (Bates et al., 2024). The covariates were modelled against three types of signature whistle production (hereafter whistle production). The response variables investigated were: 1) SWT; count of signature whistle types, effectively the number of individuals detected, 2) SW; count of total signature whistles produced, effectively the total group signature whistle production in each encounter, and 3) IWSI; inter signature whistle intervals, the temporal production rate of signature whistles for each individual. For SWT and SW full models, using a Poisson distribution, the covariates tested were group size (count), group spread (categorical), calf presence (binary presence) and surface behavioural state (Foraging (F); Milling (M); Travelling (Tr); Socialising (SOC) (categorical, Table 3.3). For the SWT and SW models, the encounter number was modelled as a random effect to remove variance based on multiple encounters within the same day. An offset variable (Recording Time ss) was set to account for different encounter durations potentially inflating whistle production. The same four covariates were used in the ISWI model, using a Gamma distribution, which is effective at modelling right-skewed time between various events where lower values are more common (i.e. inter signature whistle interval) (Crawley, 2007). In the ISWI model, both encounter number and SWT were included as random effects, as these data were analysed from SWT bouts and needed to account for variation between individuals and day.

GLMM models were tested on both the full data set and then again tested on a reduced dataset, removing any potential outliers, to test the robustness of the data set. Visual inspection of the data indicated that there was a large variation in the number of SWs detected across encounters. The interquartile range (IQR) method was applied to the SWs variable in the dataset to confirm the presence of outliers. Using the IQR, the lower and upper bounds for outlier detection were established as any values below Q1-1.5*IQR or above Q3+1.5*IQR, respectively. On review, the dispersed group spread was observed in only one encounter. Behaviourally, this category can be grouped with ‘subgroups’, and to retain the data from

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this encounter, ‘dispersed’ and ‘subgrouped’ categories were pooled in the group spread covariate for the purpose of modelling.

Collinearity was tested on all three models using a full model that included all covariates by comparing variance inflation factors (VIF), with covariates with a VIF greater than 2 being removed (R package: *car* (Fox et al., 2024).

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RESULTS

In total, 1382 whistles were identified from five study locations (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay, Richards Bay, 2016-2023, Total recording duration 19:15:10 hh:mm:ss). Across the 22 encounters, there were four encounters where no whistles were detected and three encounters where there were whistles detected but no signature whistle types detected. Out of the total number of whistles, 791 whistles had a good signal-to-noise ratio for further analysis (Table 3.4). Signature whistles accounted for 28% of total whistles detected across encounters. Group sizes across encounters ranged from 1 to 10 individuals. Calves were defined by their position in relation to their mother and identified through consistent presence at the mother's side (Gubbins et al., 1999). Calves were present in nine out of 22 encounters, and when present, the group contained one to two calves in a group. The most observed behaviour during recording sessions was milling, with no foraging behaviour observed during encounters.

Based on inter quartile range (IQR), one data point (20191016_01 (Mossel Bay)) was flagged as an outlier. The identified outlier was reviewed to assess its impact on the overall analysis. It was determined that the outlier should be excluded as there was no effect on significance seen across any covariate ($p > 0.5$) when this data point was both included and excluded, confirming the robustness of the data sets when the outlier was removed, and the models were run on 21 encounters.

With dispersed and subgroups merged, the covariate ‘group spread’ was removed from SWT (VIF = 2.1) and SWs (VIF = 3.2) best fit models, both being collinear with behaviour, which was retained as the more biologically meaningful covariate. The ISWI best-fit model removed behaviour due to a high VIF (2.7) in the full model.

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Table 3.4. Encounter information across the 22 *Sousa plumbea* encounters showing whistle counts across each encounter. Encounter number refers to the numbered encounters on that day, as some days had multiple encounters recorded. Five study sites are included (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay and Richards Bay) with the date of recording (2016-2023). For each encounter, the total number of whistles recorded, the number of high signal-to-noise ratio whistles (rated 2 or 3), the total number of signature whistles (SWs) detected, and the total number of distinct signature whistle types (SWT), representing the number of individuals, are presented. A total of 20 unique SWTs were identified across all encounters. Encounter 20191016_01 (Mossel Bay) was detected as an outlier based on whistle production and, therefore, removed from further analysis, as indicated in italics.

Region	Encounter Date	Encounter	Total Whistles	Whistles >2/3 *SNR*	SWs	SWT
False Bay	2016/05/17	01	92	29	15	4
	2016/06/23	01	61	27	7	3
	2016/11/27	01	110	68	12	3
	2017/02/23	01	1	0	0	0
	2017/03/01	01	9	5	0	0
	2017/03/01	02	0	0	0	0
	2017/03/30	01	0	0	0	0
	2017/04/18	01	0	0	0	0
	2017/04/20	01	0	0	0	0
Struisbaai	2017/03/02	01	111	65	40	1
	2017/03/04	01	5	2	4	1
	2017/03/07	01	24	12	0	0
St. Sebastian Bay	2019/09/17	01	166	89	49	8
	2019/09/17	02	13	11	5	3
Mossel Bay	*2019/10/16*	*01*	*540*	*276*	*165*	*11*
	2021/03/28	01	13	7	7	1
	2023/07/18	01	24	16	12	4
	2023/07/21	01	36	32	26	6
	2023/07/26	01	18	17	17	2
Richards Bay	2020/03/10	01	89	89	18	3
	2020/03/10	02	4	4	0	0
	2020/03/19	01	66	42	15	3
			1382 *Outlier removed 842*	***791*** *Outlier removed 515*	***392*** *Outlier removed 227*	

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1) Signature whistle type (number of individuals)

The total number of signature whistle types identified across all encounters was 20, with a median production rate of 0.03 SWT/min (IQR 0-0.06) across encounters (Figure 3.4). Across the four covariates, groups encountered without calves produced more unique SWTs compared to when calves were present (median Calves A = 3, Calves P = 1) (Figure 3.5). The number of SWT produced per encounter was greatest when dolphins were clustered in subgroups/dispersed (median subgrouped = 4) and when socialising (median behaviour Socialising = 4). However, when considered within the modelling framework, the best-fit GLMM model for SWT occurrence showed no significant effects. Further investigation into the effects of covariates was then conducted at a class level, with no significant effect seen at the class level either (Table 3.5).

2) Signature whistle production (overall acoustic production)

In total, 392 signature whistles were detected, with a median production rate of 0.14 SW/min (IQR 0-0.49) across encounters (Figure 3.4). Across the four covariates, there was very little difference between the median number of signature whistles across groups (Figure 3.6). The best fit GLMM model showed no significant effect of any covariates or any significance at class level either (Table 3.5).

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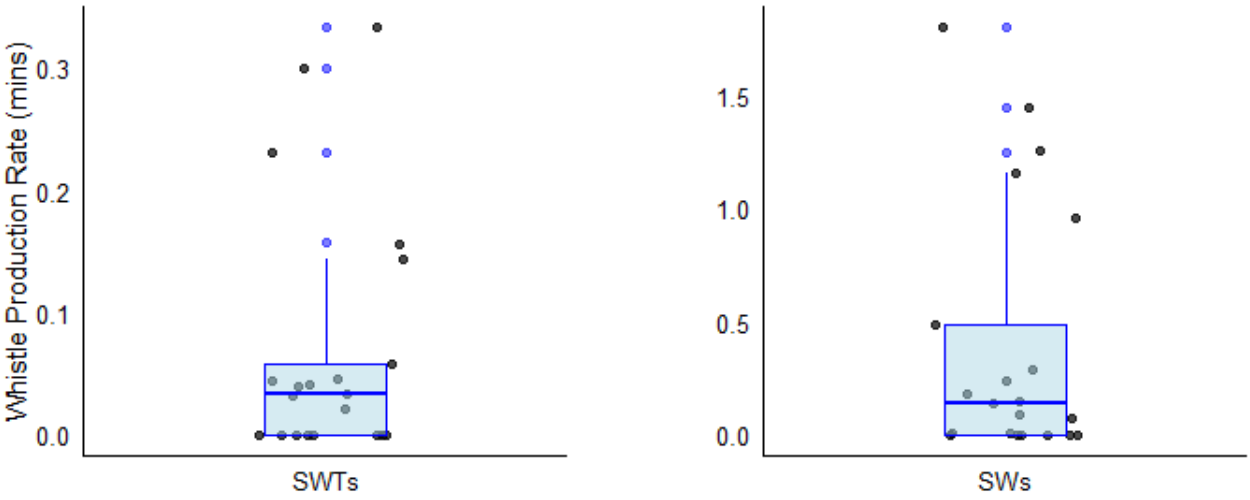


Figure 3.4. Box and whisker plot for overall whistle production rates across all *Sousa plumbea* encounters from five study locations across 22 encounters (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay; Richards Bay (2016-2023). Median production rates are indicated by the center line within the boxes (Left, signature whistle type 0.03 SWT/min IQR 0-0.06 and Right, signature whistles 0.14 SW/min IQR 0-0.49).

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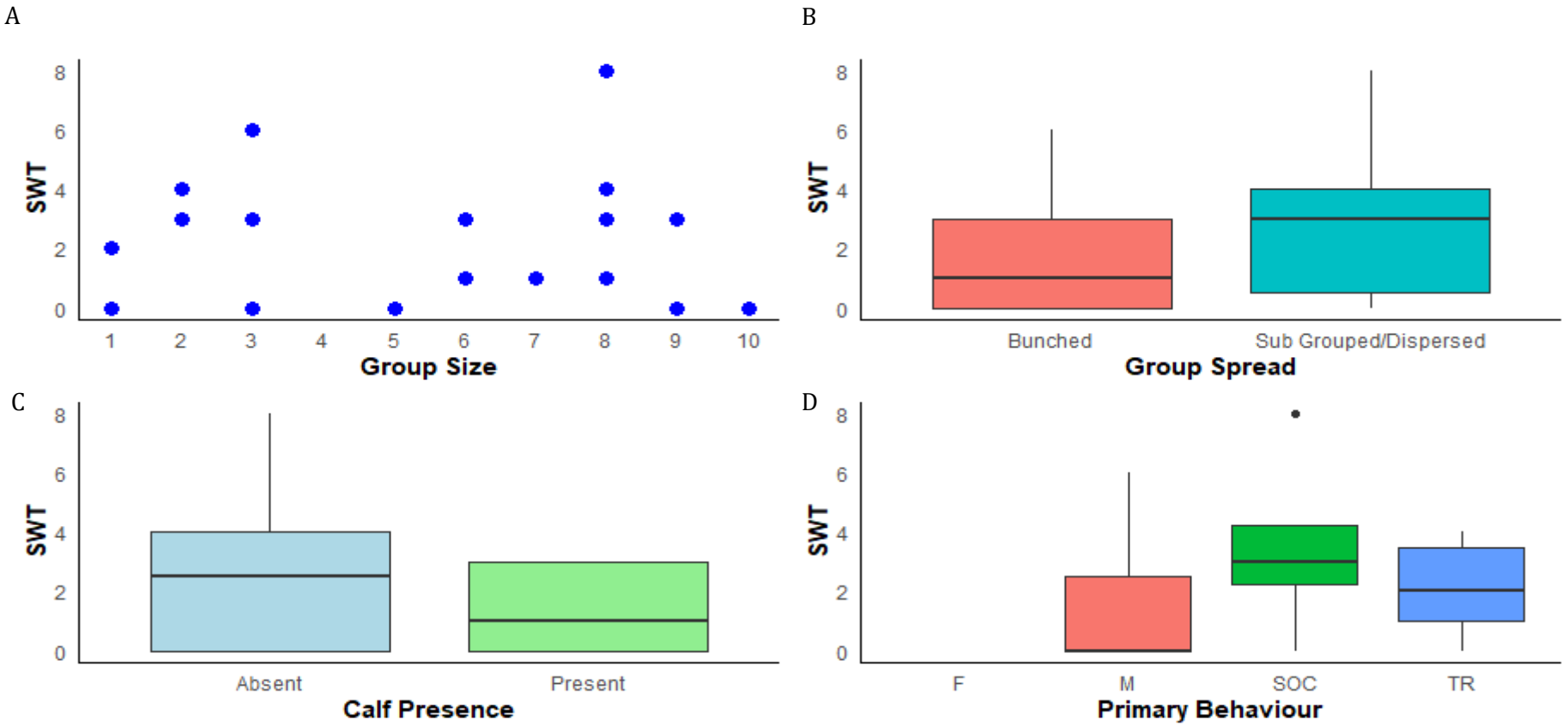


Figure 3.5. Data plots for the effect of the four covariates (Group size, Group spread, Calf presence and Primary behaviour) on *Sousa plumbea* signature whistle type production (number of individuals calling) from data across 21 encounters from five South African study sites (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay, Richards Bay, 2016-2023. For clearer data presentation, the outlier encounter is removed from these plots. A) Group sizes, represented as count data, are plotted on a scatter plot. B) Group spread is categorical and displayed using a box-and-whisker plot, which includes bunched and subgrouped/dispersed categories; dispersed and subgrouped categories are merged to reflect the modelled dataset. C) Calf presence is categorised as either absent or present and presented as a box-and-whisker plot. D) Primary behaviour is categorised into four types: foraging (f), milling (m), socializing (soc), and travelling (tr), all shown in a box-and-whisker plot as well. For plots B-D, box and whisker plots are presented with the median line present and the box representing the middle 50% of the data. The whiskers extend to 1.5 * interquartile range (IQR) to indicate the range of the data, excluding outliers.

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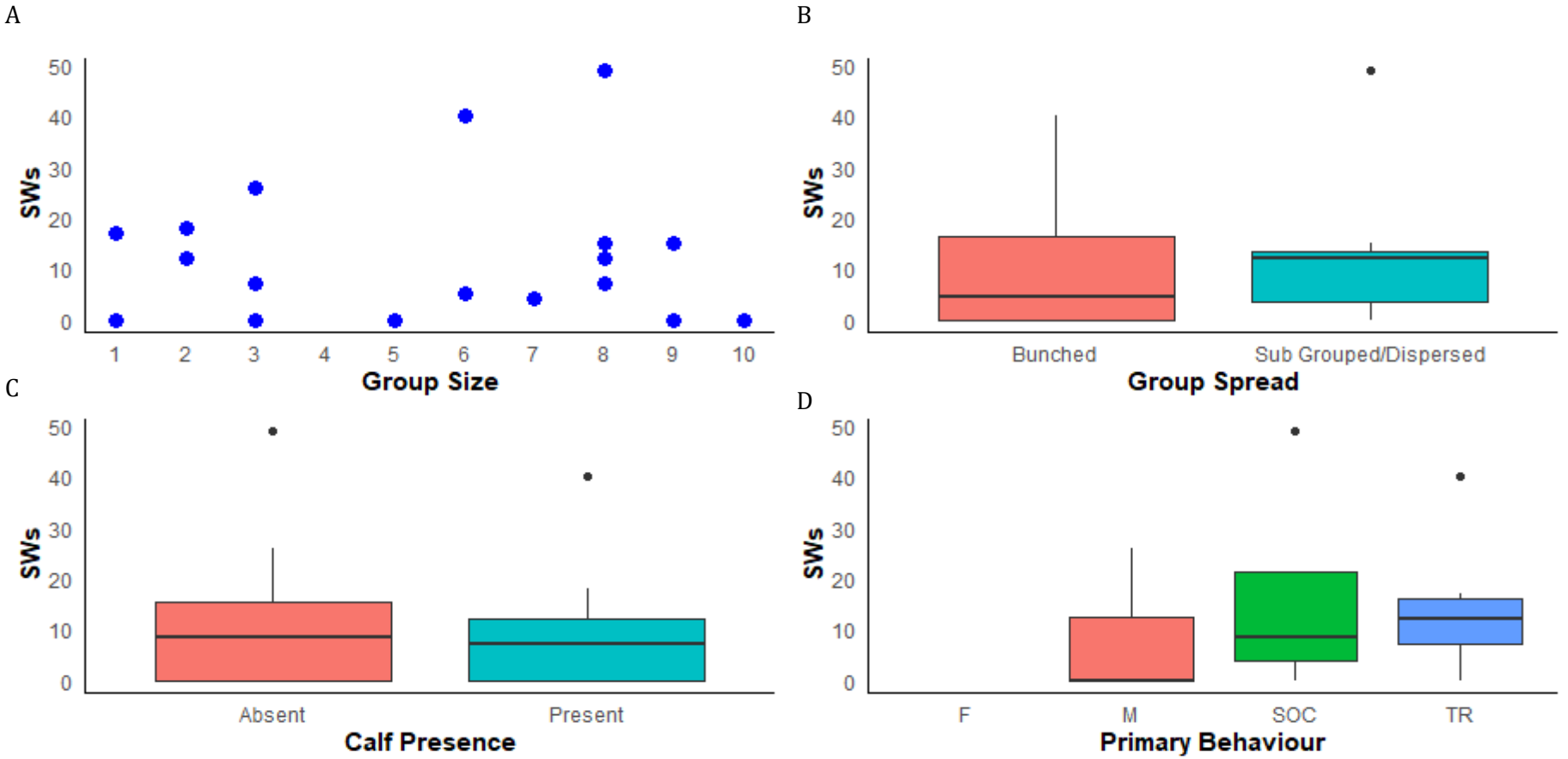


Figure 3.6. Data plots for the effect of the four covariates (Group size, Group spread, Calf presence and Primary behaviour) on *Sousa plumbea* signature whistle production (overall signature whistle count) from data across 21 encounters from five South African study sites (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay, Richards Bay, 2016-2023. For clearer data presentation, the outlier encounter is removed from these plots. A) Group sizes, represented as count data, are plotted on a scatter plot. B) Group spread is categorical and displayed using a box-and-whisker plot, which includes bunched and subgrouped/dispersed categories; dispersed and subgrouped categories are merged to reflect the modelled dataset. C) Calf presence is categorised as either absent or present and presented as a box-and-whisker plot. D) Primary behaviour is categorised into four types: foraging (f), milling (m), socializing (soc), and travelling (tr), all shown in a box-and-whisker plot as well. For plots B-D, box and whisker plots are presented with the median line present and the box representing the middle 50% of the data. The whiskers extend to 1.5 * interquartile range (IQR) to indicate the range of the data, excluding outliers.

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3) Inter signature whistle interval (temporal production)

ISWI showed a significant effect of all covariates in the best-fit model on temporal production (Table 3.5). For calf presence, the incidence rate ratio for encounters decreased ISWI by $e^{-0.61} = 0.54$ times (Figure 3.7). This indicates that temporal production of signature whistles increases by over 50% with calf presence. Sub-grouped /dispersed groups had a larger positive effect on ISWI with a higher incidence rate ratio than group size (Subgrouped / Dispersed $e^{0.43} = 1.54$, Group size $e^{0.13} = 1.14$), indicating group spread had a larger effect on a slower temporal production of SWs than group size.

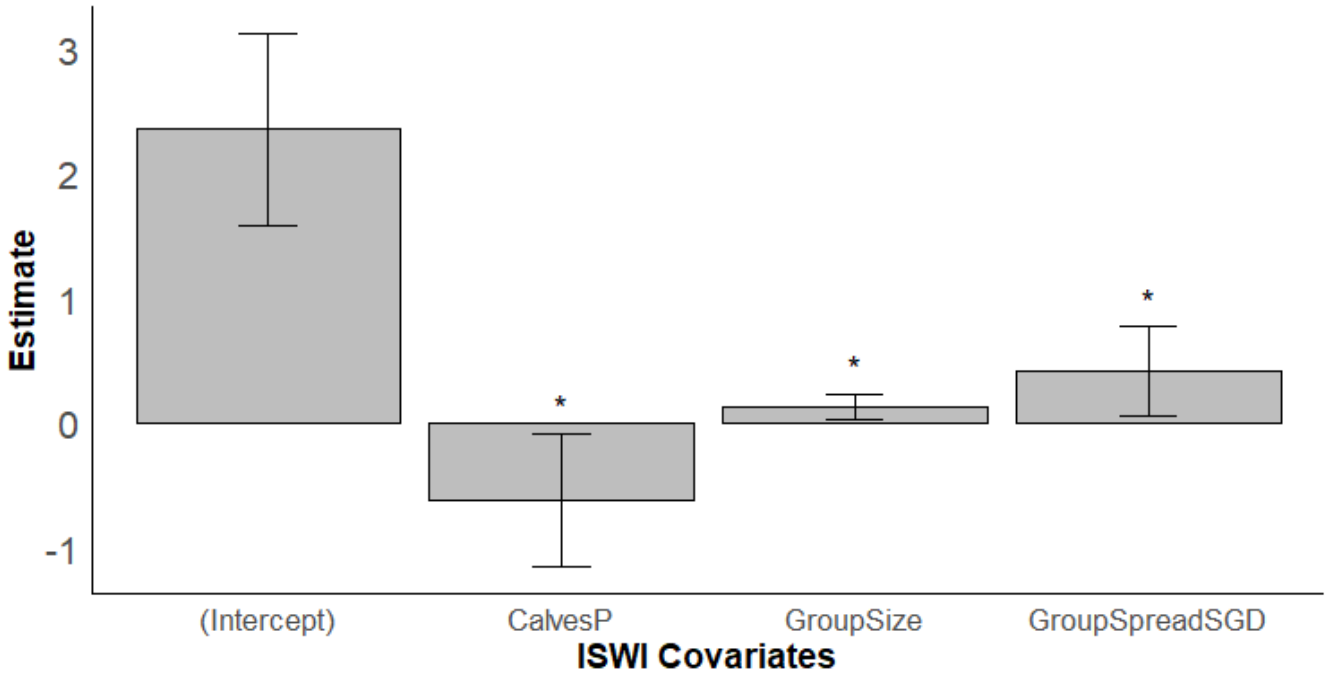


Figure 3.7. Estimated coefficients of incidence rate for generalised linear mixed models for inter signature whistle intervals in *Sousa plumbea* across 21 encounters from five South African study sites (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay, Richards Bay, 2016-2023). Significant covariates indicated with a star ($p \leq 0.01$). Covariates included in the best fit model include the presence of calves (CalvesP), group size and subgrouped and dispersed group spread (GroupSpreadSGD).

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Table 3.5. Generalised linear mixed models for signature whistle types (SWT), signature whistles (SW) and inter signature whistle interval (ISWI) in *Sousa plumbea* across 21 encounters from five South African study sites (False Bay, Struisbaai, St Sebastian Bay, Mossel Bay, Richards Bay, 2016-2023). The table includes the best fit model and individual class models that include each covariate modelled individually. The table also provides metrics to provide information about the goodness of fit and complexity of the model including Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), Log-Likelihood (LogLik), Deviance and Residual Degrees of Freedom (df residual). P values <0.01 are highlighted to show significant effects.

Model	**Distribution**	**Model type**	**AIC**	**BIC**	**LogLik**	**deviance**	**df. residual**	**Term**	**estimate**	**Standard error**	**statistic**	**p.value**
SWT ~ GroupSize + Calf Presence + Behaviour + (1 | Encounter), family =Poisson(link="log"), offset=(log(seconds))	Poisson (log)	Best	94.2	100.5	-41.1	82.2	15.0	Intercept	-7.10	0.79	-9.06	**<0.005**
								Socialising	0.70	0.85	0.82	0.41
								Travelling	-0.15	0.70	-0.22	0.83
								Group size	-0.05	0.11	-0.42	0.68
								Calves Present	-0.37	0.62	-0.60	0.55
SWT ~ Behaviour + (1 | Encounter) , Data: Encounters, Offset: (log(seconds))	Poisson (log)	Class	90.8	94.9	-41.4	82.8	17.0	Intercept	-7.56	0.54	-13.96	**<0.005**
								Socialising	0.65	0.80	0.80	0.42
								Travelling	-0.07	0.72	-0.10	0.92
SWT ~ GroupSize + (1 | Encounter), Data: Encounters, Offset: (log(seconds))	Poisson (log)	Class	89.6	92.8	-41.8	83.6	18.0	Intercept	-7.48	0.66	-11.38	**<0.005**
								Group Size	0.00	0.10	0.03	0.97
SWT ~ Calves + (1 | Encounter), Data: Encounters, Offset: (log(seconds))	Poisson (log)	Class	89.1	92.3	-41.6	83.1	18.0	Intercept	-7.27	0.42	-17.25	**<0.005**
								Calves Present	-0.45	0.63	-0.71	0.48
SWT ~ GroupSpread + (1 | Encounter), Data: Encounters, Offset: (log(seconds))	Poisson (log)	Class	89.6	92.8	-41.8	83.6	18.0	Intercept	-7.45	0.43	-17.20	**<0.005**
								Group Spread Dispersed/ SubGrouped	-0.04	0.65	-0.07	0.95
SWs ~ GroupSize + Calf Presence + Behaviour + (1 | Encounter), family =Poisson(link="log"), offset=(log(seconds))	Poisson (log)	Best	152.1	158.3	-70.0	140.1	15.0	Intercept	-7.72	1.48	-5.23	**<0.005**
								Socialising	1.34	1.61	0.84	0.40
								Travelling	1.51	1.28	1.18	0.24
								Group size	0.02	0.20	0.08	0.94

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								Calves Present	0.35	1.12	0.32	0.75
SWs ~ Behaviour + (1 | Encounter) , Data: Encounters, Offset: (log(seconds))	Poisson (log)	Class	148.2	152.4	-70.1	140.2	17.0	Intercept	-7.46	0.95	-7.87	**<0.005**
								Socialising	1.31	1.48	0.89	0.38
								Travelling	1.48	1.26	1.17	0.24
SWs ~ GroupSize + (1 | Encounter), Data: Encounters, Offset: (log(seconds))	Poisson (log)	Class	147.9	151.0	-70.9	141.9	18.0	Intercept	-6.78	1.13	-5.99	**<0.005**
								Group size	0.03	0.18	0.15	0.88
SWs ~ Calves + (1 | Encounter), Data: Encounters, Offset: (log(seconds))	Poisson (log)	Class	147.9	151.0	-70.9	141.9	18.0	Intercept	-6.73	0.76	-8.82	**<0.005**
								Calves Present	0.23	1.08	0.21	0.83
SWs ~ GroupSpread + (1 | Encounter), Data: Encounters, Offset: (log(seconds))	Poisson (log)	Class	147.9	151.0	-70.9	141.9	18.0	Intercept	-6.69	0.73	-9.23	**<0.005**
								Group Spread Dispersed/ SubGrouped	0.16	1.13	0.14	0.89
ISWI ~ GroupSize + GroupSpread + Calf Presence + (1 | Encounter) + (1 | SWT), family =Gamma(link="log")	Gamma (log)	Best	2997.1	3024.0	-1491.5	2983.1	337.0	Intercept	2.35	0.39	5.95	**<0.005**
								Group size	0.13	0.05	2.56	**0.01**
								GroupSpread Dispersed/ SubGrouped	0.43	0.18	2.35	**0.02**
								Calves Present	-0.61	0.27	-2.25	**0.02**

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DISCUSSION

Capture-recapture (CR) models provide essential estimates of a population's size, survival, and movement patterns and biases in capture probability need to be well understood (Hammond, 2009). Although these are well understood for photo ID work, the application of CR models using acoustic cues presents an opportunity for different biases, which need to be well understood. Previous work has, for example, demonstrated that broad-scale (Longden et al., 2020) and fine-scale (Fearey et al., 2022) placement of hydrophones can affect the detection of individuals and abundance estimations. Building on this, the current study was undertaken to help understand whether group behaviour, group size and composition affect signature whistle production and, consequently, whether these variables could introduce inherent bias in the detection probability of signature whistles in a capture-recapture framework in *S. plumbea*. These results, although based on limited data, indicate that behaviour and group composition have little bearing on SWT occurrence or emissions of signature whistles for *S. plumbea.* Calf presence, however, may contribute to reduced production of signature whistles and warrants further investigation.

Results from this study indicate that behaviour and group composition do not strongly affect whistle production rates, only temporal production, with overall low individual whistling rates. Acoustic behaviour in cetaceans is known to be influenced by different group compositions and contexts (Jensen et al., 2011; Visser et al., 2017). Understanding under what limits and situations calls are made or potentially not made can affect individuals' detection through acoustic monitoring (Longden et al., 2020) . Overall, and compared to other studies on dolphins (Cook et al., 2004; Hawkins & Gartside, 2010), this study did not identify a strong effect of group behaviour or group composition on SWT and SW whistle production, with only decreasing temporal effects on signature whistle production. Across other species, increasing group size is linked to increased call production (Borker et al., 2014; Payne et al., 2003; Radford and Ridley, 2008). In *Tursiops* spp., individual whistling rates increase up to a certain group size threshold, beyond which

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whistle production decreases. This suggests that there is an optimal group size for whistle production to minimize the masking of conspecific acoustic signals (Quick & Janik, 2012). In this study, group sizes were generally small, with 43% of encounters containing five or fewer individuals. This follows that of .Koper et al., (2016) and Bouveroux et al., (2018), whose studies showed significant declines in *S. plumbea* group sizes in Algoa Bay, South Africa, over a 24-year period, from mean group sizes of seven to three animals (Bouveroux et al., 2018). Kaczmarski et al. (1999) described a relatively fluid group structure with a mixture of locally resident individuals and transient groups which move further along their South African coastal range. However, more recent studies suggest a higher degree of site fidelity in *S. plumbea* in recent years (James et al., 2015), with movements of individuals averaging 120 km with long-term residency prevalent in some regions (Vermeulen et al., 2018). In dolphins, higher group stability has been shown to lead to more strongly associated individuals (Connor et al., 2001; James et al., 2015), with individuals with stronger social bonds displaying lower whistle exchange rates (Chereskin et al., 2022). With consistently small groups and evidence for long-term site fidelity from previous studies, there may be evidence for stronger bonds and more stable group dynamics seen in *S. plumbea* throughout their South African range, which would affect signature whistle production rates.

As individuals meet between groups, studies have shown whistle production to increase, consistent with evidence that suggests signature whistles function as introductory calls to communicate identity when free-ranging groups encounter one another (Quick & Janik, 2008). This was not supported in this study, as subgroups/dispersed group spreads did not significantly affect SWT and SW production. Overall, the median signature whistle production rate for *S. plumbea* across encounters (0.14 SW/min) was lower than other sympatric delphinid species (Kriesell et al., 2014, Fearey et al., 2019). Production rates are variable and context-dependent when looking at other species that occur sympatrically to *S. plumbea*. In *T. aduncus*, Gridley et al.

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(2014) found rates of 32.9 wh/min and 5 wh/min for a single stereotyped whistle from two individuals during entrapment events. This study investigated a wild free-swimming population, where signature whistle production is expected to be lower than during capture or entrapment events (Gridley et al., 2012). In free-swimming populations, Chereskin et al., (2022) found that vocal exchanges of signature whistles were more favoured between weakly associated individuals, while strongly associated individuals preferred affiliative contact to maintain social relationships. Again, in *S. plumbea*, smaller, more stable group structures, with a higher level of site fidelity, could be a reason for the lack of group size or group spread effect on whistle production, as well as account for their overall low production rates. As signature whistles function as a cohesion method, higher stability in the smaller groups would reduce the requirement to acknowledge where each other is through calls, which could be accomplished through visual contact. Within their small group sizes, *S. plumbea* may not rely on vocal communication as strongly as in larger fission-fusion groups, such as in *Tursiops* spp, with further work required in comparing regions where *Sousa* are known to occur in larger groups (See the Arabian region and China regions, Würsig et al., 2016).

The study found a significant effect of calves on temporal whistle production, with calves increasing the temporal effect of whistle production. In *Tursiops* species, under both forced separation and natural contexts, mothers and calves will increase the call rate of whistles to initiate reunions with one another, especially when out of visual range (King et al., 2016). Call rate increases specifically during the first few weeks of a calf's life to initiate maternal bonding and establish the calf's own mother in the group (King et al., 2016). Therefore, separation distances of mother and calf's do not exceed the active acoustic space of whistles to maintain consistent contact even when out of visual or olfactory distances (Quintana-Rizzo et al., 2006). However, none of the calves seen during this study were newborn or neonatal (S. Dines pers. obs. confirmed with concurrent photo ID data). In this study, although there was a temporal increase

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in whistle production, in line with other studies, there was no significant effect on the number of individuals or overall calls produced. In China, *S. chinensis*, where group sizes reach 48 individuals, nursery groups (with mother-calf pairs present) are 2-4 times larger than non-nursery groups, providing much stronger protection for calves (Liu et al., 2021). The smaller group sizes in *S. plumbea* mean they do not have a 'safety in numbers' effect, and the addition of vulnerable calves to a group will increase intraspecific and interspecific predator threats (Frainer et al., 2023; Plön et al., 2015). In these smaller groups, reliance on other methods to maintain mother-calf contact, such as visual or physical contact, might be preferred, as is seen in nursing baleen whales (Nielsen et al., 2019; Parks et al., 2019; Videsen et al., 2017).

There are notable limitations to this study, namely a small sample size. *S. plumbea* is listed as Endangered on the IUCN Red List, with low encounter rates throughout their range (Plön et al., 2021). This was highlighted in the high boat-based effort required to collect sufficient data for this study. Over a seven-year period (2016 – 2023), 207 boat-based surveys were conducted throughout South Africa. *S. plumbea* were encountered on 40% of these surveys, with acoustic recording possible on 64 occasions. A conservative approach was taken for this study whereby encounters were removed if the other species were sighted in the vicinity, and only encounters with behavioural and group composition data were used. This left the study with 22 encounters across all five study sites, where the encounters had concurrent group data and acoustic recordings, with one of those encounters further removed as an outlier for modelling. This reduced sample size is representative of other studies which have utilised *Sousa* acoustic data and highlight the difficult nature of working acoustically within this genus (This study: 22 encounters, 19 hrs, 1382 total whistles, Wang et al., 2013: 13 encounters, 40 hrs, 4630 total whistles, Hoffman et al., 2015: 8 encounters, 2.5 hrs, 1783 total whistles, Bopardikar et al., 2018: 11 encounters, 3.1 hrs, 2260 total whistles, Yang et al., 2020: four recording sessions, four *S. plumbea* click encounters). Although the range of applications of acoustic data sets differs across

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studies, an appreciation for high research effort and elusiveness of the study species limits the strength of this study's conclusions. Further work is needed to strengthen the study and improve modelling by including a greater diversity of encounters.

Other limitations to the study include *S. plumbea's* propensity to occupy regions of periodic high noise levels, including high site fidelity in harbours, river mouths and high-use coastal areas by recreational and commercial boats (Plön et al., 2021). Masking of natural sounds is known to influence acoustic behaviour in delphinids, and an understanding of ambient sound would provide greater context to how noise pollution affects this species along their narrow range (La Manna et al., 2013; Lundquist et al., 2012). Data collection methods might also have influenced acoustic behaviour. Boat-based recordings require proximity to groups to monitor and record and in some species, boat proximity influences whistling behaviour (Hawkins & Gartside, 2010). In this study, effort was made to reduce disturbance to animals, such as recording from a distance, and engines were turned off when it was safe to do so. To explore this question, future work would look at a comparison to moored data that had concurrent behaviour data, as this would gauge potential bias or identify no differences due to either methodology used. The use of moored data would require confirmed species sightings from land-based observations to ensure consistent species identification. Although there is a chance of differences between moored and boat-based collected data, a high level of bias due to the recording setup was not expected in this study. Boat-based acoustic recordings have been shown to cause minimal disturbance to focal groups during recordings, with acoustic data collection as one of the most passive observation methods (La Manna et al., 2013). During this study, individuals showed no aversion behaviours during data collection, and in most instances, the boat engines were off and were within a reasonable distance from the focal group.

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Compared to other published studies on whistle acoustics in the *Sousa* genus, there are no studies investigating the effects on whistle production rate. Signature whistles are referred to as contact calls, providing information on identity, stress and context-dependent relationships (Esch et al., 2009; Janik et al., 2006; Sayigh et al., 1995). Previous studies reporting evidence of signature whistle use in *Sousa* come from opportunistic events of isolation or injury (Cheng et al., 2017; Van Parijs & Corkeron, 2001). When monitoring an injured *S. chinensis*, Cheng et al., (2017) recorded a signature whistle production rate of 0.84 SW/min from one individual over the recording period. Although higher than the median signature whistle production rate found across encounters in this study (0.14 SW/min) direct comparison is unsuitable due to differences in study design. The outlier that was removed from the analysis had a signature whistle production rate of 2.7 SW/min across the encounter. Studies of signature whistle production rate in free-swimming populations have shown a lower signature whistle production of 38% and 70% compared to restrained animals (Janik, 2009a). In the encounter that was removed as an outlier, the number of SWTs detected, 11 (proxy for individuals) exceeded the best group size estimation (8) but did not exceed the maximum group size reported in the encounter (12), indicating that nearly every individual was acoustically recorded within the encounter. In addition, the encounter duration was not significantly longer than other encounters and there was no behaviour or context present that indicated higher whistle rates (calves absent and primary encounter behaviour travelling). Throughout the encounter, one individual dominated the signature whistle count (SWT10, 50% of SW production in the encounter). The presence of an outlier highlights that the species is not always vocally elusive, and whistle production rates can vary, especially when recording free swimming groups.

This study aimed to determine whether group composition and behavioural contexts influence signature whistle production and whether these factors could bias detection probabilities in a capture-recapture framework. Although the sample size was limited, the findings suggest that

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group composition and behaviour are not critical variables for optimizing signature whistle detection in *S. plumbea*. In passive acoustic monitoring (PAM), array designs are typically based on research objectives, equipment availability, and species behaviour. For instance, in Namibia, the highest detection probability of signature whistles in *T. truncatus* occurred in areas where dolphins patrolled for extended periods, which were linked to feeding or socializing zones (Fearey et al., 2022; Longden et al., 2020). However, with *S. plumbea*, there is only evidence that the temporal production of signature whistles is affected by these variables, indicating that other behaviour or group composition effects are either not present, sample size-dependent, or masked by the overall low whistle production rates in *S. plumbea*.

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4 THE EFFICACY OF PASSIVE ACOUSTIC MONITORING VS BOAT-BASED PHOTOGRAPHIC IDENTIFICATION OF AN ELUSIVE COASTAL DOLPHIN SPECIES

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Prepared for publication in Remote Sensing and Ecology

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ABSTRACT

Wildlife practitioners commonly apply photographic identification (photo ID) to monitor wild populations, including many dolphin species. However, for elusive species, encounter rates are low, and photo ID can be an inefficient method for detection. Dolphins develop individually unique acoustic signature whistles which convey individual identity to conspecifics. Identification of individuals through passive acoustic monitoring could allow for individual-based monitoring. To investigate the effectiveness of acoustic and photo ID methods to detect individual Indian Ocean humpback dolphins (*Sousa. plumbea)* in the population, an acoustic array was deployed in coastal waters, covering 8 km in Mossel Bay, South Africa, from 19th March – 5th April 2021. Boat-based photo-identification surveys were conducted throughout the same area and time period. Here, the two methods are compared to investigate the efficacy of the detection of dolphins using both methods over a short-term survey period. Detection characteristics (spatial variation, temporal detection and individual detection rates) were compared between methods, together with the financial and logistical constraints of each method for collecting capture-recapture data, which are key considerations for effective applied conservation programs in this species. Through automated species classifiers, 38 acoustic encounters were assigned to *S. plumbea* during 18 deployment days. Of the 226 good-quality whistles detected in these encounters, 199 signature whistles were detected, with 10 signature whistle types, SWTs (different individuals) identified. Seven SWTs were recaptured on multiple encounters, with four SWTs seen on multiple days. Acoustic detections occurred across all moorings but with substantial variation between locations. Over 80% of signature whistles were detected on non-adjacent moorings located 6 km apart (moorings 1 and 4), with three individuals accounting for 73% of all SW detections. A diurnal pattern was found with whistle detection higher overnight. In comparison, during the 13 boat-based survey days, there were encounters with *S. plumbea* on seven days (mean duration of encounter 00:46:22 hh:mm:ss (± 00:21:17 SD)). From these, 16 individuals were photographed, of which six dolphins were recaptured across different days. Throughout the survey area, dolphins were visually observed at two locations only, near to moorings 1 and 6. Acoustic methods had a greater mean detection rate of individuals (0.89 ± 1.48 SD) SWTs detected/encounter, mins) than photo ID (0.08 ± 0.05 SD) individuals photographed/encounter, mins). There was no correlation between encounter duration and individuals/SWTs detected, highlighting that longer encounters are not the determining factor of detection across both methods; rather, other factors such as environmental conditions, animal behaviour and equipment likely have a stronger determinant on the rate of detection. However, fieldwork costs "per dolphin" captured were higher for acoustic methods than for Photo ID methods when processing and data analysis times were included (Photo ID: 6400 ZAR.dolphin^{-1}; Acoustics: 12266 ZAR.dolphin^{-1}). The study recognises the value of photo ID for monitoring rare species at reduced costs. However, it highlights that continuous acoustic detection methods offer a viable alternative to visual-based approaches for elusive species such as *S. plumbea* for short-term assessments.

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INTRODUCTION

Conservation action is hindered by the ability of managers, practitioners, and policymakers to access practical information and data (Ausden & Walsh, 2020). The use of evidence in practical decision-making requires research to be translatable to implementation (Downey et al., 2022). Recent development in Conservation Evidence-based studies overcomes such hurdles by assessing scientific evidence based on the effectiveness of conservation interventions on biodiversity (Downey et al., 2022). Scientists need to consider the applications of the data they are collating and the recommendations that their results produce. This can be done by considering their studies' costs, time, logistical and analytical requirements. Monitoring is an essential tool to provide objective and sound scientific advice to decision-makers at all levels of government to implement effective management policies (Hutchings et al., 2009). Developing these monitoring programs for threatened species is often hindered by life history traits such as low numbers, low densities, and inaccessible environments (Robinson et al., 2018). Effective conservation management plans consider these logistical constraints and species-specific research to produce baseline empirical data that informs effective monitoring studies.

Available monitoring methods vary across taxa and environments, but mark-recapture is one of the most widely used research tools to assess population trends, especially on free-ranging cetaceans (Urian et al., 2015). Mark-recapture provides several population parameters including movement, residency, social relationships, population size and survivorship (Hammond, 1990; Lukacs & Burnham, 2005). Mark-recapture or capture-recapture initially relied on artificial marks (‘tags’) applied to animals, but these typically require capturing or harassment of the study animals at some level (Amstrup, 2010). Many studies, and almost all current cetacean studies, now prefer non-invasive methods, identifying individuals using naturally distinctive marks or scars and colouration patterns. This method is used across a suite of species, typically ‘capturing’ the animals photographically, and has been an effective method in monitoring cetacean species

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for over 50 years (Hammond et al., 1990; Würsig & Würsig, 1977). Long-term monitoring of animals using these non-invasive techniques, requires identifying features to be individually unique, remain stable over the individuals' lifetime and be readily identifiable in the field or in a photograph (Hammond, 1986). This method of photographic identification (photo ID) is popular due to its relatively low effort, accessible non-specialist equipment and high level of reproducibility when compared to other forms of identification methods such as physical tags. Additionally, photographic catalogues are easily contributed to by researchers, managers and citizen scientists (Cheeseman et al., 2017; Tyson & Wells, 2016; Vermeulen et al., 2018). Two key requirements of cetacean photo ID studies are the requirement for good image quality and a reasonable level of fin distinctiveness. However, this leads to an inherent bias towards the over-detection of highly marked individuals, the extent of which varies widely between focal species and even research groups (Urian et al., 2015). Standardization of methodology can reduce this bias, by applying grading scales to assess image quality and fin distinctiveness and only using a balanced subset of images (Friday et al., 2008). In the field, photo ID methods are limited by a need for daylight, boat access and environmental conditions such as glare or sea state, as well as equipment quality and photographer skill. A key assumption of capture-recapture is the homogeneity of detection across individuals, and the presence of 'trap shy/happy' or boat-averse/friendly individuals in a group which could lead to biased sampling (Hammond, 1986; Marshall & Pierce, 2012). For species with small populations and low encounter rates, the ratio of manpower and field effort to data collected can be quite poor, with elusive coastal species a prime example of this, where weeks or months of field effort can result in only 10s of animals encountered and low encounter rates (Virgili et al., 2018). For species with vocal individuality, the use of naturally occurring calls has been suggested as a more effective method of non-invasive mark-recapture monitoring (Terry et al., 2005).

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Bioacoustic monitoring of wildlife has advanced in recent years and can be used to effectively detect species and individuals across a wide geographic range. Bioacoustic monitoring has been explored in a suite of species and environments such as coral reefs (Lamont et al., 2022b), fish (Looby et al., 2022; Popper & Hastings, 2009) , cetaceans (Gridley et al., 2015), frogs (Measey et al., 2017) and even elasmobranchs (Fetterplace et al., 2022) creating a wide variety of applications. Underwater, acoustic detection by both animals and scientific instruments can occur at greater distances than visual detection, as sound transmits far more efficiently through water (Ladich & Winkler, 2017). Cetaceans produce acoustic signals which transmit well underwater, for example, in optimum conditions, dolphin whistles can be detected over more than 20 km (Janik et al., 2000) and the low-frequency tonal sounds of baleen whales can be detected up to hundreds of km (Janik, 2005), making acoustic monitoring an ideal way to passively and remotely study cetacean populations. However, for most species, acoustic monitoring is still only possible at the "species level" and not the level of individuals, as required in capture-recapture, so most passive acoustic studies focus on species presence or vocal behaviour, and generating relative density measures typically only possible with arrays of multiple hydrophones where individual callers can be localised (Hannay et al., 2013; Marques et al., 2009).

For species with small populations that are threatened, individual-level monitoring provides much richer and more useful data. Individual acoustic monitoring requires stable, stereotyped signals that can be used to identify individuals as they move throughout their range over time. Individual voice cues have been studied across many taxa, including several species of birds, as they are used to identify neighbours and kin from strangers to map territories and to select and identify mates (see Terry et al., 2005). Most species recognize other individuals through these vocal cues, where characteristics of the vocal tract produce individually identifiable signals for identification (Aubin et al., 2000; Tibbetts & Dale, 2007; Zann, 1990). However, dolphins develop unique signatures in which individual identity information occurs in the frequency modulation

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pattern, as opposed to voice cues (Sayigh et al., 2017). Termed signature whistles (SWs), these signals have high levels of distinctiveness between individual dolphins and are used to convey individual identity to conspecifics and to maintain group cohesion (Sayigh et al., 2007). Signature whistles are formed during the first year of life, developing as a contact call between mother-calf pairs and can stay stable for up to 43 years (Sayigh et al., 2022). In restrained or captive dolphins, SWs can constitute 80% - 100% of whistles produced (Janik & Slater, 1998), whilst in wild free-ranging bottlenose dolphins (*Tursiops truncatus*), SWs constitute 34% - 70% of whistles (Cook et al., 2004). Notably, SWs offer high individual distinctiveness and long-term stability and can be easily detected using noninvasive acoustic monitoring techniques, making them potentially ideal for individual acoustic-based monitoring programs (Longden et al., 2020; Terry et al., 2005).

Acoustic capture-recapture using individually distinctive signals has previously been limited by the ability to passively detect individual signals (Stevenson et al., 2022). The availability of suitable technology, such as affordable autonomous hydrophones, has only been available since the mid-2000s (Sugai et al., 2019). High financial costs have limited the capacity of researchers and organizations to utilize passive acoustic monitoring (PAM) methods (Gibb et al., 2019). Recent developments in low-cost acoustic recorders, especially for the aquatic environment have vastly increased the accessibility of these methods (Lamont et al., 2022a). In addition to logistical constraints, identifying signature whistles in the wild is challenging and often requires simultaneous visual observation or localization (Quick & Janik, 2008). Early studies of signature whistles were done on captive or temporarily held dolphins where whistles could be confidently linked to the signal producer. However, Janik et al., (2013) developed a bout analysis approach to identify signature whistles using SIGnature Identification (SIGID) from free-ranging dolphins in the wild using a single hydrophone (i.e. not through localization of calls) which has been applied successfully across multiple study populations and species (Fearey et al., 2019; Longden et al., 2020; Probert et al., 2023; Rio, 2023).

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However, when applying signature whistles in a capture-recapture context to study wild dolphins, there are several additional considerations. As signature whistles are a context-driven call type, in *Tursiops* spp., signature whistle production can vary considerably with behaviour which may be linked to specific areas or periods, so hydrophone placement within the habitat is an important consideration in previous studies (Fearey et al., 2022; Longden et al., 2020). Duty cycling is also a key consideration, as hydrophone deployment is limited by battery and storage capacity. Alternating between active and dormant periods of recording can extend deployment duration, yet missed detections may influence overall species detection as well as the ability to conduct SIGID bout analysis (Fearey et al., 2022). Additionally, almost all research on signature whistles and their potential use in capture-recapture studies are based on *Tursiops* spp., and their application to other species, where signature whistle production may be different, has not yet been tested.

Most cetacean studies to date have conducted capture-recapture using individual photographic identification and applied a range of 'open' or 'closed' models. These models can also be applied to signature whistles (for application in *Tursiops* spp., See Longden et al., 2020; Romeu et al., 2024). Longden et al., (2020) successfully compared abundance estimates from photo ID surveys to SW-type (SWT) detections from moored acoustic deployments (3 weeks per month over 6 months). The study found spatial heterogeneity in the capture of SWTs, with the majority of detections occurring in a known resting and socialising area. Temporal variability was also identified, with a majority of SWTs detected in the first 30 minutes of an encounter. The study identified a closed population of 53 SWTs, which overlapped with the known population size of 54 - 76 individuals, calculated using photo ID methods (Elwen et al., 2019). The study highlighted the potential for signature whistles to be used as proxies for individual occurrence in capture-recapture abundance models in a population of known range and abundance (Elwen et al., 2019;

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Gridley et al., 2015).

The benefit of individual acoustic capture-recapture as a monitoring tool is only truly appreciated when it can be applied to species and regions where photo ID data is lacking, or traditional photo ID methods are not logistically feasible. Compared to many other species, cetaceans are time-consuming and expensive to survey and monitor (Kaschner et al., 2012). Low densities, extensive time spent underwater, and wide home ranges mean boat-based detection or aerial surveys often need to last for many weeks or be repeated over several years to collect enough information to conduct reliable abundance and density estimates (Jewell et al., 2012). Across their coastal range in the Indian Ocean, *S. plumbea* is recognised as "Endangered" on the IUCN Red List, with monitoring of the species over the past 40 years reliant on aerial surveys, visual surveys and stranding data (Braulik et al., 2015). These costly methods have proved to be challenging for this elusive species with its low detection rates. The production of empirical data for these populations is challenging, as errors are typically high with low recaptures and sighting rates (James et al., 2015; Plön et al., 2021). Quicker and more cost-effective detection assessment methods could be invaluable in these species, allowing for more rapid identification of key regions to prioritise for continued monitoring studies, leading to greater conservation action (Braulik et al., 2017).

Throughout the whole of South Africa, the *S. plumbea* population is thought to number less than 500 individuals, split into two populations with little known exchange between the two and with an extremely nearshore range typically < 20 m water depth (James et al., 2015; Vermeulen et al., 2018). Vermeulen et al. (2018) collated available photo-identification databases over the previous 16 years to assess the movement of individuals between sites in South Africa, identifying only 247 individuals across all study sites. Individuals were shown to have a coastal range of up to 500 km with movements between False Bay and Mossel Bay (Gridley et al., Appendix 1;

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Vermeulen et al., 2018) and Plettenberg Bay and Mossel Bay (James, 2014). Mossel Bay is a well-studied location in the middle of the south coast's population range. Between 2011 and 2013, boat-based photo ID surveys identified 67 individual adult dolphins from 31 surveys over the three-year study period. A 'super-population' of 125 individuals (95% CI: 112 - 140) was calculated over the three-year period, with 32 -71 individuals estimated per year from the open population modelling (James et al., 2015). James et al. (2014) also collated survey data for all studies of *S. plumbea* in South Africa up to that point and highlighted that most studies conform to similar levels of survey effort (60 to 146 surveys over 7 to 36 months). When looking at the catalogue sizes across previous studies, *S. plumbea* were identified at a rate of 0.7 to 2.1 dolphins per encounter or 0.4 to 1.2 dolphins per survey, reflecting the low numbers and small group sizes of these dolphins, which make standard boat-based photo ID studies inefficient (James et al., 2015). Therefore, more efficient methods to detect these elusive species should be investigated with the inefficiency of boat-based detections and the fragmented efforts of long-term photo ID monitoring programs in South Africa (James et al., 2015; Vermeulen et al., 2018).

This chapter explores the application of acoustic detection of *S. plumbea* through signature whistles and compares this to simultaneously collected visual survey and photo ID data. The use of signature whistles in the *Sousa* genus has been suggested from a single-stranded *S. sahulensis* in Australia (Van Parijs & Corkeron, 2001) and an injured *S. chinensis* on the south coast of China (Cheng et al., 2017) while Gridley et al., (Appendix 1) demonstrated signature whistle production in free-ranging *S. plumbea* in South Africa. This chapter will assess individual detection using both methods. The study aims to test the efficacy of 1) spatial variation of dolphin acoustic detection, 2) temporal differences in dolphin detection levels across each method, 3) individual detection effort and 4) to discuss logistical constraints and considerations of both methods within the context of a short-term assessment period.

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METHODS

Data collection: Acoustics

Data were collected in Mossel Bay, South Africa, from 19th March – 5th April 2021. Mossel Bay is an east-facing, sheltered bay with regular sightings of marine mammals along its coastline (Levy, 2017). A short study period was chosen to reflect what is a typical requirement of rapid impact assessments, a common use of photo ID surveys for assessments of threatened species (Braulik et al., 2018). Hydrophones were deployed on moorings in an 8 km linear array following the coastline within the 15 m water depth range (Figure 4.1). Moorings were spaced ~2 km apart to ensure dolphins would not be simultaneously detected on multiple hydrophones within the array. All moorings were deployed in 8-10 m water depth, using identical setups of a ~40 kg railway track weight with instruments 3 m from the seafloor connected to a mid-water and surface buoy. Attempts were made to avoid placing moorings on or near loud reef areas to prevent overloading/excess snapping shrimp noise, but still within the suitable habitat of *S. plumbea*. All hydrophones were scheduled to start at 00:00:00 UTC on 19th March 2021 and turned off when retrieved between the 1st – 9th April 2021. Moorings were visually checked regularly during boat-based surveys throughout the data collection period to ensure no movement, animal entanglements, theft or loss of equipment.

Acoustic data was recorded using SoundTraps (Ocean Instruments Ltd., Auckland, New Zealand). The SoundTraps were set at a 96 kHz sampling rate with a continuous duty cycle of 10-minute files (this is a safety mechanism to produce individual 10-minute files over a continuous period to account for possible equipment recording faults) (Table 4.1).

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Data collection: Photographic Identification

Concurrent boat-based surveys were conducted in the same survey area when weather conditions allowed (Beaufort sea state <4 and no mist or fog to reduce visual sightings). Boat-based surveys followed the coastal tracks of the hydrophone array and were conducted from Cape St Blaize (more commonly referred to as 'The Point', the furthest southerly point in the survey area) to the Kleinbrak River mouth and back (Figure 4.1). During boat surveys, spotters were positioned on the cabin roof of the boat (~ 3 m above sea level) to search for animals, and once *S. plumbea* were observed and approached within ~ 50 m of the boat, the encounter began. Photographic data were collected from all individuals in the group where possible. The group's behavioural state (travelling, social, foraging, milling) and group size were recorded at the start and end of each encounter following Mann, (1999) and Henderson et al., (2011). The boat location (used as a proxy for group location) was also collected on a Garmin eTrex handheld GPS unit, which stores locations every minute (Figure 4.1). Encounters ended once all dolphins had been photographed or left the study area (moved out of visual range).

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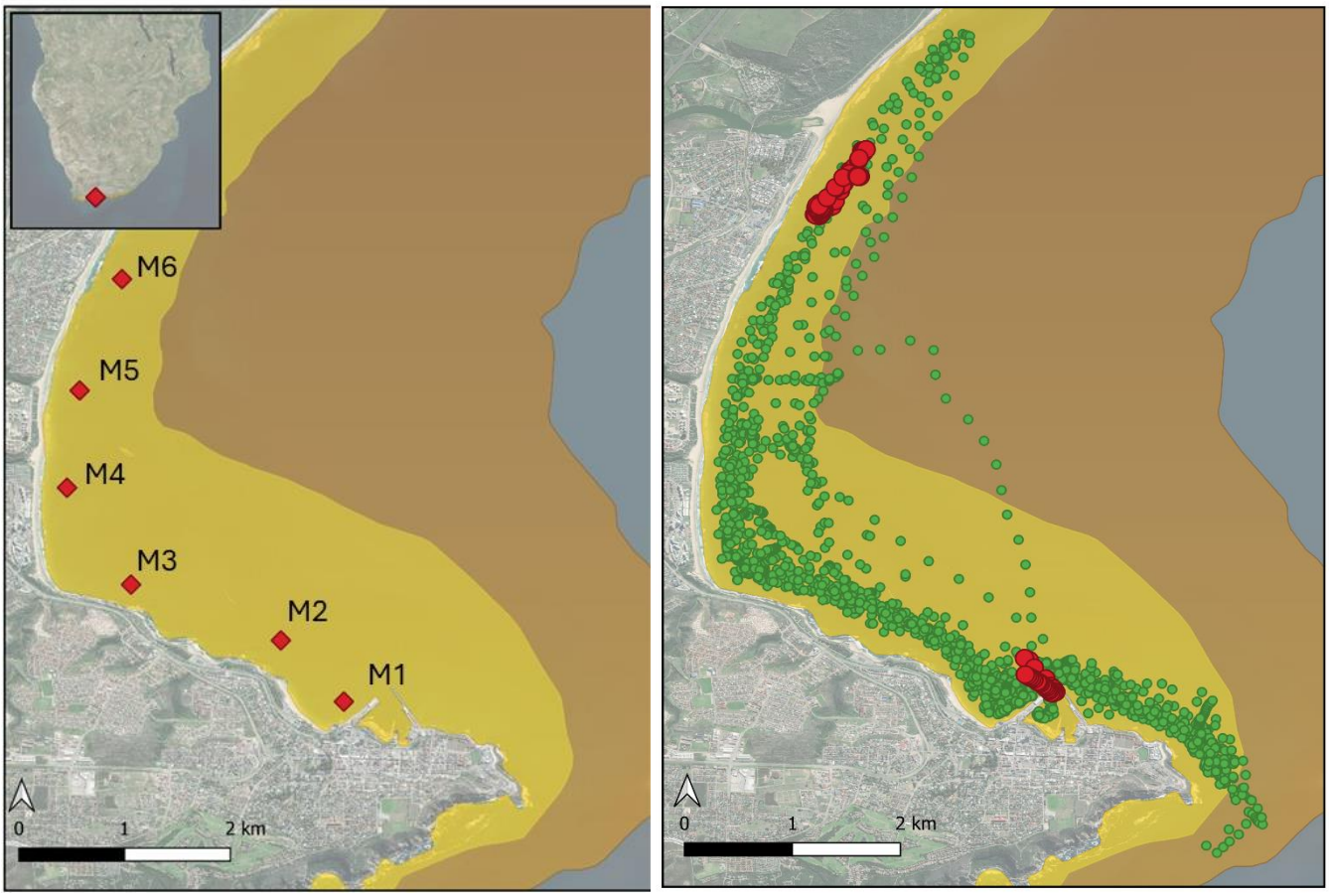


Figure 4.1. Map of moored hydrophone locations (left) and boat surveys (right) for *Sousa plumbea* in Mossel Bay, South Africa, from 19th March – 5th April 2021. Moorings were numbered from mooring 1 (south) – mooring 6 (north). Moored hydrophones are spaced equally ~2 km apart along the coastal range of *Sousa plumbea*. Boat-based surveys are shown with boat survey tracks in green (one location per minute) with tracking during *Sousa plumbea* encounters in Red. Both maps show a 15 m water depth area in orange and 20 m water depth area in brown.

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Table 4.1. Hydrophone deployment locations for *Sousa plumbea* in Mossel Bay, South Africa, from 19th March – 5th April 2021. Includes hydrophone model (SoundTrap300/500), start and end times of recordings and recording settings (Hz).

Unit Model	Mooring Code	Latitude	Longitude	Recording Start	Recording End	Sampling Rate (Hz)
ST500	M1	-34.1767	22.14221	2021/03/19 00:00:00	2021/04/09 12:37:00	96000
ST500	M2	-34.1703	22.13575	2021/03/19 00:00:00	2021/04/09 12:57:00	96000
ST300	M3	-34.1646	22.12037	2021/03/19 00:00:00	2021/04/01 13:31:20	96000
ST300	M4	-34.1446	22.1151	2021/03/19 00:00:00	2021/04/06 17:00:27	96000
ST300	M5	-34.1546	22.1138	2021/03/19 00:00:00	2021/04/03 15:00:11	96000
ST300	M6	-34.1331	22.11944	2021/03/19 00:00:00	2021/04/01 21:50:10	96000

Data analysis: Acoustics

Sousa plumbea occur sympatrically with other whistling dolphins in Mossel Bay (Indo-Pacific bottlenose dolphins *Tursiops aduncus*, occasionally common dolphins *Delphinus delphis*, and very rarely killer whales (*Orcinus orca*) (Bräger, 2011; Levy, 2017). Therefore, a robust species classifier was required to identify periods of *S. plumbea* encounters within the long-term recordings. Concurrent boat and land-based surveys during the deployment period determined *T. aduncus* as the only other cetacean to be present inshore in the bay during this period. During the study period, the only reported sightings of common dolphins were offshore, well outside the bay (reported to S. Dines as opportunistic sightings).

To identify *S. plumbea* whistles from the moored recordings, periods of dolphin encounters first had to be identified to a species level. To do so, the Whistle and Moan Detector (WMD) in the program PAMGuard was used (Gillespie et al., 2014), as well as a regionally trained, two species (*S. plumbea* and *T. aduncus*) classifier module to detect *S. plumbea* encounters from the moorings. The WMD conducts feature extraction of the whistle after filtering and noise reduction,

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eliminating clicks and background noise. These fragments were then classified using the species classifier module in PAMGuard. The classifier was built using training data from boat-based focal follows along the South African coastal range of the species (settings and training data shown in Figure 4.2). The species classifier provides a likelihood of correct species classification. The threshold for species confirmation was raised to 0.8 based on the level of correct *S. plumbea* classification in the confusion matrix (Table 4.2). Whistle outputs were grouped by encounter using custom Python code (Chromiński et al., 2021) with acoustic encounters more than 15 minutes apart (15-minute inter-whistle interval) considered to be a new group encounter, and only encounters with > 80% *S. plumbea* whistles within the encounter used for further analysis. This conservative methodology was used to ensure only single species groups were included in further analysis and removed any encounters with whistles that showed high similarity between the two species.

Table 4.2. Confusion matrix for the regionally trained species classifier using the whistle classifier module in PAMGuard for both *Sousa plumbea* and *Tursiops aduncus* based on test data throughout South Africa. The percentage of correct classification in the training model for *Sousa plumbea* is highlighted in bold.

		Prediction	
		S. plumbea	*T. aduncus*
Reference	*S. plumbea*	**81.9**	18.1
	T. aduncus	7.9	92.1

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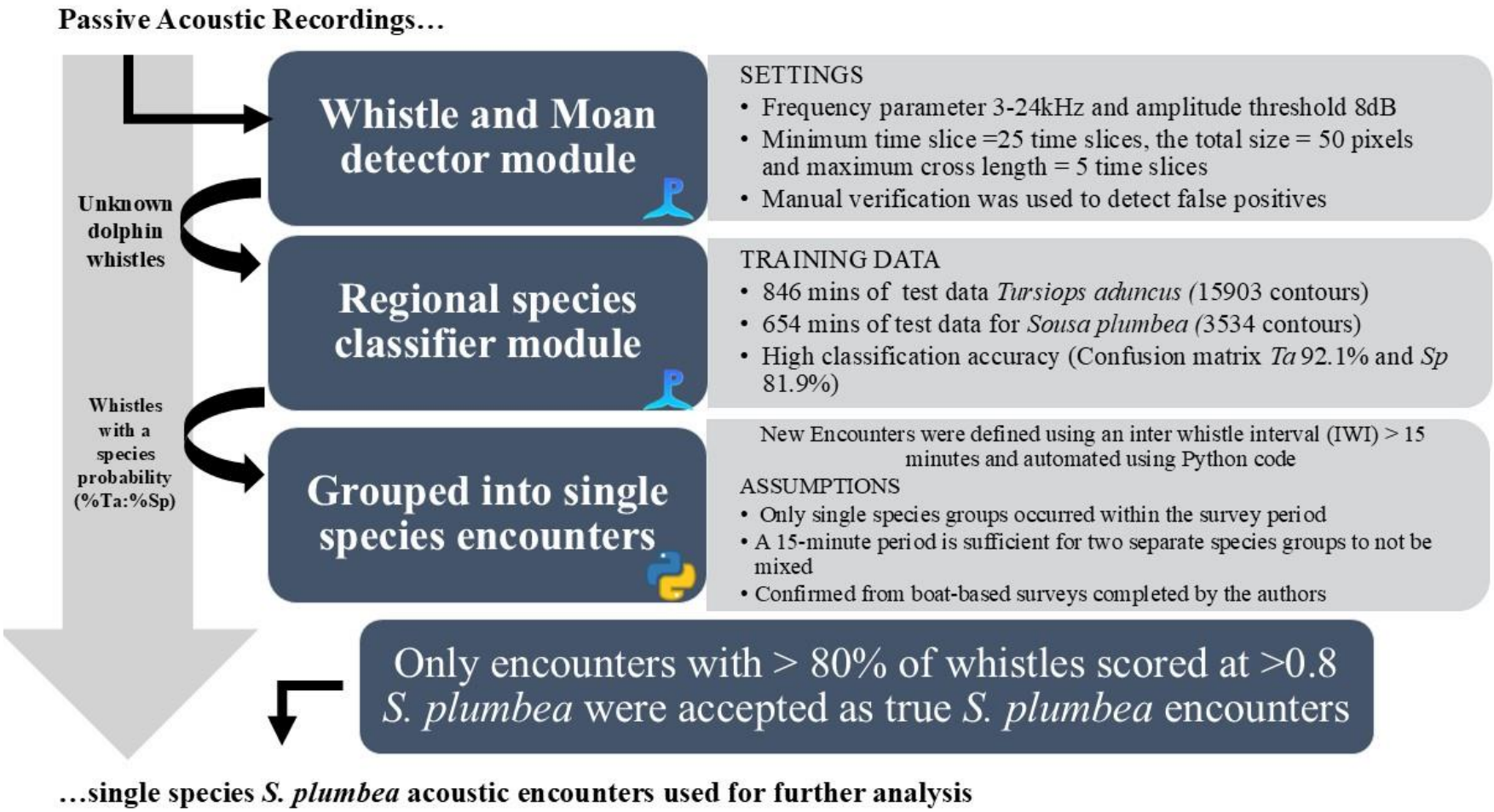


Figure 4.2. Workflow used to identify *Sousa plumbea* whistle encounters from passive acoustic recordings. The species occur sympatrically with *Tursiops aduncus* in Mossel Bay and therefore a workflow to confidently identify *Sousa plumbea* whistle encounters from passive acoustic recordings collected in Mossel Bay, South Africa was necessary. Whistle and moan detector (PAMGuard; Gillespie et al., 2014 www.pamguard.org) was used to identify dolphin whistle encounters in the recordings and a custom trained species classifier in PAMGuard used to identify *Sousa plumbea* whistle encounters.

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In confirmed *S. plumbea* acoustic encounters, whistles were manually verified and annotated in Raven Pro v1.6 (Center for Conservation Bioacoustics, 2019). For manual annotation, a whistle was defined as one continuous contour or two or more repeated contours (a contour being a continuous tonal signal) separated by a period of silence between 0.03 seconds and 0.25 seconds in duration (Kriesell et al., 2014). Each whistle was graded visually based on the signal-to-noise ratio (SNR) (Contrast 24, Brightness 45, fast fourier transformation 1024). Signal-to-noise ratio (SNR) was assigned from 1 – 3, 1) the signal was faint or partially masked but visible on the spectrogram, 2) the signal was unambiguous, and 3) the signal was prominent and dominant in the spectrogram (Gridley et al., 2015). Only good-quality whistles (those graded SNR 2 or 3) were used in further analysis.

Stereotyped whistles were categorised into possible signature whistle types using a confirmed signature whistle type (SWT) catalogue or using a bout analysis approach (SIGID, Janik et al., 2013). A catalogue of *S. plumbea* SWTs (Gridley et al., Appendix 1) used for categorisation in this study was generated from acoustic recordings collected from boat-based focal follows of *S. plumbea* along the South African coastline. In this catalogue, stereotyped whistles were visually categorised into signature whistle types (SWTs) following the SIGnature IDentification method by Janik et al., (2013). As these catalogued whistle types had previously met SIGID, in this study one capture of a whistle was required to match a whistle type in the catalogue to be classed as a signature whistle detection. New whistle types seen in this study, not present in the catalogue, could be added if they fulfilled SIGID criteria. To be considered a likely signature whistle (SW), a minimum of 3 out of 4 whistles of the same type (the same frequency modulation pattern) should occur in a bout with inter-whistle-intervals (IWIs) of 1 to 10 seconds (Janik et al., 2013). If a stereotyped whistle was present in a bout but did not match the catalogue and did not fulfil full SIGID criteria, it was classified as an unclassified SWT (SWTUC) (these can also be referred to as a repeatedly emitted whistle type (REWT)). Once the whistle was assigned to a SWT, it was

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counted as a 'capture', and the associated date, time and mooring were noted. Acoustic encounters were defined from the first whistle detected in the encounter to the last, with an inter-whistle interval of 15 minutes or more to separate encounters (Frainer et al., 2023; Lusseau, 2003). Only encounters with more than one whistle were included in further analysis to determine encounter duration (time difference from first to last whistle).

Data analysis: Photographic Identification

During boat-based focal follows, efforts were made to ensure that every individual within the encounter was photographed. Photo quality and distinctiveness were graded from 1 (poor quality) to 5 (excellent) based on clarity, focus, distance, water spray and other features that would prevent successful identification (Elwen et al., 2009). Only photographs that scored 4 or above were used for further analysis. To determine the time taken in an encounter to identify all individuals, the time-stamp of the photo was taken once all individuals had been photographed clearly (a photo grade of 4 or above, and left and right sides photographed if possible). Based on these photos, a daily ID list for each encounter was produced of the best left- and right-hand side photos for each individual within the encounter. Repeated sightings of individuals across encounters were noted, and individuals were matched to the national *S. plumbea* photo ID catalogue where possible. The national catalogue for *S. plumbea* was collated through the SouSA consortium (n = 247 individuals, Vermeulen et al., 2018) Since 2016, Sea Search Research and Conservation has maintained and updated the catalogue, with ~ 83 new individuals added (SSRC unpublished data). For this study, if an individual did not match the national catalogue, it was given a new temporary reference number.

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The efficacy of Acoustics vs Photo ID

The efficacy of photo ID surveys using boat-based focal follow methods was compared to acoustic survey methods by assessing 1) spatial variation in acoustic detection, 2) temporal differences in dolphin detection across each method, 3) individual detection rate across encounters in each method, and 4) assessing financial and logistical consideration of both methods.

Spatial variation in dolphin detection was analysed across the survey area to investigate fine-scale differences in signature whistle production throughout the array area. To analyse the effect of different mooring locations on SW counts, a generalised additive model (GAM) was fitted using a Poisson distribution and log link function. The model used the GAM function in R (version 4.0.2; R Core Team, 2020) using the *mgcv* core R package (Wood, 2017). The response variable was the count of SWs per mooring, and the predictor was mooring. Each mooring location was treated as a separate level with its own parameter estimate by including the factor variable without an intercept (factor(Mooring) - 1). The model fit was evaluated using the unbiased Risk Estimator (UBRE), commonly used to assess the goodness of fit in models with smoothing terms, with scores close to zero indicating appropriate model fit.

Temporal differences were investigated through the relationship between SWTs (number of individuals detected) and SWs (total signature whistle count) per hour over the day period. Generalised additive models were completed in R (version 4.0.2; R Core Team, 2020) using the *mgcv* R package (Wood, 2017). The distribution used was Poisson and estimated using log link function in the GAM model, as the response variable was a count. Model fit was again assessed using an unbiased risk estimator (UBRE). GAMs are nonparametric extensions of generalised linear models and were selected as they allow for flexible modelling of the nonlinear relationship between the predictor and response variables (SW or SWT) by fitting a smooth function to the predictor variable (mooring or hour) (Wood, 2006).

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Individual detection was then investigated across both methods. Descriptive statistics (mean (SD) and minimum and maximum values) were calculated for acoustic encounter duration and the time taken to detect all individuals in the encounter acoustically. For the total number of SWTs detected in an encounter both the median (IQR) and maximum value were calculated. Consistency in analysis was kept between methods with the same descriptive statistics calculated for photo ID encounters. Each method's detection rate was measured as the number of individuals in an encounter (SWTs or photo IDs) divided by the encounter duration (minutes). To compare the effects of encounter duration on the number of individuals detected, the total number of individuals detected per encounter was correlated to 1) total encounter duration (minutes) and 2) the time taken to detect all SWT or photograph all individuals within the encounter (minutes) using Pearson's correlation.

The detection of unique individuals over the deployment period was investigated using a discovery curve. Cumulative unique SWTs and photo IDs showed the detection rate of new individuals over time across both methods. For discovery curves, a population is considered open if there is a steady accumulation of newly identified individuals over time and considered closed if the accumulation of new individuals reaches a consistent plateau, indicating that all individuals in a population have been detected as capture effort/time continues (Baker et al., 2006). This study period was relatively short, with *S. plumbea* known to be able to move further east and west of the Mossel Bay study area within this time period (James et al., 2015; Vermeulen et al., 2018). Therefore, a closed population was not expected; rather, the discovery curve was conducted to monitor the degree of accumulation of detections over this short period.

The final measure of efficacy between the two methods was logistical considerations, assessing the cumulative effort in terms of equipment, boat time, research assistance, and financial costs

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attributed to both methods and including a cost-benefit analysis. This comparison is important to future researchers and managers who might be assessing the implementation of monitoring studies and helping to inform experimental design and expectations attributed to both methods.

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RESULTS

Acoustics

The acoustic array was deployed for 18 days with 151,004 minutes of total acoustic recordings across six moorings. In total, there were 305 *S. plumbea* whistles, of which 226 were good signal-to-noise ratios (SNR 2 or 3). Across all six moorings, 199 signature whistles were categorised into 10 SWTs. These signature whistles were classified into SWTs based on the *S. plumbea* SWT catalogue from Gridley et al., (Appendix 1). New signature whistle types (that were not already present in the catalogue) could be included if they fulfilled SIGID bout analysis. However, none of these new stereotyped whistles met SIGID and were classed as SWTUC (n = 27 whistles).

Whistles were detected on all moorings across 38 acoustic encounters over 11 days (including SWTUC and low SNR 1 whistles). Mooring 3 only detected low SNR whistles, which were not considered for further analysis. Of the 10 SWTs detected, four SWTs were detected on more than one mooring, with SWT66 detected on all moorings except for mooring 3 (Figure 4.3). Seven SWTs were recaptured on multiple encounters, with four SWTs seen on multiple days (Figure 4.3). Two SWTs were detected on different moorings on the same day (SWT60 and SWT66). However, the minimum time between these detections was 20 minutes, which is sufficient time for individuals to travel between moorings (Rohr et al., 2002). No SWTs were simultaneously recorded on multiple moorings. Moorings with the greatest number of different SWTs (individuals) detected were moorings 1 and 6 (> 5 SWTs detected on both moorings) (Figure 4.4). Signature whistle detection (overall whistle count) also varied across moorings, with 80% of signature whistles detected at moorings 1 and 4 (Figure 4.4). Across encounters, several individuals dominated, with SWT66, SWT60 and SWT5, accounting for 73% of total SW detections.

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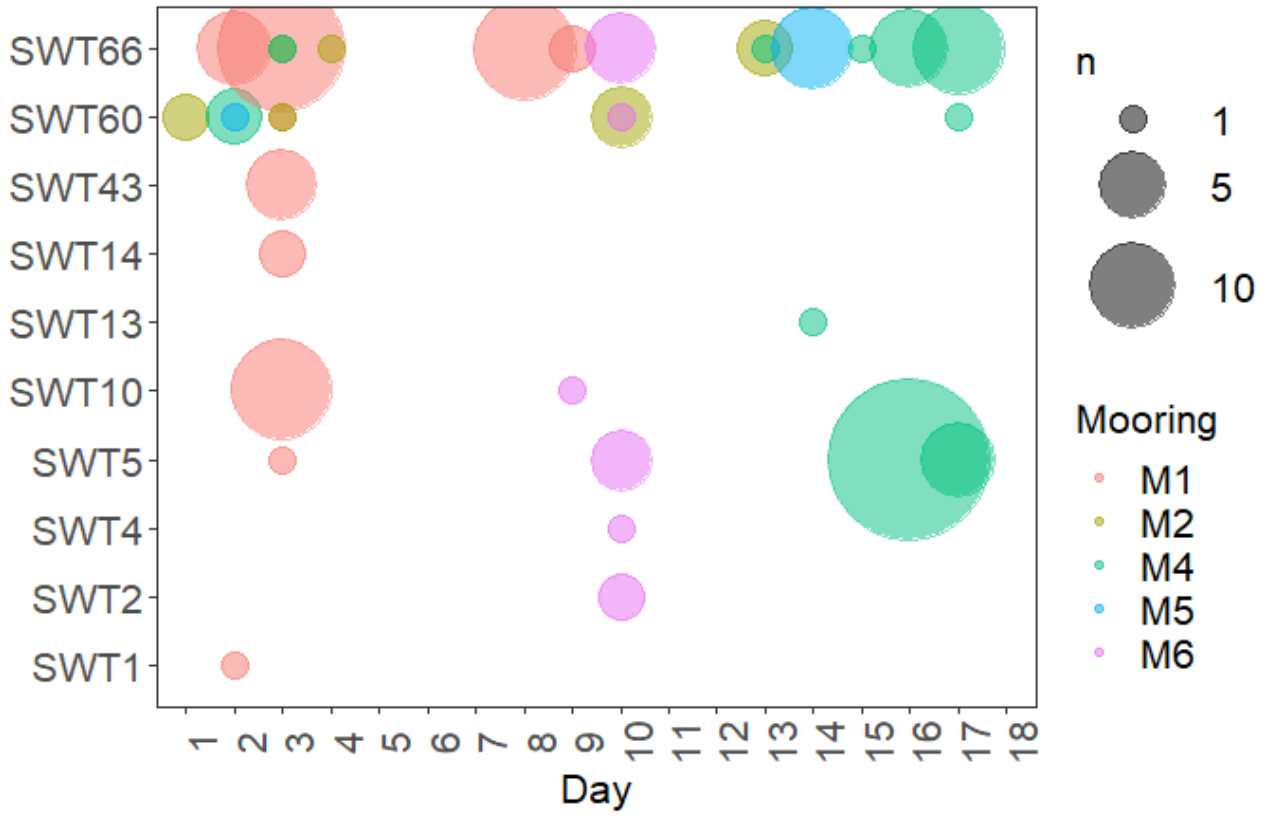


Figure 4.3. Signature whistle type detections (SWT, acoustically unique individuals) of *Sousa plumbea* in Mossel Bay, South Africa across the 18-day deployment period in Mossel Bay, South Africa, from 19th March – 5th April 2021. The moorings the SWTs are detected on are represented by different colours and the size of the dot represents the number of detections of that whistle type. Detections are grouped by 24-hour bins (days).

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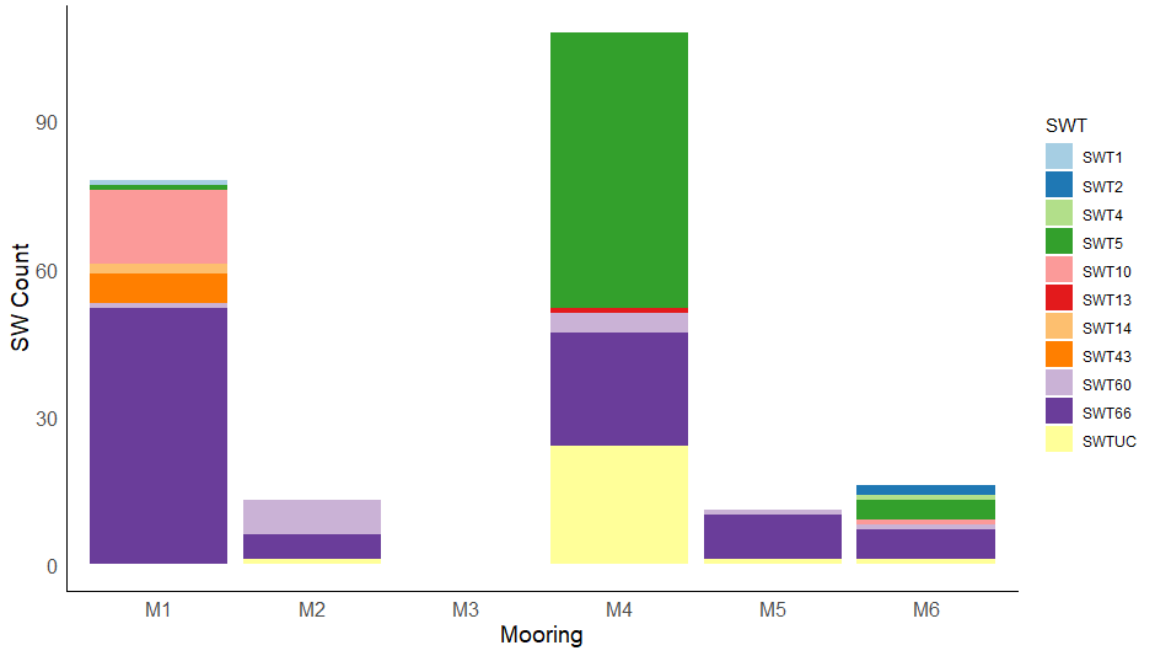
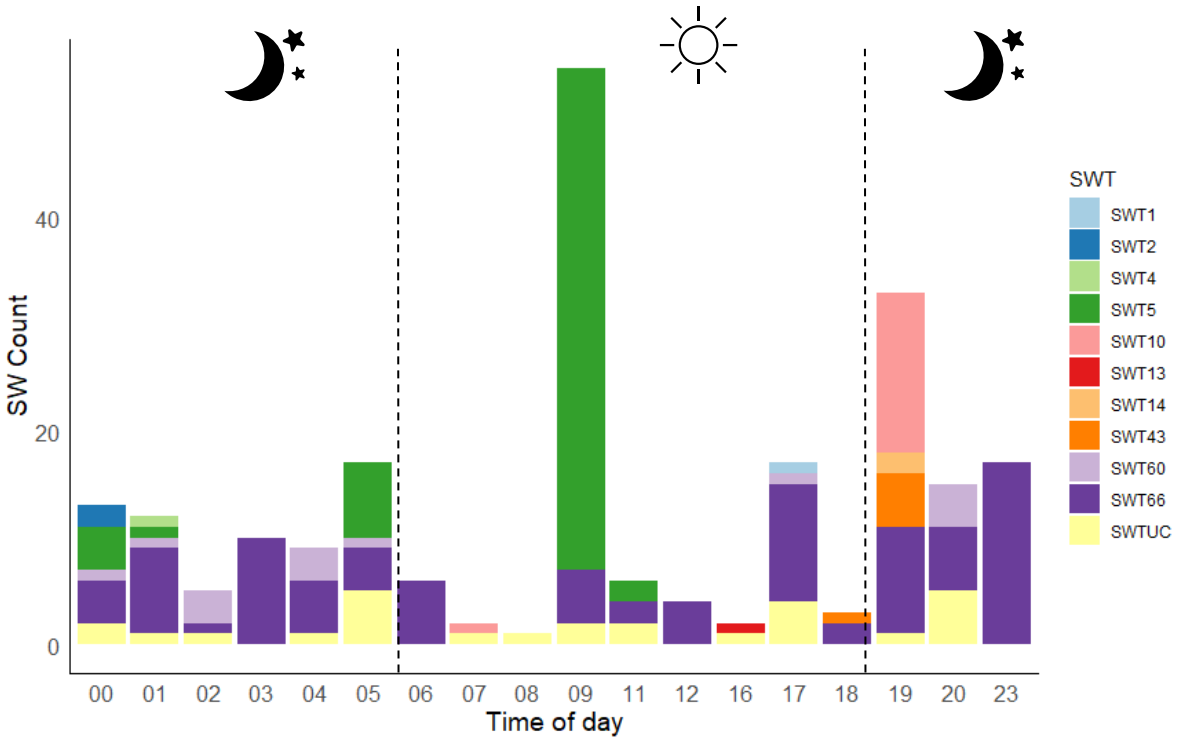


Figure 4.4. The signature whistle count (signal-to-noise ratio of 2 or 3 and including SWTUC; unclassified whistle types) of *Sousa plumbea* at each of the six moorings in Mossel Bay, along the south coast of South Africa, representing the spatial variability of dolphin detections between sites. Bars represent the signature whistle count of SNR 2 and 3 whistles at each mooring. Bars are divided by signature whistle type (SWT, proxy for individuals).

To investigate the temporal detection of individuals, signature whistle counts were analysed across a 24-hour period, split into SWTs (individuals) and overall dolphin detections (SWs), including SWTUC. The diurnal pattern for SWT production (including SWTUCs) showed a diurnal effect of SWTs (individuals) detected (proportion of SWTs detected; Day 41% Night 59%). Total dolphin call detections (SW count) also showed diurnal variability, with 62% of acoustic detections at nighttime (1800 - 0600) (Figure 4.5a). However, one encounter at 0900 accounted for a significant spike in whistle count (SWT5). Omitting this outlier and SWTUCs, 25% of SWs were produced during the day, while 75% of SWs were produced at night, showing a higher diurnal pattern in signature whistle production (Figure 4.5b).

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A



B

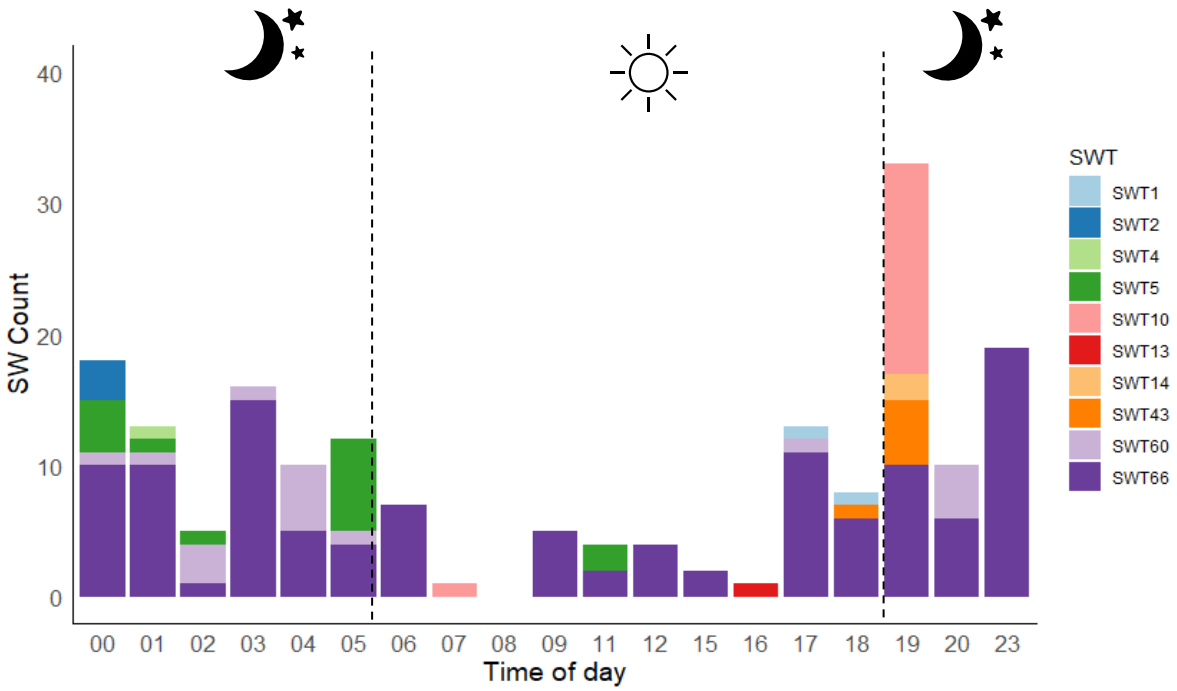


Figure 4.5. Diurnal patterns of signature whistles detected from *Sousa plumbea*, in Mossel Bay, South Africa, from 19th March – 5th April 2021, classified into signature whistle types (SWTs). The night period is specified as 1800-0600. 10 different SWTs (number of individuals) were detected throughout the deployment period. The plot shows the total signature whistle count for every SWT detected over 24-hour time bins. A) Plot includes all SWTs including unclassified SWTs (SWTUC), which are included at the bottom of the stacked bar plot (yellow) and B) Plot removes SWTUC and outlier encounters (SWT05, 0900).

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For spatial analysis, a generalised additive model (GAM) was used to analyse the relationship between the number of SWs per mooring (signature whistle count) and the mooring locations (mooring). SWTUC and outliers were removed from the modelling (SWT5, 0900 encounter). The model indicated significant differences in SW counts across moorings ($p < 0.01$, all except for mooring 3), with estimate effects highest for mooring 1 (4.58). The UBRE score was -0.94, indicating an appropriate fit for the Poisson-distributed response variable.

To explore the relationship between the number of individuals calling (number of unique SWTs) per hour and the time of day, a GAM was fitted with a smooth term for hour, again omitting SWTUC and outliers. The model showed a significant nonlinear relationship between the number of unique SWTs per hour and the time of day ($se = 0.05$, $z = 19.1$, $p < 0.005$). The smooth term for hour was significant ($edf = 7.7$, chi-sq = 59.3, $p < 0.005$), suggesting that the number of individuals detected (SWT) vary throughout the day when SWTUC and outliers are removed (Figure 4.6b). Again, the UBRE score was -0.8, indicating an appropriate model fit for the Poisson-distributed response variable. When SWTs per hour were investigated across hour time bins, significant levels of detection were seen for time bins 0000 - 0600 and 1700 -2000 ($p < 0.005$).

To investigate the relationship between overall signature whistle count per hour and the time of day, a GAM was fitted with a smooth term for hour, again omitting SWTUC and the outlier encounter. The model showed a significant nonlinear relationship between the number of SWs per hour and the time of day ($edf = 2.7$ $se = 0.02$ $z = 125$ $p < 0.005$). The smooth term for hour was also highly significant ($edf = 8.8$ chi-sq = 478.5, $p < 0.005$), again suggesting a complex, nonlinear relationship between the time of day and the number of SWs per hour (Figure 4.6b). The model's UBRE score was 0.7, indicating an appropriate fit for the Poisson-distributed response variable. When investigated across time bins, signature whistle counts per hour were

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significant throughout ($p < 0.5$). However, estimated coefficients were lowest for periods 0700-1600, showing a smaller effect on SW detection during these hours.

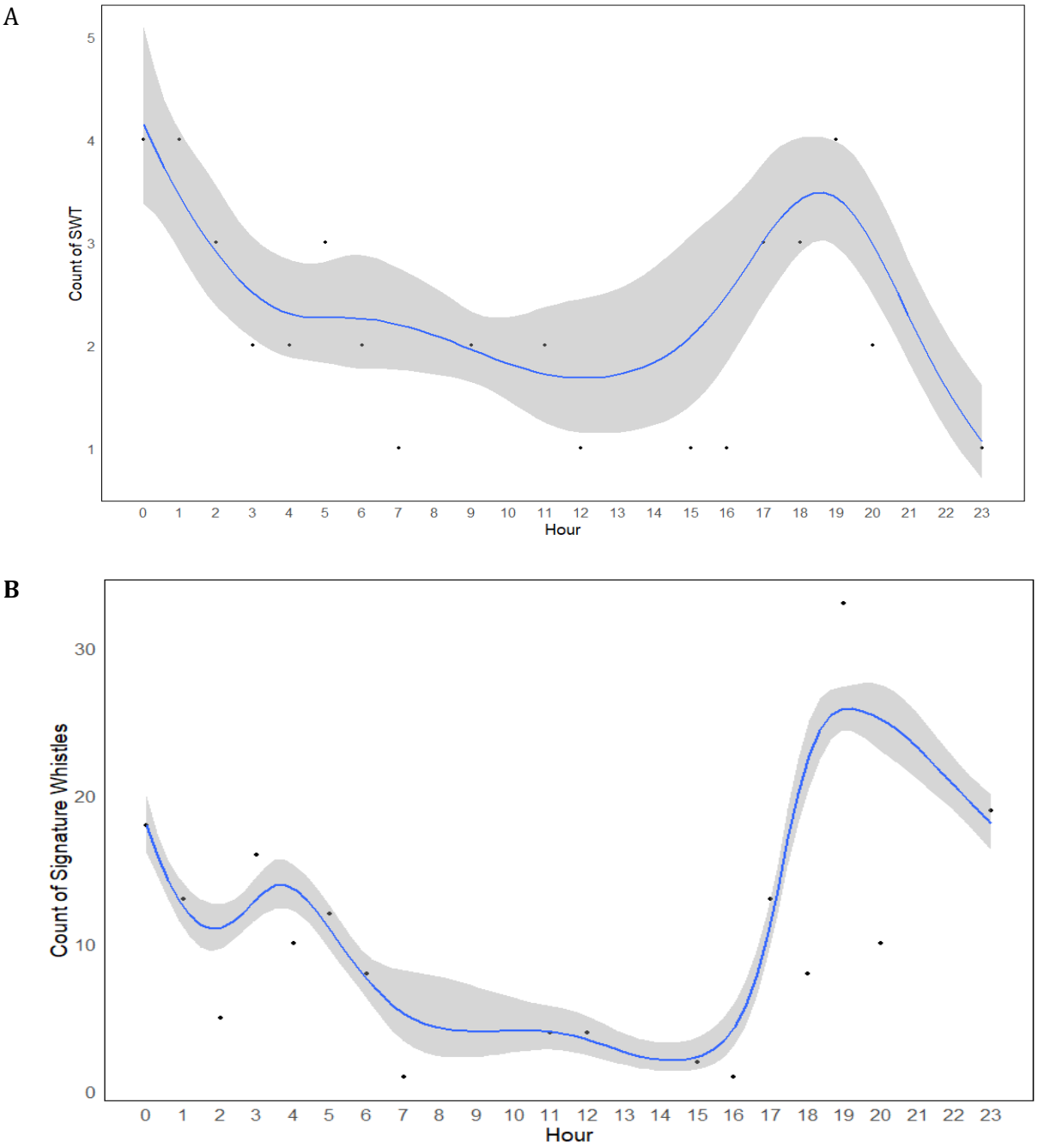


Figure 4.6. Generalised Additive Models showing the probability of detections of signature whistle type, SWT (plot A - gam(unique_SWT_per_hour ~ factor(hour) -1, family = Poisson(log))) and signature whistle count, SW (plot B - gam(SW_per_hour ~ factor(hour), family = Poisson(log))) over a diurnal period for *Sousa plumbea* in Mossel Bay, South Africa, from 19th March – 5th April 2021. The grey region represents the 95% CI region. Both plot A, signature whistle types (SWTs, count of individuals) and plot B, signature whistles (SWs, total whistle count) indicated a diurnal pattern, with increased detection over the night period 1800-0600. In both models, SWTUC and outliers were omitted.

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Individual acoustic detection was measured across the 38 signature whistle encounters (including whistles of SNR 2 or 3 only and removing SWTUC and outliers). The mean rate of SWs detected across acoustic encounters was 0.89 SW/minute (± 1.48 SD), with encounter durations ranging from 00:00:13 to 00:31:57 (hh:mm:ss). The median number of different SWTs (number of calling individuals) detected in an encounter was 2 (IQR 1 - 2) with a maximum of three unique SWTs detected in an acoustic encounter. Compared to the full duration of the acoustic deployment, the cumulative encounter duration (03:53:05 hh:mm:ss) was short, only 0.15% of the total recording effort, with each acoustic encounter lasting on average 00:06:14 (± 00:08:18 SD). There was no correlation between total SWTs detected in an encounter and 1) acoustic encounter duration and 2) the time taken to detect all the SWTs within the encounter (Pearson's correlation R = -0.03 and R = 0.17, respectively).

Photographic Identification

There was a total of 13 boat-based survey days across the 18-day acoustic deployment period (28:39 hh:mm at sea). In total, there were eight encounters with *S. plumbea* on seven different days. The mean encounter duration was 00:46:22 hh:mm:ss (± 00:21:17 SD), with a mean time taken to identify all individuals in an encounter 00:24:00 (± 00:15:34 SD). Across all encounters, the time taken to photograph all individuals was 3.7% of the total recording effort (boat-based survey time). The boat-based survey covered the inshore acoustic deployment array area with all photo ID encounters concentrated in two regions along this area, around mooring 1 (Harbour entrance) and past mooring 6 (Hartenbos river mouth) (Figure 4.1).

The median number of individuals photographed in a group was 3 (IQR 1 - 4), with a maximum of six individuals photographed in an encounter. Across the eight encounters, 16 different individuals were identified, and six individuals were seen across different days. Calves were

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visually confirmed and photographed within the group across four encounters. Six individuals were able to be matched to the national catalogue. With a dedicated photographer on board, the mean individual detection rate for photo ID was 0.08 individuals photographed/minute (± 0.05 SD) (Table 4.3). There was no correlation between the total number of individuals photographed in the encounter and 1) photo ID encounter duration and 2) the time taken to photograph all the individuals in the encounter (Pearson's correlation R = 0.27 and R = 0.30, respectively).

Table 4.3. Summary of *Sousa plumbea* detections during the 18-day deployment period in Mossel Bay, South Africa, from March 19th to April 5th, 2021. The table provides summaries of encounter durations and the time taken to detect all individuals in each encounter using both acoustic and photo ID methods, along with overall individual detection rates (total number of individuals/encounter duration in minutes). The individual detection rate was calculated by dividing the total number of individuals by the encounter duration. For each metric, the mean (standard deviation, SD), minimum, and maximum values are presented.

	Acoustics		
	Encounter Duration	**Time taken to detect all signature whistle types (SWTs)**	**Individual detection rate** *(Total number of SWTs / Enc duration, mins)*
Mean (SD)	00:06:14 (± 00:08:18)	00:01:48 (± 00:01:50)	0.89 (± 1.48)
Minimum	00:00:13	00:00:13	0.06
Maximum	00:31:57	00:06:41	5.17
	Photo ID		
	Encounter Duration	**Time taken to photograph all individuals**	**Individual detection rate** (*Total number of individuals photographed / Enc duration, mins)*
Mean (SD)	00:46:22 (± 00:21:17)	00:24:00 (± 00:15:34)	0.08 (± 0.05)
Minimum	00:22:00	00:05:58	0.17
Maximum	01:20:00	00:45:13	0.13

Acoustic methods had a slightly faster rate of cumulative detections in the first half of the recording period compared to boat-based surveys, recording more than half the total SWTs in the first half of the recording period (seven new SWTs were detected in the first nine days, 64% of total SWTs detected) (Figure 4.7). Across photo ID surveys, only half of the total individuals had been photographed after eight survey days (out of 13 total survey days). Acoustic and Photo ID unique detections both plateaued at day 3, for six and five days respectively, during which there were three boat surveys where no new individuals were photographically identified. Acoustic

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detections again plateaued toward the end of the survey period on day 15, however, this is likely limited by deployment duration.

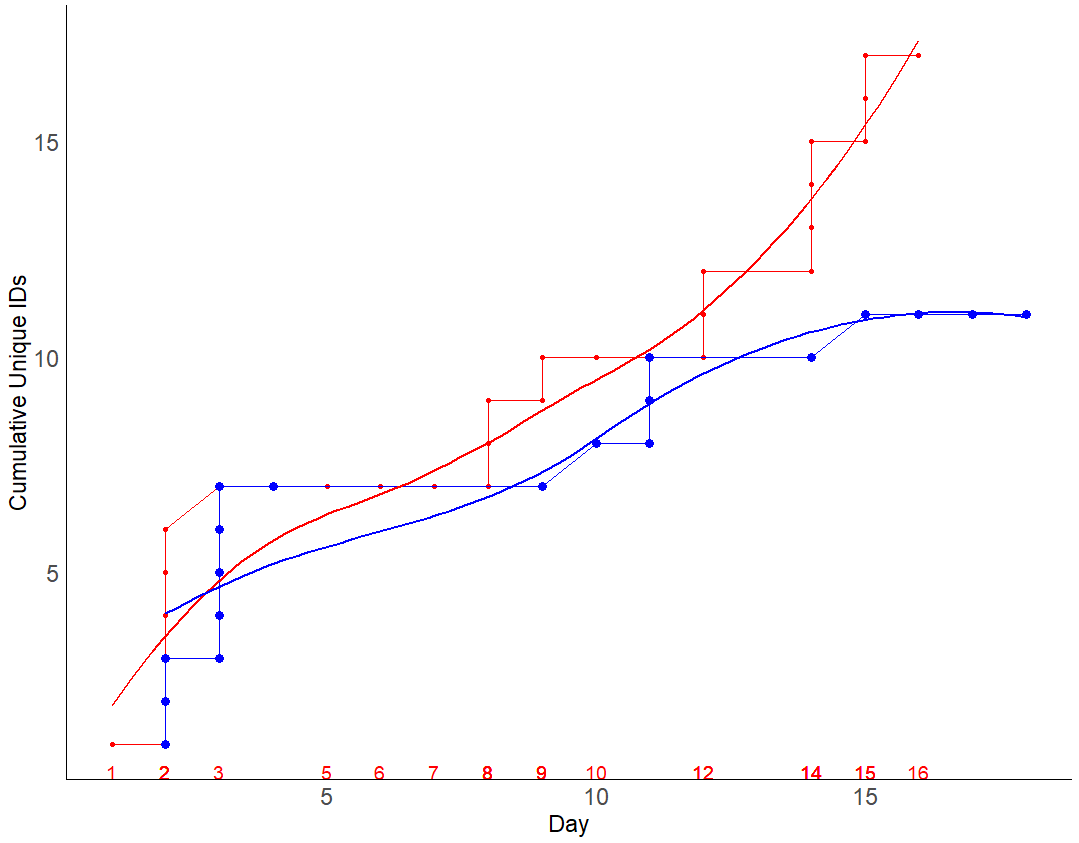


Figure 4.7. Cumulative unique IDs of individual *Sousa plumbea* detected over time through both photo ID and acoustic methods during the 18-day deployment period in Mossel Bay, South Africa, from March 19th to April 5th, 2021. Photo ID detections, red dots, indicate new IDs detected over time with a smoothed curve to show the overall discovery rate. Red numbers along the axis represent boat-based survey days (n = 13), and numbers in bold indicate boat-based surveys where new individuals were photographed (Days 1, 2, 8, 9, 12, 14, 15). Acoustic detections of signature whistle types (SWTs) over time, blue dot, indicate cumulative new SWT detections over time.

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Logistical Considerations

In this study, the survey costs for acoustic methods, including boat-based deployment and retrieval, with equipment and sustenance costs, had a total cost of 367,264 ZAR due to the inclusion of hydrophone and equipment purchasing (Table 4.4). Once these costs are removed (as hydrophones can be used across multiple projects over many years), the costs fall to 17,264 ZAR for the four days required to set up, deploy and retrieve the acoustic array, excluding personnel time. Operational costs for personnel time for the researcher conducting the deployment through set up, during deployment/retrieval and post-processing and analysis, at a daily stipend of 200 ZAR per hour, equates to operation costs of 105,400 ZAR. Equalling a total cost for the acoustic method of 122,664 ZAR.

Photo ID survey costs were calculated based on the total 18-day fieldwork period with 13 sea days. This equated to 28:39:00 hh:mm:ss of boat-based survey time of which there was a cumulative 05:52:00 (hh:mm:ss) *S. plumbea* encounter duration. The boat-based survey cost for photo ID was 161,000 ZAR, including land travel, accommodation, subsistence, and boat fuel/hire costs and skipper costs. Depending on deployment locations, these costs are also incurred during acoustic deployment and retrieval, usually a maximum of 2-4 days. This cost also included equipment costs, such as an SLR camera and a lens, removing equipment costs brings the boat-based survey costs down to 61,000 ZAR. Operational costs to include one researcher for the field work, data processing and analysis was 41,400 ZAR. Therefore, the total costs for photo ID were 102,400 ZAR for the 18-day deployment period. Most research groups will have access to camera and hydrophone equipment; therefore, these equipment costs were removed from the final cost for each method. These total costs were then calculated per dolphin captured to produce a final result of 12,266.40 ZAR.dolphin^{-1} for acoustic methods and 6,400 ZAR.dolphin^{-1} for photo ID methods.

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When evaluating the use of these methods, several factors must be considered (Table 4.4). Acoustic deployments, for instance, require time for setting up hydrophones and constructing moorings. In this study, with an array of six hydrophones, the setup time was minimal, taking only half a day. Similarly, boat-based deployment times can vary depending on the size of the array area; however, this study focused on a single bay, allowing deployment to be completed in just a few hours, with half a day allocated for both deployment and retrieval. Although processing acoustic data can be time-consuming, the relatively small scale of this study, limited to six hydrophones over an 18-day deployment, meant that processing time was shorter compared to larger acoustic studies. The most time-consuming aspect was the analytical process, which involved extracting, annotating, and categorising whistles into an existing SWT catalogue, which spanned several weeks.

In this study, boat-based surveys were conducted over a two-week period to align with the acoustic deployment period. This required having a boat on standby throughout the period, ready to be deployed during suitable weather conditions. The setup time for boat-based surveys was minimal, as trained personnel only needed to ensure that equipment was charged, and data sheets were prepared before launching. The primary time commitment for boat-based surveys was the surveys themselves. Post-processing of the data was also relatively quick, with daily photo IDs completed within a few hours after each survey day. A direct comparison of costs and logistical considerations between the methods is provided in Table 4.4, while a general SWOT (strengths, weaknesses, obstacles and threats) analysis of both methods is presented in Table 4.5.

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Table 4.4. Fieldwork and operational costs and total project hours incurred across both acoustic and boat-based recording methods for *Sousa plumbea* during during the 18-day deployment period in Mossel Bay, South Africa, from March 19th to April 5th, 2021. Costs per individual *Sousa plumbea* were calculated as operational costs (total time X researcher time) plus total fieldwork costs (minus equipment) and based on a total of 10 signature whistle types (SWTs) and 16 Photo ID individuals detected during this survey period using either method.

	Hours	**Cost (ZAR)**
ACOUSTIC DEPLOYMENT		
Survey Costs (2 days Deployment/Retrieval)		
Boat and skipper hire (incl Fuel)		12264
Commercial diver day rate		5000
Researcher cost per hour		200
Equipment (5 x ST300 unit cost 70000 ZAR)		350000
Time		
Project Set Up	05:00:00	
Boat Based Time for deployment/retrieval	08:00:00	
Data Processing Time	10:00:00	
Data Analysis (3 weeks)	504:00:00	
Total Fieldwork Time	**13:00:00**	
Total Time	**527:00:00**	
Total field work costs		367264
Total field work costs (hydrophone removed)		17264
Operational costs		105400
Total Costs for Acoustics		**122664**
PHOTO ID		
Survey Costs (18 days with 13 days at sea)		
Logistics		41000
Boat hire/fuel/skipper		20000
Researcher cost per hour		200
Equipment		100000
Time		
Project Set Up	01:00:00	
Boat Based survey Time	28:39:00	
Data processing Time	05:00:00	
Data analysis (1 week)	168:00:00	
Total Fieldwork Time	**29:39:00**	
Total Time	**207:33:00**	
Total field work costs		161000
Total field work costs (camera equipment removed)		61000
Operational costs		41400
Total Costs for Photo ID		**102400**
COSTS PER DOLPHIN		
Acoustic (10 SWTs)	53 hrs.dolphin^{-1}	12266.40 ZAR.dolphin^{-1}
Photo ID (16 Photo ID individuals)	13 hrs.dolphin^{-1}	6400 ZAR.dolphin^{-1}

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Table 4.5. SWOT (Strengths, Weaknesses, Obstacles and Threats) analysis of both acoustic and photo ID methodologies to detect *Sousa plumbea* during the 18-day deployment period in Mossel Bay, South Africa, from March 19th to April 5th, 2021.

	Acoustics	**Photographic Identification**
Strengths	• Long continuous deployment time of weeks to months (battery limited) • No animal interaction • Broader range of study area (limited by equipment access) in comparison to daily boat survey areas • High initial costs offset long-term by repeated equipment use and minimal data collection costs	• Relatively low-cost data collection • Accessibility to data through the use of tourism boats as a platform, and the contribution of citizen scientists • Additional information on gender, health, and behaviour can be collected. • Computer-assisted matching of fin IDs rapidly improving
Weaknesses	• High cost of equipment purchase • Requires deployment and retrieval. Use of commercial divers or pop-up acoustic moorings (added costs)	• Requires boat access • Requires extended periods near animals • With *S. plumbea*, they have a low encounter rate and low encounter duration • Restricted range, limited by boat range over a day. • Greater long-term costs with boat time and personnel costs
Obstacles	• Procurement of hydrophones • High computational power and storage required. • Acoustic analysis requires acoustic technicians. • Requires custom regional species detector models • Instrument failure is undetected until recovery	• Boat-based surveys are limited by single study locations; therefore, it is difficult and time-consuming to document the whole range. • *S. plumbea* occupy very inshore shallow areas, dangerous working conditions for boat-based researchers
Threats	• High reliance on hydrophones. High risk of equipment damage, theft, failure	• Reliance on opportunistic boat surveys – ecotourism groups provide minimal access to groups with low encounter durations. These periods are often not long enough to photograph all individuals in a group. • Requires multiple encounters over a greater length of time to access *S. plumbea* to collate data.

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DISCUSSION

The study aimed to investigate the efficacy of both photo ID surveys and passive acoustic monitoring methods to detect and monitor an elusive coastal delphinid species in South Africa. For threatened species with low population sizes, boat-based focal follow surveys for photo ID can be a time-consuming and expensive research method requiring boat access and hire, as well as repeated surveys over several weeks and repeated over multiple years to generate sufficient data for managers and policymakers (Braulik et al., 2018). South Africa lacks long-term consistent studies over time and regions, with all available studies localised and never exceeding 3 years (James et al., 2015, Table 7). In this study, encounter rates were relatively low across both methods, indicative of this species' low population size throughout the coastline, with a lower encounter rate using photo ID methods, but comparable to those of previous studies on the species in South Africa (James et al., 2015, Table 7). The study found clear temporal differences with acoustic activity greater during the night period, showing high dolphin activity during a period unavailable to boat-based surveying. Spatially, both methods detected high levels of dolphins around the Harbour entrance (mooring 1), with boat-based encounters also high around Kleinbrak River mouth areas (mooring 6) and acoustic detections also high around Hartenbos (mooring 4). Boat-based encounters only occurred in two areas of the bay, whereas acoustic detections across the whole array show dolphins moving throughout the deployment area. Financial considerations showed that due to a higher analysis time and the lower total number of individuals detected, acoustic methods had a greater cost per dolphin than photo ID methods. It is important to understand the differences between these methods and how variability in both field-based differences and logistical considerations affects visual and acoustic detection when deciding to use either methodology for effective data collection.

An important consideration for this study is the short survey period, as this was designed to reflect a realistic time frame in which field trips to distant locations or environmental impact assessments are often made. Environmental impact assessments and rapid biological

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assessments are often required in marine spatial planning, harbour construction and marine construction (Mubanga & Kwarteng, 2020). Rapid environmental assessments for threatened species are important as many are underrepresented in published studies with ranges and movements poorly understood (Becker et al., 2021). Effective methods to collect broad-scale baseline data on threatened species over a limited temporal period are key to enabling effective conservation policy (Braulik et al., 2017). In this short-term study, understanding how acoustic monitoring can be best used to detect the presence/absence of *S. plumbea* under a limited time frame is an important consideration if this is to be proposed as a viable monitoring method. This study's key result was the spatial and temporal variability in dolphin detection using both methods. Acoustically, dolphins were detected throughout the 8 km acoustic array across both day and night periods, with higher acoustic activity at night, indicating a broader scale of detection than if only boat-based encounter data was used.

Survey design includes deciding where to deploy hydrophones and recording set-up (duty cycling). This is often a tradeoff between maximising deployment time and conserving battery and storage space. The study found no correlation between individuals detected (SWTs) and acoustic encounter duration. If correlated, this metric could influence future duty cycling settings, requiring extended recording duration in future experimental designs. However, average *S. plumbea* acoustic encounters were less than 10 minutes (mean 00:06:14 ± 00:08:18 SD). Therefore, the standard duty cycles often used for cetaceans, with periods of 10 or more minutes off, would mean whole encounters would be missed for this species. Ensuring the periods of dormancy within duty cycles are short enough not to miss encounters requires the focal species' acoustic behaviour to be taken into account (Fearey et al., 2022). When designing rapid assessments and field studies, these temporal impacts on detection are important to understand. In South Africa, there is a growing need for minimum data requirements to determine basic empirical data such as species presence, abundance, and distribution. These are typically left up to the environmental assessment practitioner and often left out of research studies (See Evidence

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Conservation – Downey, 2022). Optimum requirements of a continuous recording schedule in vocally elusive species, such as *S. plumbea,* is important information to enable threatened species of cetacean to be included in global and regional initiatives. This allows for these species to be included in assessments when identifying important biodiversity areas such as ecologically or biologically significant marine areas (EBSA), important marine mammal areas (IMMAS) or key biodiversity areas (KBA) as well as feed into the environmental impact assessment process.

Spatially, there was a variability in dolphin detection between the two methods. Boat-based surveys covered the whole array area, but *S. plumbea* encounters occurred in two areas, around moorings 1 and 6. This could be due to a bias in boat surveys occurring in the morning (during good weather periods), i.e. aligning to when dolphins rested or socialised in these areas, or a strong fine-scale distinction of higher use in only these two areas with little to no occupancy elsewhere. The latter is not the case through acoustic data, and dolphins move throughout the array area, with the highest level of acoustic activity occurring during the night. This has previously been reported in other studies, where diurnal activity has been detected in dolphin species in False Bay, South Africa, with higher levels of acoustic behaviour at night, indicating inshore feeding areas, previously unknown to researchers (A. Prinsloo unpublished). Land-based observations in Mossel Bay by James (2014) observed mixed habitat use by *S. plumbea* throughout the Mossel Bay coastline, with observed surface behaviours strongly linked to substrate type. Substrate type varies throughout the array area, from fine-grained sand to wave-cut rocky platforms. Moorings were purposefully deployed around a sandy substrate for easy retrieval and minimal noise from reefs or snapping shrimp that would mask or overload the hydrophone recordings (Aubauer et al., 2000). James (2014) found socialising behaviour predominantly on sandy beach habitats, a behaviour which is linked to higher signature whistle production in other delphinid species (Cook et al., 2004), while the primary observed surface behaviour of *S. plumbea* in Mossel Bay was travelling. Previous chapters showed behaviour and group composition to have a minimal effect on signature whistle production in *S. plumbea;*

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therefore, the effect of mooring location is likely not to have affected signature whistle detection in this study. Boat-based encounters are behaviour-dependent and are primarily focused on non-travelling behaviour to safely stay with the focal group, occurring in one area while photo ID can be collated. Therefore, there is a bias to non-travelling behaviour during boat-based surveys. This bias is removed using acoustic arrays, as highlighted through the continuous acoustic spatial and temporal detection of *S. plumbea* throughout the array.

Alongside spatial and temporal efficacy, individual detection was used as a key determinant of an effective monitoring method. The individual detection rate was measured to directly compare individual detection across both methods. Measured as the total number of individuals detected in an encounter, divided by encounter duration, acoustic detection methods were faster. Detection probability is a key bias in capture-recapture studies using signature whistles (Fearey et al., 2022). Naturally occurring sounds (SWs) and marks on an animal (photo ID) have inherent biases regarding their use in mark-recapture. An assumption of acoustic detection through signature whistles is the requirement for every dolphin in a group to signal to conspecifics using their signature whistle. *S. plumbea* are known to have small group sizes and low signature whistle production rates (Chapter 3). Therefore, it is reasonable to assume that in *S. plumbea*, group sizes are underrepresented using only acoustic methods of individual detection. Yet, all dolphins can ultimately be detected over time, given the constraints of animal behaviour and hydrophone deployments. In contrast, many animals can never be identified in photo ID studies due to a lack of marks or scars. This applies both within groups (so group size is not typically estimated from photos in most studies but by visual estimation) and across whole studies, resulting in the need to extrapolate mark-recapture estimates of marked animals upwards to account for these unmarked animals (see Elwen et al., 2019; Urian et al., 2014). In addition, the observed acoustic asymptote at the end of the sampling period is likely due to the limited survey period and is more likely a plateau often seen over sampling periods. The actual asymptote would be reached once a greater number of individual whistles have been detected, following the idea that all dolphins

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would be acoustically detected given a long enough sampling period. Therefore, the use of a discovery curve in this study demonstrates the rate of cumulative individuals detected over time rather than providing any results on the reliability of an asymptote due to the relatively short sampling period.

The strength of utilising acoustic methods is the versatility of the data acquired and the large amounts of acoustic data it produces over time. Both photo ID and acoustic methods can detect non-target species if in the vicinity during the survey; however, the strength of acoustic detection is the wider array of sound types that can be recorded. Photo ID is limited to species that surface near the boat during target species focal follows. A moored acoustic array will record all species that vocalise, even when the target species is not present. This study focused on *S. plumbea* detection, yet other whistling delphinid species (including clicks), baleen whales, fish and anthropogenic sounds were also detected in the recordings. The high initial costs of acoustic tools (such as equipment costs) can, therefore, be shared between multiple researchers and organisations, increasing the applicability of the dataset. For example, a key use of these datasets is the contribution to soundscapes and monitoring of anthropogenic noise. Until 2022, no soundscape studies had been published in South Africa (Schoeman et al., 2022). Soundscape monitoring is developing as an important tool in identifying long-term changes in the contribution of anthropogenic noise to underwater sound environments. Mossel Bay is a working harbour with several single-point mooring buoys situated 800 m – 1 km offshore, used for refuelling large tanker ships moored in the bay. Local port authorities have included Mossel Bay as a priority area for future development, which would directly impact *S. plumbea*, as seen in this study, due to their high use of the area near the harbour entrance (Bloom, 2019). At current levels, anthropogenic noise levels vary across the bay, and although these weren't measured as part of this study, there is the potential for this to increase in the future with continued coastal development (See planned Operation Phakisa, Findlay, 2018). The use of large acoustic datasets for long-term soundscape modelling has previously been hindered by computational power for

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signal extraction. The recent development of machine learning and automated processing tools means these large datasets can be processed and analysed more rapidly (Frainer et al., 2023; Gibb et al., 2019). The future direction of this data and our understanding of *S. plumbea* habitat threats requires an understanding of baseline soundscape metrics to assess future impacts on this threatened species adequately.

While acoustic datasets are incredibly versatile, photographic data can also provide additional data on a population equally useful for researchers and policymakers. Humpback whales (*Megaptera novaeangliae*) are one of the most widely photographed cetaceans, and the recent development of open-source, online fluke-matching programs has rapidly advanced the reach of these programs (Cheeseman et al., 2017; Patton et al., 2023). Providing increasingly more information on the global migration routes of humpback whales, such as tail fluke matches between South Africa and Brazil, matches that would have previously taken a great deal of international collaboration and work to find (Ramos et al., 2023). Across cetacean species, photo ID methods have provided additional information on killer whale scarring in humpback whales (Elwen et al., 2014), entanglement rates (Robbins & Mattila, 2004), sex assignment (James et al., 2022), intraspecific aggression (Felix & Novillo, 2015), and skin disease (Kiszka et al., 2009). Photo ID methods have also provided important data on *S. plumbea* populations, namely calving intervals and injuries (Frainer et al., 2023; Plön et al., 2021). In gregarious species with a high occurrence of surface visibility, photo ID methods are still a useful tool in species monitoring. However, when using photo ID as a tool for detection, in species that are rarely observed surfacing, such as Blainville beaked whales (*Mesoplodon densirostris*), who spend a predominant amount of time in deep water feeding, it has been estimated that it would require 25 - 30 years of photo ID data to detect a decline of 5%, further highlighting the ineffective use of this method across all species (Marques et al., 2019). Yet, when available, and often through opportunistic sightings, the use of photo ID for threatened species is still an important method to provide

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additional data to population studies, especially in species where targeted boat-based surveys are infrequent.

Logistical evaluation between the two methods provides ways to reduce the costs involved to allow both methods to be accessible to a wide range of researchers and managers. Field-work costs considered for the acoustic deployment cover only the travel to and from the study site and the four days for deployment and retrieval of the acoustic array. Some variables are constant in both methods, such as travel to and from the study site (potentially more for multiple acoustic deployments or complications), daily subsistence, stipends and accommodation costs. The total costs for the acoustic deployment are greater than boat-based surveys due to the greater start-up costs of the hydrophone array, with one SoundTrap hydrophone costing ~70,000 ZAR and greater analysis time required for acoustic data. Excluding equipment purchasing from acoustic methods, the cost of deploying and retrieving an acoustic array vastly decreases. In this study, the 'cost per dolphin' calculated throughout the two-week data collection period using acoustic methods is nearly twice that of photo ID detection methods, mainly due to the high analytical time required for processing and analysing acoustic data. These measures are subject to change in the future as machine learning methods are rapidly improving both fields, in particular methods to automate the classification of vocalisations (Frainer et al., 2023; Jiang et al., 2019; Patton et al., 2023).

The main limitation of photo ID studies is the high effort in boat and personnel time required, which accounts for a large portion of the costs incurred during visual-based surveys, especially in elusive species with low encounter rates, as seen in this study. The optimum requirement of at least a minimum of three people to conduct boat-based surveys (data, photo ID, skipper) considerably increases the costs of photo ID studies. Additionally, effort is highly dependent on camera equipment, camera lens quality, skipper skill, and skill of the photographer and ultimately requires finding the focal species and spending extended repeated periods with the animals to

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collect data. As with camera equipment, some research groups will have access to hydrophones, utilised across multiple research projects.

The application of acoustic methods in the future will require careful consideration of financial constraints, particularly the need for multiple hydrophones to extend the detection range, which remains a key limitation of this approach. The development of automated whistle extractions and species classifier software now requires far less computational power (Frainer et al., 2023). In this study, the costs and logistical considerations were comparable because they were conducted simultaneously. It should be noted that in future applications of either method, there are caveats that can both reduce these factors. Photo ID can be collected from tourism boats if access for researchers is free or through the use of public or citizen science data, as has been seen to be effective through the online free-to-use platform, HappyWhale, for humpback whales (*Megaptera novaeangliae*) (Cheeseman et al., 2017). Equipment costs such as cameras and hydrophones are often already covered by research groups or collaborators and may not be required in set-up costs. Boat hire, in general, is one of the most expensive costs of any field work and was heavily subsidised in this study. This should be factored in for larger deployments at considerable additional cost. However, with the potential for cost reduction in future applications, the increased affordability of research methods is a hopeful prospect for the field of cetacean research and conservation.

In conclusion, while photo ID is a valuable and accessible tool, passive acoustic monitoring provides a useful alternative method due to its broad temporal and spatial coverage and ability to replicate a study exactly over time. Once initial high start-up costs are removed, passive acoustic monitoring becomes an accessible method for researchers and conservation managers due to its ease of data collection, versatile application of acoustic datasets, and continuous animal detection capabilities. Advances in post-processing tools and data management further ensure that passive acoustic monitoring will continue to be a valuable approach for gathering data

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relevant to various species and research questions. In this study, acoustic detection methods provided useful information on species movement throughout the array, including increased acoustic activity at night and a greater spatial range of detection than boat-based methods. Ultimately, both methods can be used alongside each other and often complement one another. However, in a targeted short-term assessment, continuous acoustic detection methods proved to be a reliable alternative to photo ID, offering reduced field effort, while effectively monitoring spatial and temporal movements of *S. plumbea*.

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5 SWORD: SIGNATURE WHISTLES FOR OCCURRENCE, RECAPTURE AND DENSITY FOR INDIAN OCEAN HUMPBACK DOLPHINS, *SOUSA PLUMBEA* IN SOUTH AFRICA

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ABSTRACT

Capture-recapture (or mark-recapture) is a key tool in conservation for studying population trends, survival, and habitat use. Passive acoustic monitoring, using signature whistles as proxies for individual detection, offers a less invasive and time-consuming alternative to standard visual survey detection methods. This framework of using signature whistles for individual detection and monitoring has been termed SWORD – signature whistle for occurrence, recapture and density. This proof-of-concept study deployed a 120 km array of five hydrophones for 42 days across three high-use bays (Mossel Bay, Buffelsbaai and Plettenberg Bay) within the central part of the range of south coast population range of Indian Ocean humpback dolphins (*Sousa plumbea*). Across all five moorings, 1282 total signature whistles were detected from 25 different signature whistle types (SWTs) or individuals. Occurrence was seen across the array with high detections on two moorings at Mossel Bay 1 and Buffelsbaai 7 (77% of total detections). Closed (Huggins) models were used to estimate the number of animals using each of the three bays during the study period (Mossel Bay: 8 (95% CI 8-8); Buffelsbaai: 19 (95% CI 14-53); Plettenberg Bay: 22 (95% CI 4012 19-33). For the combined dataset, "open" population models (POPAN JCS) and spatially explicit mark-recapture models (SECR) were applied to obtain baseline population assessments for the south coast region. A range of SECR models explored different mask resolutions, truncated datasets, and mooring configurations. In all models, density, D, was very low (D ranged from 0.000061 - 0.000439 animals.ha^{-2}). Across the different SECR models, abundance was extrapolated by multiplying the predicted D estimate by the effective study area. These different models all resulted in similar estimates to those from the capture-recapture open population model (SECR abundance model estimates ranged from 23-26; POPAN Jolly and Seber super population estimate of 26 individuals (95% CI 23-30). These estimates from the combined study sites indicated a high number of recaptures throughout the survey period and little effect on abundance estimates from fine-scale trap configuration. Therefore, this suggests that when limited by deployment time and equipment, expanding the geographic range of the acoustic array is more effective for increasing individual unique recaptures than increasing the density of recorders within a smaller area. A growing preference for density estimates in conservation management due to the useful inclusion of spatial range demonstrates that SECR and population abundance models are adequate for assessing population estimates using individual acoustic monitoring in elusive coastal dolphins.

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INTRODUCTION

Effective wildlife management strategies rely on accurate estimates of population occurrence, abundance and density (Marques et al., 2013). Detection of individuals within a capture-recapture framework is beneficial for assessing small, at-risk populations, where the loss of one individual can significantly affect population health (Lacy, 2000; Parra et al., 2018). Producing a population estimate of animal abundance over multiple years can be challenging. However, these estimates are beneficial for management as it helps identify trends in population size and how the use of the study area might change over time, even if the estimate produced is not the total population abundance and variation around these estimates is high (Bejder et al., 2006; Elwen et al., 2009). Density is often seen to be a more meaningful metric for wildlife management than abundance (Forsyth et al., 2019; Glen & Dickman, 2005). Calculating the number of animals within a known area describes a population within its habitat, or sections thereof, enabling more effective management strategies (Forsyth et al., 2019). Both abundance and density estimates are helpful, but with the development of spatially explicit capture-recapture models. (Efford, 2011), estimating density has become more accessible, and both of these methods are beneficial for managers and conservation (O'Brien et al., 2005).

The most widely used index of conservation status is the IUCN Red List assessment, which guides global conservation priorities and policy (Marsh et al., 2023). When assessing risk criteria for a species or population, IUCN assessments consider various factors, including abundance, distribution or isolation, and anthropogenic and environmental threats. Some of the more critical risk criteria are knowledge of geographic range, population size and density, and incidents of population fragmentation across known habitats (Marsh et al., 2023). The extent of a species' occurrence, recapture rates, and density require empirical temporal and spatial data. These metrics are difficult to ascertain for many species, especially in the marine environment. Despite challenges, advancements in capture-recapture methods have made it possible to collect this

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data, and future progress towards these methods is necessary to develop conservation efforts with more valuable estimates.

Acoustic production is highly prevalent in marine mammals and is used for navigation, foraging, and communication with conspecifics (Lammers et al., 2006). In species where monitoring is hindered using traditional visual observation methods, such in the marine environment, acoustic monitoring has become widely accepted as a method for cetacean detection (Mellinger et al., 2007), enabling a range of new study designs and methods (Marques et al., 2013; Ryder et al., 2023). In marine mammals, traditional visual surveys, which require boat-based effort, favour gregarious species with frequent surfacing behaviour. These methods can be costly and are limited by weather and daylight (Chapter 4). Cetaceans are highly vocal, and passive acoustic survey methods, such as moored hydrophones or towed arrays, provide a far more continuous dataset, enabling the development and applications of long-term acoustic datasets. This is exemplified by studies on the highly elusive beaked whales (*Ziphius* spp.) or sperm whales (*Physeter macrocephalus*), that vocalise almost exclusively during deep foraging dives but are primarily silent during their ascent and their short surface intervals (Johnson et al., 2004). Passive acoustic monitoring (PAM) provides an alternative study method for detecting species that would be overlooked due to “historical" logistical constraints. Methodologically, the versatility of PAM utilises a range of sensors to record underwater sound, from single-dipping hydrophones (Fearey et al., 2019) to autonomous gliders (Cauchy et al., 2023) and even real-time 3-dimensional detections of porpoises (Macaulay et al., 2022). In recent years, improvements in computational power, battery life, and storage space have allowed large sets of data to be collected, allowing for post-processing analyses and in-situ processing to detect, classify and localise marine mammals (Gillespie et al., 2013). The exponential growth of scale within PAM has provided new opportunities for scientific research. These developments have especially benefited Critically Endangered species, such as the North Atlantic right whale (*Eubalaena glacialis*). Large multi-disciplinary PAM projects have been conducted throughout the North Atlantic right whale

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migratory corridors, combining static moorings and autonomous gliders. These studies have supported recommendations on seasonal adjustments to shipping lanes to safeguard the migratory corridors of these Critically Endangered species (Durette-Morin et al., 2022).

Although cetaceans have been acoustically recorded since the ~1950s (e.g. Watkins sound library, Sayigh et al., 2017), PAM as a monitoring tool has only been used for research since the early 1990s when towed arrays were used to detect sperm whales (Leaper et al., 1992). More recently, PAM has been used in a variety of ways, with hydrophones deployed from vessels, free-floating or moored and used to estimate abundance, investigate habitat ranges and density, or localisation for mitigation techniques (reviewed in Cato et al., 2006; Sousa-Lima et al., 2013; Zimmer, 2011). More recently, PAM has been used to estimate abundance by identifying signature whistles in *T. truncatus* within a capture-recapture framework (Fearey et al., 2022; Longden et al., 2020; Romeu et al., 2024). Signature whistles are stereotyped whistles produced by an individual, usually in bouts. The long-term, stereotyped tonal signals function to convey the identity and presence of the whistling individual to conspecifics, as well as facilitate group cohesion and assist in mother-calf reunions (Janik & Slater, 1998; King et al., 2016; Smolker et al., 1993). Signature whistles can account for 38 -70% of whistles produced in free-ranging dolphins, with their production behaviourally and contextually dependent in *Tursiops* spp. (Cook et al., 2004). Applying the same capture-recapture framework widely developed for photo ID (Rogan et al., 2000; Vermeulen & Bräger, 2015) to signature whistle detections from long-term PAM can overcome limitations of spatial and temporal resolution. The high production rate of these individual signals offers a viable alternative to boat-based photo-ID methods (Fearey et al., 2021). Signature whistle detection using PAM only requires detection on a single hydrophone. Therefore, with a multiple hydrophone array, density metrics can also be estimated using spatially explicit capture-recapture models (Borchers & Efford, 2008).

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The incorporation of a spatial point process to capture-recapture models allows for an extension of previous methods as they incorporate spatial information on where captures are made and where they are absent (Borchers & Marques, 2017; Efford et al., 2009; Noss et al., 2012; Royle et al., 2009). Spatially explicit capture-recapture (SECR) models estimate density using an array of traps (Efford, 2004). SECR has become a robust application in many fields (Kalle et al., 2011; Marques et al., 2013; Sollmann et al., 2011) and enables the estimation of home ranges and density outside of the survey area (Borchers & Efford, 2008). Applications range from cage trapping possums (Efford et al., 2005), acoustically "trapping" birds using song and time-of-arrival (Dawson & Efford, 2009) and camera trapping tigers, identifying individually unique marks (Royle et al., 2009). Within these applications, many different detectors for "capture" are possible, such as physical traps, which are limited to capturing one individual per occasion, to proximity detectors, such as camera traps, or microphones/hydrophones, which can record multiple individual numerous times within a single capture occasion. Spatially explicit capture-recapture allows density to be explicitly modelled and estimates the home range directly (Borchers, 2012). SECR provides an estimate of the spatial extent of the survey range from detectors deployed, improving the effectiveness of results for managers and researchers to implement appropriate conservation measures. SECR has previously been applied to non-individual calls, such as minke whale 'boing' sounds (Marques et al., 2012), as well as being used in other species, such as owls (Martin et al., 2024) and frogs (Stevenson et al., 2015) through identifying individuals based on time of arrival. Individual identification of SWTs has successfully been shown in *T. truncatus* to produce comparable estimates similar to photo-ID studies (Longden et al., 2020). This chapter builds on these earlier successes. Rather than arrival time of calls, uses SECR based on recaptures of signature whistles of *S. plumbea* across single hydrophones in an SECR framework to calculate animal abundance and density.

In South Africa, *S. plumbea* is a highly coastal delphinid and the most threatened marine mammal. (Plön et al., 2016). Recent studies have suggested that the population is fragmented along the

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southern African coastline and appears to have reduced substantially since the late 1990s (Plön et al., 2021; Vermeulen et al., 2018). There is no national abundance estimate for this species in South Africa, but photo ID estimates have been published for several regions. (Conry, 2017; James et al., 2015), with Karczmarski, (1996), extrapolating local abundance estimates from Algoa Bay to suggest a national abundance of fewer than 1000 individuals in the late 1990s. Regional estimates from the south coast population have provided catalogues of 24 – 96 distinct individuals (Plettenberg Bay: 41 Greenwood, 2013, 63 Jobson, 2006; Mossel Bay: 67 James et al., 2015; Algoa Bay: 50 Bouveroux et al., 2018, 70 Karczmarski & Cockcroft, 1998). Closed model estimates on these distinct catalogues vary, with estimated abundances ranging from 42 (SE:42-55) to 112 (SE: 75-133 Greenwood, 2013; Jobson, 2006). Open population models for the south coast estimate higher abundances of 84 (SE 72 - 115) – 466 (SE 447- 485) individuals (Conry, 2017; Karczmarski & Cockcroft, 1998; Karczmarski et al., 1999). Across these regional studies, high boat-based effort and a low encounter rate limit the range of study areas and the extrapolation of these estimates.

Vermeulen et al., (2018) was the first study to attempt a regional assessment of how these populations and abundance estimates were related, collating 16 years of photo ID data from various organisations and researchers across South Africa. The study identified many coastal movements between sites used in previous matches, with a clear separation of a south and east coast population, and ultimately identified 247 well-marked individuals. Non-aligned timelines for the data sets meant capture-recapture models were not possible. Still, the study suggested an update to the Karczmarski, (1999) estimate of fewer than 500 individuals (Vermeulen et al., 2018).

The use of naturally occurring, individually distinct acoustic markers, such as signature whistles, has been suggested as an effective tool for species where other monitoring methods are inconsistent and difficult to obtain (Janik et al., 2013; Terry et al., 2005). This thesis has explored

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aspects that might affect detection using individual acoustic monitoring with passive acoustic monitoring demonstrated to be an effective tool for individual acoustic detection using signature whistles (Chapter 4). In this chapter, I aim to build on these previous chapters, providing a proof-of-concept using the SWORD (signature whistles for occurrence, recapture and density) framework (Fearey et al., 2021). The chapter investigates both fine-scale deployment location and organisation of the array. It presents a proof of concept for PAM's effectiveness, using capture-recapture and SECR models across both individual bays and cumulatively, to assess population dynamics based on a central portion of the south coast population of *S. plumbea* in South Africa.

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METHODS

Data collection

A preliminary investigation was conducted into the efficacy of using signature whistles (signature whistle type, SWT, to denote an individual) as the identifying feature of occurrence. SWTs were used to create capture histories for capture-recapture (CR) models to estimate abundance, investigating both open and closed models within each bay and for the entire range. SWTs were also used in SECR models to estimate density and home range and extrapolate abundance over the study area.

Data were collected from five moorings deployed in three bays within the center of *S. plumbea*'s south coast range (Mossel Bay, Buffelsbaai and Plettenberg Bay) (Figure 5.1), covering ~120 km of coastline. Where two receivers were placed in the same bay (i.e., Mossel Bay and Plettenberg Bay), the two receivers were spaced far enough apart (3 km) to ensure that individuals could not be detected on both hydrophones simultaneously, eliminating the possibility of pseudo-replication. The two deployment locations in Mossel Bay were based on previous deployment locations in 2021 and redeployed at locations 1 and 3 (M1_Mossel Bay Harbour and M3_Mossel Bay Diaz), chosen as one high-use mooring location and one low-use mooring location, respectively (Chapter 4 methods) to investigate detection differences within the models. One mooring was deployed at Buffelsbaai (M7), a known hotspot for *S. plumbea* based on a visual survey with no previous acoustic behaviour data available (Conry, 2017), and two were deployed in Plettenberg Bay (M8, M9), in known hotspots from previous visual and acoustic surveys (Hlati, 2018; G Penry, Pers. Comm.). Mooring ID codes from Chapter 4 were used to ensure continuity within the data: M1 – Mossel Bay 1, M3 – Mossel Bay 2, M7 – Buffelsbaai, M8 – Plettenberg Bay 1, M9 – Plettenberg Bay 2 (Figure 5.1).

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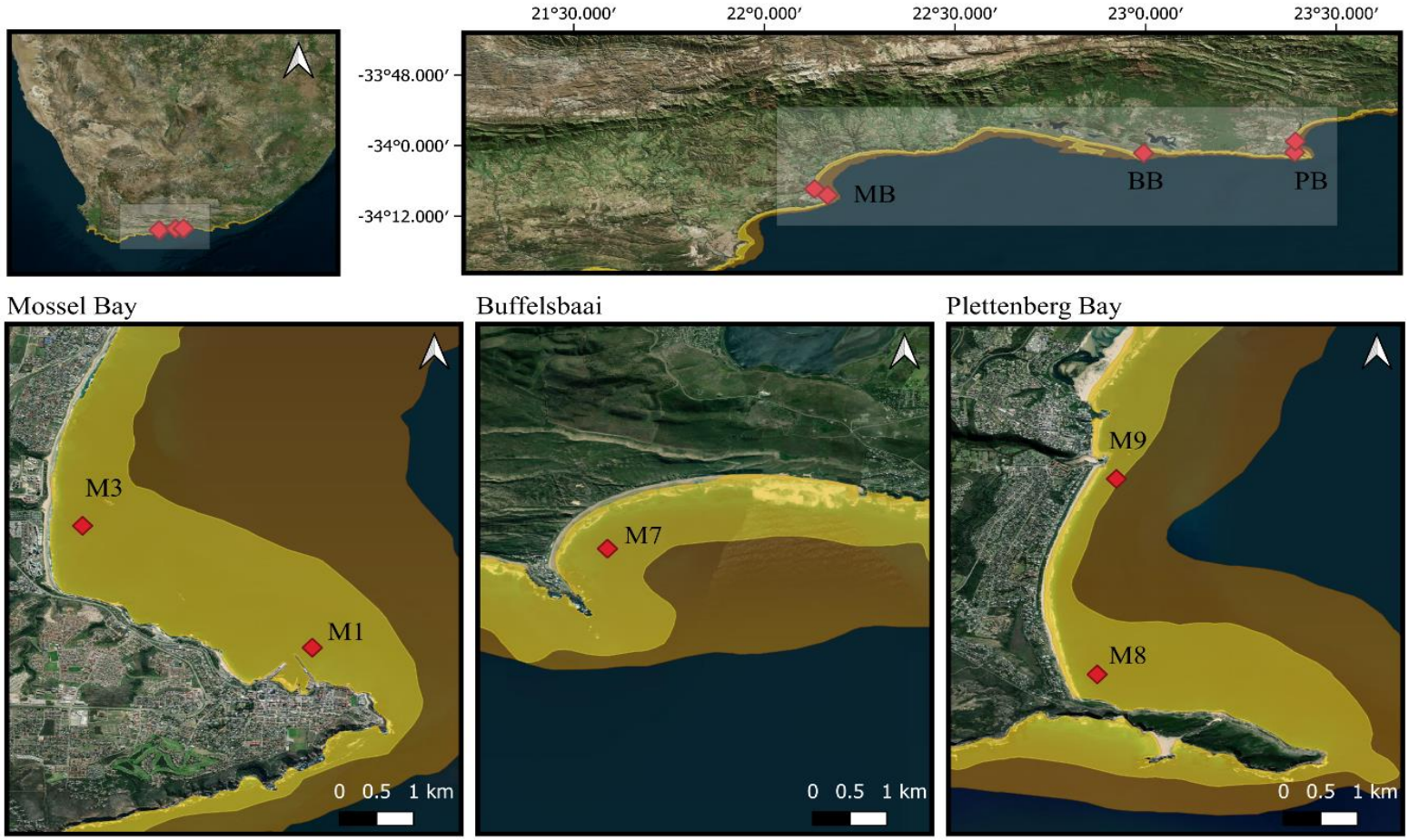


Figure 5.1. Mooring locations across the three bays to acoustically detect *Sousa plumbea*. Hydrophones were deployed along a 120 km central portion of the south coast of South Africa (Mossel Bay, Buffelsbaai and Plettenberg Bay). Five mooring locations were selected across three bays. Mossel Bay was located furthest west, and Plettenberg Bay was furthest east. All moorings were deployed within 15 m water depth (orange range indicating 15 m water depth and brown indicating 20 m depth).

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Hydrophones were all set to a continuous recording cycle with a sample rate of 96 kHz (High Gain), using three models of SoundTrap hydrophones (ST300, ST400, ST500 SoundTraps; Ocean Instruments Ltd., Auckland, New Zealand). Due to the recording settings, deployment time for the ST400 was limited by battery length to ~ 23 days; therefore, these devices were placed in Mossel Bay and Plettenberg Bay, where there were two moorings, alongside ST500 units that had a longer battery time of 45 days. Hydrophones were deployed in 8 - 10 m water depth with sub-surface buoys secured to ~ 40 kg weights, secured 3 m from the seafloor. In this deployment, pop-up acoustic releases (Vemco, Innovasea Systems Inc.) were used for ease of retrieval (as used in the ATAP array, Cowley et al., 2017). Attempts were made to avoid loud reef systems to mitigate masking (e.g., snapping shrimp noise, environmental noise), but still within the detection range of the highly coastal *S. plumbea* habitat.

Data processing

To identify *S. plumbea* from the long-term acoustic recordings, whistle encounters were identified using the whistle extraction and species identification method described in Chapter 4. The process used both PAMGuard's whistle and moan detector (WMD) and a custom species classifier to identify single species *S. plumbea* encounters (Gillespie et al., 2009). Encounters with more than 80% confidence for *S. plumbea* were selected and confirmed. Each encounter was visually scanned, and all *S. plumbea* whistles were manually annotated and extracted for each mooring.

Signature whistle identification

A whistle was defined as a frequency-modulated tonal signal with a unit of one continuous contour or two or more repeated contours (contour – continuous tonal signal), separated by a period of silence between 0.03 seconds and 0.25 seconds in duration (Kriesell et al., 2014). Only high-quality whistles (SNR 2 or 3) were used in further analyses. Whistles were compared to the

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Sousa SWT catalogue from Gridley et al., (Appendix 1) to identify likely signature whistles. Stereotyped whistles that exhibited highly similar frequency modulation patterns occurring in bouts but did not match a signature whistle type in the catalogue were added if they met SIGID (SIG*nature* ID*entification*) criteria (Janik et al., 2013). The SIGID method consists of two requirements based on the well-studied bout-production nature of signature whistles (Janik et al., 2013; Sayigh et al., 2007). Both criteria must be met to identify signature whistle types in free-ranging populations confidently. The first criteria requires the identification of four repetitions of highly stereotyped whistle contours, and the second requires that on at least one occasion, three out of four stereotyped whistles occur within 1-10 s of the previous stereotyped signal (Janik et al., 2013). New signature whistles were only included and added to the *Sousa* SWT catalogue if they met SIGID in this study's acoustic data. Signature whistle types that were previously catalogued only required one signal to be counted as a capture in this study. As new SWTs were catalogued, successive and previous whistles were checked to include new SWTs throughout the whole dataset. All whistles classed as a "capture" were recorded, with the associated date, time and location, for the construction of capture histories.

Data analysis: Occurrence

Occurrence was investigated through spatial and temporal recaptures of SWT over the whole survey period. Discovery curves were used to investigate SWT detections at each mooring, as well as the cumulative detections for the entire array (i.e., all five receivers).

Data analysis: Capture-Recapture for Abundance

To investigate the SWT capture-recapture data in a similar framework to what might be done for a photo ID study (e.g. James et al., 2015), which typically concentrates research effort into a single bay over an extended period, data from each bay was analysed separately, using closed capture-

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recapture models. Days were initially used as capture occasions, but overall recapture rates were low, so data were combined to make each recapture occasion one calendar week. Bays with multiple moorings were combined into a single capture history; thus, mooring 1 and mooring 3 in Mossel Bay were combined, and mooring 8 and mooring 9 in Plettenberg Bay with detection on either or both hydrophones within an occasion being considered as a capture.

Closed population capture-recapture models were run for each of the three study areas separately, and then for all areas combined as one population capture to provide estimates of the number of individuals using each study area as well as for the whole study range. The mark-recapture analyses were conducted in the programme MARK (White & Burnham, 1999) using the RMark interface (Laake, 2013). For closed models (within the three bays independently, and for the whole study area, 'çombined') mixture models developed by (Huggins, 1989) were used, which are recommended because of their flexibility and consider heterogeneity within the data to account for variation in capture probability ((Elwen et al., 2019; Manly et al., 2010). In the absence of covariates, Huggins and full-likelihood models provide equivalent results (Lukacs, 2012). As there is no biological reason to expect a behavioural change between initial captures and recaptures (as in the case of physically trapped animals), c (recapture probability) was set equal to p (initial-capture probability) throughout (Nicholson et al., 2012; Parra et al., 2018). Similarly, capture probability was assumed to vary between occasions because environmental conditions were not constant, and only a portion of the population was captured on any occasion. Thus, models with constant capture probability were not considered, with the exception of running a null model for comparative purposes. Although animals almost certainly moved out of the study area during the study, violating some of the model assumptions, the results of the models can be interpreted as 'the number of animals using the study area over the study period', which is a useful metric for management.

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Population abundance estimates generated by the closed population mixture model for the whole study area as one population was compared to an open population model which takes into consideration the entering and exiting of individuals into and out of the population (immigration, emigration, births, deaths). This was done using the POPAN formulation of the Jolly-Seber (hereafter PJS) mark-recapture approach using the package "marked" (Laake, 2013), calculating population estimates using the RMark interface (Laake, 2013). PJS models estimate total population size ($\widehat{N}$) using four parameters: survival probability ("phi" (φ)), detection/capture probability ("p"), the total number of individuals able to enter the population ("N", superpopulation), and the probability of entry into the population ("pent"). A subset of candidate PJS models was run and compared, subsequently selecting the model that minimises the Akaike Information Criterion (AIC) and the negative log-likelihood (neg2lnl). Within this subset, parameters Phi, p and pent were set as constant, assuming a constant survival rate, capture probability, and probability of entry. The parameter N was set as constant, assuming the superpopulation does not change over time during this short study period. Upon model selection, $\widehat{N}$ was derived from the estimated parameters within the selected model using RMark.

Data analysis: Spatially Explicit Capture-Recapture for Density

Density throughout the south coast range was investigated using spatially explicit mark recapture using models fitted using the R package: 'secr v4.5.10' (Efford, 2016). Preliminary models were fit to evaluate mask size and data efficacy. Given the duration of the study period and the aim of estimating density across the home range, captures were grouped into a single occasion, modelling constant density. All models were fit using a hazard exponential detection function (HEX), with the three default null model parameters: density, lambda and sigma. Only a null model was fit, and no further parameterisation was investigated due to inherent limitations in the trap design. The purpose of this analysis was to determine if SWT detections from moored acoustic receivers were viable identifying markers for SECR density and sigma estimates.

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Logistical limitations prevented a more viable trap design. However, using the null model, three different model types were investigated 1) capture history truncations, 2) mask objects, and 3) trap configurations.

1. *Capture history truncation*

Initial model investigation resulted in NA values for all parameters in the null model. This issue was determined to be caused by one individual (SWT13) which was detected 482 times over the study period, a significantly higher recapture rate than all other individuals. By truncating the data to limit the number of recaptures to 200, the model was able to converge. Consequently, the data was further truncated to 100 recaptures as well, for further investigation into this effect on density and sigma estimates.

2. *Masks*

To generate density, SECR models require a mask, or geographic area, in which to estimate the model parameters. These can either be automatically generated using the trap locations or can be manually limited based on various parameters (e.g., spacing, buffer), and constrained using .shp files. SECR models are able to estimate home ranges and density outside of the mask constraint based on the size of the buffer. The mask area represents the area in which the detectors are operating and usually does not extend to an entire population. For investigation into survey design and effectiveness of model inputs, three different masks were tested. These included a 1) basic mask (Mask.Def) with no unsuitable habitat excluded (i.e., land), 2) a low resolution mask (Mask.LR) and 3) high resolution mask (Mask.HR), covering their coastal habitat up to 30 m water depth (Figure 5.2). Mask.LR and Mask.HR included coastal range 100km both west and east from M1 and M9, based on *S. plumbea* known recaptures to these distance

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(Bouveroux et al., 2019; Vermeulen et al., 2017). A buffer of 10 000 m was determined, after performing mask diagnostics, functions: suggest.buffer() and mask.check() in the R:secr package.

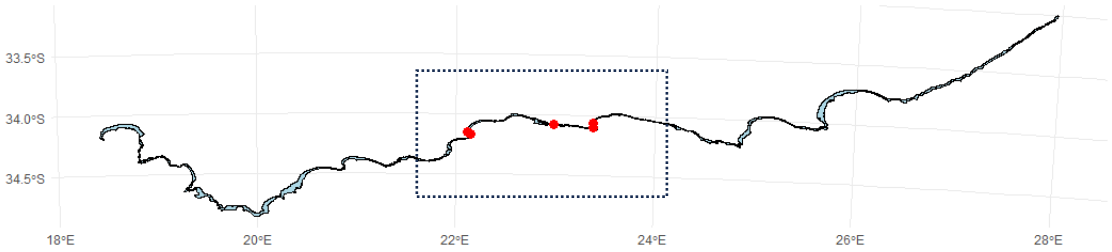
3. Trap Configurations

Given the study design, with two locations containing multiple moorings (Mossel Bay and Plettenberg Bay), two possibilities to improve the model were investigated:

1. *CaptHist.Rem* – Removed the two least active receivers from the dataset (M3 and M8)
2. *CaptHist.Comb* – Combined the captures from the two receivers within the same bay (M1+M3, M8+M9).

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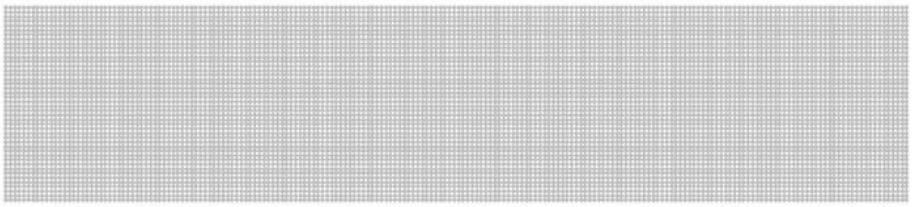
A



B

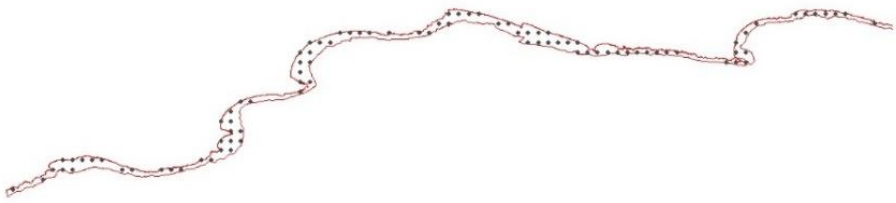
1 Mask.Def

Defatult mask using trap locations with a buffer of 10 000 m;



2 Mask.LR

Low resolution mask (nx = 100, ny = 10), bounded by *S. plumbea* home range with a 10000 m buffer



3 Mask.HR

High resolution mask (nx = 1000, ny 100), bounded by *S. plumbea* home range with a 10000 m buffer.



Figure 5.2. A) Locations of the three bay locations for acosutic detection of *Sousa plumbea* along the south coast, South Africa. Mossel Bay and Plettenberg Bay (furthest west and east) contain two moorings (spaced 3 km apart) and Buffelsbaai (middle) contains one mooring. Masks were made to represent different levels of habitat resolution. B1) Default mask with non suitable habitat (land) included. B2) Low resolution mask along coastal distribution up to 30 m depth (N cells = 91). B3) High resolution mask using the same spatial habitat (N cells = 8633).

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With these three tests (1. capture history truncations, 2. mask objects, and 3. trap configurations), various combinations thereof were fitted to a null SECR model. The null model contained three parameters: Density, Lambda, and Sigma. Density (D) is an estimate of the number of individuals over the effective sampling area. Lambda is an estimate of the proportion of individuals that are captured during a study period. Sigma is an estimate of the average distance an individual moves between capture occasions and over the study period, representative of an individual or population's home range. Given the null model, using one capture occasion, density can be interpreted as a uniform density of all individuals in the study area, and sigma is the average distance all individuals moved during the study period. To help the model converge, starting values were provided: D = 0.0001 (starting density to get model to converge), lambda0 = 20 (based on lower boundaries of individuals detected in the study), sigma = 15000 (starting estimation from literature ~15 km) (M. Efford, Pers. Comm.).

Estimates from all these model combinations were then evaluated by extrapolating abundance by multiplying the predicted D estimate by the effective study area or total area. These were compared to the open capture-recapture estimates for the combined survey area.

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RESULTS

Over the full 42-day period, individual hydrophone recording periods ranged from 19 – 42 days based on battery usage, memory, and planned retrieval times (Table 5.1) Moorings 3 (MB_Diaz) and 8 (Plett_RP) recorded for around 20 days and then ended due to battery failure. Moorings 1 (MB_Harbour) and 7 (Buffelsbaai) both recorded for around 30 days, whilst mooring 9 (Plett_BI) had an extended recording period of 42 days due to delayed retrieval of the unit (Table 5.1). Spatial and temporal detection of SWTs was conducted on the full 42-day survey period to investigate overall occurrence throughout the study. However, the data was truncated to 32 days (five weeks) for further analysis of occurrence (discovery curves) and recapture (abundance models). SECR models were able to utilise the full dataset due to the use of a single capture occasion (see below).

Table 5.1. Deployment summaries for the five moorings deployed throughout the central portion of the south coast of South Africa to acoustically detect *Sousa plumbea*. Mooring codes follow from the nomenclature in Chapter 4. Therefore, Mossel Bay moorings are M1 and M3, and Buffelsbaai continues from M7. Recording start dates were all scheduled at the same time, and recording ends varied based on battery and retrieval dates. Rec days indicate the total number of days the recorders were recording throughout.

Mooring code	**Location**	**Distance from west neighbouring receiver (km)**	**Recording start**	**Recording end**	**Rec days**
M1	Mossel Bay_Harbour	0	26/08/2022 00:00:00	28/09/2022 14:10:42	33
M3	Mossel Bay_Diaz	3.8	26/08/2022 00:00:00	14/09/2022 22:50:40	19
M7	Buffelsbaai	80.4	26/08/2022 00:00:00	27/09/2022 18:20:41	32
M8	Plettenberg Bay_Robberg Point	36.3	26/08/2022 00:00:00	15/09/2022 00:50:40	20
M9	Plettenberg Bay_Beacon Isle	3.4	26/08/2022 00:00:00	07/10/2022 20:10:53	42

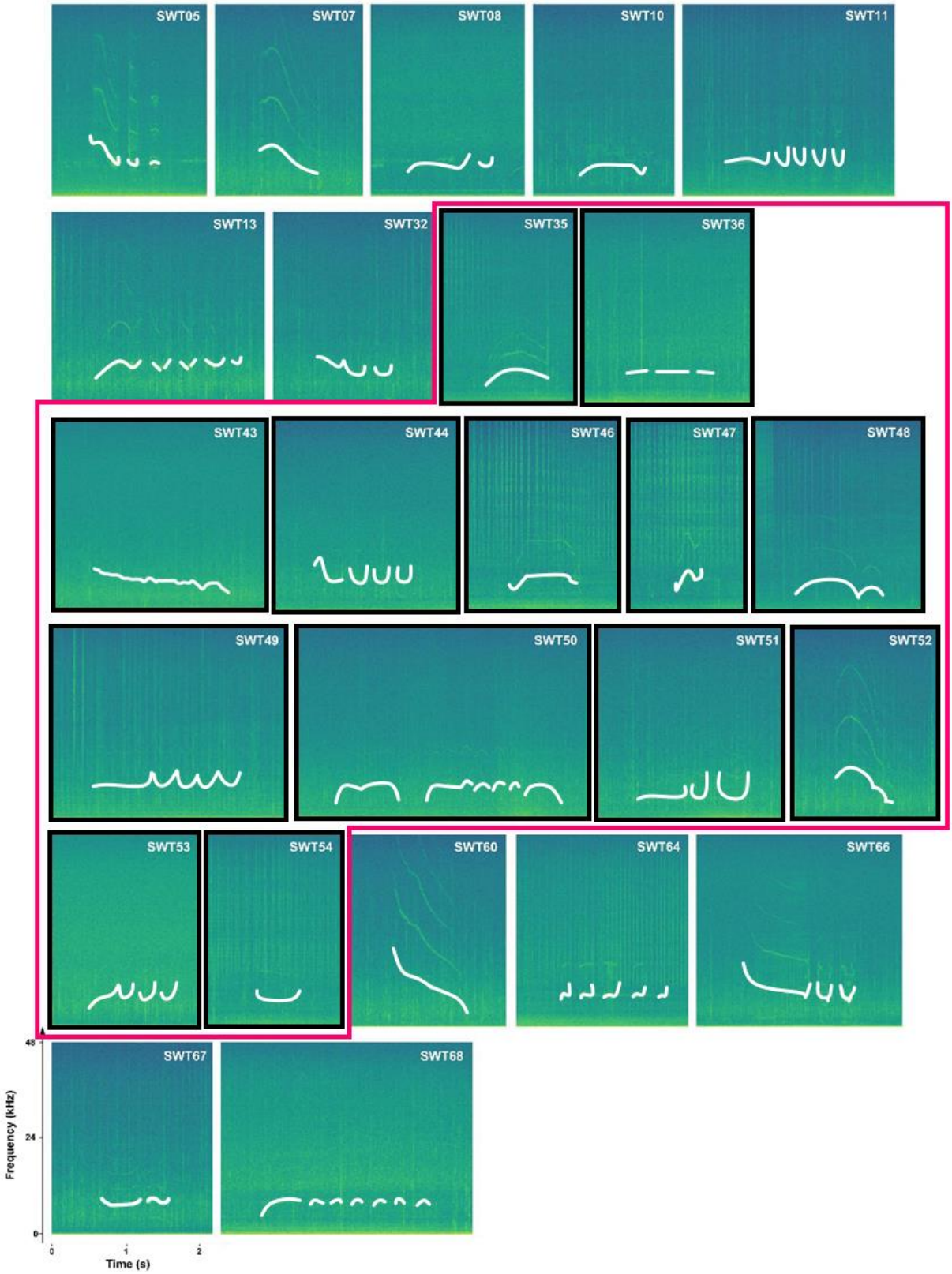
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A total of 4425 whistles were annotated from all recordings in the acoustic array (total recording duration; 3559.44 hr) of which 2946 were considered SNR 2/3 (Table 5.2). Across all five moorings, 1282 total signature whistles were detected from 25 different signature whistle types (SWTs), or individuals. Of these 25 SWTs, 12 matched the catalogue with 13 new whistle types added from detections on moored data using the SIGID bout analysis method (Figure 5.3).

Table 5.2. Summary table of *Sousa plumbea* whistles detected across the acoustic array, with a total number of signature whistles (SW) and signature whistle types (SWTs, individuals) detected at each mooring. The proportion of whistles analysed (using a signal-to-noise ratio of 2 or 3) was calculated from total detections at each mooring and presented as a percentage.

	Total detections (SNR 1-3)	**Whistles (SNR 2,3)**	**% of whistles analysed**	**Number of SWs**	**Number of individuals captured (SWTs)**
M1(MB_Harbour)	612	458	75	272	7
M3 (MB_Diaz)	139	94	68	59	2
M7 (Buffelsbaai)	2070	1532	74	719	13
M8 (Plett_RP)	409	192	47	87	9
M9 (Plett_BI)	1195	670	56	145	15
Total	4425	2946		1282	25

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Figure 5.3. All *Sousa plumbea* signature whistle types (SWTs) detected within this study (n = 25). SWTs that matched the existing catalogue are not framed (n = 12) (Gridley et al., Appendix 1). New SWTs added to the catalogue during this study all fulfilled SIGID and are boxed within the red lines (n = 13). SWTs are traced with a white line for clarity of catalogue entry.

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Occurrence

Over the full 42-day survey period, geographic and temporal variability in SWT detections was seen across moorings. Across moorings, the majority of total signature whistle detections occurred on mooring 1 (MB_Harbour = 7 days of SWT detections) and mooring 7 (Buffelsbaai = 5 days of SWT detections), detecting 77% of the total signature whistles (n = 991). Mooring 3 and mooring 8 had lower detection rates, with mooring 3 only having SWT detections on two days and M8 having SWT detections on three days (n = 548). However, these hydrophones also had shorter recording periods.

When looking at the number of captures of SWTs over both time and space, over half the SWTs were detected on more than one mooring (16 of the 25 SWTs) (Figure 5.4). However, the number of recaptures varied, with a majority of SWT detections occurring across two moorings only (52%), with only SWT50 and SWT66 recaptured on four of the five moorings (Figure 5.4). No SWTs were detected on all five moorings. Over the full six-week survey period, capture occasions were classed as week periods, with 56% of SWTs detected on one or two capture occasions, 40% encountered across three different occasions, and 4% of SWTs encountered on four occasions. No SWTs were detected across all capture occasions (Figure 5.5).

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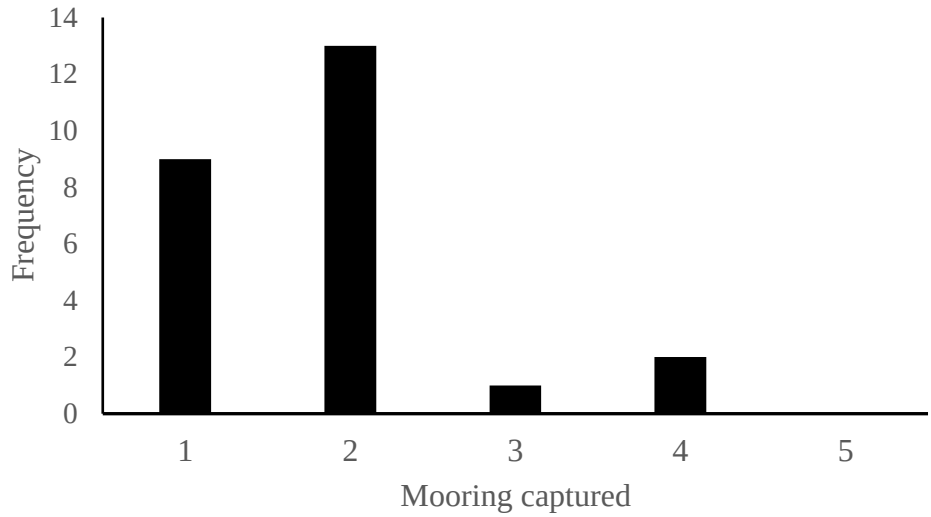


Figure 5.4. Frequency distribution showing the number of different moorings each *Sousa plumbea* signature whistle type (SWT) was captured on over the full 42-day deployment period. The highest number of different moorings SWTs were detected on was two, while no SWTs were detected on all five moorings.

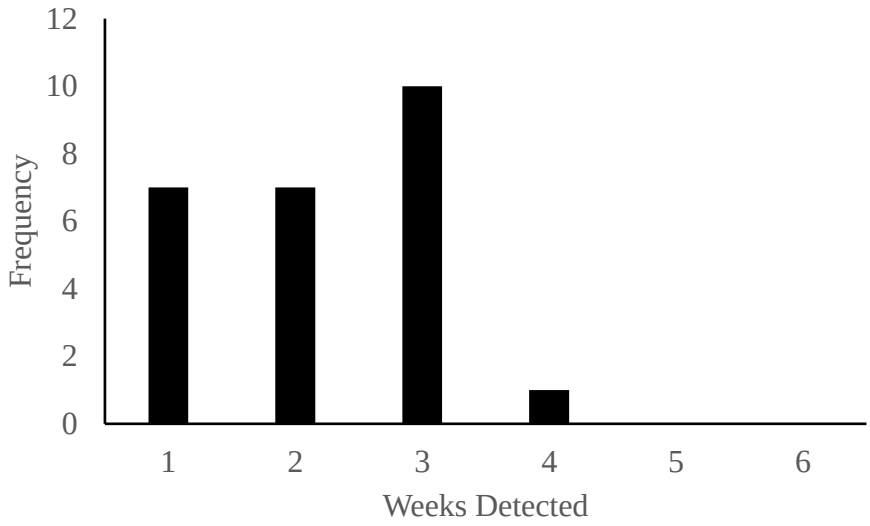


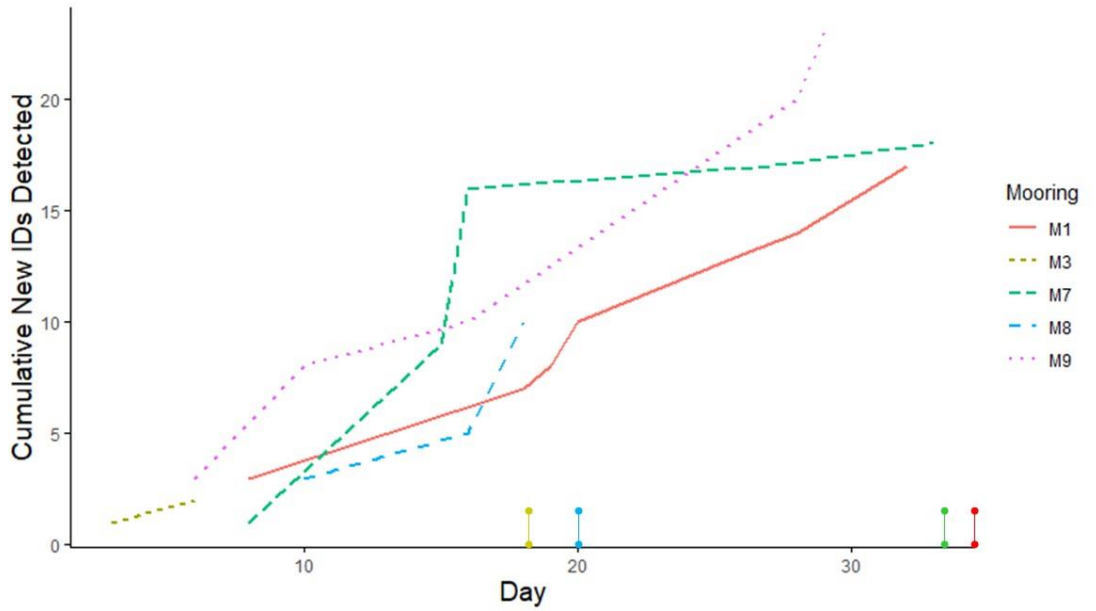
Figure 5.5. Frequency distribution of the number of capture occasions *Sousa plumbea* signature whistle types (SWTs) were detected across all moorings combined over the full 42-day deployment. One capture occasion = one calendar week. The highest number of weeks on which SWTs were detected was three, with no SWTs detected across more than four weeks.

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When investigating the cumulative detection of new SWTs over time, the extended recording duration of mooring 9 in Plettenberg Bay created a bias in the dataset, so the dataset from Plettenberg Bay was truncated at day 32 (five weeks). Only one encounter occurred over week six on this single hydrophone and was therefore lost in this truncation. The rate of cumulative detection of unique signature whistle types varied across all five sites (Figure 5.6a). Low detection rates at mooring 3 and mooring 8, coupled with their shorter deployment duration, meant that a limited number of SWTs could be detected at these moorings. No complete asymptote was reached at any of the five moorings (Figure 5.6a) but when combined across all five moorings, the rate of new detections reached an asymptote from day 28 of 32 onwards (Figure 5.6b).

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A



B

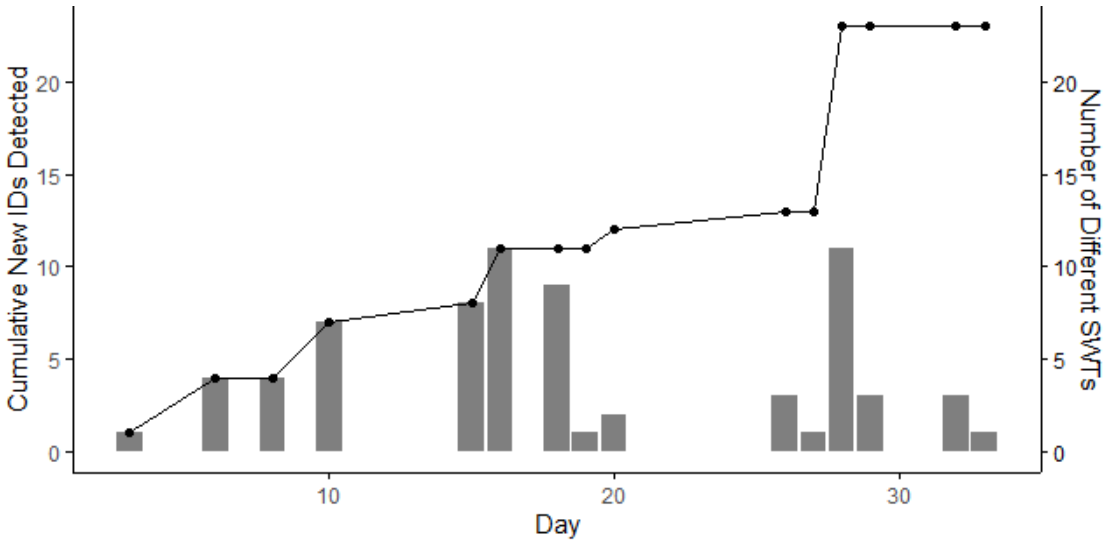


Figure 5.6. Discovery curves for cumulative detection of unique *Sousa plumbea* signature whistle types, (SWTs, unique IDs) over five weeks (five capture occasions) throughout the study area. A) For individual moorings, coloured marks along X axis indicate when the mooring ended recording. B) All moorings pooled to show total cumulative new SWTs (unique IDs) detected over time across the array. Bars representing total SWTs (unique IDs) detected per day throughout the study period.

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Recapture

Population models were run on the truncated dataset of five capture occasions as only one hydrophone recorded in week 6 (mooring 9, see above). In Mossel Bay, the best-fit model was pi(~1)p(~1)c() and in Buffelsbaai and Plettenberg Bay, the best-fit model was 'pi(~1)p(~time)c()'. Closed population models produced the following regional estimates $\widehat{N}$; Mossel Bay: 8 (95% CI 8-8); Buffelsbaai: 19 (95% CI 14-53); Plettenberg Bay: 22 (95% CI 19-33) (Table 5.3). For the combined closed population model, the best-fit model was pi(~1)p(~time)c(). This combined model produced the best regional estimate of 26 (95% CI 25-32) (Table 5.3).

Model selection for the open population POPAN Jolly-Seber models showed that model Phi(~1)p(~time)pent(~1)N(~1) best fitted the data, and assumed a change in capture probability over time, and a constant probability of survival rate, entry of individuals into the population and population size as births or deaths were not expected over this period. This model estimated a superpopulation size of 26 individuals (95% CI 23 – 30) (Table 5.3).

Table 5.3. Estimated population size ($\widehat{N}$) of *Sousa plumbea* for both closed (Huggins) models for each bay individually and combined estimates as well as open (POPAN Jolly-Seber) model for the combined population estimate for *Sousa plumbea* throughout the south coast subpopulation range . Number of parameters (npar), standard error (SE), lower and upper limits of the 95% confidence interval (Lower CI and Upper CI).

Model type (region)	**Number SWTS**	**Best Model**	**npar**	$\widehat{N}$	**SE**	**Lower CI**	**Upper CI**
Closed model (Mossel Bay)	7	pi(~1)p(~1)c()	2	8	0.00	8	8
Closed model (Buffelsbaai)	13	pi(~1)p(~time)c()	6	19	7.34	14	53
Closed model (Plettenberg Bay)	18	pi(~1)p(~time)c()	6	22	2.95	19	33
Closed model (combined)	25	pi(~1)p(~time)c()	6	26	1.33	25	32
Open PJS model (combined)	25	Phi(~1)p(~time)pent(~1)N(~1)	8	26	1.3	23	30

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Density

A null model was used to estimate density across the full survey area and duration. Models were tested with three types of combinations 1) capture truncations, 2) three distinct spatial masks with varying resolutions and spatial constraints, and 3) three trap configurations to investigate survey design and the viability of using signature whistle types in an SECR framework. The SECR model used investigates captures between 'traps' but ignores time variation effectively using a single capture occasion for the entire study period, hence the full deployment period was used.

Truncating the data to reduce detections of SWT13, helped to improve model convergence. Multiple levels of truncation (Trunc) were investigated (200, 150, 100, 50), but only Trunc200 and Trunc100 are reported, as the effect on the output were the same. At Trunc200, SWT13, which had 482 detections, was limited to 200 detections; only individual SWT13 was affected. At Trunc100, three individuals were affected, SWT13, SWT44 and SWT66. Across these truncated data sets, the number of detections was reduced, but the total number of individuals remained at 25, for the single capture occasion.

The SECR design configurations reported (Table 5.4) include the full capture history and high-resolution mask (Full.HR), combinations of the two truncation levels with both the high and low resolution mask (200.HR, 100.HR, 200.LR, 100.LR), and the trap configurations adjusted by removing the two least active traps (200.HR.Removed) and combining the traps that were placed in the same bay (200.HR.Combined). The predicted estimates for both density (D) and sigma throughout these model combinations were very comparable. Sigma (the average distance an individual moves between capture occasions) was estimated to be between 10 km and 16 km and likely constrained by detector distribution throughout the array than overall movement of

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animals. Abundance estimates, extrapolated from the SECR models were all close to 25 individuals, for the study area and period, as seen in Table 5.5.

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Table 5.4. Full model outputs across different model combinations to estimate density for *Sousa plumbea* throughout the south coast survey area, South Africa. Model combinations are highlighted in grey. Combinations include: 1) Capture history truncations (Full detections and capped at 200 and 100 detections); 2) Mask objects (High, low and default resolutions); and Trap configurations (Full moorings - *CaptHit.Full*, Removed two least active receivers – *CaptHist.Rem*, Combined the two moorings from the same bay – *CaptHist.Comb*). Model parameters and estimates are reported, as well as the detection function used and Akike Information Criterion (AIC) scores. Bolded values are best estimates of density (D) and sigma (σ).

Model Data	Capture Truncation	Trap Combination	Mask	Parameter	Detect Fn	Log Lik	AIC	AICc	Coef (beta, log link)	Predicted	SE Estimate	Lower Confidence	Upper Confidence
Full.HR	Full	M1, M3, M5, M8, M9	HR	D	HEX	-1.00E+10	2.00E+10	2.00E+10	NA	NA	NA	NA	NA
				Lambda					NA	NA	NA	NA	NA
				Sigma					NA	NA	NA	NA	NA
200.HR	200	M1, M3, M5, M8, M9	HR	D	HEX	-559.48	1124.95	1126.09	-7.731	**0.0**	0.0	0	0
				Lambda					4.995	147.7	8.5	132	165
				Sigma					9.446	**12659.2**	350.7	11990	13365
100.HR	100	M1, M3, M5, M8, M9	HR	D	HEX	-504.57	1015.15	1016.29	-7.739	**0.0**	0.0	0	0
				Lambda					4.435	84.4	6.0	73	97
				Sigma					9.612	**14936.8**	517.9	13956	15987
200.LR	200	M1, M3, M5, M8, M9	LR	D	HEX	-597.74	1201.47	1202.61	-7.782	**0.0**	0.0	0	0
				Lambda					4.982	145.8	7.5	132	161
				Sigma					9.421	**12348.3**	313.2	11749	12978
100.LR	100	M1, M3, M5, M8, M9	LR	D	HEX	-526.92	1059.84	1060.98	-7.784	**0.0**	0.0	0	0
				Lambda					4.317	75.0	5.4	65	86
				Sigma					9.567	**14291.1**	519.1	13309	15345
200.HR.Removed	200	M1, M5, M9	HR	D	HEX	-350.17	706.34	707.61	-7.827	**0.0**	0.0	0	0
				Lambda					5.103	164.5	10.2	146	186
				Sigma					9.666	**15778.7**	683.1	14496	17175
200.HR.Combined	200		HR	D	HEX	-368.9	743.79	744.94	-7.731	**0.0**	0.0	0	0
				Lambda					5.225	185.8	12.5	163	212

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		(M1+M3), M5, (M8+M9)		Sigma					9.554	**14103.5**	524.6	13112	15170
200.Def	200	M1, M3, M5, M8, M9	Default	D	HEX	-670.56	1347.12	1348.27	-9.711	**0.0**	0.0	0	0
				Lambda					4.859	128.9	8.7	113	147
				Sigma					9.286	**10782.3**	311.6	10189	11411

Table 5.5. Predicted estimates of density and abundance for *Sousa plumbea* throughout the south coast survey area are shown for each of the model combinations. Estimates were calculated from the *estimated density * study area*. Sigma is an estimate of the average distance an individual moves between capture occasions and over the study period. All values were transformed from the log scale used by the model to real estimates.

Model	**Predicted sigma (m)**	**Predicted density (*animals.ha*$^{-2}$)**	**area (ha)**	**Predicted abundance**
Full.HR	na	na	na	na
200.HR	12659.23	0.000439	58501.7	26
100.HR	14936.80	0.000435	61666.3	26
200.LR	12348.33	0.000417	58501.7	26
100.LR	14291.08	0.000416	61666.3	26
200.HR.Removed	15778.73	0.000399	58501.7	23
200.HR.Combined	14103.47	0.000439	58501.7	26
200.Def	10782.32	0.000061	412539.5	25

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DISCUSSION

This study provides a proof of concept on the effectiveness of PAM at detecting individual dolphins within this small population over a multi-bay deployment. Detections across the survey period suggested a high capture rate of a portion of the total population within the survey area. The discovery curve for the combined dataset showed a clear asymptote, indicating that over the full survey area, nearly all available individuals using the study area over the study period were detected. This was confirmed in the abundance estimates across both the CR and SECR model estimates, resulting in values not much higher than the total number of identified SWTs. All capture-recapture models run on the various sets of data produced similarly low abundance estimates and density estimations for the survey area.

Abundance estimates in this study were based on a continuous acoustic deployment (3559.44 hrs) with five capture occasions (weeks) included in the final abundance modelling, producing a combined abundance estimate of 26 in both the closed and open combined abundance models across the whole survey area (Closed 95% CI 25-32, Open 95% CI 23-30). Comparatively, in Mossel Bay, James et al. (2015) produced a closed abundance estimate across years of 2011 – 71 ind. (95% CI: 30–167); 2012 –28 ind. (14–55); and 2013 – 43 ind. (19–98) but over a much longer timeframe of 8 to 12 months of survey effort per year (Table 5.6). Although these estimates are higher, the number of *S. plumbea* encounters throughout the three-year survey period in James et al., (2015) was very low, with 0 - 3 encounters per month across this period. When further looking at abundance estimates in this study across the three bays separately, the number of SWTs captured, and abundance estimates produced across the three bays in this study are comparable to similar local photo ID assessments in Plettenberg Bay over short time periods (Table 5.6). In this study, in Plettenberg Bay, 18 individuals were acoustically detected, producing closed abundance estimates with 95% CI of 19 to 33 individuals throughout the area. By comparison, Conry, (2017) encountered 33 individuals over 39 survey days within a 25-month

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study period in the same area. Jobson, (2006) detected 63 individuals and estimated an abundance of 112 (95% CI 75 - 133) during a 24-month study in Plettenberg Bay. Additionally, Greenwood, (2013) identified 41 individuals with a closed population estimate of 42 (95% CI 42-55) during a 17-month study period. The lower estimates within this study are likely reflective of the shorter study duration in comparison to these previous studies, the limited movements of individuals within this study period, and the caveat of signature whistles captures occurring at lower rates than photographic capture. Direct comparison between studies is difficult due to different methodologies and study efforts. One of the key differences within this study was the reduced survey period of weeks in comparison to months and even years in previous studies.

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Table 5.6. Summary of study effort and capture-recapture results obtained from previous studies on *Sousa plumbea* in southern South Africa: Mossel Bay (James et al., 2015; 2011-2013), Goukamma (Conry, 2017; 2014-2015), Plettenberg Bay (Jobson, 2006; 2002-2003), Plettenberg Bay (Greenwood, 2013; 2012-2013) and this study: Mossel Bay, Buffelsbaai, Plettenberg Bay, combined south coast (five weeks, 2022). Values not modelled are reported as na.

Study site	**Duration of study (weeks)**	**Number of surveys**	**Catalogue size (individuals identified**	**Closed model abundance estimate (ind. [95% CI])**	**Open model abundance estimate (ind. [95% CI])**	**Year of study**	**Reference**
Mossel Bay	224	81	67	46 (38-56)	125 (61-260)	2011-2013	James et al,. 2015
Goukamma - Tsitsikamma MPA	112	101 (231 hr)	65	84 (72-98)	84 (72-115)	2014-2015	Conry, 2017 (Thesis)
Plettenberg Bay	168	87 (289 hr)	63	112 (75-133)	727	2002-2003	Jobson, 2006 (Thesis)
Plettenberg Bay	119	20 (38 hr).	41	42 (42-55)	41 (28-54)	2012-2013	Greenwood, 2013 (Thesis)
Mossel Bay	5	Continuous recording period (3559 hr)	7 SWT	8 (8-8)	na	2022	This study
BuffelsBaai	5		13 SWT	19 (14-53)	na		
Plettenberg Bay	5		18 SWT	22 (19-33)	na		
South Coast (combined sites)	5		25 SWT	26 (25-32)	26 (22-30)		

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In using signature whistles as markers in capture-recapture models, it is important to consider that their production can be behaviourally dependent and context-driven, and when used as an individual marker for detection, they can have inherent biases due to these behavioural effects (Fearey et al., 2022; Longden et al., 2020; Romeu et al., 2024). One of the assumptions of mark-recapture is capture (or tag) homogeneity between sampling periods and consistent identification. Signature whistles show stability over time, highlighting their suitability to be used in this context (Sayigh et al., 2022), but their production is context-dependent, and dolphins can be present in an area and not produce these specific calls (Fearey et al., 2022). In other capture-recapture studies, the use of hair traps or camera traps provides a more homogenous rate of detection between individuals, although they are still vulnerable to behavioural variations, such as the amount of 'hair captured, the angle of animal in the photo, the general quality of the photograph etc. In relative density studies, the use of call detections in species such as birds (Dawson & Efford, 2009) and frogs (Stevenson et al., 2015) is simplified as these species have uniform call rates between individuals, which is shaped by basic fitness and evolutionary drivers. These factors provide an overall detection probability that is homogenous over a population. In other dolphin species, group size, composition and behaviour highly influence individual whistle production (Kriesell et al., 2014; Quick & Janik, 2008). In *S. plumbea*, these contextual effects have been demonstrated to have less influence on signature whistle production rates, potentially producing a more homogenous detection rate amongst individuals (Chapter 3). Although based on a small available sample size., low overall whistle production rates are unlikely to have affected detection as only one detection within a week-long sampling occasion is required for an individual to be classed as 'captured'. As the *Sousa* SWT catalogue grows, the strength in detecting individuals will increase, as seen in other populations with larger, more extensive SWT catalogues (Longden et al., 2020).

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In this study, abundance estimates in both closed and open models were similar to but below abundances from previous studies in the region and highlighted that nearly all individuals were captured in the region. When looking at the number of recaptures across the survey period, five capture occasions over a five-week period produced a high number of recaptures across two and three capture occasions, and very few animals were caught only once. This implies a high number of available individuals were detected throughout the survey period, as opposed to a higher number of single captures seen in mark-recapture studies for 'open' and very large populations (Reisinger & Karczmarski, 2010). The use of weekly capture occasions likely contributes to the higher number of recaptures within this study's models. The decision to use weeks as capture occasions rather than days was made to ensure a sufficient period of mixing and movement of the population between sampling occasions based on known residency and resighting rates (Vermeulen et al., 2018) and serve to reduce the number of zeroes in the data set which can affect model convergence. Vermeulen et al., (2018) found the shortest time span between two resighting's of an individual was one day (between Knysna and Plettenberg Bay), across a 45 km area. However, for distances > 100 km, the shortest time span between resighting was 24 days. With the maximum array distance between mooring 1 and mooring 9, ~120 km and the deployment duration limited to around six weeks, weekly capture occasions were deemed the most reliable period to ensure sufficient mixing of the population, considering the levels of extended site fidelity seen in previous studies and ensuring likely detection of all available dolphins within the deployment period. (Vermeulen et al., 2018). The estimates produced within this study are likely a reliable estimation of the available individuals within the 120 km area throughout the survey period, highlighting the limited movement of individuals entering or exiting the survey area. The limitations of centring the array within a 120 km region of the south coast will have led to uncaptured individuals as they moved beyond Mossel Bay or Plettenberg Bay or didn't enter into this region within the survey period. To increase capture rates, future efforts should prioritise expanding study sites along the south coast range with greater distances between moorings as fine-scale coverage of bays did not increase capture rates within this study.

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Optimising for a larger geographic scale and weekly capture occasions would enhance the potential for new captures and encourage increased population mixing.

The final aspect of the SWORD framework is density estimation, conducted through SECR in this study, which found extremely low densities across the survey area. Across all SECR models tested, density estimates were very low, and movement estimates (Sigma) were limited by detector location. Survey design requires consideration of both logistical constraints and the budget available, as this will affect the scale and scope of the study's ability to detect and estimate animals, especially within spatially explicit models. The variation in the scale of PAM survey designs is large, with the US Navy heavily investing in passive acoustic arrays to investigate the impacts of sonar on marine mammals at both the individual and population levels (Mellinger et al., 2007). Selecting array dimensions is especially important if investigating call rates or density estimations based on time of arrival (Marques et al., 2013), with greater distances between hydrophones increasing the magnitude of the acoustic time delay of multiple detections of the same sounds (Ryder et al., 2023). The use of SWT in this study removed these aspects from the survey design, with single hydrophone detection providing individual captures of individuals as long as individuals called as they moved throughout the survey area. This study was restricted in scope through the number of hydrophones available, hence the investigation into placing multiple hydrophones in single bays or a single hydrophone in a bay. The manipulation of trap design, through the combination or removal of these within-bay traps, did not affect estimates, indicating that SECR is robust to limited survey efforts or elusive species that are difficult to capture. Using a single hydrophone in bays provides a greater spatial range to be covered, with reduced effects from potential variation in production rates in *S. plumbea*. This would enable a more useful estimate for Sigma that is less biased from hydrophone location with a greater range of movement within the model. Yet, regardless of mask size, accuracy, or resolution, density estimates had estimates of 23-26 individuals – in line with the combined closed and open population models.

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In producing population trends, the rate of population change, natural variation in population size and the frequency of surveys across geographic and temporal scales will be species-specific to the degree of these effects (Thompson et al., 2000). With naturally small populations, *S. plumbea* are more vulnerable to genetic problems such as loss of genetic variability and inbreeding depression (Avery & Hill, 1977). They can also be driven towards extinction by natural variations in individual reproductive and mortality rates (demographic variation) (Stacey, 1992). These small populations can be more vulnerable to anthropogenic or natural changes in environmental conditions. This is likely already occurring in *S. plumbea* off the KwaZulu-Natal coastline, which has recorded the highest levels of PCBs and DDT of any delphinid globally (Gui et al., 2016). Problems from such pollutants include reproductive abnormalities and impaired testosterone production (Duinker et al., 1979; Subramanian et al., 1987), the full effect of which has not yet been monitored in this population. Overall, for *S. plumbea,* throughout their whole global range, there has been little empirical evidence on population trends, but available studies have shown a worrying degree of regional decline throughout southern Africa. On the south coast of Zanzibar, *S. plumbea* population estimates declined by 63% since 2002 for non-calf individuals, with an abundance of 19 (95% CI 14-25) in 2015, meeting the threshold for Critically Endangered on the IUCN Red List criteria (Sharpe & Berggren, 2019). In Plettenberg Bay, South Africa, the population also declined by around 50%, from 93 (95% CI 72–114) to 41 (95% CI 28–54) individuals between 2002-03 and 2012-13 (Greenwood, 2013). In comparison, these studies' estimate of 22 (95% CI 19-25) individuals in Plettenberg Bay is lower but, with a combined closed estimation of 26 (CI 95% 25-31) for the survey area, indicates that Plettenberg Bay forms are a large proportion of the studies detected individuals. Expanding the geographic ranges of the survey area further east in future deployments would provide a more robust estimate for the whole south coast that could produce more comparable trends with a higher degree of repeatability to monitor more useful trends to population size.

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This study used signature whistles as individually unique acoustic markers in a capture-recapture framework to provide abundance estimates for *S. plumbea* across both individual bays and a combined range along the south coast of South Africa. This method provided low estimations across models, similar to that of detected SWTs, indicating that the majority of acoustically available individuals in the area were detected throughout the study period. Further scope for improving these estimates in future deployments includes expanding the geographic range of the acoustic array, rather than increasing the density of recorders in the array, to optimise captures of unique SWTs throughout the south coast.

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6 CONCLUSION

The field of acoustic ecology is emerging globally and has undergone an exponential increase within the African continent in recent years (Becker et al., 2022). This is in part due to the development of more accessible and cost-effective equipment, along with the recent development of the African Bioacoustics Community fostering capacity-building and training opportunities (Becker et al., 2022; Lamont et al., 2022). Passive acoustic monitoring (PAM) has proven to be an effective method for coastal studies and applied acoustic ecology, with great potential for generating much-needed knowledge of low-density and threatened species (Rogers et al., 2013).

In 2015, the Indian Ocean humpback dolphin, *Sousa plumbea*, was listed as "Endangered" by both the International Union for the Conservation of Nature (IUCN; Braulik et al., 2015) and the Red List of Mammals of South Africa (S Plön et al., 2016), making them the first and only officially endangered marine mammal resident in South Africa. The preference for shallow habitats by *S. plumbea* coincides with the region's most heavily utilised waters, affected by coastal fishers, the shipping industry, coastal construction, and agricultural run-off (Braulik et al., 2015). These anthropogenic pressures on the coastal regions of South Africa overlap with *S. plumbea's* range, however, single, quantifiable threats are difficult to identify, as highlighted by Plön et al., (2021). Given the small number of individuals remaining in the population and the widespread threats they face, individual-based monitoring is particularly important. The understanding of home ranges for South African populations of *S. plumbea* is currently limited by the ability to adequately study them using traditional boat-based surveys and photographic capture-recapture techniques. Given the threats to the remaining population, this thesis explores how individual passive acoustic monitoring using signature whistles could be used as a cost-effective and non-invasive method to detect *S. plumbea* throughout their range which could best inform managers and policymakers.

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Chapter 2 presents the first description of whistle contour features of *S. plumbea* in South Africa, with an analysis of geographic variation between local regions within South Africa and across ocean basins (comparisons made with a population of *S. plumbea* in Maharashtra, India). The study showed a degree of increasing level of acoustic variation with increasing geographic isolation throughout the study range with the least variation demonstrated amongst the four south coast sites in South Africa, which is likely attributed to the regular mixing of individuals throughout this range. However general whistle features overlap. which supports the use of whistle characteristics for acoustic monitoring, in particular species classification.

One of the novel results from this chapter was the presence of ultrasonic whistle contours throughout *S. plumbea's* South African range. The ability to produce ultrasonic whistle contours (> 20 kHz) has been demonstrated across the genus, but the presence of whistles over 40 kHz throughout South Africa provides the first evidence of these high-frequency whistles consistently throughout a region. The production of high-frequency whistles is a morphological adaptation of the sound-generating structure present in *Sousa* spp. (Frainer et al., 2019), which can also be seen across several other high-frequency whistling delphinids across separate evolutionary lineages (Lammers & Au, 2003; Rasmussen et al., 2006). Morphologically, *Sousa plumbea* has a specialised structure next to the melon's small left posterior branch, which is associated with higher frequencies and directionality of sound (Frainer et al., 2019). This structure is present in several other species that also feature high-frequency characteristics, including Spinner dolphins *Stenella longirostris*, Risso's dolphins *Grampus griseus*, white-beaked dolphins *Lagenorhynchus albirostris*, and Franciscana dolphins *Pontoporia blainvillei* (Frainer et al., 2019). Equipment limitations and limited encounter durations in previous studies may mean these upper bandwidth capabilities are likely widespread within this genus but were either not recorded (Schultz & Corkeron, 1994; Van Parijs & Corkeron, 2001) or investigated across other studies (Sims et al., 2012; Zbinden et

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al., 1977). These features of the whistle repertoire within *S. plumbea* are important markers that can be used for species monitoring, for example within the development of regional species classifiers. Regionally specific species classifiers were developed for this thesis to extract signature whistles from long-term moored acoustic data with high overlap in whistle features seen throughout the South Africa study sites. These classifiers were employed to extract *S. plumbea* whistles from long-term moored acoustic data and are now available for use by other researchers and organizations to identify *S. plumbea* from passive acoustic recordings.

Comparison between the south and east coast sites indicated that Richards Bay was more distinct than the south coast sites with little misclassification of Richards Bay whistle contours to south coast study sites. This evidence supports the two-population hypothesis, suggesting greater acoustic variation between the south and east coast populations than within the south coast population due to increased geographical isolation. Lampert et al., (2021) used mtDNA to show that these two populations were historically a single population with further range contraction and habitat fragmentation causing these populations to become geographically distinct, with little mixing today (Caputo et al., 2017; Vermeulen et al., 2017). Acoustic variation between the south and east coast populations may be explained by the acoustic adaptation hypothesis, where signals are adapted to environmental factors such as water depth, temperature, and salinity to maximize transmission and minimize degradation (Ey & Fischer, 2009). Richards Bay is an industrialised harbour area with distinct habitat differences to the south coast sites. In addition to habitat differences, boat activity has been shown to cause dolphins to modify their vocalizations and acoustic behaviour (May-Collado & Quiñones-Lebrón, 2014; Perez-Ortega et al., 2021). Boat activity has been observed to cause both an upward shift in whistle frequencies (Guerra et al., 2014; Heiler et al., 2016) and lower frequencies with fewer frequency modulations in areas of high ambient noise (Morisaka et al., 2005). This follows the acoustic niche hypothesis, which states that cetaceans shift their communication sounds to be below or above ambient noise and

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was developed from studies of the gray whale (*Eschrichtius robustus,* Dahlheim & Fisher, 1983) and the bottlenose dolphin (*Tursiops* spp., Gao et al., 1995). The high use of the harbour environment and habitat characteristics, such as high ambient noise, likely contributed to the distinctiveness between the east and south coast populations. The use of *S. plumbea* classifiers (Chapter 4), with the future inclusion of clicks (Yang et al., 2020), could then utilise long-term acoustic arrays throughout the east coast region to further explore *S. plumbea* range movements. Richards Bay is a study site 220 km from the southern Mozambique border. Presently, there is no knowledge of individuals moving between these two countries. With *S. plumbea* listed as locally Endangered on the IUCN Red List of mammals for South Africa, there is little additional protection in Mozambique (Plön et al., 2016). Bycatch has been highlighted as a key threat to *S. plumbea* populations further along their Indian Oceans range (Amir et al., 2002; Stensland et al., 2006), and Mozambique is a country where the threat of bycatch is high (Kiszka et al., 2009). The next available study on the species in Mozambique is in Maputo Bay, 100 km north of the southern border, where transient individuals were high throughout the area (Guissamulo & Cockcroft, 2005). Boat-based surveys demonstrating connectivity between these areas have not been previously attempted, but acoustic deployments covering this stretch of coastline throughout iSimangaliso wetland park, south of the Mozambique border and into Ponto do Ouro reserve, in southern Mozambique could provide the data required to understand individual movement across this border using PAM. Therefore, providing evidence to extend conservation priorities into Mozambique, if movement of these individuals were shown to occur throughout this region.

Chapter 3 extends these findings by examining individually distinct whistles, known as signature whistles (SWs), and explores individual acoustic behaviour and how variables such as group composition and behavioural states might affect whistle production rates. In recent years, using individually distinct calls for acoustic monitoring has been proposed as an alternative to tagging or photographic methods for capture-recapture studies (Longden et al., 2020; Terry et al., 2005).

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However, using natural vocalizations as detection markers comes with biases, such as varying detection probabilities across species, groups, and individuals. Understanding what influences acoustic production is key when designing the timing and locations of long-term passive acoustic arrays to detect animals using these passive acoustic methods.

The results from this chapter demonstrated a relatively low signature whistle production rate in free-swimming *S. plumbea*, supporting vocally elusive behaviour. It was hypothesized that clear effects on signature whistle production would be observed from group size, calf presence, or group behaviour, as suggested by existing literature from other, more widely studied species (Cook et al., 2004; dos Santos et al., 2005). However, across the 21 encounters included in the study (with the removal of one outlier group), behaviour and group composition do not strongly affect whistle production rates, only temporal production. Passive Acoustic Monitoring (PAM) can theoretically be conducted continuously, but practical constraints such as battery life and storage capacity currently limit the duration of deployments (Riera et al., 2013; Thomisch et al., 2015). Duty cycling, which alternates between periods of active recording and dormancy, can extend deployment time, by conserving both battery and storage, yet will miss detections when dormant. When selecting an optimal recording period for acoustic recorders, it is important to consider the vocal behaviour of the species, the production rates of the species, and the study's overall research questions (Riera et al., 2013). Therefore, with little effect from selecting mooring locations around specific activity centres, it is more important to consider appropriate recording schedules to optimise signature whistle detection. With short acoustic encounters and low production rates, duty cycling needs to consider this, especially when small group sizes further influence limited detection opportunities. In *S. plumbea* continuous recording periods were deemed most effective for optimising detection and therefore used throughout the thesis during long-term acoustic moorings. Where possible, continuous recording schedules are

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recommended, or if required, very short periods of dormancy within a duty cycle when *S. plumbea* is the focal species.

Across all the covariates tested, the presence of calves in a group increased the temporal effect of whistle production and although there was a small relationship between SWT (individuals) and SW counts during encounters without calves present, there was no significant effect on the number of signature whistles produced. Mothers and calves have been shown to increase the call rate of signature whistles to initiate reunions with one another, especially when out of visual range (King et al., 2016). The low whistle rates in mother-calf groups in this study indicate that either mother-calf pairs were not moving apart from one another during encounters, (this was later confirmed through boat-based photo ID S. Dines pers. obs.), or that mother-calf pairs do not utilise vocal communication in the same way as other species do. In other species, other forms of contact are often recorded when vocal communication is less preferred, such as in nursing baleen whales, who use visual or physical contact with calves, to avoid predator detection (Nielsen et al., 2019; Parks et al., 2019; Videsen et al., 2017). Across *S. plumbea's* coastal range, the possibility of inter- or intraspecific detection is high, which could be dangerous for the calf (Koper & Plön, 2016). In China, *S. chinensis* group sizes reach 48 individuals, and nursery groups (with mother-calf pairs present) are 2-4 times larger than non-nursery groups, providing much stronger protection for calves (Liu et al., 2021). The consistently smaller group sizes in *S. plumbea* in South Africa mean they do not have a 'safety in numbers' effect (Lehtonen & Jaatinen, 2016), and the addition of vulnerable calves to a group will increase intraspecific and interspecific predator threats (Frainer et al., 2023; Koper & Plön, 2016). It is known that within cetaceans mother-calf pairs are generally in very close proximity (Gubbins et al., 1999) and within these smaller groups, reliance on other methods to maintain mother-calf contact, such as visual or physical contact, might be preferred. In future studies, calf presence on vocal production is certainly one aspect of vocal behaviour that warrants further inspection.

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Behavioural states have been demonstrated to have some of the strongest effects on whistle production, with increased arousal linked to increasing levels of production in both wild and captive populations (Cook et al., 2004; Luís et al., 2016; Probert et al., 2021). However, in this study, no greater effects on whistle production were seen during socialising contexts than when travelling or milling behaviours were observed. Signature whistles function as a contact call and a stronger effect during social events would have been expected (Janik, 2009). Socialising behaviours such as tail slaps, events that are associated with high levels of arousal, were not observed during encounters (Luís et al., 2021). Longden et al., (2020) found clear, behaviourally linked capture heterogeneity for signature whistle detections of *T. truncatus*. A key element for optimal acoustic detection in other species is prior knowledge of habitat use and behaviours within the study area, to maximise capture probability (dos Santos et al., 2005; Fearey et al., 2022). Passive acoustic monitoring for *S. plumbea* is novel throughout the south coast with little information available on acoustic behaviour from moored hydrophones throughout this region. The lack of strong behavioural or group composition effects on signature whistle production in *S. plumbea* is an important result for optimising passive acoustic detection in this species, suggesting that group composition and behaviour are not critical variables for optimizing signature whistle detection in *S. plumbea*.

Chapter 4 investigated individual detection through both traditional boat-based surveys with moored acoustic deployments during a short-term assessment in Mossel Bay. The reason for focusing the chapter on a short-term rapid assessment was the need to investigate the effectiveness of alternative methods for individual detection that could be invaluable in these threatened species (Braulik et al., 2018). The rapid identification of key regions to prioritize for continued monitoring studies has been highlighted as a requirement for greater conservation action (Braulik et al., 2018). Effective conservation management plans consider logistical

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constraints and species-specific research to produce baseline empirical data that informs effective monitoring studies (Downey et al., 2022). For acoustic monitoring to be suggested as an effective method, spatial and temporal differences in individual detection were investigated using concurrent PAM and boat-based surveys, including an evaluation of the logistical and financial constraints and considerations of both methods within the context of a short-term deployment period.

The strength of the acoustic array was in the broad spatial scale it covered and its continuous recording capabilities. Static passive acoustic moorings detected animals across all six sites, but detection levels varied, with mooring three only detecting low signal-to-noise ratio calls (and therefore excluded from analysis), and moorings one and six showed the greatest diversity in individual detection (> 5 SWTs). Boat-based surveys were conducted throughout the bay with photo ID encounters concentrated in two locations within this array area. The high encounter rate at these locations would have indicated that these were high-use areas for these dolphins from the boat-based assessments. The bias of these surveys to good weather conditions and a short survey period, usually in the middle of the day, limits any conclusions on habitat use that can be made from boat-based encounters alone. One of the inherent limitations of boat-based surveys is they are constrained by good weather and daylight (Baird et al., 2005). In addition, boat-based encounters are behaviour-dependent and in the presence of boats, dolphins are less likely to stay socialising and more likely to shift to travelling (Kassamali-Fox et al., 2020). Through passive acoustic detections, it was apparent that *S. plumbea* moved throughout the whole survey area, with higher levels of vocal activity during the night. The broader spatial component of acoustic surveys provides a stronger level of data on species occurrence, including nocturnal activity, as demonstrated in this chapter. Mossel Bay is one of the harbours that are planned for development over the next 25 years (Bloom, 2019). As part of Operation Phakisa, the South African government plans to develop the country's blue economy, and coastal development,

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namely harbour expansions, will be implemented over the next 25 years (Guerreiro, 2022). If these developments are to go ahead, the inclusion of *S. plumbea* activity levels and fine-scale habitat use throughout this area in environmental impact assessments (EIAs) could dissipate the impact on the population. The inclusion of recommendations, such as suggesting construction outside of high activity periods or avoiding high-use areas altogether, could be provided through these PAM datasets. This has been done effectively in Walvis Bay, Namibia, to investigate the spatial use of *T. truncatus* around planned port developments and led to the inclusion of a dolphin priority area in the marine spatial planning process prior to coastal development (Finke et al., 2020).

Overall, when comparing the two methods over a short period, both had low detection rates yet successfully recaptured individuals during the study period. When evaluating 'cost per dolphin' (CPD), while the costs associated with fieldwork were lower for acoustic methods, photo ID proved to be a more cost-effective approach overall. This conclusion was largely due to the significant amount of time required for data processing and analysis in acoustic methods, which offset their initial cost advantages. When considering these methods, especially in the context of long-term monitoring for managers, the inclusion of desk-based time is an important factor. Time-based measures for analytical processes are subject to change in the future as machine learning methods are rapidly improving processing capabilities (Frainer et al., 2023; Jensen et al., 2024; Jiang et al., 2019). Moreover, the broader application of acoustic studies to include soundscape analysis and other species detection widens the applicability of these surveys. Furthermore, as data storage continues to become cheaper, these hurdles will continue to optimize passive acoustic monitoring as an option for a wider number of organisations. Photo ID serves as a valuable and accessible method, ultimately proving to be more cost-effective for this short-term study. However, in a targeted short-term assessment, continuous acoustic detection methods proved to be a reliable alternative to photo ID.

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Chapter 5 builds on the work from previous chapters and covers a 6-week static acoustic deployment across three study locations throughout the central portion of the south coast of South Africa, covering ~ 120 km of the south coast range. The chapter aimed to provide a proof-of-concept using the SWORD (signature whistles for occurrence, recapture and density) framework (Fearey et al., 2021) to present an analysis of PAM's effectiveness, using both capture-recapture and SECR models in *Sousa plumbea*.

Capture-recapture and SECR models provided reliable estimates for the survey area and highlighted the limited movement of individuals throughout the survey period. The three sites used in this study were known as high-use areas based on previous photo ID studies and were successful as proof-of-concept deployments. The deployment region for this chapter was selected as central within the south coast population range, incorporating sites where individuals were known to move between and previous regional estimations had been made (Conry, 2017; James et al., 2015; Vermeulen et al., 2017). Buffelsbaai, a newly established site for acoustic deployment, accounted for 77% of all SW detections and was monitored using a single hydrophone. This demonstrates that future moorings can effectively utilize bays with little to no prior behavioural or acoustic data, offering valuable movement insights in previously unexplored areas along the greater south coast. In addition, the study found nearly all the available individuals in the region were acoustically detected, as shown through the low population estimates, high number of recaptures, and asymptote seen in the combined moorings discovery curve. Over this period, the movement of individuals beyond the acoustic array was limited. Through trap configuration investigation, the chapter showed that the increased density of units within a bay did not increase estimates, and therefore future priority should be given to expanding the linear range of the array, utilising single moorings in locations and increasing the spacing between units to increase the number of unique SWTs captured across the south coast.

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In using signature whistles as markers in CR models, it is important to consider that their production can be behaviourally dependent and context-driven, and they can thus have inherent biases (Fearey et al., 2022; Longden et al., 2020; Romeu et al., 2024). In *S. plumbea*, behavioural and contextual effects have been demonstrated to have less influence on signature whistle production rates, potentially producing a more homogenous detection rate amongst individuals (Chapter 3). One of the assumptions of capture-recapture is capture (or tag) homogeneity between sampling periods and consistent identification. Unlike other CR studies, where seasonality affects population estimates (Elwen et al., 2019), no consistent seasonality patterns were seen over a three-year photo ID survey conducted in Mossel Bay (2011-2013, James et al., 2015). Therefore, removing the possible effect of seasonality on passive acoustic monitoring studies in this species. Signature whistles show stability over time, highlighting their suitability to be used in this context (Sayigh et al., 2022). Low overall whistle production rates by individuals were unlikely to have affected detection rates as only one detection within the weeklong sampling occasion was required to be classed as present. As the *Sousa* SWT catalogue grows, the strength in detecting individuals will increase, as has been seen in other populations with larger, more extensive SWT catalogues (Longden et al., 2020).

In conclusion, the work presented in this thesis provides information on whistle features, geographic variability, acoustic behaviour, and production rates of signature whistles in *S. plumbea* in South Africa. The thesis increases our understanding of how *S. plumbea* acoustically communicate along their South African coastal range, leading to the first population and density estimates for *S. plumbea* along a portion of their south coast range based on individual acoustic detection. This work has highlighted challenges and logistical considerations when monitoring an acoustically elusive species, with recommendations moving forward. The nature of PAM provides large datasets, and previously, species identification has been a limiting factor in the

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application of these datasets, as well as data storage. Over the course of this thesis, custom-trained classifiers were used to identify *S plumbea* from moored recordings, a first for the south coast region and available for future studies. The development of machine learning will further advance these analytical processes, with Frainer et al., (2023) recently developing a four-species classifier based on convolutional neural networks that provide additional applications of these datasets. The thesis also shows that PAM for *S. plumbea* requires a continuous recording schedule for optimal detection that considers their vocally elusive nature and intermittent encounter durations. Final recommendations show that expanding the south coast array to cover *S. plumbea's* full range, will provide a more reliable population estimate for *S. plumbea*, due to their limited movement during the survey period.

The conclusions presented here using individual acoustic detection through signature whistles provide a baseline dataset with recommendations for future studies. With the considerations and recommendations implemented, passive acoustic monitoring can provide a tool that can support managers and policymakers in developing effective conservation efforts for this threatened species with scope to strengthen these results in the future.

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APPENDIX 1

Using acoustic signatures to understand the endangered Indian Ocean humpback dolphin (*Sousa plumbea*)

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ABSTRACT

When species are on the brink of extinction, individual rather than population, based monitoring is both feasible and necessary, proving crucial information for conservation management. The Indian Ocean humpback dolphin (*Sousa plumbea*) is South Africa's most endangered marine mammal. Due to their elusive nature, data on population demographics are minimal. Here we investigate whether capture-recapture of individuals is possible over time and space, using naturally occurring signature whistle types and photographic identification data collected during focal group follows. Contours were visually grouped into whistle types of similar frequency modulation pattern. The temporal production of whistles was similar to that of bottlenose dolphins (*Tursiops truncatus*) and likely signature whistles were identified through a bout analysis approach applied to whistle types. Overall, 410 whistles were grouped into 21 whistle types, resulting in 14 likely signature whistles following bout analysis. Acoustic and photographic recaptures were identified through catalogue matching. We provide evidence of acoustic recapture of signature whistle types over distances spanning 480 km and eight years. By presenting strong evidence for signature whistle use and in demonstrating the feasibility of capture-recapture using individually distinctive acoustic signatures, we provide the foundations for an alternative, non-invasive method for individual based monitoring of this endangered species.

KEY WORDS

humpback dolphin, signature whistle, mark recapture, AIID, identity

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INTRODUCTION

Traditionally, wildlife conservation and management action has been based on patterns observed at the population and species level. However, for small populations of endangered species, every individual counts and individual-based monitoring becomes necessary. Understanding individual differences can improve and enable adaptive management of small populations and is possible through individually unique identifiers such as man-made tags or naturally occurring scars and marks. In Capture-Recapture (CR) individuals are identified by these marks and re-captured over time and space (Hammond, 1986) generating a capture history which enables calculation of individual ranging patterns (e.g. Vermeulen et al., 2018) and survival rates (Koivuniemi et al., 2019), as well as social associations (Lusseau et al., 2006). Population abundance and density can be estimated by applying various modelling approaches to capture histories over time and within a spatial framework (spatial capture recapture, SCR, Efford et al., 2009). A fundamental assumption of capture-recapture is that marks must be individually distinctive and stable to persist over time (Hammond, 1986). Historically, CR relied on artificial marks or physically marking animals, e.g. though ear clipping, freeze branding or paint marking or attaching tags. (Hammond, 1986). However, financial, logistical and ethical considerations have promoted natural non-invasive markers. For example, pigmentation patterns (Koivuniemi et al., 2019), nicks or scars are recaptured through photographs and hair snares enable recapture through genetic identification.

Recently passive acoustic methods have been proposed for use in CR of individuals (Marques et al., 2013) through individual acoustic identification (AIID, Terry et al., 2005; Klenova et al., 2009; Janik et al., 2013). AIID can be achieved by recognition of subtle but consistent differences in the time or spectral parameters of the same call types which is common across taxa, mammals(e.g., Charrier et al. 2003), birds (e.g. Mathevon, 1997), reptiles (e.g. Mathevon et al., 2013), and may arise from inter-individual differences in sound production morphology (i.e. by-product

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distinctiveness, Boughman & Moss, 2003). Alternatively, the communication system of some species is based on call types with high levels of inter-individual distinctiveness (Boughman & Moss, 2003), enabling AIID though individually distinct acoustic labels (Longden et al., 2020).

Some dolphin species produce whistles which encode identity information in their distinct frequency modulation pattern (reviewed by Janik & Sayigh, 2013). These whistle types, termed signature whistles (Caldwell & Caldwell 1968; Caldwell et al., 1990) are the strongest example of a designed individual signature in the animal kingdom (Janik et al., 2006; Sayigh et al., 2007). Signature whistles are structurally stable over time (Sayigh et al., 1990; Sayigh et al. 2022), commonly exchanged to communicate between individuals (Quick and Janik 2012; King and Janik 2013) and used to maintain group cohesion (Janik and Slater, 1998). Although best studied in common bottlenose dolphins (*Tursiops truncatus*) (reviewed in Janik & Sayigh, 2013), several other dolphin species including the Indian Ocean bottlenose dolphin (*Tursiops aduncus*, Gridley et al., 2014; King et al., 2018) and the common dolphin (Delphinus delphis, Caldwell & Caldwell, 1968; Fearey et al., 2019a) likely also use signature whistles to communicate identity information.

Due to their long term stability (up to 43 years, Sayigh et al., 2022) and high inter-individual distinctiveness, dolphin signature whistles have been identified as candidates for long term individual acoustic monitoring (Terry et al., 2005; Janik et al., 2013). Kriesell et al., (2014) demonstrated recaptures of wild bottlenose dolphin (*T. truncatus*) from coastal Namibia over time and space through acoustic monitoring of signature whistles. They applied a bout analysis approach (SIGID, Janik et al., 2013) to acoustic recordings made during focal group follows, to generate a catalogue of likely signature whistles to which future recordings could be matched and recaptures determined. Results were ground-truthed with simultaneously collected photo-identification data demonstrating acoustic (with photographic) recapture over time and space

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The first CR population abundance estimate individually distinctive signature whistles (Longden et al., 2020) were similar to photo-identification estimates of the same population (Elwen et al., 2019). Combined, these studies demonstrate the application of signature whistle monitoring for generating CR data on coastal dolphin populations, including generating empirical information on individual ranging patterns as well as population abundance estimates. Extending this, spatial capture-recapture using dolphin signature whistles to generate information on habitat usage and even density estimates has been explored, with promising results (Romeu et al., 2024).

Humpback dolphins (species from the genus *Sousa*) are obligate coastal species inhabiting waters less than 30 meters in depth, estuaries and even rivers (Jefferson & Curry 2015). This habitat makes them especially vulnerable to human impacts, which are greatest in the near-shore environment. Of the four species in the genus, the two which occur in African waters are considered the second (*S. teuszii,* IUCN Critically Endangered) and eighth (*S. plumbea,* IUCN Endangered) most 'at risk' cetacean species globally (Jefferson, 2019). However, no formal assessment of absolute species abundance exists for either species. For *S. plumbea,* the best available estimates are in the low 10's of thousands, with small populations spread thinly and discontinuously across their 27,000 km range between South Africa and India (Braulik et al., 2015).

Within South Africa, *S. plumbea* is the only endangered marine mammal resident in national coastal waters, an area where human activities have substantially altered the ecosystem which is undergoing a possible regime shift. Sighting rates are declining, group sizes are reduced and their distribution range is shrinking, indicating reductions in overall population size (Plön et al., 2015). Direct mortality through entanglement in bather protection nets is unsustainable (Atkins et al., 2013; Atkins et al., 2016; Plön et al. 2015). Concentrations of harmful organic pollutants in bycaught dolphins are among the highest for any dolphin globally (Gui et al., 2016) and 12% of

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those recently observed suffered rostrum abnormalities. They are naturally predated on by sharks and face competition for resources from other dolphin species, notably the Indian Ocean bottlenose dolphin (*T. aduncus, Schoeman and Plön 2016*). Fewer than 500 individuals likely remain in South African waters (Vermeulen et al., 2018), split between two sub populations, further increasing their susceptibility to local depletion, inbreeding depression or catastrophic events. Urgent conservation action is therefore required to prevent further declines in *S. plumbea* in coastal South Africa (Plön et al., 2021a).

Up to date scientific information on *S. plumbea* ranging patterns, abundance and density is essential to inform managers and contribute to regional biodiversity action plans (BAPs), (Plön et al., 2021a). However, most work on *S. plumbea* took place over a decade ago with limited temporal and spatial scope. Key data are missing, especially for the south coast population, where their inaccessible habitat and low density have made studies of the population and individual health metrics close to impossible. Their near shore distribution, often located within the surf zone where boat-based research is dangerous, make data acquisition of *S. plumea* problematic throughout their range, but particularly so along this stretch of exposed coastline. Specimens strand infrequently and when found are often in a degraded state and hard to reach location, further preventing comprehensive genetic or health assessments of beach-cast animals.

Acoustic monitoring offers benefits beyond boat-based photo-identification surveys for CR, in that data can be collected near to shore (where swells can make boat based photo-identification hazardous), 24 hrs a day and over multiple weeks or longer (reviewed in Zimmer, 2011). Therefore, signature whistle CR is increasingly being explored as a valid tool for monitoring coastal dolphin populations (Longden et al., 2020; Fearey et al., 2022). A programme of long term, individual monitoring of *S. plumbea* along the SA coastline has the potential to generate important information on individual behaviour and ranging patterns over time and space, population

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abundance and even density, essential data for conservation management of the species. This study explores the feasibility of individual based passive monitoring acoustic monitoring of *S. plumbea* using signature whistles. By combining acoustic and simultaneously collected photo-identification data from several sites in South Africa we investigate the potential of individual CR using passive acoustic monitoring of stereotyped whistle contours. We highlight some of the prominent characteristics of *S. plumbea* acoustic behaviour, which may reflect acoustic adaptation for signal propagation and localisation within the coastal zone, but present challenges to automated identification of signature whistles. We discuss the strength and weakness of this acoustic method for understanding endangered *S. plumbea* in Southern African waters.

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METHODS

Field data collection:

Continuous recordings of *S. plumbea* acoustic signals were made during focal follows of single species groups from multiple locations along South Africa's south coast (SC) and in Richards Bay, on the east coast (EC). Recordings were made whenever possible when dolphins were within 200 meters, using the equipment and recording set-ups summarised in Table 1. When using the single element HTI-96 MIN hydrophone this was mounted to a 1cm diameter steel chain. The SoundTrap (Ocean Instruments Technology) was attached to a bungee and weighted with a divers' weight. In both instances, the weighting helped to hold the hydrophone vertically in the water column, even when there was current, and the hydrophones were tethered to the research vessel and deployed to a depth of 2-3 meters. A concerted effort was made to generate recordings with the research vessel's engines switched off, however as *S. plumba* were frequency located close to rocky shore, within the surf zone, this was not always possible. Some recordings contained intermittent periods of high engine noise during manoeuvring of the research vessel or unusable periods when the hydrophone was temporarily lifted out of the water, neither of which were included in the analysis. During data collection, standard information on the group size, composition and distance from the research vessel were collected at the start and end of each encounter. High quality dorsal fins images were taken of dolphins within each group encountered using a Canon digital SLR (200-400 mm lens), to assess the individuals present, using natural nicks, scars and dorsal hump shape (Hammond 1989; Vermeulen et al., 2017). Photographic data collection was focussed to the start of each encounter and continued throughout recording periods when animals were within range.

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Data Analysis: Terminology

The most basic unit in delphinid whistle communication is the frequency modulated whistle contour, defined as a continuous narrow band tonal sound exceeding 100 ms in duration (Janik et al., 2013). Contours can be categorised into stereotyped whistle types, containing all whistles of a particular frequency modulation pattern (Janik & Sayigh, 2013). Signature whistles may be composed of a single continuous contour with a unique frequency modulation pattern (single element signature whistle, SE) or as a looped form where the contour is unbroken but produced with a repetitive pattern (continuous multi-looped signature whistle, CML). Alternatively, several contours may be repeated in a sequence with a short, stereotyped period of silence separating them, which in *T. truncatus* is around 100 ms but can last up to 250 ms in (disconnected multi-loop, DCML). In the latter case, the constituent contours, termed 'loops' can have an identical or non-identical frequency modulation pattern (Esch et al., 2009b). A simple transition analysis may be applied to identify likely DCML and non-identical DCML in acoustic recordings of dolphins (Janik & Slater 1998; Gridley, 2011). Operationally, signature whistles can be identified as the most commonly produced whistle type produced by an individual when it is in isolation (Caldwell et al., 1990) or temporarily restrained (Sayigh et al., 2022). Alternatively, a bout analysis approach, termed SIGID can be used on recordings of free ranging dolphins, whereby whistles of a type repeated within 1 to 10 secs are highly likely to be a signature whistle (Janik & Sayigh, 2013). The minimum threshold of three out of four whistles occurring within 1 to 10 secs has been verified for *T. truncatus* (Janik et al., 2013) and is commonly used (Fearey et al., 2022). SIGID has been used to identify likely signature whistles from free ranging dolphins (Quick and Janik, 2012, King and Janik, 2013, Kriesell et al., 2014; Fearey et al. 2019a; Fearey et al., 2022), where tagging or isolation is not logistically feasible.

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Acoustic Analysis

Acoustic data from single species groups of *S. plumbea* were standardised through decimation and removal of stereo tracks, to ensure that all recordings were mono, 96 kHz sample rate and saved at 16 bit. The entire acoustic dataset was then batch loaded to Raven Pro v1.6 (Center for Conservation Bioacoustics, 2019) viewed in the spectrogram display using the page function (10 sec window), facilitating contour assessment and classification cross-checking across the entire recording dataset. Recordings were scanned in the spectrogram display (settings: FFT : 1024, Hann window, 50% overlap) to detect whistle contours. Once found, contours were boxed using the selection function which automatically logs the signal start and end time from the selected contour as well as other user defined measurements. Each selected contour was graded according to a visual assessment of the signal to noise ratio (SNR) following Gridley et al., (2012) - 1: signal is faint and barely visible on the spectrogram, 2: signal is clear and unambiguous, 3: signal is prominent and dominates.

Both visual and automated methods can be used to verify categorisation of bioacoustic signals (Janik 1999, Deecke and Janik 2006). Signature whistles can be produced with multiple loops and the number of loops does not preclude matching to the signature whistle catalogue (Sayigh et al., 2007; Janik and Sayigh, 2013) when matching is done visually. However, whistle production with extensive looping (whereby a contour is produced with a short, stereotyped period of silence) can pose a contour extraction and classification challenge using automated categorisation involving methods such as contour overlapping or spectrographic cross correlation or image comparison (Frainer et al., 2023). Even the application of dynamic time warping (Deecke and Janik, 2006) within automated categorisation through neural networks struggles with how to approach these looped whistles and their constituent periods of silence. Our dataset of whistle contours recorded from *S. plumbea* was not extensive but contained relatively complex whistle types with multiple disconnected loops, often with a non-identical loop structure. Consequently,

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we used visual categorisation to identify whistle types, which is a reliable method for signature whistle identification in *Tursiops* spp. (Janik, 1999). On scanning the data, a whistle catalogue was created using clear examples of stereotyped whistle contours, and this was used to classify whistles during analysis. All high-quality whistles (i.e. SNR 2/3, not masked) were either assigned as 1) a whistle type in the catalogue and given the associated unique ID, 2) a variable whistle i.e, unique frequency modulation pattern which does not match the catalogue) or 3) a copy (overlaps in time with another contour with very similar frequency modulation pattern, Kriesell et al., 2014). Poor quality whistles (rated SNR 1 and/or masked) were left 'unclassified'. Throughout the analysis, a high proportion of the poor-quality whistles could be assigned to a whistle type with reasonable confidence, particularly when considering the context of recording and if they were produced within a whistle bout. Where possible, these were assigned to the whistle type within an exploratory 'best guess' sub-category which is reported on but not included within any of the formal analysis.

The grouping of whistles into discrete types was reviewed multiple times by two authors (TG/SD) until consensus on categorisation decisions was made. In instances of discrepancy, a conservative approach was taken and whistles were grouped into one type, rather than split. Once the catalogue was finalised, we assessed the reliability of the whistle grouping. Five naive observers were presented with the catalogue of whistle types composed of a single, clear example of each whistle type in the dataset and asked to match 5 examples of each whistle to the catalogue. Where categories exceeded 6 whistle repeats (i.e. the catalogue whistle and 5 examples to classify), a subsample of five whistles were selected at random from the available pool of clear whistles in that type. Whistles used in the classification trial were saved as electronic spectrogram images (.png) with standardised time/ frequency axis before being presented to the observers in a randomised order for the classification trial. Inter-rater reliability in the classification of the

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images was statistically assessed using Fleiss' kappa statistics to assess the degree of agreement over chance.

For types containing 4 or more whistles, the inter-whistle interval between whistles of the same type was then calculated – from the end of the whistle to the start of the following whistle of the same type. The temporal patterning of whistle production was explored using log survivorship plots of inter-whistle-intervals (IWI) from 0.25 to 60 sec. This threshold was chosen, as intervals longer than 60 sec might reflect movement of dolphins out of the detection radius or recording interruption, rather than the true temporal pattern in whistle production.

To explore these features of the humpback dolphin acoustic behaviour, all good quality (SNR 2/3) whistles of each type were classified as one of four categories (SE, CML, DCML, NID-DCML) and the number of loops counted. Loop duration and the inter-loop interval (from the end of the loop contour to start of next one) were calculated for all disconnected multi-looped whistles. We explored the characteristics and variability in whistle production, including the duration and structure of the introductory loop (loop 1) compared to subsequent loops and the distribution of inter-loop intervals.

We applied the SIGID criteria to identify likely signature whistles (signature whistle types, SWTs) using the sequence application of the SIGID bout criteria (Janik et al., 2013). To be considered a likely signature whistle, a minimum of 3 out of 4 whistles of the same type (the same frequency modulation pattern) should occur in a bout with IWIs of 1 to 10 sec (Janik et al., 2013). Whistle copies are rare (Janik, 2000) and can be identified as a) variable examples of the catalogued signature whistle (King et al., 2013), b) similar whistle contours overlapping in time (Janik 2000, Kriesell et al. 2014). Where the production of stereotyped whistles did not meet the SIGID criteria,

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the type was classified as a repeatedly emitted whistle type (REWT) and the occurrence of SWTs and REWTs was considered in the following analysis. Reporting on the characteristics of REWTs is relevant when assessing signature whistles in new species, as the distinction between a SWTs and REWTs may be related to sample size effects (Janik et al., 2013) or species differences in whistle temporal production (i.e. IWI, Fearey et al., 2019b) and understanding these constraints or bias can help inform future PAM effort.

Capture-Recapture using acoustic and photo-graphic data

The encounter data of SWTs and REWTs were converted into a capture matrix of whistle types over time and space. Evidence for acoustic recapture events was considered depending on whether the following criteria were met. There was strong evidence for signature whistle recapture if, on the successive encounter > 4 whistles of the same type were detected and independently matched SIGID production criteria. Should recapture locations differ, they were over likely ranges for individual *S plumbea* movement as previously demonstrated through photo-identification (mean 140 km, max 500 km, Vermeulen et al., 2017). There was moderate evidence of acoustic recapture when one to three whistles of a type were recaptured, and recapture locations were over likely ranges for individual *S plumbea* movement. However, if one to three whistles were recaptured and recapture locations were beyond *S plumbea* home ranges, this was considered weak evidence for recapture.

Associated photo-ID data of *S. plumbea*, based on natural marks on dorsal fins and dorsal humps, were processed using standard approaches (Elwen et al. 2019) to confirm the identity of dolphins present during each encounter. The number of unique individuals present from photo-ID data was taken as the minimum group size per encounter. However, animals can be present during an encounter but not adequately photographed (particularly if the group is located within the surf zone), therefore if best estimates of group size made in the field were greater that photographic

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counts of individuals present, the field estimate was used. The count and identity of individuals present was compared across the acoustic and photographic capture matrices. Assuming no two dolphins have exactly the same signature whistle (Janik & Sayigh, 2013), we compared the number of individuals captured photographically and acoustically. Given that individual dolphins can be present but their signature whistle not captured in recordings if a) it did not produce its signature whistle, b) the recording of was not of sufficient quality to facilitate matching to the catalogue, c) the dolphin whistled but was out of (clear) detection range, and that whistle copying is rare (Janik, 2000), we expected the number of whistle types to be less than or equal to the best group size estimate per encounter.

ETHICS STATEMENT

Data were collected under University of Stellenbosch and University of Cape Town ethics permits (reference: 2020/V11/TG) awarded to TG and University of Pretoria ethics permit awarded to SE/SA. Research was conducted under a series of research permits issued by the Department of Environment Forestry and Fisheries (and its predecessors) to SE/EV/SA

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RESULTS

Recording effort was spread over 8 years (2016 – 2023) with 21 recording encounters over 27 days. When encountered, *S plumbea* were in small group sizes ranging from 1 to 12 individuals and close to shore, estuaries, and harbours. Groups were recorded across a range of group sizes and compositions, but whistles were not detected in all recording sessions. Overall, we detected 1551 whistle contours from 17 encounters spanning 20 recording hrs with *S. plumbea* in coastal South Africa. The whistles of *S. plumbea* were highly stereotyped. Of the 591 good quality whistles recorded (clearly detected, SNR 2/3, not masked), 410 (69%) were grouped into one of 23 whistle types, containing between 2 and 100 whistle emissions per type. A further 276 whistles of low recording quality (SNR 1 or masked) could be allocated to these types with moderate confidence (i.e. considered a 'best guess') but to ensure standards of data quality, were not included within subsequent analysis. After removing types with fewer than four whistles, 21 whistle types remained which were identified over 13 encounters with *S. plumbea* (Table 1). Visual categorisation of signature whistles was well supported, demonstrating highly stereotyped groupings.

The distribution of IWIs and log survivorship plot (Figure 2) demonstrate a clear pattern in the temporal production of whistles in the 21 types, with a peak at 1 to 3 sec, and 65% of whistles in a bout occurring within 0.25 to 10 sec of another whistle of the same type (Figure 2). By applying the SIGID criteria to this temporal production data, two thirds of the whistle types identified (n = 14) were confirmed as likely signature whistles (SWTs). Notably, the size of each whistle type category, i.e., the number of whistles grouped together, strongly influenced whether a type met or exceeded the SIGID bout criteria threshold. All whistle types where ten or more whistle repeats were grouped together were subsequently identified by SIGID as signature whistles. However, only one type (SWT13), with 15 whistle emissions failed to meet the SIGID production criteria.

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This suggests an influence of sample size on whether whistle types will subsequently meet the bout criteria thresholds which enable them to be assigned as signature whistles with confidence.

Most whistle types were produced in both single element and disconnected multi-looped forms (12 of the 14 SWT, Figure 3). Recording SWT with up to 4 loops was common (at least once in 9 of 14 SWT) with a maximum of 13 loops recorded (SWT5). When produced with loops, the single element version matched the frequency modulation of the introductory loop of the DCML version. Of the loop sequence, the introductory loop was usually the longest in duration and most complex in terms of frequency modulation pattern (Figure 7). From loop two onwards, these repeated loops were short contours and simplistic in form (e.g. Figure 2, SWT5, 6 and 11). The two SWT which were never recorded in a DCML form were recorded relatively infrequently, i.e. SWT1 (n = 12) as a connected multi-loop and SWT61 (n = 10) as a single element. With greater recording effort of these whistle types, a higher degree of variability in the whistle form may become apparent.

The recapture of dolphins through whistle monitoring is indicated in Table 2. Over the eight years with recording effort and 14 SWT, most were recaptured at least once - i.e. recorded on separate occasions. From these, five were recaptured over multiple years and five over multiple locations. In rare cases, SWTs were recorded over multiple years and locations. For example, SWT11 was recorded three times in False Bay in 2016, in St Sebastian Bay in 2019 and then in Mossel Bay in both 2019 and 2023 (Figure 8).

There is a weakly positive linear relationship between the number of individual animals identified through photographs during encounters and the number of whistle types (grey, SWT + REWT) and signature whistle types (red, SWT). The relationship is stronger (R^2 = 0.11 compared

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to $R^2 = 0.06$) when all whistle types are considered compared to when only signature whistle types are considered.

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DISCUSSION

Species extinction is rapid within the Anthropocene Epoch with an increasing number of species recognised as threatened, vulnerable or endangered. Conservation actions for *Sousa* species have been limited and until recently they received almost no management attention (Jefferson & Curry 2015). However, the IUCN, International Whaling Commission (IWC) and Convention on Migratory Species (CMS) are now united in calling for urgent global action to conserve *Sousa* species and prevent them following other cetaceans species, such as the Baji and *vaquita* on the path to irreversible extinction. Generating empirical data on *Sousa plumbea* is difficult. However, here we demonstrate that a novel acoustic approach can be used to determine animal presence and ranging patterns, with individuals re-captured over time (up to 8 years) and space (up to 480 km). Passive acoustic monitoring is becoming a common tool in conservation biology used to document the occurrence and distribution of severely threatened species such as the critically endangered Vaquita (Rankin et al., 2009), Maui's Dolphin (Rayment et al., 2011) and the recently extinct Baji (Wang et al., 2006) as well as *Sousa* species (Caruso et al., 2020). However, this is the first attempt to use individually distinctive acoustic signals in the conservation of an endangered marine mammal.

Acknowledging individual behavioural differences can help avoid sampling and experimental bias and increase explanatory power. Where actions and responses differ at the individual scale, such variation provides the raw material for natural selection and therefore understanding personality differences may help predict the effects of environmental change on wild populations (Roche et al., 2016). Only recently are we starting to understand a) the extent of individual variation in behaviour within and between species b) the methods by which this variation can be measured, c) the impact this variation has on individual survival, reproductive success and wellbeing, and d) the impact this variation has on the evolution of populations. Our results indicate that certain individuals within the South African Sousa population are ranging along the

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coastline, over several hundred km. This was evidenced through both the photo-graphic data and the acoustic data collection. In fact, the recapture through acoustic data, indicated a range greater than for the Photo-ID, whereby there was confident recapture of SWT11 between False Bay and Mossel Bay. Assuming this is reflective of the individual present these acoustic data are providing important insight into both the occurrence and ranges of individual humpback dolphins. These data can therefore be used within a risk analysis approach, to determine the exposure of individual to likely threats along the south African Coastline.

In addition to encoding identity information, the general form of whistles within a repertoire (i.e. single element, connected multi-loop, disconnected multi-loop or non-identical disconnected multi-loop) may encode population or species information. Additionally, variability in loop duration and / or of the silent gap between loops (the inter-loop interval) in disconnected whistles could encode additional identity or arousal cues (Esch et al., 2009a, Briefer 2012; Probert et al. 2021).

Although the relationship between whistle types and animals present was far from 1:1, it was linear and positive. As signature whistles function as cohesion calls (Janik & Slater, 1998), their production is strongly context dependent (Cook et al. 2004; Quick and Janik 2012; Probert et al. 2021). Consequently, with the limited focal group follow recording effort presented, it is unsurprising that the number of photographically identified animals usually exceeded that identified through acoustic methods. Longer recording effort, has in other species, show string alignment to photographically determined estimates of population size (Longden et al. 2020). That the relationship was stronger when all whistle types are considered compared to when only signature whistle types are considered s provides some support to the proposal that REWTs are putative signature whistles but undefined as such due to sample size restrictions. In rare cases, the number of photo-identified identified individuals exceeded the number identified

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acoustically. However, on assessment of the field best estimates, this could be explained by the incomplete photo-identification coverage, due to the inshore distribution of the group and the need to be stationed safely further from shore to collect acoustic data. Furthermore, on the date when the largest discrepancy occurred, the field estimate of group size exceeded the number of animals identified, compared to only 6 animals identified. On this day, it is apparent that the photo-identification data was not a complete representation of the group present.

In recognition of the challenges facing *S. plumbea*, a nationwide, 'SouSA Consortium' was established in 2016, involving 18 individuals from 13 institutions (Plön et al. 2021b) with the goal of collaborating to share knowledge and improve conservation actions. The consortium has already produced several peer reviewed articles (Vermeulen et al. 2017; Plön et al. 2021b;Plön et al. in press), IWC reports and successful funding applications. This study uses data from consortium members and collaborations generated across the geographic range of the species in South Africa. By combining local knowledge, resources and data the SouSA consortium the consortium is finding innovative ways to generate important information which will facilitate species conservation management. The results of this initial investigation demonstrate the applicability of passive acoustic monitoring for this endangered species upon which future studies can be based.

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TABLES

Table 1: Summary of recording effort, recording equipment and sampling occasions for *Sousa plumbea* data collection

Location	Dates	Days, Years	Recording Equipment	Sample Rate (kHz)
False Bay	2016: 17/05, 27/11, 6/23 2017: 23/02, 30/03, 18/04, 20/04	7, 2	SoundTrap & HTI-96 MIN	96, 576
Struisbaai	2017: 01/03, 02/03, 04/03, 07/03	4,1	HTI-96 MIN	96
St. Sebastian Bay	2019: 17/09	1	SoundTrap	576
Mossel Bay	2019: 30/09, 16/10, 2020: 06/03 2021: 17/03, 18/3,24/03,28/03 2023: 04/07, 07/07,14/7, 18/07, 21/07 ,26,07	4,4	HTI-96 MIN	96
Richards Bay	2020: 10/03,19/03	1, 3	SoundTrap	576

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Table 2: Summary of recording encounters and re-capture information of whistle types over time and space. Signature whistle types are highlighted in **bold**.

	Year	2016	2016	2016	2017	2017	2019	2019	2021	2023	2023	2023	2020	2020	Recaptures			
Type ID	Region	FB	FB	FB	StrB	StrB	StSeb	MB	MB	MB	MB	MB	RB	RB	Whistles	Total	Years	Region
swt 1		**7**	**2**				**3**								12	3	2	2
rewt 2		2	1		1			1							5	4	3	3
swt 4							**23**								23	1	1	1
swt 5				**6**			**1**	**26**					**2**		35	4	2+1	3+1
rewt 6							7						1		8	2	1+1	1+1
rewt 7				5				2			1				8	3	3	3
swt 8					**39**			**1**							40	2	2	2
rewt 9						4	1								5	2	2	2
swt 10		**1**					**2**	**94**		**2**		**1**			100	5	3	3
swt 11		**5**	**4**	**2**			**1**	**4**		**7**					23	6	3	3
rewt 13								15							15	1	1	1
swt 16														**14**	14	0	1	1
swt 30														**12**	12	0	1	1
swt 32													**17**		17	0	1	1
swt 61											**10**				10	1	1	1
swt 64								**2**			**8**				10	2	2	1
swt 66								**1**	**7**	**1**	**3**	**16**			28	5	3	1
swt 67							**10**								10	1	1	1
rewt 68								9							9	1	1	1
rewt 69							7								7	1	1	1
swt 70								**14**							14	1	1	1
Total Whistle Types		4	3	3	2	1	9	11	1	3	4	2	3	2				
Signature Whistle Types		3	2	2	1	0	6	7	1	3	3	2	2	2				
Repeatedly Emitted Whistle Types		1	1	1	1	1	3	4	0	0	1	0	1	0				

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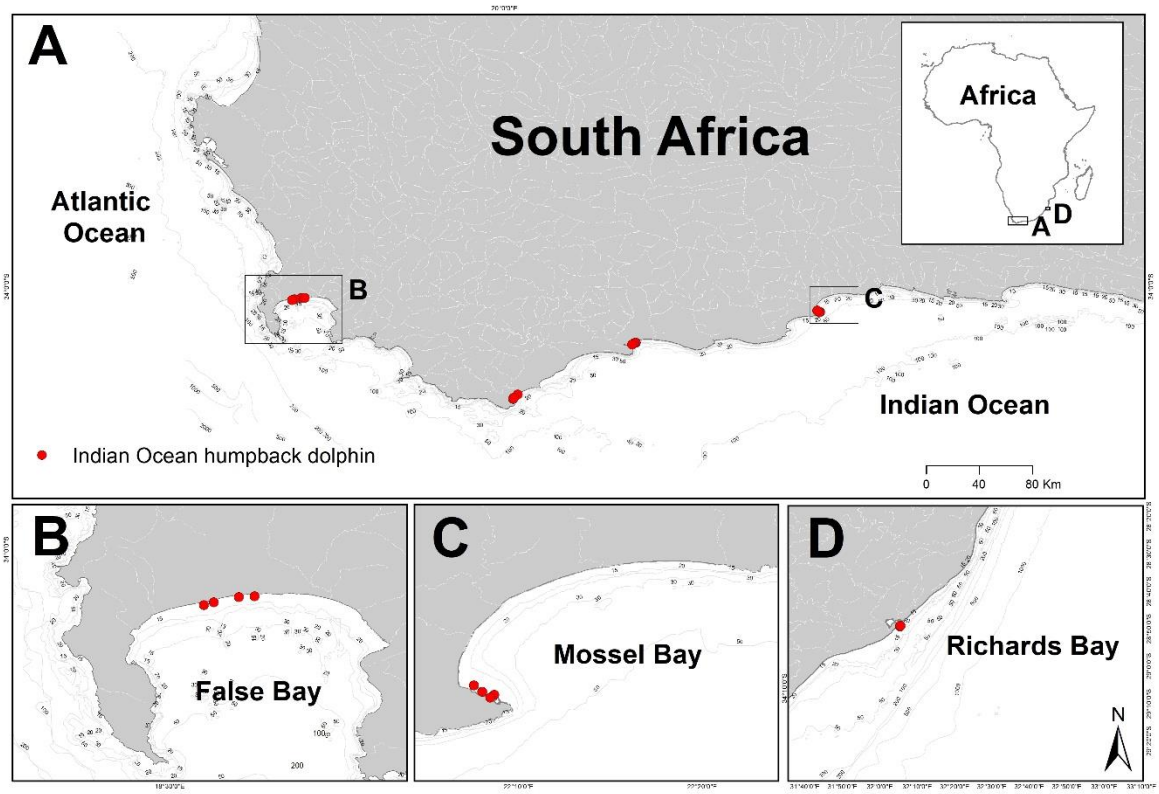


Figure 1: Map of study area showing the recording sites of *S.plumbea* along the South Africa coast-line

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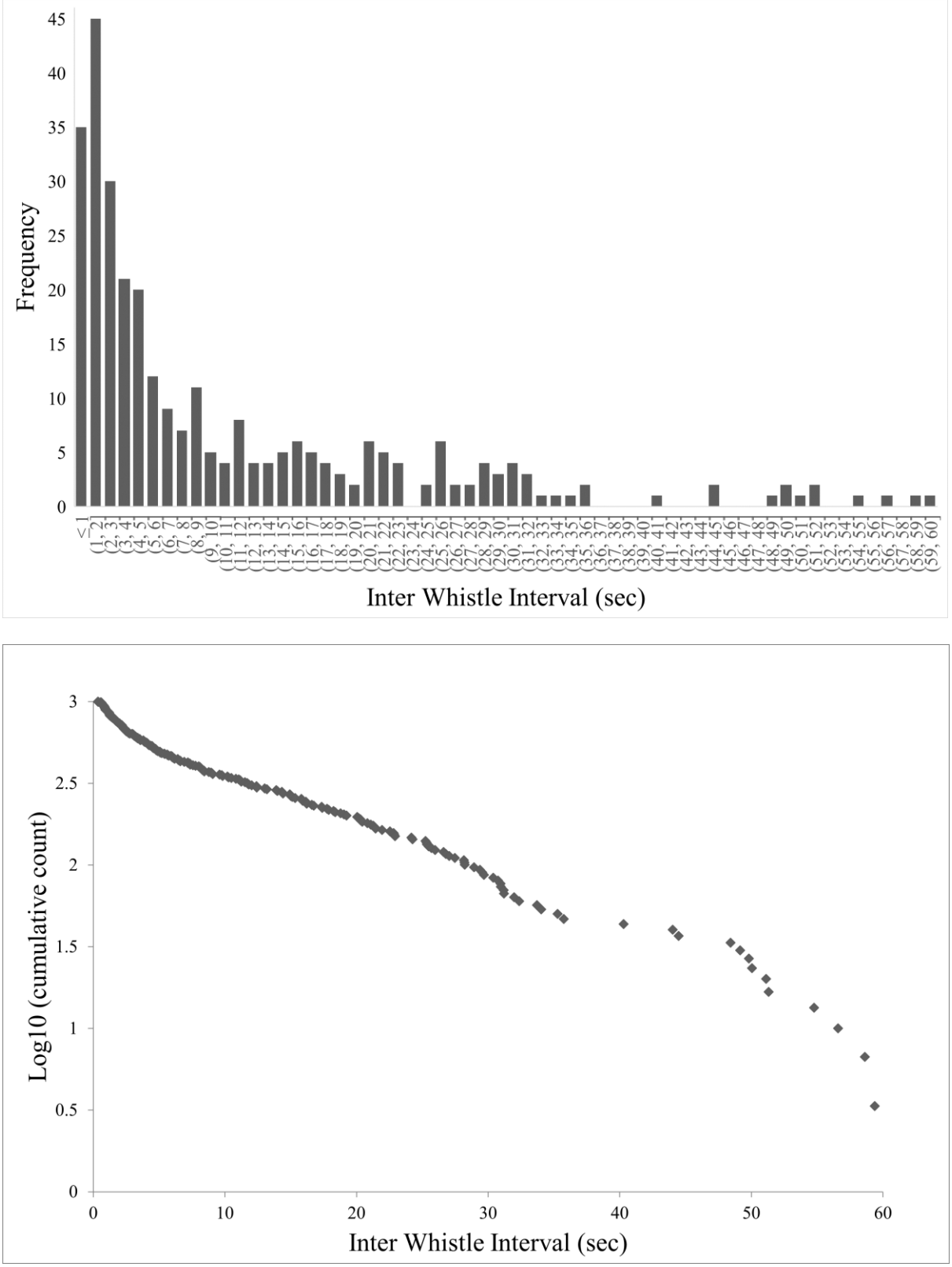


Figure 2: Distribution (a) and log survivorship (b) of inter-whistle-intervals between 0.25 and 60 sec for whistle types identified in *Sousa plumbea.* A peak in IWI is apparent from one up to 10 sec, aligning with temporal production of signature whistles in bottlenose dolphins.

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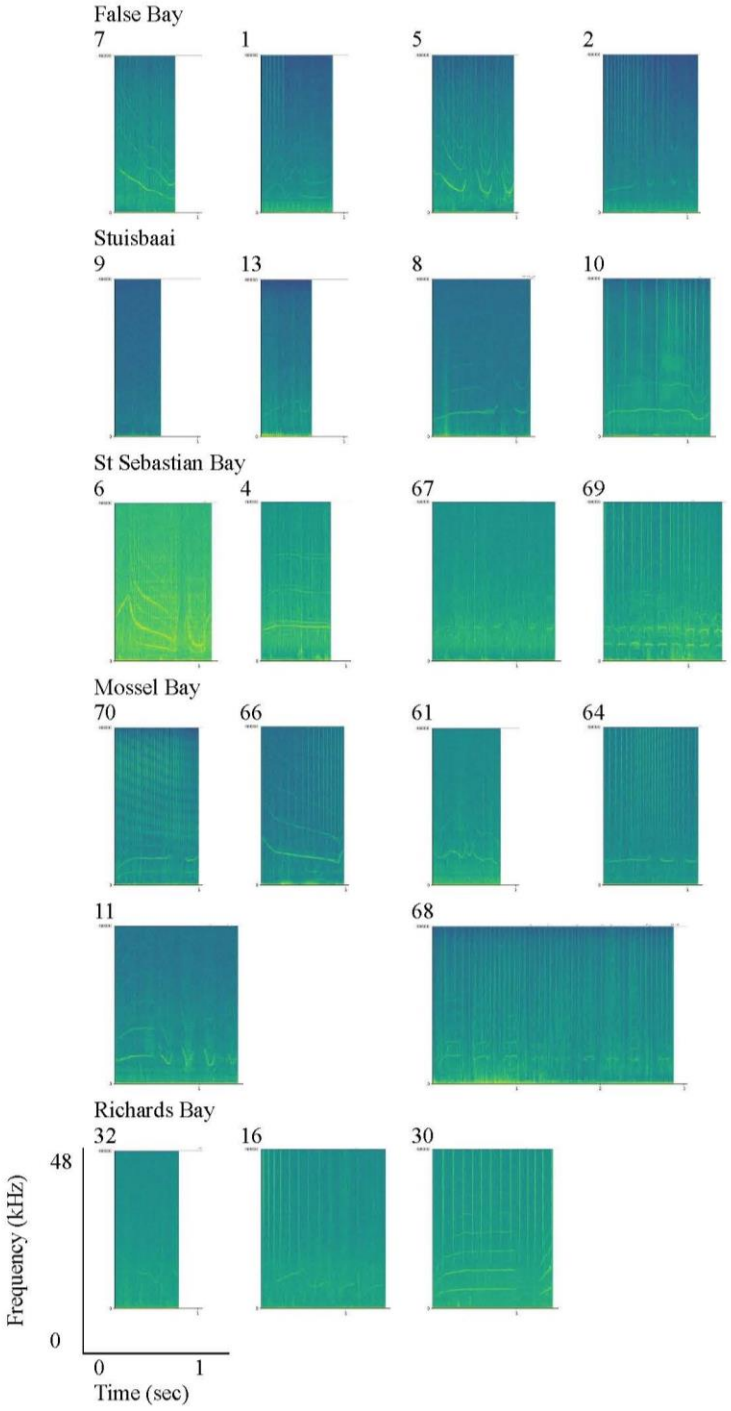


Figure 3: Representative spectrograms of whistle types (SWT and REWT) identified in focal follow acoustic data from *S. plumbea* in South Africa. Types are shown with a unique, numeric ID and displayed per region they were identified. However, given recaptures occurred between regions, this is not exclusive of where the whistle was recorded. (Spectrogram Settings: FFT 1024, Hop size 512, Hann window).

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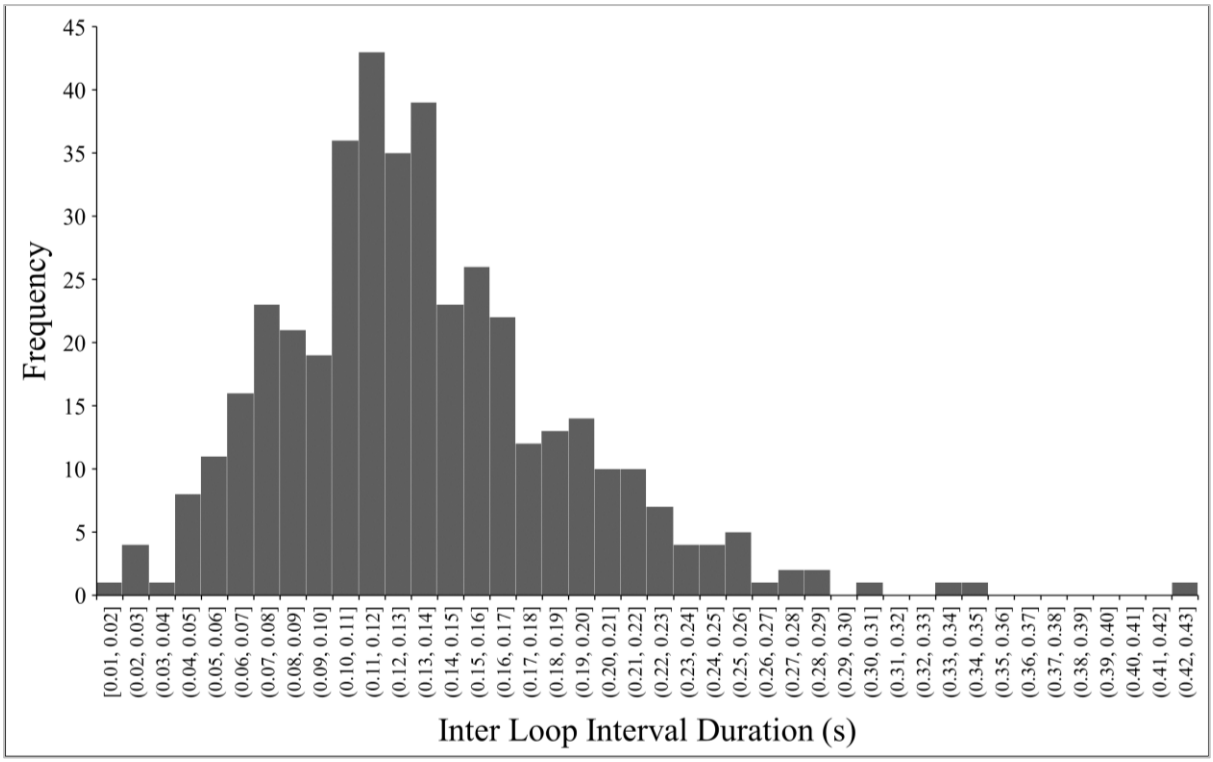


Figure 4: Distribution of the inter-loop intervals found in *S. plumbea* signature whistles demonstrating a distinct peak at 0.1 to 0.14 sec (i.e. 100 to 140 ms). This aligns well with the temporal production of looped whistles bottlenose dolphins.

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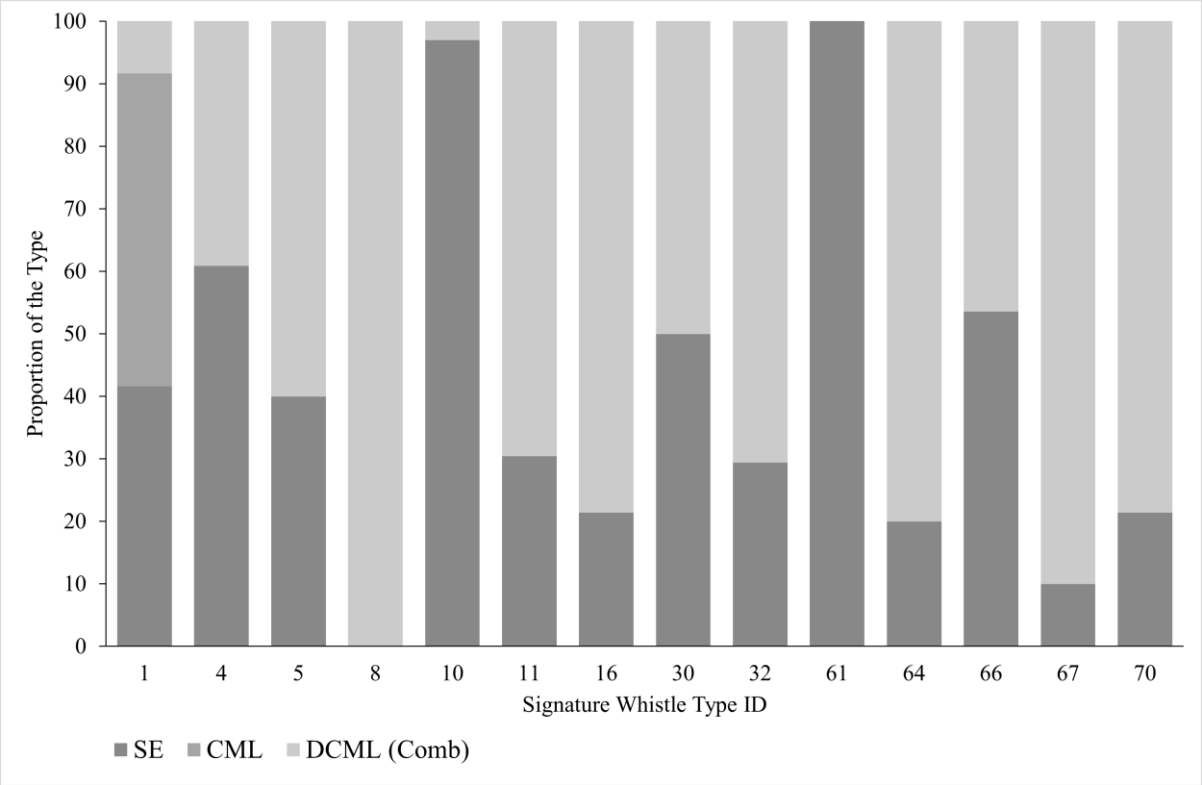


Figure 5: Forms of *S. plumbea* signature whistle displayed as the proportion of each SWT recorded in single element (SE), connected multi-loop (CML) or disconnected multi-looped forms (DCML). Most whistle types were produced in both single element and disconnected multi-looped forms.

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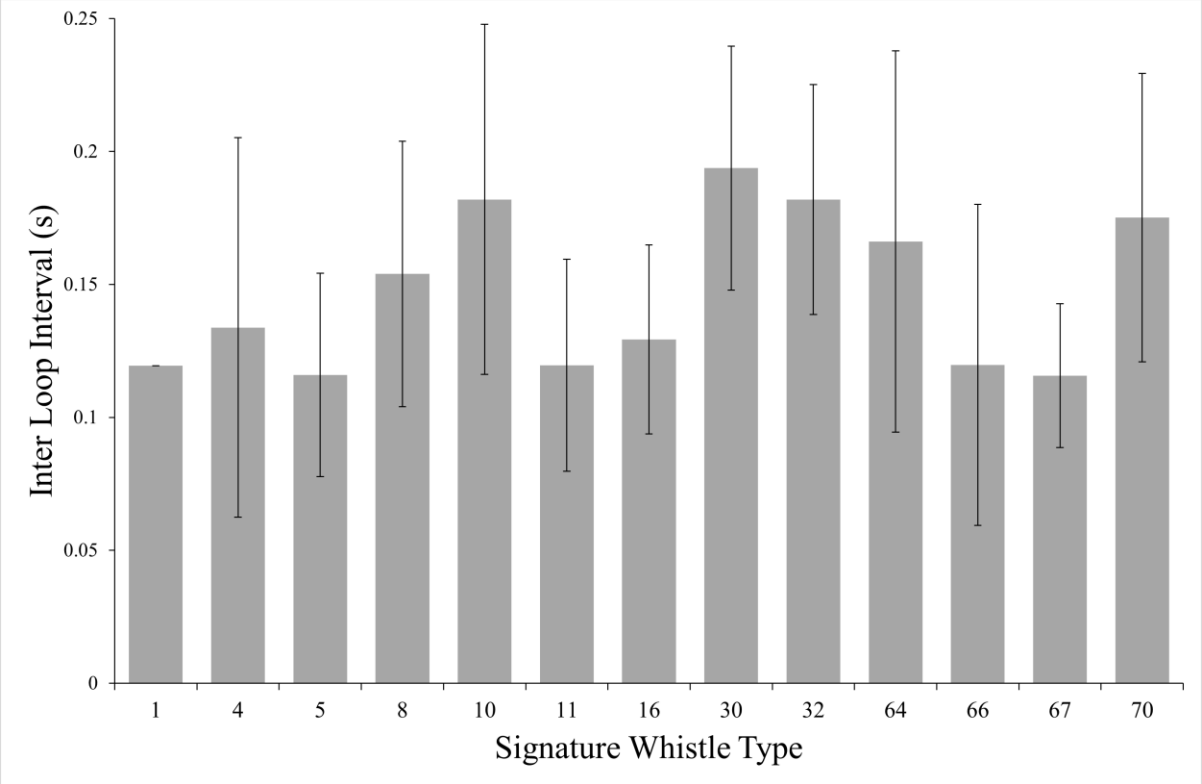


Figure 6: Mean (+/- SD) of the inter-loop intervals for different SWT identified in *S. plumbea,* demonstrating consistent, individual differences in the periods of silence incorporated into more complex SWT.

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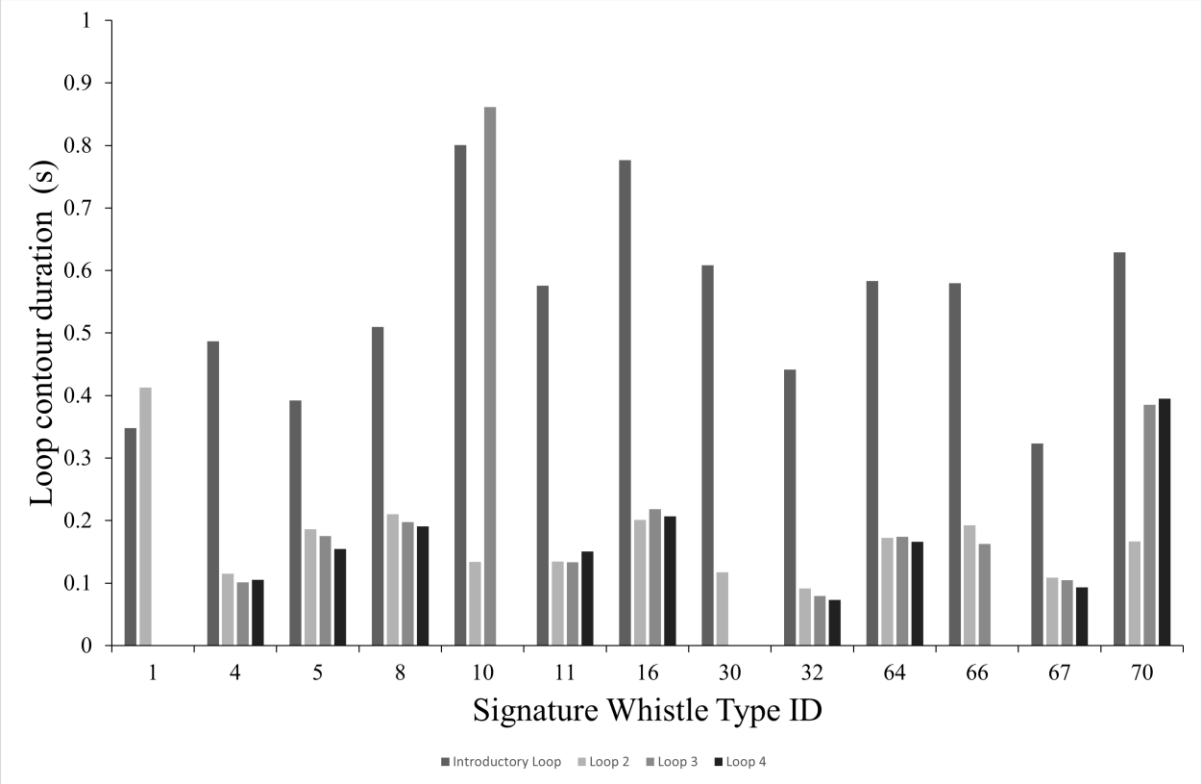


Figure 7: Comparison of duration of introductory loops to subsequent loops (2-4) demonstrating that in most case the introductory loop is the longer and subsequent loops are shorter and more simplistic (refer to Figure 1).

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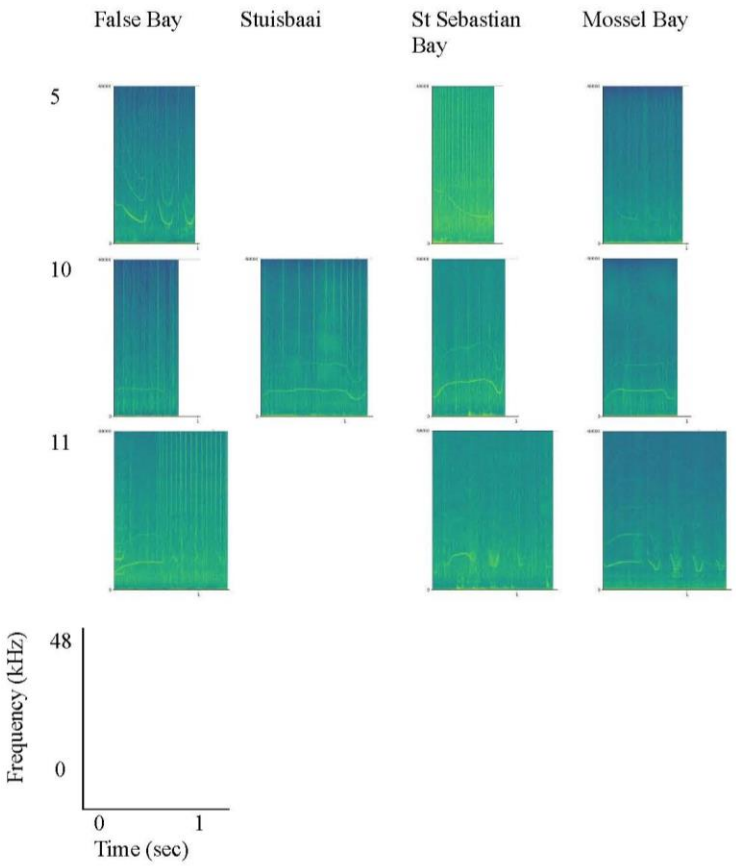


Figure 8: Representative spectrograms of signature whistle type recaptures between the four south coastal recording sites of South Africa. (Spectrogram Settings: FFT 1024, Hop size 512, Hann window).

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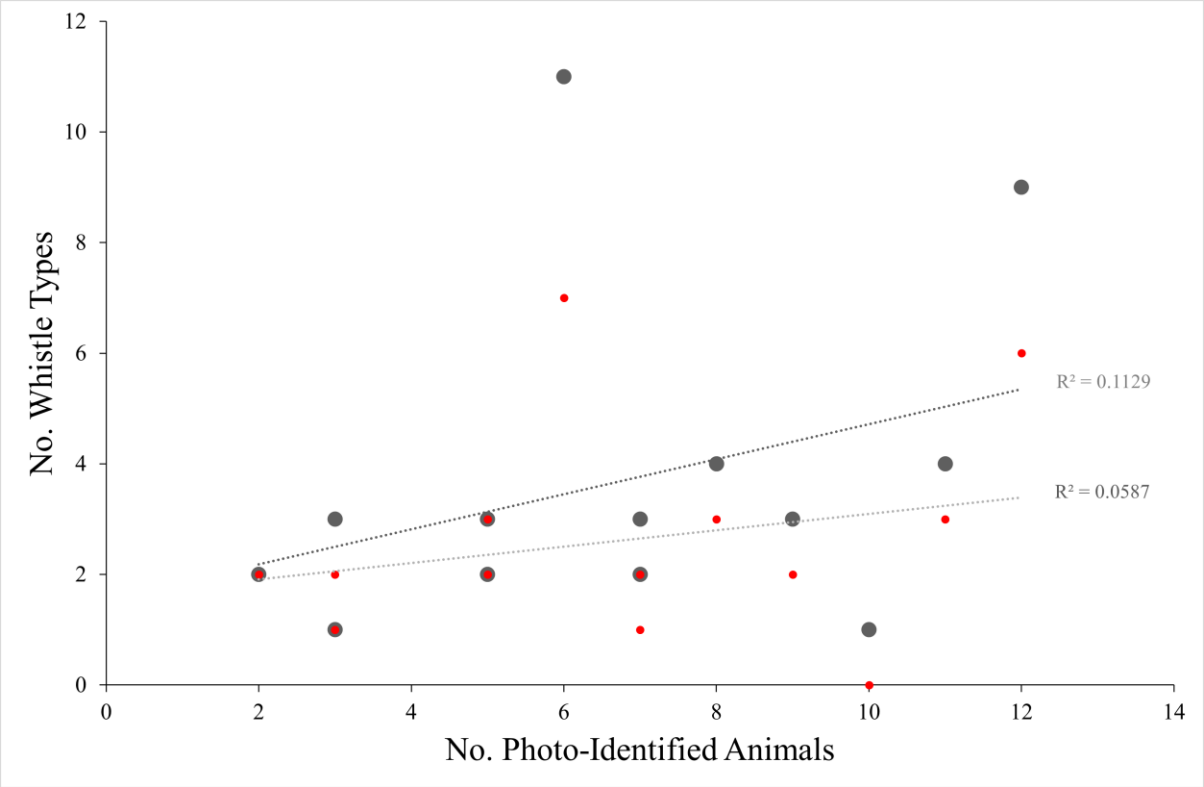


Figure 9: Comparison between the number of individual animals identified through photographs during encounters to the number of whistle types (grey, SWT + REWT) and signature whistle types (red, SWT). There is a weakly positive linear relationship, which is stronger when all whistle types are considered, compared to when only signature whistle types are considered.

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APPENDIX 2

Case study: Evidence of long-term stability in a stereotyped whistle in a single free-ranging humpback dolphin (*Sousa plumbea*) found in sympatry (*Tursiops aduncus*)

Submitted JASA Express Letters

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This 11-year case study describes the acoustic behaviour of a resident Indian Ocean humpback dolphin, *(Sousa plumbea*), during commercial swim-with-dolphin activities in Mozambique. Combining data collected using low-cost action cameras with full bandwidth hydrophone recordings, a temporally stable stereotyped whistle contour was identified that met the SIGID bout criteria used to identify signature whistles in other dolphin species. This whistle was produced with potential information-enhancing features (biphonation and subtle variations in frequency modulation). Despite evidence in other *Sousa* species, signature whistle-based communication has not been described in *S. plumbea*. This case study provides evidence for a single, stable, stereotyped call type. from a single individual in a mixed species group, contributing to the growing body of evidence for possible signature whistle use in the *Sousa* genus.

Keywords: Acoustic, Sound, Stereotyped whistle, *Sousa*, biphonation

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Introduction

The acoustic repertoire of *Sousa* spp. comprises acoustic signals common in delphinids, typically grouped into echolocation clicks, burst pulses and whistles (Herzing, 2014; Van Parijs et al., 2000; Yang et al., 2020). Whistles are frequency modulated tonal calls strongly associated with social contexts (Tyack, 1997), and humpback dolphins seem to present anatomical adaptations for high frequency communication sounds (Frainer et al., 2019). Whistle frequency parameters vary across the genus with Atlantic humpback dolphins (*S. teuszii*) producing whistles with the lowest fundamental frequency across the genus, with a mean maximum of 8.2 kHz (Weir, 2010). Australian humpback dolphins (*S. sahulensis*) produce whistles with a higher mean maximum of 10.58 kHz and a maximum recorded fundamental frequency of 22 kHz (although this maximum frequency may have been limited by recording equipment, Van Parijs & Cockeron., 2001). The whistle repertoire of Indian Ocean humpback dolphins (*S. plumbea*) has only been reported from India where mean minimum and maximum frequencies ranged from 7.6 – 10.2 kHz, respectively, and an absolute maximum frequency of 33 kHz was reported (Bopardikar et al., 2018).

Humpback dolphins (*Sousa* spp) are listed as endangered on the IUCN Red List (Braulik et al., 2017), and effective methods to monitor populations at an individual level are needed. Passive acoustic monitoring (PAM) is a well-established research tool for the study of wild cetacean populations through the identification of species-specific calls in long-term acoustic recordings. More recently, the use of PAM for studying delphinids has been extended to individual capture-recapture using 'signature whistles' (Erbs et al., 2017; Fearey et al., 2022; Longden et al., 2020; Romeu et al., 2023). These are individually distinctive calls used to communicate identity information and maintain group cohesion and are well demonstrated in bottlenose dolphins (*Tursiops truncatus*) (Janik & Sayigh, 2013). Since the 1960s, evidence has been presented that supports likely signature whistle use across other odontocete species (*Tursiops aduncus* Gridley et al., 2014 *Stenella longirostris* Rio, 2023, *Stenella plagiodon* Perin 1973, *Delphinus delphis* Fearey

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et al., 2019, *Sousa chinensis* Van Parijs & Corkeron, 2001, *Pseudorca crassidens* Rio & Rosales-Nanduca, 2023, *Legenorhynchus obliquidens* Caldwell and Caldwell 1970, *Sotalia guianensis* Lima & Le Pendu, 2014, and *Monodon monoceros* Shapiro, 2006). Evidence for signature whistle use has been presented in *Sousa* spp. under heightened instances of stress, including an injured Indo-pacific humpback dolphin (*S. chinensis*) (Cheng et al., 2017) and from a stranded Australian humpback dolphin (*S. sahulensis*, formally described as *S. chinensis*) (Van Parijs & Corkeron, 2001). Evidence for signature whistle use has not yet been demonstrated in free swimming Indian Ocean humpback dolphins (*S. plumbea*).

To reliably communicate identity information, the acoustic signal should have high inter-individual (i.e., distinctive) and low intra-individual (i.e., stereotyped) variability (Janik et al., 2006; Sayigh et al., 1990, Charrier et al 2003). This can be achieved through inter-individual differences in the vocal apparatus, resulting in by-product distinctiveness (so called voice features) (Taylor & Reby, 2010). Provided that these differences remain stable over time and the receiver can perceive these differences, identity information can accurately be encoded and decoded by the signaler and receiver, respectively. Alternatively, and more cognitively complex, is use of 'designed individual signatures' which can develop through a process of production learning. For bottlenose dolphins, there is strong evidence that signature whistles are developed through a process of vocal production learning (Janik et al., 2006), where the acoustic environment experienced by the dolphin can influence the overall frequency modulation pattern of the 'crystalised' signature whistle (Fripp et al., 2005, Janik & Sayigh, 2013). Signature whistles are established in calves as early as the first two months of life, with high production rates in mother-calf pairs who use signature whistles to facilitate reunions after short separation periods (Smolker et al 1993; Janik, 2009; King et al., 2016). As cohesion calls, signature whistles are produced at elevated rates during separation or isolation events and high arousal states including times of stress (Caldwell et al., 1990; Esch et al., 2009a; Probert et al., 2021, 2023a) In these

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contexts, emissions may become less stereotyped and more variable in nature, occasionally with the production of non-linear phenomena (Sportelli et al., 2023).

Like other species within the genus, the habitat preference of *S. plumbea* includes estuaries and murky rivers, and they are rarely held in captivity, therefore underwater observations of humpback dolphins are rare and the context of sound production often unknown. It is generally accepted that the acoustic signals produced by dolphins originate from the monkey-lips dorsal bursa complex (Cranford et al, 1996), and that bi-phonation is possible whereby dolphins can produce whistles and pulsed calls (clicks and burst pulses) simultaneously (Madsen et al., 2013; Ridgway et al., 2015; Kriesell et al., 2014). Unlike other wild dolphin populations, information on individual acoustic behaviour, including signature whistle production, and biphonation, is restricted (Herzing, 1996; Papale et al., 2015). However, Ponta do Ouro, Mozambique, is one of the few places where long term in-water interactions with this species have been recorded. Using historical data, the acoustic behaviour of *S. plumbea* was investigated using a case study of a resident individual in Ponta do Ouro, Mozambique, named Herme, using in-water video and acoustic recordings.

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Methods

Study site and species

Ponta do Ouro Partial Marine Reserve (26.84°S, 32.88°E) is Mozambique's southernmost conservation area. Boat-based research and dolphin tourism involving swim-with-dolphin experiences have been taking place in Ponta do Ouro since 1994, mainly targeting inshore Indian Ocean bottlenose dolphins (*Tursiops aduncus*). Over 250 bottlenose dolphins have been catalogued within this population, of which 60 are considered 'resident', and live sympatrically with *S. plumbea* (AG pers. comm). Comparatively, only a small number of *S. plumbea* (n = 6) have been identified and catalogued during this time (AG pers. comm). One humpback dolphin, locally named Herme, was born in 2006 is the most commonly sighted humpback dolphin in the study area and is exclusively encountered with *T. aduncus* groups.

Data Collection

Acoustic and video data were opportunistically collected between 2009 and 2019 during commercial swim-with-dolphin activities in the Ponta do Ouro bay. During 2020, additional acoustic data were collected with a focus on *S. plumbea*. Acoustic data were collected by permitted tour operators and/or under research permits from an 8 m dive boat with two 90 hp four stroke engines. Encounters were limited to 30 minutes and a maximum of two in-water encounters were permitted per launch (up to three launches per day).

Once dolphins were sighted and before they were approached, the group were observed for 5 minutes and swimmers did not enter the water if the dolphins showed restless behaviour, if they had newborns in the group, or if they were travelling. If conditions were suitable, dolphins were close (< 20 meters) and it was considered safe to do so, swimmers entered the water to the side or ahead of the group according to strict regulations to minimise disturbance. During underwater encounters, audio and visual data were collected in close proximity to both *T. aduncus* and *S. plumbea* using the equipment summarised in Table 1.

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Table 1. Equipment and recording summary for video and acoustic data collected on the individual humpback dolphin from 2009-2020. Data collection included authors Angie Gullen (AG), Sasha Dines (SD) and Rachel Probert (RP).

Dates	**Recorder**	**Recording Equipment**	**Sample Rate (kHz)**	**Frequency Response**	**Total Recording Duration**	**Days (encounters)**
2009-2019	AG	Sony HD FDR F×7video camera in an Ikalite housing or GoPro Hero4 video camera	48 kHz	15 kHz roll off	01:29:49	30
2020	SD,RP	SoundTrap hydrophone ST300 GoPro Hero4 video camera	576 kHz (16-bit, mono)	288 kHz	00:15:55	1

Data processing and whistle extraction

Videos of encounters with Herme present were confirmed by author AG through visual confirmation of known individual identifying features. The videos were scanned for periods of whistles produced by Herme, whereby the individual was the only humpback dolphin present within the mixed species group, and the dolphin was both audibly and visually (through the production of concurrent blowhole movements and/or bubblestream emissions) calling. The acoustic tracks of the video files from 2009-2019 were extracted as mono .wav files (see Table 1) using the behavioural software BORIS (Behavioural Observation Research Interactive Software, Friard and Gamba, 2016). Acoustic data from the SoundTrap (2020) was matched to the

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concurrent GoPro video using time stamps, and then decimated to 96 kHz to facilitate ease of whistle analysis.

Whistles were manually annotated and visually classified using Raven Pro v 1.6 (Center for Conservation Bioacoustics, 2019). All whistles of a particular frequency modulation pattern or contour (see Janik et al., 2013) were manually grouped into whistle types (SWT) by two observers (TG, SD). The inter-whistle interval (IWI) of stereotyped contours was measured, and a bout analysis was applied by identifying periods where at least 3 out of 4 whistles are produced in a bout with IWI of 1-10 seconds (for the SIGID method see Janik et al., 2013). This method was applied to investigate whether the temporal production of this SWT was similar to that observed in common bottlenose dolphins (*T. truncatus*). Due to data limitations (i.e., the duration of continuous recording), SIGID was only applied to calls recorded during research effort in 2020, comprising 15 mins and 55 secs of continuous recording with dolphins (Table 1). Standard time-frequency parameters of the whistles were measured (start frequency – startF, end frequency – endF, minimum frequency – minF, maximum frequency – maxF, and duration), together with simultaneous burst pulse calls using the selection function in Raven Pro. The number of inflection points were manually counted. For the burst pulse signals, the peak frequency (peakF) was calculated within a constrained bandwidth between 2 to 5 kHz, as this was where most of the energy was focused and prevented erroneous measurement due to the simultaneous high amplitude whistles which occupied the frequency bandwidth above 5 kHz.

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Results

Across the 31 encounters with Herme over the 11-year period, he produced a stereotyped whistle (Figure 2a) on 13 encounters and this whistle was only recorded in encounters with Herme present. Over 110 emissions of this whistle were documented, with a relatively stable frequency modulation pattern over the time span (Figure 2a).

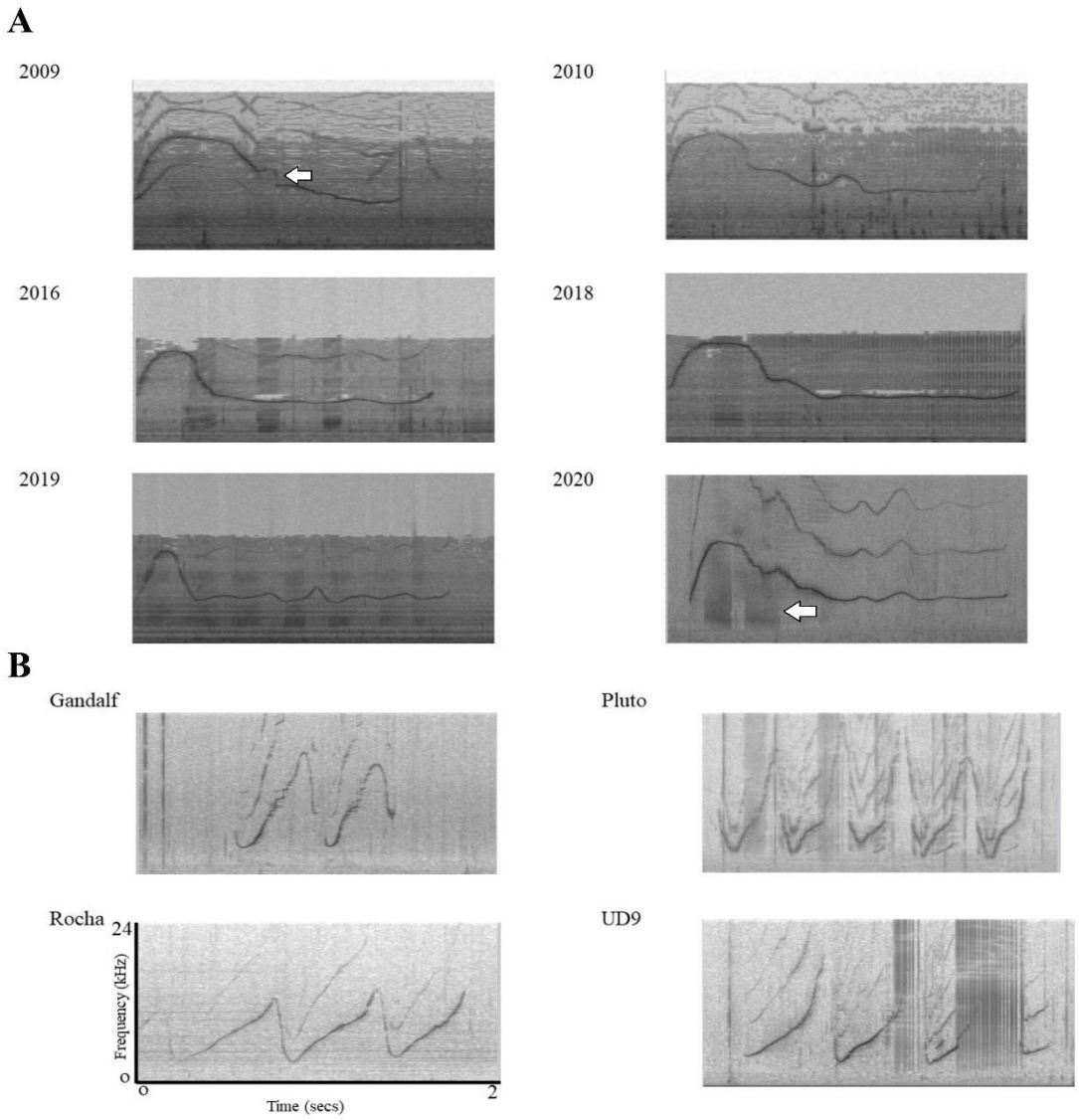


Figure 2.A. Long-term stability in the stereotyped whistle type produced by Herme from 2009 - 2020. Spectrograms shown on axis 0- 24 kHz and 0-2 secs. Spectrograms also show periods of nonlinear phenomenon (biphonation and frequency jumps, see arrows) 2.B. Examples of signature whistle types from sympatric *Tursiops aduncus* in Ponta do Ouro, showing high similarity amongst SWTs (*T. aduncus* SWT repertoire from Probert et al., 2023).

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This stereotyped whistle was commonly produced in bouts and met the criteria of SIGID on seven out of eight bouts within the 2020 encounter (Janik et al., 2013). The distribution of inter-whistle intervals and log survivorship plot demonstrated a clear pattern in the temporal production of whistles. Across the 21 types, with a peak at 1 to 3 seconds, with 65% of whistles in a bout occurring within 0.25-10 seconds of another whistle of the same type (Figure 2).

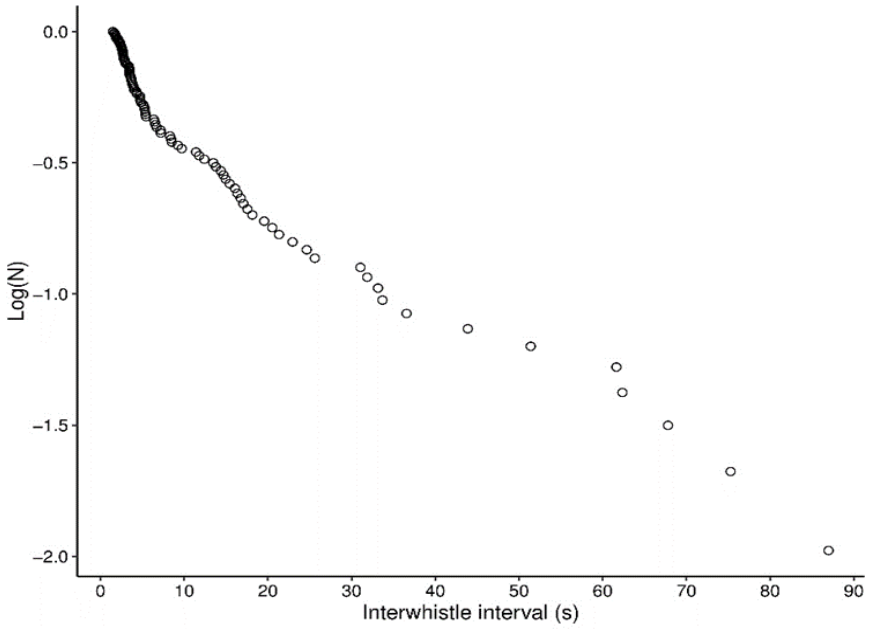


Figure 2. Log survivorship plot of inter-whistle intervals of Herme's SWT within the 2020 dataset.

The SWT produced during the *S. plumbea* encounters was higher in minF (5.94 ± 1.00 kHz), lower in mean maxF (14.64 ± 1.28 kHz), and longer in duration (1.7± 0.7 secs) compared to other whistles recorded from the same encounter, which were most likely emitted by *T. aduncus*. The contour shape of Herme's SWT was highly distinctive from the other whistle types, and from *T. aduncus* in the area (Figure 1b), which in the most part produced SWT which conform to an upsweep multi-loop pattern (see Probert et al., 2023b). Hermes' SWT was more complex, having multiple inflection points (mean 9.2 ± 4.9, compared to *T. aduncus* mean 3.6 ± 3.3) and an

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introductory loop which differed from successive loops, with distinct periods of biphonation and other non-linear phenomena such as frequency jumps (see Figure 1a).

Although this whistle was stereotyped, there were variations in time-frequency components over the 11-year time frame (coefficient of variation, CoV: minF = 17, maxF= 7, dur = 42, Frange = 16, inflection points = 53). The introductory loop was most consistently produced, with greater variability in the occurrence, duration, and number of inflection points of the subsequent tail of the whistle contour (Figure 1a). The whistle was mostly emitted as a continuous contour, with a maximum duration of 3.6 secs. However, a disconnected version of the whistle, including periods of silence, was produced on three occasions.

Non-linear phenomena, such as frequency jumps and biphonation, were often seen in Herme's SWT (63/110, 57%). Frequency jumps were present in 31 whistle emissions. One to seven discrete burst pulse sounds were recorded during 43% (47/110) of Herme's SWT emissions, providing evidence of biphonation. These burst pulses were highly stereotyped in terms of duration (0.2 ± 0.07 secs, CoV = 33) and peakF (3.43 ± 0.43 kHz, CoV =13). On only one occasion was a burst pulse (with the same characteristics) produced immediately preceding the SWT. In all other emissions, the burst pulses were produced at intervals separated by 0.1 – 0.2 secs throughout the whistle duration.

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Discussion

This case study provides insight into the acoustic behaviour of a single *Sousa plumbea* individual encountered exclusively in a mixed species group with *Tursiops aduncus* over an 11-year period.

This case study provides evidence of a single stereotyped call type produced in bouts by Herme, showing individual distinctiveness and temporal stability. In addition, the study presents evidence of inter-specific acoustic communication, non-linear phenomena, and variability in whistling behaviour possibly linked to arousal.

In high arousal contexts, where signalling becomes strained or tremulous, signal production (Esch et al., 2009; Probert et al., 2021) and variability in call structure (Briefer, 2012) can increase. The combination of human activity (boat presence, engine noise, swimmers in the water) and mixed species assemblage will have likely increased arousal of the dolphins during the recording context (Scarpaci et al., 2000). Australian bottlenose dolphins (*T. australis*) are known to change their behavioural pattern after being exposed to 'swim-with-dolphin' tourism occasions (Peters et al., 2013; Filby et al., 2017) and even after the interactions, behaviour patterns did not return to the same levels they were before the interaction (Peters et al., 2013). Mixed species assemblages can elicit behavioural modifications, for example, Guyana dolphins (*Sotalia guianensis*) change the frequency of their whistles when interacting with bottlenose dolphins (*T. truncatus*) at the southern Caribbean coast of Costa Rica (May-Collado, 2010), although these changes are linked to harassment behaviour from bottlenose dolphins. Aggressive behaviour between Herme and *T. aduncus* individuals was not observed in this study, although he was often observed on the fringes of the social group and rarely involved in social behaviour with *T. aduncus* (Figure 3). Humpback dolphins in Algoa Bay, South Africa, and Zanzibar Island, Tanzania, tend to avoid bottlenose dolphins, potentially to minimise costly interactions (Stensland et al., 2003; Koper and Plön, 2016). Thus, the variations observed in Herme's call type

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could be related to behavioural changes due to unknown, potentially complex interspecific interactions or to the nature of data collection (i.e., during swim-with-dolphin activities).



Figure 3. Stills from video footage during 2020 data collection to show the social organization within these mixed species encounters. Herme (*S. plumbea*) circled in a mixed species group with *T. aduncus*. Example of social organization of the mixed species group with *S. plumbea* in the company of *T. aduncus* but not directly engaged in social activity.

Biphonation of whistles and burst pulses were prevalent in Herme's vocal repertoire indicating that the stereotypic use of biphonation might be a purposeful addition of information encoding (Papale et al., 2015; Sportelli et al., 2022). Biphonation can be categorized as the production of two fundamental frequencies, harmonically unrelated to each other, from a single sound source (Kriesell et al., 2015; Papale et al., 2015), or the production of two call types using different vocal structures, which is often a frequency modulated whistle with an overlapping burst pulse (Lilly and Miller, 1961; Caldwell & Caldwell, 1967). In other species, the production of two fundamental frequencies has been suggested to increase the probability of caller identification (Volodina et al., 2006; Filatova et al. 2009; Filatova et al. 2012). These can be used as markers of a pod or a matriline association (Filatove et al., 2009) or vocal cues for mate-mate and/or parent-chick

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recognition (Aubin et al., 2000; Lengagne et al., 2001; Aubin and Jouventin, 2002; Volodina et al., 2006). As burst pulses are more directional than whistles, the addition of these to signature whistles might help to strengthen individual identification through directionality and amplification. Alternatively, this could be a result of simultaneous navigation and communication (multitasking) (Sportelli et al., 2022).

Opportunistically collected acoustic data using low-cost action cameras has previously been validated (Chapuis et al., 2021; Probert et al., 2023b). Although most of the extracted acoustic data had a maximum usable frequency of 15 kHz, the proximity of the animals to the videographer combined with visual cues of whistle production makes this a useful dataset. Although the use of visual cues, particularly bubble stream emissions, to identify call producer has been critiqued (see Fripp, 2005), this method has previously been used to identify call producers in other dolphin species (Miles and Herzing, 2003; Slack, 2018; van der Woude 2020), including long-term signature whistle stability in *T. aduncus* in the same region (Probert et al., 2023b). Furthermore, Probert et al., (2023b) suggests that bubble stream production is an unbiased cue for identifying both signature and non-signature whistles in *T. aduncus*.

In conclusion, the study documents a single stereotyped whistle that dominated recordings during encounters with a single free-swimming humpback dolphin over 11 years, whilst in a mixed-species group. Through interrogation of the acoustic behaviour displayed, this case study could support further evidence that signature whistles are produced by *Sousa* spp. And provides temporal evidence of stereotyped whistle use in a free-swimming *S. plumbea*. However, although this case study contributes to the growing body of evidence for possible signature whistle use, further data is required to fully confirm this, including single species recordings and warrants further investigation.

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Acknowledgements

Authors would like to thank the Maputo National Park (previously known as the Ponta do Ouro Partial Marine Reserve), the Department of Conservation Areas (ANAC) in Mozambique, Natural History Museum Maputo and previous volunteers and guests with DolphinEncountours.org.

This research was funded by the National Research Foundation Marine and Coastal grant (2019 – 2021) awarded to author Tess Gridley.

Author Declarations

Conflict of Interest

The authors declare that the research was conducted in the absence of any conflict of interest.

Ethics Approval

All research was passive and observational, and all research activities were within the scope of normal activities carried out by Dolphin Encountours Research Center's memorandum of understanding with the Department of Conservation Areas (ANAC). Acoustic monitoring of these animals was non-invasive and carried out under ethics permits issued by Stellenbosch University, University of KwaZulu-Natal, University of Cape Town and Ponta do Ouro Partial Marine Reserve

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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APPENDIX 3

Rostrum abnormalities in the endangered Indian Ocean humpback dolphin (*Sousa plumbea*) in South Africa

Integrative Zoology 13th October 2022

https://doi.org/10.1111/1749-4877.12685

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ORIGINAL ARTICLE

INTEGRATIVE ZOOLOGY

Rostrum abnormalities in the endangered Indian Ocean humpback dolphin (*Sousa plumbea*) in South Africa

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Abstract

Morphological abnormalities in wild animals can be indicators of the underlying health of a population and may be determined through routine photographic surveys. Here, we assess unusual rostrum conditions in Indian Ocean humpback dolphins (*Sousa plumbea*) inhabiting South African coastal waters to understand the rate of prevalence of abnormal rostrums and formulate hypotheses on potential causes. Photographic data were collated from systematic boat surveys and opportunistic sightings, obtained between April 1998 and March 2021 in various regions along the South African coast. Overall, 31 unique individuals were found with abnormal rostrum conditions, varying from slight misalignments to severe wounds and/or aberrant morphologies. In most cases, injuries were likely caused by natural events during the animal's life history such as interactions with sharks and/or reef-associated hunting strategies. Mark–recapture data indicated that individuals had survived with these injuries for up to 10 years. This study reports the highest incidence of rostrum abnormalities in the species. As numbers reflect only those that have survived their injuries, they are considered a minimum estimate. A better understanding of the cause(s) of these injuries is important given the endangered status of this species.

Key words: endangered species, human impacts, injuries, mark-recapture , natural history

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APPENDIX 4

Outreach and awareness

With threats to *S. plumbea* increasing along their southern African coastline, a large part of this project's aim was to increase their profile across these regions. It endeavoured to develop capacity building throughout southern Africa.

Over the past four years, I have undertaken fieldwork across southern Africa to investigate *S. plumbea*'s acoustic behaviour and also work with researchers to develop capacity building in team members, providing training in acoustic methods and humpback dolphin awareness.

Capacity building (training, equipment sharing, mentoring)

Oceans Research Institute, Mossel Bay
Dolphin Encounters Mozambique
All out Africa, Tofo, Mozambique
Fire Island Conservation, Ilha do Fogo, Mozambique

Science Communication

- I produced a documentary in collaboration with Homebrew Films called 'The Sound of Hope: The Humpback Dolphin,' which focused on my research and aimed to raise the profile of humpback dolphins. I organised an online world premiere, including a live Q and A with an invited expert panel of researchers. I was also invited to present the documentary at the African Youth Summit (August 2021) and invited to be on a live panel of experts to discuss the research and film
- 50/50 on SABC (South African TV Channel). Dedicated featurette on my research.
- Experience across a range of media highlighting myself and my research. Featured in Cape Times, The Mercury, Daily Mail, The Conversation, Mongabay Podcast, BBC Radio 1, RFI, RiverWise, Eyethu Magazines.
- Personal profiles published with Leading Women in Ocean Science and Girls in Ocean Science.
- Marine Mammal Science podcast – bioacoustics of endangered humpback dolphins.
- RTE German TV news channel feature (June 2024).

Invited speaker

Hal Whitehouse Lab Dalhousie University, Canada (January 2021) Using acoustic methodology to monitor endangered dolphin species in southern Africa.

Ecology and Evolution Lab Leeds University, UK (January 2021) Whistling Humpbacks

Blackpool and Clyde College, University of Lancaster, UK (March 2022) L6 third year Marine Biology BSc 'Eavesdropping on the ocean; an introduction to bioacoustics'